

Straight-line certificates for Jones’s Theorem 5 (1980): the reported count 100, and a reduction from 129 to 89

Satellite article to the corrected edition of *Undecidable Diophantine Equations*

Satellite to the corrected edition, 14 September 2026

Abstract

Jones’s 1980 announcement states that for the equations of its Theorem 3, “modified to do away with $q = b^{5^{60}}$ ”, the total number of indicated arithmetical operations is $o = 100$, and derives from this Theorem 5: every theorem of an effectively axiomatizable theory has “another proof consisting of 100 additions and multiplications of integers”. Earlier revisions of the corrected edition retained 100 as the author’s reported count. This satellite article settles what the number is and proves Theorem 5 in a precise form.

(i) *Where 100 comes from.* The seventeen polynomial equations of Theorem 3 (all equations except $q = b^{5^{60}}$) contain exactly 100 addition, subtraction and multiplication signs when each sign is counted once as printed and exponentiation is not counted. The same rule applied to the fourteen equations of Theorem 2.12 of Jones, Sato, Wada and Wiens (1976) gives exactly their reported count 87, and applied to the thirty-six equations (1.3) of Jones (1978) gives 241 against the reported 243. This is the only counting rule we found that reproduces the published numbers.

(ii) *An explicit certificate.* We write down the modified system: the equation $q = b^{5^{60}}$ is replaced, by Lemma 2.26 of Jones’s 1982 paper, by three polynomial equations in four new unknowns, the exponent 5^{60} entering only as an integer constant. We prove that the modified system still defines $x \in W_{\langle z, u, y \rangle}$, and we give a straight-line program, checked symbolically, that verifies all twenty equations with 129 additions and multiplications of integers (75 multiplications, 54 additions, subtractions being checked as additions). Taking certificate length as the complexity measure, we reduce this to a **89-instruction certificate** using 47 multiplications and 42 additions for 22 equations in 34 positive unknowns, with three fixed index parameters besides the input x . The reductions combine shared arithmetic, exact Pell index recovery, and a coefficient compiler with an affine radix, signed carry resets, and two copied inputs. An odd-index polynomial identity reduces the auxiliary Pell parameter; a positive product bound and binary codes remove two coding operations and a redundant supplied witness. Narrower packing and a carry-exclusion argument remove one more multiplication while preserving the same fixed index and raw input. Every power is built by multiplication; fixed numerals have no cost. A finite support layout encodes L by the fixed parameter $\psi_2(L)$. We give complete instruction tables and positive-domain proofs for the successive constructions, with no lower bound or optimality claim.

(iii) *Theorem 5.* With these certificate bounds, Theorem 5 follows from Theorem 3 by the standard passage from an effectively axiomatizable theory to a recursively enumerable set of Gödel numbers; we spell this out and state exactly what the “proof” of Theorem 5 is and what it is not.

(iv) *How the count arose.* Quotations from the 1976, 1978, 1980 and 1982 papers show that o was defined as the number of operations “necessary to write the system”, “very closely associated with the number of symbols”, that is, as typographical complexity; the proof-theoretic reading was attached to that number afterwards and the convention was inherited from the 1976 primality bound, where the two counts coincide.

1 The claim and its context

The 1980 announcement [3] presents, without proof, three universal systems of Diophantine equations and reports several size measures for them. On its last page it introduces a further measure:

A different measure of size and complexity of a system of diophantine equations is the total number of indicated arithmetical operations, o , i.e. the number of additions, subtractions and multiplications necessary to evaluate or determine the correctness of a proposed solution. For the equations of Theorem 3, modified to do away with $q = b^{5^{60}}$, it can be shown that $o = 100$.

and states

THEOREM 5. For any effectively axiomatizable theory T and any proposition P , if P has a proof in T , then P has another proof consisting of 100 additions and multiplications of integers.

The full paper [4] repeats both statements verbatim (“By an arithmetical operation we shall mean an addition or multiplication. We will consider subtraction as merely a signed form of addition. . . . it can be seen that $o = 100$ ”), and adds that the corresponding count for the 36-equation system of [2] was $o = 243$. Neither paper explains the count, and neither writes down the “modified” system. The corrected edition therefore carried the footnote that the exact count 100 is the author’s reported count, not independently reconstructed. This article removes that caveat.

The system in question is Theorem 3 of [3] (identical to Theorem 3 of [4]), in the corrected form of the present edition:

$$elg^2 + \alpha = (b - xy)q^2, \tag{E1}$$

$$q = b^{5^{60}}, \tag{E0}$$

$$\lambda + q^4 = 1 + \lambda b^5, \tag{E2}$$

$$\theta + 2z = b^5, \tag{E3}$$

$$\ell = u + t\theta, \tag{E4}$$

$$e = y + m\theta, \tag{E5}$$

$$n = q^{16}, \tag{E6}$$

$$r = \left[g + eq^3 + \ell q^5 + (2(e - z\lambda)(1 + xb^5 + g)^4 + \lambda b^5 + \lambda b^5 q^4)q^7 \right] (n^2 - n) \\ + \left[q^3 - b\ell + \ell + \theta\lambda q^3 + (b^5 - 2)q^8 \right] (n^2 - 1), \tag{E7}$$

$$p = 2ws^2r^2n^6, \tag{E8}$$

$$p^2k^2 - k^2 + 1 = \tau^2, \tag{E9}$$

$$4(c - ksn^2)^2 + \eta = k^2, \tag{E10}$$

$$k = r + 1 + hp - h, \tag{E11}$$

$$a = (wn^2 + 1)rsn^2, \tag{E12}$$

$$c = 2r + 1 + \varphi, \tag{E13}$$

$$d = bw + ca - 2c + 4a\gamma - 5\gamma, \tag{E14}$$

$$d^2 = (a^2 - 1)c^2 + 1, \tag{E15}$$

$$f^2 = (a^2 - 1)i^2c^4 + 1, \tag{E16}$$

$$(d + of)^2 = \left((a + f^2(d^2 - a))^2 - 1 \right) (2r + 1 + jc)^2 + 1. \tag{E17}$$

The unknowns are the 28 positive integers

$$a, b, c, d, e, f, g, h, i, j, k, \ell, m, n, o, p, q, r, s, t, w, \alpha, \gamma, \eta, \theta, \lambda, \tau, \varphi;$$

the parameters are x and the admissible coding triple $\langle z, u, y \rangle$ of [4, (4.1)], with $\delta = 4$ and $\nu = 58$. We have numbered the equations in the printed order, giving the exponential equation the label (E0) so that the seventeen polynomial equations keep the labels (E1)–(E17) throughout.

Where 5^{60} comes from. In [4, §4] the base is $B = b^{\delta+1} = b^5$ and the code length is $q = B^{(\delta+1)^{\nu+1}} = B^{5^{59}}$, hence $q = b^{5 \cdot 5^{59}} = b^{5^{60}}$. The exponent is a fixed integer; it is not built up inside the system.

2 Two counting conventions

The phrase “number of additions, subtractions and multiplications necessary to evaluate or determine the correctness of a proposed solution” admits two readings, and the two give different numbers for Theorem 3. We define both.

Definition 2.1 (indicated operations). The *indicated operation count* of a system of polynomial equations is the number of addition, subtraction and multiplication signs in the printed system, each occurrence counted once: a binary $+$ or $-$ counts one, a multiplication sign or a juxtaposition of two factors counts one, a numeral coefficient such as $2z$ or $4a\gamma$ counts one multiplication per factor, and an exponent x^k is not counted. Repeated subexpressions are counted each time they are printed.

Definition 2.2 (straight-line certificate). A *straight-line certificate* for a system of polynomial equations in the variables $v_1, \dots, v_m$ consists of integer values for the variables, integer values for finitely many auxiliary names, and a finite list of instructions, each of the form $t = u + v$ or $t = u \cdot v$ where u, v are variables, auxiliary names, earlier instruction targets, or integer numerals, followed by a finite list of equality tests between names. It is *valid* if every instruction and every equality test holds for the supplied values. Its *length* is the number of instructions. A subtraction $t = u - v$ is recorded as the instruction $t + v = u$, so subtraction costs one operation and no subtraction primitive is needed. Equality tests, reading the supplied integers, and (in the basic form) writing down numerals are free. A certificate *verifies* the system if the validity of the certificate implies that the supplied values satisfy every equation of the system, and every solution of the system extends to a valid certificate.

Definition 2.3 (certificate complexity). The complexity $\mathcal{C}(\Sigma)$ is the least length of a *uniform certificate scheme* verifying Σ in the preceding sense. Its finite list of names, instruction operands and equality tests is fixed before the values of the input and witnesses are supplied. Witness domains are part of the scheme. The integer constants and any fixed admissible index are specified when the represented set is chosen; an existential witness cannot stand in for an unchecked encoding parameter.

Definition 2.2 is the “calculator” reading: a calculator that can only add and multiply integers, variables with fixed values, any auxiliary integers typed for free, intermediate results written down and reused at will. It is the convention under which the corrected 1976 edition reproduces the 87-operation primality certificate of [5, Theorem 5], and it is the convention that makes “a proof consisting of o additions and multiplications of integers” literally true: the proof is the list of instructions. Definition 2.1 is the “as written” reading of “indicated operations”: it charges nothing for the exponentiations that a calculator would have to perform by repeated multiplication, and it charges twice for a subexpression printed twice.

The two conventions coincide for a system in which every power is a square of a variable that is not reused and no subexpression repeats. They diverge for Theorem 3, which contains 44 exponentiations, among them q^{16} , q^8 , q^7 , q^5 , q^3 , six occurrences of b^5 , n^6 , c^4 and a fourth power of a trinomial.

3 Where the number 100 comes from

Proposition 3.1. *The seventeen polynomial equations (E1)–(E17) of Theorem 3 contain exactly 100 indicated operations in the sense of Definition 2.1: 33 additions, 15 subtractions and 52 multiplications. The 44 exponentiations are not counted, and the equation (E0), $q = b^{560}$, contributes nothing.*

Proof. Count each equation as printed. The per-equation totals are

equation	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	E11	E12	E13	E14	E15	E16	E17
signs	6	3	2	2	2	0	32	4	3	5	4	5	3	10	3	4	12

with sum 100. For instance (E1) has elg^2 (two multiplications), $+\alpha$, xy , $b - xy$ and $(\cdot)q^2$: six signs; (E7) has 32 signs, of which the first bracket contributes $g + eq^3$ (2), $+lq^5$ (2), $+2(e - z\lambda)(1 + xb^5 + g)^4$ (8), $+\lambda b^5$ (2), $+\lambda b^5 q^4$ (3), $(\cdot)q^7$ (1), and the rest is $(n^2 - n)$ (1), the product (1), the second bracket $q^3 - b\ell + \ell + \theta\lambda q^3 + (b^5 - 2)q^8$ (9), $(n^2 - 1)$ (1), the product (1) and the final sum (1). The companion program `round4_1980_operation_count.py` parses the transcribed equations and returns the same totals. $\square$

Proposition 3.2 (calibration). *Under Definition 2.1:*

1. *the fourteen equations of Theorem 2.12 of [5], from which the authors “calculated” the bound 87 of their Theorem 5, contain exactly 87 indicated operations (35 additions, 17 subtractions, 35 multiplications; 24 exponentiations not counted);*
2. *the thirty-six equations (1.3) of [2], for which the author reports 243, contain 241 indicated operations (111 additions, 31 subtractions, 99 multiplications; 54 exponentiations not counted);*
3. *the seventeen polynomial equations of Theorem 3 of [3] contain 100 indicated operations, the reported value.*

Proof. Direct counts, reproduced by the companion program from transcriptions of the three systems, which were checked against the scans. $\square$

Remark 3.3 (reading the evidence). Two of the three published counts are reproduced exactly and the third to within two signs; we could not locate the two missing signs in (1.3) and record the discrepancy. No other rule we tried comes close on all three systems. In particular, under Definition 2.2 with every power built by repeated multiplication and every repeated subexpression computed once, a direct shared-expression evaluation of the printed Theorem 3 without (E0) uses 127 operations before anything replaces the exponential equation, and the initial certificate of Section 5 uses 129 with the replacement; for [2] the same convention gives 242, and for [5] the naive count is 99, which the corrected 1976 edition brings down to 87 only by algebraic rewriting. The natural conclusion is that Jones’s o is the count of indicated operation signs of Definition 2.1, with exponentiation regarded as notation rather than as an operation, and that “modified to do away with $q = b^{560}$ ” means that the exponential equation is set aside and not that its polynomial replacement has been counted. One small inconsistency should be recorded: [2] remarks that combining its 36 equations into one by transposing and summing squares “would increase this number to 350”, and $350 - 243 = 107 = 36 + 36 + 35$ counts one subtraction, one squaring and (for 35 joins) one addition per equation; there the squaring is charged, whereas the powers inside the printed equations are not.

4 The modified system

To make Theorem 5 a statement about a polynomial system we must actually do away with (E0). The 1982 paper provides the tool: Lemma 2.26 of [4] defines an exponential $Q = B^{B^1}$ on top of the Pell

machinery that Theorem 3 already contains, at the cost of three conditions and four new unknowns, with the exponent B_1 entering only as an integer constant. We recall it in the notation of Theorem 3.

Lemma 4.1 ([4, Lemma 2.26, primed form]). *Let R, N, b satisfy the hypotheses of [4, Lemma 2.25] ($R \geq 8, N \geq 8, R \geq b, N \geq b > 0$), and let B, Q, B_1 be positive integers with $3 < 3B_1 \leq B \leq Q \leq N \leq R$. Then $N^2 \mid \binom{2R}{R}$, $b \text{ pow } 2$ and $Q = B^{B_1}$ hold together if and only if the equations (B1)–(B14) of Lemma 2.25 hold with (B8) replaced by $C = C_1 + B' + \varphi$ and the following three conditions adjoined, for some positive integers C_1, D_1, Δ :*

$$D_1 \equiv Q + C_1(A - B) \pmod{2AB - B^2 - 1}, \quad (\text{C1}')$$

$$(A^2 - 1)C_1^2 + 1 = D_1^2, \quad (\text{C2}')$$

$$C_1 = B_1 + \Delta(A - 1). \quad (\text{C3}')$$

In Theorem 3 the quantities of Lemma 2.25 appear as $R = r, N = n, A = a, C = c, K = k, P = p, M = rsn^2, U = wn^2, Y = n^2s, W = bw, V = 2, B' = 2r + 1$; this dictionary was verified equation by equation. We now take $B := b, Q := q$ and $B_1 := 5^{60}$ in Lemma 4.1, write $\kappa, \mu, \rho, \Delta$ for C_1, D_1 , the multiplier of the modulus in (C1'), and Δ , and obtain:

Definition 4.2 (the modified system Σ). Σ consists of the seventeen equations (E1)–(E17) with (E13) replaced by

$$c = 2r + 1 + \kappa + \varphi \quad (\text{E13}')$$

and the three equations

$$\mu = q + \kappa(a - b) + \rho(2ab - b^2 - 1), \quad (\text{E18})$$

$$(a^2 - 1)\kappa^2 + 1 = \mu^2, \quad (\text{E19})$$

$$\kappa = 5^{60} + \Delta(a - 1), \quad (\text{E20})$$

in the 32 unknowns $a, \dots, \varphi, \kappa, \mu, \rho, \Delta$, all ranging over positive integers, and the parameters x, z, u, y .

The constant 5^{60} in (E20) is a numeral, exactly as 2, 4 and 5 are numerals in (E3), (E10) and (E14). Nothing in Σ has degree larger than that of (E17); the degree of Σ in the unknowns is $\max(16, \deg(\text{E7}), \deg(\text{E17}))$ and no longer 5^{60} .

Theorem 4.3. *For every admissible coding triple $\langle z, u, y \rangle$ and every positive integer x : $x \in W_{\langle z, u, y \rangle}$ if and only if Σ has a solution in positive integers.*

Proof. The proof of Theorem 3 in [4, §4] shows that (E1)–(E17), including (E0), are equivalent to the conditions (U1)–(U24) of §4 together with $b \text{ pow } 2$ and $n^2 \mid \binom{2r}{r}$, the latter two being expressed by (B1)–(B14) of Lemma 2.25 with (B14) replaced by (Q1)–(Q3) and (B13) in the χ -form. In that proof the hypotheses of Lemma 2.25 are established from the other equations: $8 \leq r, 8 \leq n, b \leq r, 0 < b \leq n$; the two bounds on r are obtained there through the block bounds $0 \leq S_i, T_i$, whose derivation uses (E0). We check these two bounds and the additional hypotheses of Lemma 4.1 with $B = b, Q = q, B_1 = 5^{60}$, in both directions, using only the equations of Σ or the parameters.

$3 < 3 \cdot 5^{60} \leq b$. From (E1), $(b - xy)q^2 = \ell g^2 + \alpha \geq 2$, hence $b > xy \geq y$. For an admissible triple, by [4, (4.1)] with $\delta = 4, \nu = 58, z \geq 2$ and $z + P_{0,\dots,0} \geq 1$,

$$y \geq (z + P_{0,\dots,0})(2z)^{5^{59}} \geq 2^{5^{59}} > 3 \cdot 5^{60},$$

the last inequality because $\log_2(3 \cdot 5^{60}) < 141 < 5^{59}$.

$b \leq q$. From (E2), $\lambda(b^5 - 1) = q^4 - 1$ with $\lambda \geq 1$, so $q^4 \geq b^5 > b^4$ and $q > b$.

$q \leq n \leq r$, with $8 \leq r$ and $b \leq r$. $n = q^{16} \geq q$ by (E6). The block bounds $0 \leq S_i, T_i$ of [4, §4] cannot be invoked here: the derivation of $0 \leq S_3$ there uses $xb^5 \leq q$, which comes from (E0). Instead write S and T for the two brackets of (E7), so that $r = S(n^2 - n) + (T + 1)(n^2 - 1)$ factors as

$$r = (n - 1)(Sn + (T + 1)(n + 1)).$$

Since $r \geq 1$ and $n - 1 \geq 1$, the second factor is a positive integer, and it equals 1 only if $(T + 1)(n + 1) \equiv 1 \pmod{n}$, that is, only if $n \mid T$. But $0 < T < n$: by (E5), (E3), $m \geq 1$ and $y \geq 2z$ (from the bound on y above) we have $e \geq y + \theta \geq b^5$, so (E1) gives $b^5 \ell \leq e \ell g^2 < (b - xy)q^2 < bq^2$, hence $\ell < q^2$ and $\ell(b - 1) < bq^2 < q^3$, which makes $T > q^3 - \ell(b - 1) > 0$; and $\theta\lambda < q^4$ because $\theta < b^5 - 1$, so $T < q^3 + q^7 + b^5 q^8 < q^{16} = n$. Therefore the second factor is at least 2, and $r \geq 2(n - 1) \geq n$; in particular $r \geq 8$ and $r > b$.

Sufficiency. Let the 32 unknowns satisfy Σ . Lemma 2.25 and Lemma 4.1 apply, the unprimed conditions (B1)–(B14) being (E8)–(E17) with $c = 2r + 1 + \kappa + \varphi$ in the role of $C = C_1 + B' + \varphi$, and (E18)–(E20) being (C1')–(C3'): (E18) with $\rho \geq 1$ implies the congruence (C1'). Hence $b \bmod 2$, $n^2 \mid \binom{2r}{r}$ and $q = b^{5^{60}}$. So (E0) holds, and the original proof of Theorem 3 gives $x \in W_{\langle z, u, y \rangle}$.

Necessity. Let $x \in W_{\langle z, u, y \rangle}$. The necessity proof of Theorem 3 supplies positive values of the 28 original unknowns satisfying (E0)–(E17), with $q = b^{5^{60}}$, $a = A$ the Pell modulus of Lemma 2.25 and $c = \psi_a(2r + 1)$. Put $\kappa = \psi_a(5^{60})$ and $\mu = \chi_a(5^{60})$, the solutions of $\mu^2 - (a^2 - 1)\kappa^2 = 1$; this is (E19). Since $\psi_a(k) \equiv k \pmod{a - 1}$ and $\psi_a(k) > k$ for $a \geq 2, k \geq 2$, we have $\kappa = 5^{60} + \Delta(a - 1)$ with an integer $\Delta \geq 1$; this is (E20). Lemma 2.22 of [4] (the χ -form of [4, Remark after Lemma 2.28]) gives $\mu \equiv q + \kappa(a - b) \pmod{2ab - b^2 - 1}$, so $\mu - q - \kappa(a - b) = \rho(2ab - b^2 - 1)$ for an integer ρ . To see $\rho \geq 1$ note that $2ab - b^2 - 1 = (a^2 - 1) - (a - b)^2 > 0$ since $1 < b < a$, that $\mu > \sqrt{a^2 - 1}\kappa > (a - 1)\kappa = (a - b)\kappa + (b - 1)\kappa$, and that $(b - 1)\kappa \geq \kappa = \psi_a(5^{60}) \geq (2a - 1)^{5^{60} - 1} > a > q$, the last step by the estimate $Q^3 < A$ in [4, (13)]; hence the left side is positive and $\rho \geq 1$. Finally φ must be re-chosen for (E13'): $c - 2r - 1 - \kappa = \psi_a(2r + 1) - (2r + 1) - \psi_a(5^{60}) > 0$ by [4, (14)], which is exactly the inequality $\psi_A(B_1) + 2R + 1 + 2^{2R+1} < \psi_A(2R + 1)$ proved there for $3B_1 \leq R < A$. Thus all 32 values are positive integers satisfying Σ . $\square$

Remark 4.4. One may equally take $B := b^5$ and $B_1 := 5^{59}$, which is the literal reading of [4, §4]; then (E18) reads $\mu = q + \kappa(a - b^5) + \rho(2ab^5 - b^{10} - 1)$ and (E20) has the constant 5^{59} . Since b^5 is already computed, the modulus identity used below gives the same basic instruction count for both choices. The hypothesis $3 \cdot 5^{59} \leq b^5$ is checked in the same way. The unprimed group (C1)–(C3), which uses the congruence $(B^2 - 1)QC_1 \equiv B(Q^2 - 1)$, costs more operations and is not used.

5 An explicit straight-line certificate

Theorem 5.1. *The system Σ of Definition 4.2 is verified, in the sense of Definition 2.2, by the straight-line certificate of the Appendix A, which has 129 instructions: 75 multiplications and 54 additions (37 additions and 17 subtractions, each subtraction recorded as an addition), followed by 20 equality tests. If the numerals 2, 4, 5 and 5^{60} must be generated from 1 by additions and multiplications, ten further instructions suffice ($2 = 1 + 1$, $4 = 2 + 2$, $5 = 4 + 1$ and the addition chain 1, 2, 3, 6, 12, 15, 30, 60 for 5^{60}), for a total of 139.*

Proof. For each of the 20 equations of Σ the certificate computes both sides (or one side, when the other is a single variable) and the program `round4_1980_operation_count.py` verifies, with SymPy, that the difference of the two compared names is, as a polynomial in the 36 variables of Σ , equal to the left side minus the right side of the source equation, up to sign. Hence the equality

tests hold exactly when the equations hold, so a valid certificate proves the equations, and every solution extends to a valid certificate by evaluating the instructions. The counts are read off the list. The residual signs, the operation histogram and the count of the numeral chain are written to `round4_1980_operation_count.json`. $\square$

The economies used, all of them plain integer identities, are the following. Powers are shared: $q^2, q^3, q^4, q^8, q^{16}$ (five multiplications, the shortest chain containing 2, 3, 4, 8, 16) and b^2, b^4, b^5 (three). The first bracket of (E7) is evaluated by Horner's rule as $g + q^3(e + \ell q^2 + q^4(2(e - z\lambda)C^4 + \Lambda(1 + q^4)))$ with $C = 1 + xb^5 + g$ and $\Lambda = \lambda b^5$ taken from (E2), which removes the need for q^5 and q^7 ; C^4 is two squarings. In (E8)–(E12) the products wn^2 , sn^2 and rsn^2 are computed once and $p = 2ws^2r^2n^6 = 2(wn^2)(rsn^2)^2$. (E9) is evaluated as $(p^2 - 1)k^2 + 1$, (E11) as $r + 1 + h(p - 1)$, (E14) as $bw + c(a - 2) + \gamma(4a - 5)$, (E16) as $(a^2 - 1)(ic^2)^2 + 1$, and $a^2 - 1$, d^2 , f^2 , c^2 , $r + 1$, $2r + 1$ are shared between (E15), (E16), (E17), (E19). The modulus of (E18) is $2ab - b^2 - 1 = (a^2 - 1) - (a - b)^2$, the identity that the corrected 1976 edition uses for the 1976 count. All intermediate integers, including the signed ones such as $e - z\lambda$ and $b^5 - 2$, are supplied with the certificate.

Remark 5.2 (the count is an upper bound). Neither the printed sign count nor a particular evaluation schedule gives a lower bound for all certificates. In particular, computations may be shared across equations, previously checked equalities may be reused, and the underlying Diophantine encoding may be changed. The 129 instructions above are a reproducible starting point for these optimizations. The historical interpretation of 100 rests on the evidence in Remark 3.3, not on an assertion that smaller certificates are impossible.

Theorem 5.3 (sharing a product). *The same system Σ has a certificate with 128 instructions: 74 multiplications and 54 additions, followed by the same 20 equality tests. Generating the numerals from 1 raises this bound to 138.*

Proof. Put $U = wn^2$, $Y = sn^2$, $M = rY$ and $V = UM$. Then the two equations

$$a = (U + 1)M, \quad p = 2UM^2$$

can be checked as $a = V + M$ and $p = 2(VM)$. Computing $U, Y, M, V, V + M, VM, 2(VM)$ takes seven instructions. These replace the eight instructions 58–65 of Appendix A and preserve both residual polynomials exactly. The companion program checks all twenty residuals again for this modified schedule and records the result as `shared_product_certificate`. $\square$

6 Reducing certificate length to 120

We now take straight-line certificate length, rather than indicated operation count, as the complexity measure to minimize. The preceding 129-instruction schedule is a useful starting point, but it is not a minimum. Two shared computations reduce the certificate for exactly the same system to 127. A positive Pell-solution gap removes a redundant inequality and gives an equivalent system with a 126-instruction certificate. Finally, changing the admissible index and packing its two coefficient congruences gives a universal system with a certificate of length 120. The last construction still has 32 positive unknowns, twenty equations, and three fixed index parameters besides the input x .

6.1 Two savings without changing the system

Put $U = wn^2$ and $M = rsn^2$, quantities already computed in the baseline schedule. Equations (E8) and (E12) require

$$p = 2UM^2, \quad a = (U + 1)M.$$

Compute $J = UM$ once and instead check

$$a = J + M, \quad p = 2JM.$$

This replaces five instructions, after U, M are available, by four: one multiplication is saved.

For the second saving, temporarily write $B = b^5$. Equations (E2), (E3) give the useful identity

$$1 + \theta\lambda = \lambda + q^4 - 2z\lambda. \quad (6.1)$$

Compute $Z_\lambda = (2z)\lambda$, and write the coefficient $2(e - z\lambda)$ in the first bracket of (E7) as $2e - Z_\lambda$. This costs the same three instructions as before. Its second bracket then becomes

$$q^3(\lambda + q^4 - Z_\lambda) - (b - 1)\ell + (B - 2)q^8.$$

The quantity $\lambda + q^4$ is already available from (E2), and Z_λ is now shared between the two brackets. Eight instructions replace the previous nine.

Theorem 6.1. *The original modified system Σ has a straight-line certificate with 127 instructions: 73 multiplications and 54 additions, with subtractions recorded as additions, followed by twenty equality tests.*

Proof. Apply the two independent savings to the archived 129-instruction schedule. The shared-product identities are polynomial identities. The second saving uses equations already checked, so its exact verification must record that dependence. If F_i denotes the source residual, left side minus right side, of (Ei), the residual evaluated by the new (E7) schedule is

$$F_{7,\text{new}} = F_7 + (\lambda F_3 - F_2)q^3(n^2 - 1).$$

Thus it vanishes exactly when F_7 does, once (E2) and (E3) have been checked. The companion program verifies this identity symbolically, along with the other nineteen residuals and the instruction count. $\square$

6.2 Removing a redundant Pell gap

The strengthened equation (E13') asks for $c > \kappa + 2r + 1$. The two Pell equations already ensure that $c > \kappa$ implies this stronger inequality. The following elementary argument does not need the general classification of Pell solutions.

Lemma 6.2. *Let $a \geq 2$ and suppose positive integers c, d, κ, μ satisfy*

$$d^2 - (a^2 - 1)c^2 = \mu^2 - (a^2 - 1)\kappa^2 = 1, \quad c > \kappa.$$

Then $c - \kappa \geq 2a - 1$.

Proof. Write $D = a^2 - 1$, $t = \mu c - d\kappa$ and $v = d\mu - Dc\kappa$. Since

$$\frac{d}{c} = \sqrt{D + c^{-2}} < \sqrt{D + \kappa^{-2}} = \frac{\mu}{\kappa},$$

the integer t is positive. Also $v > 0$, because $d > \sqrt{D}c$ and $\mu > \sqrt{D}\kappa$. Direct multiplication gives

$$v^2 - Dt^2 = 1, \quad c = \mu t + \kappa v.$$

As $t \geq 1$, the first identity implies $v \geq a$; likewise $\mu \geq a$. Consequently

$$c - \kappa = \mu t + \kappa(v - 1) \geq a + \kappa(a - 1) \geq 2a - 1. \quad \square$$

Let Σ_{flat} replace (E13') by $c = \kappa + \varphi$, with φ still positive, and retain the other nineteen equations.

Theorem 6.3. *The system Σ_{flat} defines the same set of inputs as Σ , with an explicit bijection of their slack witnesses, and has a 126-instruction certificate: 73 multiplications and 53 additions.*

Proof. The other equations give $b \geq 2$, $q \geq 2$, $n \geq 2$, and

$$a = (wn^2 + 1)rsn^2 \geq 20r.$$

By Lemma 6.2, the new equation gives $c - \kappa \geq 2a - 1 > 2r + 1$. Therefore

$$\varphi_{\text{old}} = \varphi_{\text{new}} - (2r + 1) > 0, \quad \varphi_{\text{new}} = \varphi_{\text{old}} + 2r + 1$$

are inverse maps between solutions. Since $2r + 1$ is already computed for (E17), checking $c = \kappa + \varphi$ saves one addition over checking $c = 2r + 1 + \kappa + \varphi$. Combine this with Theorem 6.1. $\square$

6.3 A larger radix supplied by the index

Put

$$L = 5^{59}, \quad M = 5^{58}, \quad L = 5M.$$

Let (z, u, y) be an admissible index of the original construction and write $Z = 2z$. Choose a power of two H so large that

$$H > \max\{2Z^{4L+1}, Z60^4, 3L\}. \quad (6.2)$$

The new radix will be $B = Hb^4$. Constructing H is part of constructing the index of the represented r.e. set. It is an explicitly declared fixed parameter; it is not an unrestricted existential witness supplied by a purported certificate. This distinction is necessary for soundness.

First retain both u and y , so the admissible index has four components (Z, u, y, H) . Define Σ_{rec} by the following replacements in Σ :

$$\begin{array}{lll} b = x + \beta, & elg^2 + \alpha = q^2, & \text{in place of (E1),} \\ \theta + Z = B, & & \text{in place of (E3),} \\ c = \kappa + \varphi, & & \text{in place of (E13'),} \\ \mu = q + \kappa(a - B) + \rho(2aB - B^2 - 1), & & \text{in place of (E18),} \\ \kappa = L + \Delta(a - 1), & & \text{in place of (E20).} \end{array}$$

Replace every occurrence of b^5 by $B = Hb^4$, and replace $2(e - z\lambda)$ in (E7) by $2e - Z\lambda$. All remaining equations and positive domains are retained, and β is an additional positive unknown. Thus Σ_{rec} has 33 unknowns and 21 equations.

Theorem 6.4. *For every admissible (Z, u, y, H) , the system Σ_{rec} defines the same r.e. set as the original index $(Z/2, u, y)$. Its certificate uses 123 instructions: 70 multiplications and 53 additions.*

Proof. The radix $B = Hb^4$ is computed in three multiplications, the same number as b^5 . The split (E1) costs five instructions instead of seven. Using Z directly saves the separate computation of $2z$; its occurrence in $2e - Z\lambda$ costs the same three instructions as the old coefficient. The remaining three savings are those of Theorems 6.1 and 6.3. Hence the count is $129 - 2 - 1 - 2 - 1 = 123$.

For the mathematical equivalence, the original proof uses $b > xy$ to ensure $b > x$, a sufficiently large radix for Lemma 2.9, and the coefficient bound needed to prevent carries. Here $b > x$ is explicit, while (6.2) gives

$$B > 2Z^{2L+1}, \quad z60^4b^4 < B/2.$$

These are the required radix and coefficient bounds. The original uncombined inequality $elg^2 < q^2$ gives $e, \ell < q^2$ and $g < q$. The retained congruence equations and $u, y \geq Z$ also give $e, \ell \geq B$, hence $q^2 > elg^2 \geq B^2$ and $q > B$ before using the intended exponential relation. Therefore $\lambda = (q^4 - 1)/(B - 1) > q^3 > e$, and $0 < D_0 = z\lambda - e < z\lambda$.

For completeness, the positivity of the third block can also be checked before invoking the exponent lemma. With $C = 1 + xB + g$,

$$C \leq xB + q < (b + 1)q, \quad (b + 1)^4 < 6b^4 \quad (b \geq 2).$$

Since $H > 12z$, this gives $2C^4 D_0 < B\lambda q^4$, so

$$0 < S_3 = -2C^4 D_0 + B\lambda(1 + q^4) < 2q^8 < q^9.$$

The other block bounds follow from $e, \ell < q^2$, $g < q$ and $b < B < q$. Thus $n = q^{16} \leq r$, and $3 < 3L \leq B < q \leq n \leq r$. The Pell gap restores (E13'); Lemma 4.1 applies with base B and exponent L , yielding b pow 2 and $q = B^L$. Since H is a power of two, all required radices are powers of two. The original digit-extraction and no-carry proof now applies with the new radix and the same coefficient index.

Conversely, choose a power-of-two bound b for an actual solution and use its intended codes. The source estimates

$$e < 2zB^L, \quad \ell < 2B^M, \quad g < bB^M$$

give $elg^2 < 4zb^2 B^{8M} < B^{10M} = q^2$, so the new α is positive; $\beta = b - x > 0$. The remaining digit and Pell witnesses are supplied by the original necessity construction, now using base B and exponent L . Positivity of its congruence multipliers and of the new $\varphi = c - \kappa$ follows as in Theorem 4.3. $\square$

6.4 Packing the coefficient congruences and sharing a square

There is a further useful change to the encoding. Put

$$V = y + uZ^{2L}. \tag{6.3}$$

Call (Z, V, H) admissible if it is obtained this way from an admissible $(Z/2, u, y)$ and satisfies (6.2). This restores three index parameters, so the complete parameter list is x, Z, V, H .

Define Σ_{pack} from Σ_{rec} by writing $C = 1 + xB + g$, replacing its inequality equation by

$$elC^2 + \alpha = q^2, \tag{6.4}$$

and replacing (E4), (E5) together by

$$e + \ell q^2 = V + t\theta. \tag{6.5}$$

Delete the now-unused unknown m . The quantities B, C are abbreviations whose arithmetic is charged, not free auxiliary definitions. The resulting system again has 32 positive unknowns and twenty equations.

Theorem 6.5. *For every admissible packed index (Z, V, H) , the system Σ_{pack} defines the same recursively enumerable set as its original Jones index. It has a straight-line certificate with 120 instructions: 68 multiplications and 52 additions, followed by twenty equality tests.*

Proof. We first establish the bounds needed to use the Pell lemmas, without assuming $q = B^L$. Equation (6.4) and positivity give $e, \ell < q^2$ and $C < q$. Since $b > x \geq 1$ and $C > B$, we have

$q > B > b \geq 2$ and $g < q$. As before, $\lambda > q^3 > e$ and $0 < D_0 = z\lambda - e < z\lambda$. Now the stronger bound $C < q$ directly gives

$$2C^4 D_0 < 2z\lambda q^4 < B\lambda q^4, \quad 0 < S_3 < 2q^8 < q^9.$$

The remaining block bounds are

$$\begin{aligned} 0 < g < q^3, \quad 0 < q^3 - 1 - (b-1)\ell < q^3, \\ 0 < e + \ell q^2 < q^4, \quad 0 < (B-Z)\lambda < (B-1)\lambda = q^4 - 1, \\ 0 < (B-2)q < q^2 < q^9. \end{aligned}$$

For the second use $\ell < q^2$ and $b < q$; for the third use integer $e, \ell \leq q^2 - 1$. Thus the three blocks lie within their widths q^3, q^4, q^9 , and the definition of r gives $r \geq n^2 - 1 \geq n$. We obtain $3 < 3L \leq B < q \leq n \leq r$ entirely from the polynomial equations. Lemma 6.2 restores the stronger gap needed by Lemma 4.1. The Pell lemmas therefore give b pow 2, $q = B^L$ and the central-binomial divisibility. The resulting powers of two allow the three no-carry conditions to be unpacked.

It remains to justify that the single congruence loses none of the coefficient information. Let $e_0(B), \ell_0(B)$ denote the intended digit polynomials of (4.5)–(4.6) in [4], and put

$$F(B) = e_0(B) + \ell_0(B)B^{2L}.$$

The first summand has degree at most L . The second occupies positions $2L + 5^i$ for $1 \leq i \leq 58$, all exceeding $2L$ and at most $2L + M < 4L$. Their digit positions are disjoint, all digits lie in $[0, Z]$, and $F(Z) = V$.

Apply [4, Lemma 2.9] to $Y = e + \ell q^2$, digit base Z , and $m = k = 4L$. Its congruence hypothesis is (6.5); its size hypothesis follows from $Y < q^4 = B^{4L} < ZB^{4L}$; and its mask hypothesis is precisely the second no-carry condition

$$\tau_2(Y, (B-Z)\lambda) = 0.$$

The radix hypothesis $2Z^{4L+1} \leq B$ follows from (6.2). The lemma gives $Y = F(B)$. Both the supplied and the intended e, ℓ lie in $[0, q^2]$, so uniqueness of quotient and remainder upon division by q^2 gives $e = e_0(B)$ and $\ell = \ell_0(B)$. The first no-carry condition now identifies the witness code g , and the third gives the vanishing coefficient, with the bound $z60^4 b^4 < B/2$ preventing carries. This proves soundness.

For necessity, choose a power-of-two b bounding a solution, take $B = Hb^4$, $q = B^L$, and use the intended codes. Since $x \leq b-1$, $1 + xB \leq bB \leq bB^M$, and hence $C < 2bB^M$. Therefore

$$e\ell C^2 < 16zb^2 B^{L+3M} = 16zb^2 B^{8M} < B^{8M+1} < B^{10M} = q^2.$$

Here $H > 16z$ and $2M > 1$ justify the last two inequalities. Thus the new α is positive. The polynomial F has nonnegative digits and positive degree, so $F(B) > F(Z) = V$; its difference is divisible by $B - Z$, giving the positive integer t in (6.5). All other witnesses follow from the same digit and Pell necessity construction.

Finally, C^2 was already computed to obtain C^4 in (E7), so (6.4) removes the separate squaring of g . Equation (6.5) needs one multiplication and one addition instead of the two of each in (E4), (E5); its left side was already computed as the second S-block. These three savings reduce 123 to 120. The program schedules the inequality check after C^2 has been computed, verifies all twenty residuals, and records the explicit dependence on (E2), (E3) for the transformed (E7). $\square$

The progression distinguishes a shorter certificate for the same system from a change of slack variable and from a change of encoding:

Construction	Multiplications	Additions	Total
Archived certificate for Σ	75	54	129
Shared product UM	74	54	128
Also reuse (E2), (E3): same Σ	73	54	127
Pell gap removed: Σ_{flat}	73	53	126
Radix index and split bound: Σ_{rec}	70	53	123
Packed index and shared square: Σ_{pack}	68	52	120

These are proved upper bounds under Definition 2.2; no minimum is asserted. In particular, the successful reductions replace the earlier suggestion that algebraic rewriting cannot materially close the gap. The 120-operation result uses free integer numerals and admissible index parameters. It does not include the computation of the index (Z, V, H) , nor claim a 120-operation bound when every such integer must itself be generated from 1. If the parameters x, Z, V, H remain supplied but the numerical constants 2, 4, 5, L must be generated from 1, the checked addition chain for $L = 5^{59}$ gives eleven additional instructions and a total of 131.

The independently executable companions are

```
round4_1980_optimized_certificate.py,
round4_1980_recoded_certificate.py,
round4_1980_packed_certificate.py.
```

The receipt of the second of these records the 126-instruction count of the recoding alone; the 123-instruction count of Theorem 6.4 is the entry `recoded_system_123` of the first receipt. They preserve the original 129-step schedule, verify the new schedules by exact symbolic identities, and write their operation histograms and residual checks as JSON receipts. The positive-domain arguments above supply the mathematical implications that polynomial expansion alone cannot certify.

7 Further reductions in the Pell block

Two changes to the Pell block save four operations independently of the digit encoding: a shorter parameter in the final Pell equation saves two, and a pair of linear interval equations saves another two. All unknowns remain positive integers. Signed intermediate expressions retain the auxiliary domain of Definition 2.2.

7.1 Factoring the final Pell parameter

Write $J = 2r + 1$. The remark after Lemma 2.28 of [4] explicitly permits replacing the parameter $a + f^2(d^2 - a)$ by

$$G_0 = a + f^2(f^2 - a).$$

This already saves one operation: if $X = (a^2 - 1)(ic^2)^2$, then (E16) asserts $f^2 = X + 1$, and

$$G_0 - 1 = (f^2 - 1)(f^2 - a + 1).$$

The four instructions

$$T = f^2 - (a - 1), \quad G_- = XT, \quad G_+ = G_- + 2, \quad D_G = G_- G_+$$

replace five instructions for $G_0^2 - 1$; both X and $a - 1$ are already needed elsewhere.

This replacement preserves positive witnesses constructively. Once $c = \psi_a(J)$ and $d = \chi_a(J)$, keep positive f, i satisfying (E16). For either the original parameter or G_0 , one has $G \equiv a \pmod{f}$, $G \equiv 1 \pmod{c}$, and $G \geq 2a + 1$. Set $I = \chi_G(J)$ and $H = \psi_G(J)$. The Pell recurrence polynomials give $I \equiv d \pmod{f}$, $H \equiv J \pmod{c}$, while growth gives $I > d$ and $H > J$. Thus $o = (I - d)/f$ and $j = (H - J)/c$ are positive integers. This explicitly supplies the required positive first coordinate; it does not assume that an arbitrary signed square root in Lemma 2.28 is positive.

A signed congruence permits a further reduction. Replace (E17) by

$$(of - d)^2 = (G^2 - 1)(J + jc)^2 + 1, \quad G = 1 + (a + 1)(f^2 - 1). \quad (7.1)$$

The variable o is positive; only the expression $of - d$ may have either sign.

Lemma 7.1. *For $a > 1$ and $0 < k \leq m$, the congruence $\chi_a(n) \equiv \pm \chi_a(k) \pmod{\chi_a(m)}$ implies $n \equiv \pm k \pmod{2m}$.*

Proof. The positive-sign case is the usual χ step-down lemma quoted after Lemma 2.28 of [4], with its stronger modulus $4m$. For the negative sign, the Pell addition identities give

$$\chi_a(n + 2m) \equiv -\chi_a(n) \pmod{\chi_a(m)},$$

because $\chi_a(2m) = 2\chi_a(m)^2 - 1$ and $\psi_a(2m) = 2\chi_a(m)\psi_a(m)$. Apply the positive-sign case at $n + 2m$ and reduce modulo $2m$. $\square$

Proposition 7.2. *Under $a > 1$ and $1 < J < c$ with J odd, equations (E15), (E16), and (7.1) are existentially equivalent to $c = \psi_a(J)$, with every system unknown positive.*

Proof. For sufficiency, write

$$d = \chi_a(k), \quad c = \psi_a(k), \quad f = \chi_a(m), \quad ic^2 = \psi_a(m).$$

The standard Pell divisibility fact used in the proof of Lemma 2.28, $\psi_a(k)^2 \mid \psi_a(m) \Rightarrow \psi_a(k) \mid m$, gives $c \mid m$; in particular $0 < k < c \leq m$. Our parameter satisfies

$$G > 1, \quad G \equiv -a \pmod{f}, \quad G \equiv 1 \pmod{c}.$$

Put $I = |of - d|$ and $H = J + jc$. The last Pell equation gives a positive index n with $I = \chi_G(n)$ and $H = \psi_G(n)$. Since $I \equiv \pm d \pmod{f}$ and $\chi_{-a}(n) = (-1)^n \chi_a(n)$, Lemma 7.1 gives $n \equiv \pm k \pmod{2m}$, hence modulo c . On the other hand, $G \equiv 1 \pmod{c}$ gives $H \equiv n \pmod{c}$, so $J \equiv \pm k \pmod{c}$. The positive-sign case yields $J = k$. The other case yields $J + k = c$, impossible because $c = \psi_a(k) \equiv k \pmod{2}$ and J is odd.

For necessity, choose a positive Pell denominator divisible by c^2 , as in the construction of Lemma 2.28 of [4]; this supplies positive f, i satisfying (E16). Put $I = \chi_G(J)$, $H = \psi_G(J)$. Since J is odd, $I \equiv -d \pmod{f}$ and $H \equiv J \pmod{c}$. Therefore $o = (I + d)/f > 0$ and $j = (H - J)/c > 0$ supply the desired witnesses. $\square$

For its certificate, compute $a_- = a - 1$, $a_+ = a + 1$, and $A = a_-a_+$ in place of the three existing instructions jointly needed for $a^2 - 1$ and $a - 1$. The parameter coefficient then takes only

$$G_- = a_+X, \quad G_+ = G_- + 2, \quad D_G = G_-G_+.$$

Thus (7.1) saves two operations relative to the original five-instruction parameter computation. Substituting $X = f^2 - 1$ uses (E16): the verifier records the exact triangular correction

$$F_{17}^{\text{calc}} = F_{17}^{\text{source}} + F_{16}(a + 1)(G_{\text{source}} + G_{\text{calc}})(J + jc)^2.$$

7.2 Replacing the quadratic approximation by an interval

The factor 4 in (E10) is unnecessary for these encodings, and the square can subsequently be removed too. The proof must establish the evenness used in rounding before appealing to binomial divisibility. We give that argument explicitly.

Use the notation

$$\begin{aligned} N = n, \quad R = r, \quad Y = sN^2, \quad U = wN^2, \quad M = RY, \\ A = a = M(U + 1), \quad P = p = 2UM^2, \quad K = k, \quad C = c, \quad J = 2R + 1. \end{aligned}$$

The encoding, independently of the Pell block, establishes

$$3 < 3L \leq B < q \leq N \leq R, \quad R, N \geq 8, \quad 0 < b \leq N.$$

Moreover $B = Hb^4$ is even because admissibility fixes an even H . This does not yet assert that q or N is even. Both $N = q^{16}$ and the shorter encoding $N = q^8$ satisfy the argument below.

Lemma 7.3. *Replace (E10) temporarily by $(C - KY)^2 + \eta = K^2$ with $\eta > 0$, retaining the other equations. Then all conclusions of the original Pell block still hold, and in fact $|C/K - Y| < 1/2$.*

Proof. We have $U, Y \geq 64$, $M \geq 512$, $A \geq 33280$, and

$$|C/K - Y| < 1. \tag{7.2}$$

Equation (E11) gives $K \geq R + 2$, so $C > (Y - 1)K \geq 63(R + 2) > J$. Therefore either the source Pell construction or Proposition 7.2 gives $C = \psi_A(J)$. By (E9), (E11), and the elementary Pell congruence,

$$K = \psi_P(R + 1 + v(P - 1)), \quad v \geq 0.$$

If $v \geq 1$, Lemma 2.19 of [4] gives

$$\frac{C}{K} \leq \frac{(2A)^{2R}}{(2P - 1)^{R+P-1}} \leq \left(\frac{2A}{2P - 1} \right)^{2R} < \frac{1}{2},$$

contradicting $C/K > Y - 1 \geq 63$. Hence $K = \psi_P(R + 1)$.

Set $\xi = (U + 1)^{2R}/U^R$. The elementary estimates in the proof of Lemma 2.25 now apply to these exact Pell indices and give

$$\xi \left(1 - \frac{R}{M(U + 1)} \right) < \frac{C}{K} < \xi \left(1 + \frac{R}{2M^2U} \right). \tag{7.3}$$

In particular $C/K > \xi/2 > U^R/2 > U^{R-1} + 1$. Combining this with (7.2) yields $Y > U^{R-1}$ and therefore

$$A = RY(U + 1) > RU^R. \tag{7.4}$$

Put $W = bw$. We have $2^{3J} \leq 8N^{2R} \leq RU^R < A$ and $W^3 \leq U^R < A$. Equation (E14), with $C = \psi_A(J)$ and $d = \chi_A(J)$, is exactly the χ congruence of Lemma 2.22. Thus $W = 2^J$, which implies that b is a power of two and

$$U = N^2w \geq Nbw = NW \geq 8 \cdot 2^J > 4 \cdot 2^{2R}. \tag{7.5}$$

Next, $\kappa < C$, (E19), and (E20), together with $0 < L < J < A$, give $\kappa = \psi_A(L)$ by Lemma 2.23. By (7.4),

$$B^{3L} \leq B^B \leq N^N \leq U^R < A, \quad q^3 \leq N^N \leq U^R < A.$$

Equation (E18) and the χ form of Lemma 2.22 therefore give $q = B^L$. Since B is even, q and N are now even. These two exponential arguments precede the rounding step; they do not invoke Lemma 2.26 with a weakened hypothesis.

From $\xi/2 < C/K < Y + 1$ we get $\xi < 2Y + 2 < 3Y$. Equation (7.3) therefore gives

$$\left| \frac{C}{K} - \xi \right| < \frac{R\xi}{M(U+1)} = \frac{\xi}{Y(U+1)} < \frac{3}{65} < \frac{1}{4}.$$

The binomial expansion of Lemma 2.24 and (7.5) give

$$0 \leq \xi - \lfloor \xi \rfloor < \frac{1}{4}, \quad \lfloor \xi \rfloor \equiv \binom{2R}{R} \pmod{U}.$$

Hence $|Y - \lfloor \xi \rfloor| < 3/2 < 2$. Both integers are even: $Y = sN^2$, $U = wN^2$, and $\binom{2R}{R} = 2\binom{2R-1}{R-1}$. Thus $Y = \lfloor \xi \rfloor$. This proves $N^2 \mid \binom{2R}{R}$, and the last two error estimates also give $|C/K - Y| < 1/2$, restoring the original stronger bound. $\square$

Proposition 7.4. *In the admissible encodings considered here, (E10) may be replaced by*

$$c = ksn^2 + \eta, \quad k = \eta + \zeta, \quad \eta, \zeta > 0. \quad (7.6)$$

This replacement saves two operations and adds one positive unknown and one free equality test.

Proof. The two equations give $0 < C/K - Y < 1$, so Lemma 7.3 proves sufficiency. For necessity, an original solution satisfies the weaker bound too: $K^2 - (C - KY)^2$ is a positive integer and supplies its slack. Applying that lemma gives $Y = \lfloor \xi \rfloor$ and $\xi < Y + 1/4 < 2Y$. The lower estimate in (7.3) implies

$$\xi - \frac{C}{K} < \frac{\xi}{Y(U+1)} < \frac{2}{U+1} < \frac{2}{U}.$$

The positive term of order $1/U$ in the binomial expansion gives

$$\xi - Y \geq \frac{\binom{2R}{R-1}}{U} \geq \frac{2R}{U} > \frac{2}{U}.$$

Consequently $C/K > Y$. Together with $|C/K - Y| < 1/2$, this proves that $\eta = C - KY$ and $\zeta = K - \eta$ are positive integers. The certificate computes ksn^2 , adds η , and computes $\eta + \zeta$: three operations in place of the previous five. $\square$

The proofs above preserve the other positive witnesses; j, o may be re-chosen for the new Pell parameter, and η, ζ supply the new interval. The residual verifier checks all changed equations and the two triangular identities exactly. Positivity and the equivalence of the encodings depend on the mathematical arguments, not on polynomial expansion alone.

8 A coefficient normalization allowing shorter masks

The 120-step system still pays for two powers that can be removed by changing the coefficient encoding. Its first coefficient code has a nonzero digit in position L , and the packing shifts the second code by q^2 . We construct a quartic representation whose digit in position L is zero. Both coefficient codes can then be bounded by q , their packing uses a shift by q , and the block widths can be reduced to q, q^2, q^5 . Their product is q^8 ; only q^2, q^4, q^8 need be computed.

The normalization is part of the admissible index, not an assumption about an arbitrary polynomial. In particular, simply deleting the leading digit of the old code would change the coefficient being tested. The construction below makes that digit zero while preserving the represented set.

8.1 Making the leading coefficient digit vanish

Start with the quadratic-system representation underlying the universal pair (58, 4), and shift its witnesses to nonnegative coordinates. Write its residuals as $F_i(X, z_1, \dots, z_{58})$, with constant coefficients c_i . Introduce a nonnegative witness h and put

$$F_i^h = F_i - c_i + c_i h, \quad S = \sum_i (F_i^h)^2 + (Xh - X)^2.$$

The polynomial S has degree at most four and zero constant coefficient. For $X > 0$, $S = 0$ forces $h = 1$ and hence is equivalent to the original system. Introduce a further nonnegative witness σ and choose a power of two $z \geq 4$. The polynomial to be encoded is

$$R = 24S + z(\sigma - 1). \quad (8.1)$$

For nonnegative integral witnesses, $R = 0$ is equivalent to $S = 0$ and $\sigma = 1$: values $\sigma \geq 2$ make R positive, while $\sigma = 0$ would require $24S = z$, impossible because 3 does not divide a power of two. Thus this remains a universal quartic representation for positive inputs. It now uses $\nu = 60$ witness coordinates, including h, σ . These are coordinates encoded in g , not additional unknowns of the final twenty-equation system. The forced coordinate $\sigma = 1$ ensures that the actual witness code g is positive.

For multiindices $I = (i_0, \dots, i_\nu)$ of total degree at most four, put

$$c_I = \frac{4!}{i_0! \cdots i_\nu! (4 - \sum_j i_j)!}, \quad R = \sum_I c_I P_I X^{i_0} z_1^{i_1} \cdots z_\nu^{i_\nu}.$$

Every c_I divides 24. Consequently the contribution of $24S$ to every P_I is integral; the new linear coefficient of σ contributes $z/4$, also integral. Choose z so large that the fixed contributions from $24S$ have absolute value less than z . The variable σ is fresh and hence these contributions do not depend on z . We obtain

$$P_0 = -z, \quad -z < P_I < z \quad (I \neq 0). \quad (8.2)$$

The convention here encodes R directly as $\sum c_I P_I$ times the monomials. There is no further factor of 24 in the extracted coefficient.

Set

$$Z = 2z, \quad M = 5^{60}, \quad L = 5^{64} = 625M, \quad w(I) = i_0 + i_1 5 + \cdots + i_\nu 5^\nu.$$

Then $w(I) \leq 4M < L$, and distinct I have distinct weights by uniqueness of base-five digits. Define

$$\begin{aligned} \ell_0(B) &= \sum_{j=1}^{\nu} B^{5^j}, \\ e_0(B) &= \sum_I (z + P_I) B^{L-w(I)}, \\ V &= e_0(Z) + \ell_0(Z) Z^L. \end{aligned}$$

The leading digit of e_0 vanishes by (8.2), so $\deg e_0 \leq L - 1$. All its other digits lie in $[0, Z)$; the two parts of the packed polynomial $e_0(B) + \ell_0(B) B^L$ have disjoint positions.

Choose a power of two H with

$$H > \max\{2Z^{2L+1}, Z(\nu + 2)^4, 3L, 64\}. \quad (8.3)$$

An index (Z, V, H) is now called admissible if constructed by these rules from a quadratic-system representation. Its three components remain fixed parameters. The complete parameter list is x, Z, V, H , as in the preceding packed system, although this is a different class of admissible indices.

8.2 The shortened system and its certificate

Keep the 32 positive unknowns of Σ_{pack} and the abbreviations $B = Hb^4$, $C = 1 + xB + g$. Replace the coding equations by

$$b = x + \beta, \quad e + \ell C^4 + \alpha = q, \quad (8.4)$$

$$\lambda + q^2 = 1 + \lambda B, \quad \theta + Z = B, \quad (8.5)$$

$$e + \ell q = V + t\theta, \quad n = q^8. \quad (8.6)$$

Use the following abbreviations in the equation defining r :

$$\begin{aligned} S_3 &= (2e - Z\lambda)C^4 + B\lambda(1 + q^2), \\ S &= g + q(e + \ell q + q^2 S_3), \\ T + 1 &= q - (b - 1)\ell + \theta\lambda q + (B - 2)q^4, \\ r &= S(n^2 - n) + (T + 1)(n^2 - 1). \end{aligned}$$

Retain the remaining Pell equations, including $c = \kappa + \varphi$, and use the new fixed exponent L in $\kappa = L + \Delta(a - 1)$. The exponent-replacement congruence still uses base B . There are twenty equations; all abbreviations are charged in the certificate.

Theorem 8.1. *For every admissible index just defined, the shortened system represents the same recursively enumerable set as the quadratic system from which the index was constructed. It has a certificate of 118 instructions: 65 multiplications and 53 additions, followed by twenty equality tests.*

Proof. We first verify the hypotheses needed to apply the Pell lemmas without assuming the exponential equation. Equation (8.4) gives

$$e < q, \quad \ell < q, \quad C^4 < q, \quad (b - 1)\ell < \ell C^4 < q.$$

Since $C > B > b \geq 2$, it follows that $q > B$ and $g < q$. Equation (8.5) gives $\lambda > q$, so, writing $z = Z/2$,

$$0 < D_0 = z\lambda - e < z\lambda, \quad 2C^4 D_0 < 2z\lambda q < B\lambda q.$$

The blocks therefore satisfy

$$\begin{aligned} 0 &< g < q, \quad 0 \leq q - 1 - (b - 1)\ell < q, \\ 0 &< e + \ell q < q^2, \quad 0 < (B - Z)\lambda < (B - 1)\lambda = q^2 - 1, \\ 0 &< S_3 < B\lambda(1 + q^2) = \frac{B(q^4 - 1)}{B - 1} < 2q^4 < q^5, \\ 0 &< (B - 2)q < q^2 < q^5. \end{aligned}$$

Thus their respective widths are $N_1 = q$, $N_2 = q^2$, $N_3 = q^5$; their product is $n = q^8$. The equation for r gives $r \geq n^2 - 1 \geq n$, and (8.3) yields $3 < 3L \leq B < q \leq n \leq r$. The elementary Pell gap restores the stronger inequality required by Lemma 4.1. The Pell lemmas now give b pow 2, $q = B^L$, and the central-binomial divisibility. Since H is a power of two, the block radices are powers of two, and the three separate no-carry conditions follow.

Apply [4, Lemma 2.9] to $Y = e + \ell q$ with digit base Z and $m = k = 2L$. The polynomial

$$F(B) = e_0(B) + \ell_0(B)B^L$$

has degree at most $L + M < 2L$, digits less than Z , and $F(Z) = V$. The congruence in (8.6), the bound $Y < q^2 = B^{2L}$, the mask $(B - Z)\lambda$, and (8.3) supply the lemma's hypotheses. Hence $Y = F(B)$. The supplied and intended e, ℓ all lie in $[0, q)$, so quotient-and-remainder uniqueness gives $e = e_0(B)$ and $\ell = \ell_0(B)$. The first mask identifies the witness digits of g and bounds them by b .

To identify what the last mask tests, regard B temporarily as a formal base. The coefficient of B^L in

$$C^4(e_0(B) - z\lambda)$$

is exactly R evaluated at the encoded tuple. Indeed, the monomial exponent $w(I)$ in C^4 has coefficient c_I times its monomial value; the complementary digit of $e_0 - z\lambda$ is P_I . This includes $I = \mathbf{0}$, where that digit is $-z$, precisely the constant coefficient in (8.1).

Every coefficient of $e_0 - z\lambda$ has absolute value at most z . Thus every coefficient of the product has absolute value less than $z((\nu + 2)b)^4 < B/2$. Its degree is at most $2L - 1 + 4M < 3L$. The offset $(B/2)\lambda(1 + q^2)$ supplies the digit $B/2$ at every position below $4L$, preventing carries throughout. The third no-carry condition therefore tests, by [4, Lemma 2.8], whether the coefficient R is zero. The equivalence following (8.1) proves soundness.

Conversely, take a solution of the original quadratic system, set $h = \sigma = 1$, and choose a power-of-two b greater than the input and all sixty witnesses. Form the intended codes using $B = Hb^4$ and $q = B^L$. Their bounds are

$$e_0(B) < ZB^{L-1}, \quad \ell_0(B) < 2B^M, \quad C < 2bB^M.$$

Because $B > 2Z$, the first is less than $q/2$. Also

$$\ell C^4 < 32b^4 B^{5M} < \frac{1}{2} B^{5M+1} \leq \frac{1}{2} q,$$

using $H > 64$ and $L = 625M \geq 5M + 1$. Consequently the new $\alpha = q - e - \ell C^4$ is positive. The packed polynomial F has nonnegative digits and positive degree, so $F(B) > F(Z)$ and $t = (F(B) - F(Z))/(B - Z) > 0$. The coordinate $\sigma = 1$ ensures $g > 0$. All remaining witnesses follow from the same digit and Pell necessity constructions with the new base and exponent.

For the count, replace the three instructions computing $e\ell$, $(e\ell)C^2$ and addition of α by the three instructions computing ℓC^4 , addition of e , and addition of α . This exchanges one multiplication for one addition but preserves the length. All packing formulas retain their previous instruction counts. The power chain now consists only of q^2, q^4, q^8 ; the instructions for q^3 and q^{16} disappear. This reduces 120 to 118.

The exact verifier checks all twenty residuals. Its shared (E7) calculation records the adapted identity

$$F_{7,\text{new}} = F_7 + (\lambda F_3 - F_2)q(n^2 - 1),$$

where F_2, F_3 now refer to (8.5). Thus the symbolic check accounts for the dependence on the preceding equations as well as the operation count. $\square$

Corollary 8.2. *With the same admissible index (Z, V, H) , the shortened system together with the signed Pell replacement of Proposition 7.2 and the linear ratio replacement of Proposition 7.4 has a 114-instruction certificate: 63 multiplications and 51 additions, followed by 21 equality tests. It has 33 positive unknowns and the four parameters x, Z, V, H , and defines the same recursively enumerable set.*

Proof. Both Pell replacements apply to the shortened system: the relevant equations retain their form, and the bounds $q > B > b \geq 2$, $n = q^8 \geq 2$ and $a \geq 20r$ supply their hypotheses. Their positive-domain equivalences therefore compose with Theorem 8.1. The final certificate is independently checked by `round5_1980_certificate.py`; its exact residual checks and instruction histogram give the asserted

114 instructions and 21 tests. The linear ratio replacement adds one positive unknown and one equation. $\square$

For comparison, applying the three independent changes in this order gives:

Construction	Multiplications	Additions	Total
Packed encoding of Theorem 6.5	68	52	120
Shorter masks	65	53	118
Also the signed Pell parameter	65	51	116
Also the positive interval	63	51	114

This is a further verified upper bound, not an assertion of minimality. The choice $L = 5^{64}$ also makes the fixed numeral inexpensive to generate: after $2 = 1 + 1$, $4 = 2 + 2$, $5 = 4 + 1$, six successive squarings give $5^2, 5^4, \dots, 5^{64}$. Thus, if x, Z, V, H remain supplied parameters while $2, 4, 5, L$ must be generated from 1, nine additional instructions suffice, giving a total of 123.

9 Testing quadratic residuals with one mask

The sum of squares used to obtain a quartic polynomial is not needed for coefficient testing. We can test the original quadratic residuals at separate digit positions, using the already decoded variable mask as the test mask. Extra digit coordinates make this possible without extra unknowns in the final polynomial system.

9.1 A fixed layout for all quadratic systems

Start from the quadratic-system representation in 58 existential variables underlying the universal pair $(58, 4)$ in [4, §5]. Shift its positive variables to nonnegative variables $X_1, \dots, X_{58}$, and put $X_0 = x$. The integer quadratic residuals have fixed coefficients for the represented r.e. set. Their rational linear span has dimension at most

$$m = \binom{61}{2} = 1830,$$

the number of monomials of degree at most two in 59 variables. Choose a basis from the original integer residuals and pad it with zero polynomials to obtain $F_0, \dots, F_{m-1}$. Simultaneous vanishing is unchanged: every original residual is a rational linear combination of the chosen basis, and every chosen residual is original.

Use the following fixed positions:

$$\begin{aligned} v_i &= 1 + 3(6729i + i^2) \quad (0 \leq i \leq 58), & v_* &= 1180939, \\ D &= 4v_* + 1, & T &= (m + 1)D + 2v_*, \\ t_j &= T + jD \quad (0 \leq j < m), & t_* &= T + (m - 1)D. \end{aligned}$$

Let $d_* = T - 2v_*$. The inequalities

$$D > 2v_*, \quad d_* > \max(v_*, t_* - T), \quad L = 5^{16} = 152587890625 > 3t_* + 1 = 51873937495 \quad (9.1)$$

are exact integer inequalities; the companion program checks them. The positions v_i are for the input and the 58 actual variables. The positions t_j are for 1830 additional, unconstrained digit coordinates and will also be the equation-testing positions.

For a multiindex $I = (I_0, \dots, I_{58})$ with $|I| \leq 2$, put

$$w(I) = \sum_{i=0}^{58} I_i v_i, \quad c_I = \frac{2!}{I_0! \cdots I_{58}! (2 - |I|)!} \in \{1, 2\}.$$

Write $F_j = \sum_I a_{j,I} X^I$, and define $P_{j,I} = 2a_{j,I}/c_I \in \mathbb{Z}$. Choose a power of two $z \geq 2$ strictly larger than every $|P_{j,I}|$, and put $Z = 2z$. Define the polynomial with possibly negative coefficients

$$\mathcal{D}(B) = \sum_{j=0}^{m-1} \sum_{|I| \leq 2} P_{j,I} B^{t_j - w(I)}.$$

Its exponents are nonnegative. Within one row, distinct multiindices have distinct weights. Indeed $v_i \equiv 1 \pmod{3}$, so the weight determines the total degree, which is at most two. Degree-one weights are strictly increasing. For a degree-two weight $v_i + v_j$, subtracting 2 and dividing by 3 gives $6729(i+j) + i^2 + j^2$. Since $0 \leq i^2 + j^2 \leq 2 \cdot 58^2 < 6729$, division by 6729 recovers both $i+j$ and $i^2 + j^2$. These determine the unordered pair $\{i, j\}$. The supports of different rows lie in disjoint intervals $[t_j - 2v_*, t_j]$, by (9.1). Thus every coefficient of $\mathcal{D}$ has absolute value less than z , and its support lies in $[d_*, t_*]$.

The two nonnegative digit polynomials are

$$\begin{aligned} \ell_0(B) &= \sum_{i=1}^{58} B^{v_i} + \sum_{j=0}^{m-1} B^{t_j}, \\ e_0(B) &= z \sum_{h=0}^{L-1} B^h + \mathcal{D}(B). \end{aligned}$$

Every coefficient of e_0 lies in $[0, Z)$, and its degree is $L-1$. The mask ℓ_0 has coefficients zero or one and degree $t_* < L$. The admissible index is (Z, V, H) , where

$$V = e_0(Z) + \ell_0(Z)Z^L, \quad H > \max\{2Z^{2L+1}, Z(1890)^2, 3L, 16\} \quad (9.2)$$

and H is a power of two. There are $58 + 1830 = 1888$ non-input digit coordinates; their number explains the factor $1890 = 1888 + 2$. They are encoded in one integer g and are not new unknowns of the final polynomial system. All these choices are effective index construction. The certificate parameters remain x, Z, V, H .

9.2 The changed equations and the bounds before decoding

Retain the signed Pell parameter and the linear interval equations proved above. Put $B = Hb^2$ and $C = 1 + xB + g$, with both abbreviations computed by charged instructions. Replace the bound and the third testing block by

$$\begin{aligned} e + \ell C^2 + \alpha &= q, \\ S_3 &= (2e - Z\lambda)C^2 + B\lambda(1 + q^2), \\ T_3 &= (B - 2)\ell. \end{aligned}$$

The other coding equations remain

$$b = x + \beta, \quad \lambda + q^2 = 1 + \lambda B, \quad \theta + Z = B, \quad e + \ell q = V + t\theta, \quad n = q^8.$$

The first two pairs of blocks are

$$S_1 = g, \quad T_1 = q - 1 - (b - 1)\ell, \quad S_2 = e + \ell q, \quad T_2 = \theta\lambda,$$

with widths $N_1 = q$, $N_2 = q^2$, $N_3 = q^5$. Accordingly,

$$\begin{aligned} S &= g + q(e + \ell q + q^2 S_3), \\ T + 1 &= q - (b - 1)\ell + \theta \lambda q + (B - 2)\ell q^3, \\ r &= S(n^2 - n) + (T + 1)(n^2 - 1). \end{aligned}$$

These blocks have the required signs and widths before either b being a power of two or $q = B^L$ is known. Indeed, positivity gives $e, \ell, \mathcal{C}^2 < q$, while $\mathcal{C} > B > b \geq 2$ gives $q > B$. The geometric equation yields $\lambda > q$, so $0 < z\lambda - e < z\lambda$. Moreover

$$2\mathcal{C}^2(z\lambda - e) < 2z\lambda q < B\lambda q, \quad 0 < S_3 < B\lambda(1 + q^2) < 2q^4 < q^5.$$

Since $(b - 1)\ell < \ell\mathcal{C}^2 < q$, we have $0 \leq T_1 < q$. Likewise $0 < T_3 = (B - 2)\ell < \ell\mathcal{C}^2 < q$. The familiar integer bounds give $0 < S_2 < q^2$ and $0 < T_2 < (B - 1)\lambda = q^2 - 1$, while $0 < g < q$. Consequently $r \geq n^2 - 1 \geq n$, and admissibility gives

$$3 < 3L \leq B < q \leq n \leq r, \quad r, n \geq 8, \quad 0 < b \leq n.$$

As B is even already, the noncircular Pell and interval arguments of Section 7 apply. They give b a power of two, $q = B^L$, and $n^2 \mid \binom{2r}{r}$. Thus the three separate no-carry conditions follow from Lemmas 2.16 and 2.11 of [4].

9.3 Decoding and excluding unwanted coefficients

Apply Lemma 2.9 of [4] to $Y = e + \ell q$, with digit base Z and $m = k = 2L$. Its fixed digit polynomial is $e_0(B) + \ell_0(B)B^L$, of degree $L + t_* < 2L$, with coefficients in $[0, Z)$. Equation (9.2) supplies its value at Z ; the packed congruence, $Y < q^2 = B^{2L}$, the mask $\theta\lambda$, and $B > 2Z^{2L+1}$ supply the other hypotheses. Therefore $e + \ell q = e_0(B) + \ell_0(B)q$. Since all four components lie in $[0, q)$, quotient-and-remainder uniqueness gives $e = e_0(B)$ and $\ell = \ell_0(B)$.

The first no-carry condition now decodes

$$\mathcal{C}(B) = 1 + xB + \sum_{i=1}^{58} X_i B^{v_i} + \sum_{j=0}^{m-1} Y_j B^{t_j}, \quad 0 \leq X_i, Y_j < b.$$

The Y_j are the unconstrained extra coordinates. They do not affect the equation tests. To see this, regard B as a formal variable and write

$$e_0 - z\lambda = \mathcal{D} - z \sum_{h=L}^{2L-1} B^h.$$

The tail starts above every testing position, since $t_* < L$. The remaining factor $\mathcal{D}$ starts at degree d_* . Any monomial of $\mathcal{C}^2$ involving a Y_j has degree at least T , so its product with $\mathcal{D}$ has degree at least $T + d_* > t_*$. Thus no extra coordinate contributes to any testing position. Also $d_* > v_*$, so the coefficients at the low positions v_i are automatically zero.

At a high position t_j , a contribution from row k and true-variable monomials I, J requires

$$t_k - w(I) + w(J) = t_j.$$

Both weights lie in $[0, 2v_*]$, so the row spacing forces $k = j$. Uniqueness of the quadratic monomial weights then forces $I = J$. Therefore

$$[B^{t_j}] \mathcal{C}(B)^2 (e_0(B) - z\lambda(B)) = \sum_I c_I P_{j,I} X^I = 2F_j(x, X_1, \dots, X_{58}). \quad (9.3)$$

This proves both kinds of separation: other rows cannot interfere, and the extra coordinates at the row positions cannot interfere.

Every coefficient of $e_0 - z\lambda$ has absolute value at most z . The sum of coefficients of $\mathcal{C}^2$ is less than $(1890b)^2$. Hence every coefficient of their product has absolute value less than

$$z(1890b)^2 < B/2.$$

Its degree is at most $2L - 1 + 2t_* < 3L$. The offset $(B/2)\lambda(1 + q^2)$ supplies the digit $B/2$ at every position below $4L$, so all resulting digits lie strictly between zero and B . Now S_3, T_3 are even, and their third no-carry condition is equivalent to that for

$$\frac{S_3}{2} \quad \text{and} \quad \frac{T_3}{2} = (B/2 - 1)\ell_0(B).$$

The latter mask tests exactly the low positions v_i and the high positions t_j . At each such position it requires the digit of $S_3/2$ to equal $B/2$. The low tests hold automatically; by (9.3), the high tests are equivalent to $F_j = 0$ for every j . This proves soundness.

9.4 Positive witnesses and the certificate count

Conversely, choose a nonnegative solution of the quadratic system and assign the extra coordinates arbitrarily in $[0, b)$, with one of them equal to 1. Choose the power of two b larger than x and all these coordinates. The extra value 1 guarantees $g > 0$ even if every true existential coordinate is zero. Take the intended B, q, e, ℓ, g . Their bounds are

$$e < ZB^{L-1} < q/2, \quad \ell < 2B^{t_*}, \quad \mathcal{C} < 2bB^{t_*}.$$

Using $H > 16$ and $L > 3t_* + 1$, we obtain

$$\ell\mathcal{C}^2 < 8b^2B^{3t_*} < \frac{1}{2}B^{3t_*+1} \leq q/2.$$

Thus $\alpha = q - e - \ell\mathcal{C}^2$ is positive, as is $\beta = b - x$. The packed quotient t is a positive integer because its fixed nonnegative digit polynomial has positive degree and $B > Z$. All decoded no-carry conditions hold by the preceding coefficient calculation. The remaining positive witnesses follow from the Pell necessity constructions and the positive interval proof. This proves necessity with all domains preserved.

Theorem 9.1. *For each r.e. set, an effective admissible index (Z, V, H) as above gives a system in 33 positive unknowns and 21 equations such that, for positive input x , membership is equivalent to a valid certificate with **113 arithmetic instructions**: 62 multiplications and 51 additions. Equality tests and supplied parameters are free. Generating the fixed numerals 2, 4, 5, L from 1 adds seven instructions, giving 120 under that convention.*

Proof. Relative to the 114-instruction certificate, $B = Hb^2$ saves the multiplication constructing b^4 , and $\mathcal{C}^2$ saves the one constructing $\mathcal{C}^4$. Computing the changed third mask in the packing expression costs one extra multiplication; the other counts are unchanged. The $T + 1$ computation may be organized as

$$T + 1 = q(\lambda + q^2 - Z\lambda) + (B - 2)((\ell q)q^2) - (b - 1)\ell,$$

using the geometric and θ equations and the already computed product ℓq . The exact schedule and its triangular residual corrections are verified by `round6_1980_quadratic_certificate.py`. The source system has the same 33 positive unknowns and 21 equations; the additional digit coordinates are encoded inside g . The equivalence is proved above. Starting from 1, three additions produce 2, 4, 5 and four successive squarings of 5 produce $L = 5^{16}$, giving the seven-instruction numeral construction. $\square$

10 Shifting the first Pell parameter

One more operation can be removed by shifting the first Pell parameter. Use the notation of Section 7, and put

$$X = 2UM^2 = 2ws^2r^2n^6, \quad P = X + 1.$$

Delete the unknown p and its defining equation (E8), and replace (E9) and (E11) by

$$X(X + 2)k^2 + 1 = \tau^2, \quad k = r + 1 + hX. \quad (10.1)$$

Every other equation is retained, including the positive interval $c = ksn^2 + \eta$, $k = \eta + \zeta$. The mathematical parameter P need not be computed: its coefficient is $P^2 - 1 = X(X + 2)$, and its index modulus is $P - 1 = X$.

Lemma 10.1. *Under the initial encoding bounds*

$$3 < 3L \leq B < q \leq N \leq R, \quad R, N \geq 8, \quad b \leq N,$$

with B even and $N = q^8$, the replacement (10.1) preserves existential equivalence with positive witnesses. It re-chooses only k, τ, h, η, ζ and removes or restores p . It applies both to $B = Hb^4$, $L = 5^{64}$ and to the direct quadratic encoding $B = Hb^2$, $L = 5^{16}$.

Proof. The initial bounds give $U, Y \geq 64$, $M = RY \geq 512$, and $A = M(U + 1) \geq 33280$. Write $C = c$, $K = k$, $J = 2R + 1$ as before. The positive interval says $0 < C/K - Y < 1$. Equation (10.1) gives $K \geq R + 2$, so $C > (Y - 1)K > J$. Proposition 7.2 therefore gives $C = \psi_A(J)$ and $d = \chi_A(J)$, before any approximation is used. The other Pell equation and its elementary index congruence give

$$K = \psi_P(R + 1 + vX), \quad v \geq 0.$$

If $v \geq 1$, the Pell growth bounds imply

$$\frac{C}{K} \leq \frac{(2A)^{2R}}{(2P - 1)^{R+X}} \leq \left(\frac{2A}{2P - 1} \right)^{2R} < \frac{1}{2},$$

contrary to $C/K > Y - 1 \geq 63$. Here $X \geq R$ and $2A/(2P - 1) = 2M(U + 1)/(4UM^2 + 1) < 1/2$. Thus $K = \psi_P(R + 1)$.

Put $\xi = (U + 1)^{2R}/U^R$. At these exact indices the same growth bounds give

$$\frac{C}{K} \leq \frac{(2A)^{2R}}{(2P - 1)^R} = \xi \left(1 + \frac{1}{4UM^2} \right)^{-R} < \xi, \quad (10.2)$$

$$\frac{C}{K} \geq \xi \left(1 - \frac{1}{2A} \right)^{2R} \left(1 + \frac{1}{X} \right)^{-R} > \xi \left(1 - \frac{R}{A} - \frac{R}{X} \right). \quad (10.3)$$

The last inequality follows from Bernoulli's inequality, with both factors positive. Since $A/X = (U + 1)/(2UM) < 1$, the error coefficient is

$$\varepsilon = \frac{R}{A} + \frac{R}{X} < \frac{2}{Y(U + 1)} < \frac{1}{2}.$$

Consequently $C/K > \xi/2 > U^R/2 > U^{R-1} + 1$, while $C/K < Y + 1$. Hence $Y > U^{R-1}$ and $A > RU^R$.

This last bound permits the two exponent arguments in their original order. First,

$$2^{3J} \leq 8N^{2R} \leq RU^R < A, \quad (bw)^3 \leq U^R < A,$$

so (E14) and the χ form of Lemma 2.22 of [4] give $bw = 2^J$, hence

$$U \geq Nbw \geq 8 \cdot 2^J > 4 \cdot 2^{2R}. \quad (10.4)$$

Next, $\kappa < C$, (E19), (E20), and $0 < L < J < A$ give $\kappa = \psi_A(L)$. The bounds $B^{3L}, q^3 \leq N^N \leq U^R < A$ and (E18) give $q = B^L$ by the same exponent lemma. Since B is even already, q and N are now even. Neither argument has used parity rounding or central-binomial divisibility.

From $\xi/2 < C/K < Y + 1$ we have $\xi < 2Y + 2 < 3Y$, so (10.2)–(10.3) imply

$$0 < \xi - C/K < \xi\varepsilon < \frac{6}{U+1} \leq \frac{6}{65} < \frac{1}{4}.$$

The binomial expansion, using (10.4), gives

$$0 < \xi - \lfloor \xi \rfloor < \frac{1}{4}, \quad \lfloor \xi \rfloor \equiv \binom{2R}{R} \pmod{U}.$$

Thus $|Y - \lfloor \xi \rfloor| < 3/2 < 2$. Both integers are even, since Y, U are even and $\binom{2R}{R}$ is even. Therefore $Y = \lfloor \xi \rfloor$, so $N^2 \mid \binom{2R}{R}$ as required.

It remains to justify strict positivity when changing witnesses. Start with an old solution and set

$$K_{\text{new}} = \psi_{X+1}(R+1), \quad \tau_{\text{new}} = \chi_{X+1}(R+1), \quad h_{\text{new}} = \frac{K_{\text{new}} - R - 1}{X} > 0.$$

The quotient is integral by the Pell congruence and positive by strict growth. The old solution has $Y = \lfloor \xi \rfloor$ and $\xi < 2Y$. Thus the new ratio estimates give $0 < \xi - C/K_{\text{new}} < 4/U$, whereas the positive fractional tail gives

$$\xi - Y \geq \frac{\binom{2R}{R-1}}{U} \geq \frac{2R}{U} > \frac{4}{U}.$$

Consequently $Y < C/K_{\text{new}} < \xi < Y + 1/4$, and $\eta_{\text{new}} = C - K_{\text{new}}Y$, $\zeta_{\text{new}} = K_{\text{new}} - \eta_{\text{new}}$ are positive integers.

Conversely, restore $p = X$ and choose $K_{\text{old}} = \psi_X(R+1)$, $\tau_{\text{old}} = \chi_X(R+1)$, and $h_{\text{old}} = (K_{\text{old}} - R - 1)/(X - 1) > 0$. The elementary old-base estimates at these exact indices give

$$\xi(1 - R/A) < C/K_{\text{old}} < \xi(1 + R/X), \quad |C/K_{\text{old}} - \xi| < 2/U.$$

The same positive tail yields $C/K_{\text{old}} > Y$, while its upper bound gives $C/K_{\text{old}} - Y < 1/4 + 2/U < 1$. The corresponding old η, ζ are therefore positive as well. Every other witness is preserved. The proof uses only the stated initial bounds and evenness of B , which were established for both encodings independently of the Pell block. $\square$

Theorem 10.2. *For each r.e. set, the effective admissible index (Z, V, H) of Theorem 9.1 gives a system in 32 positive unknowns and 20 equations such that, for positive input x , membership is equivalent to a valid straight-line certificate with **112 arithmetic instructions**: 62 multiplications and 50 additions. Equality tests and supplied parameters are free. Constructing $2, 4, 5, L = 5^{16}$ from 1 adds seven instructions, giving 119 under that convention.*

Proof. Apply Lemma 10.1 to the 113-operation direct quadratic system of Theorem 9.1. The shared value X was already computed by the old (E8) block. Replace the two instructions for $p^2 - 1$ by

$$X_+ = X + 2, \quad D_P = XX_+,$$

and use hX in place of $h(p - 1)$. This deletes one subtraction, the positive unknown p , and its free defining equality. All other counts are unchanged. The complete primitive schedule, equality list, declared domains, and exact residual checks are in `round6_1980_certificate.py` and its JSON receipt. The standalone shift and its full positive-witness proof are also recorded in `round5_1980_shifted_pell_certificate.py` and `PELL_SHIFTED_BASE_PROOF.md`. $\square$

11 Reversing the packed codes

The inequality $e + \ell C^2 < q$ costs a multiplication and two additions, including its positive slack. The already computed packed quantity can replace the product, but its two components must still be recovered uniquely. Reversing their order and shortening the coefficient code makes the third mask enforce that uniqueness. The resulting bound costs two additions.

11.1 The revised fixed index and system

Retain the quadratic layout, the polynomial $\mathcal{D}$, and the mask ℓ_0 from Section 9. Put $K = t_* + 1$ and keep $L = 5^{16}$. The fixed numerical layout satisfies

$$L > 3K + 2. \quad (11.1)$$

Shorten the dense part of the coefficient polynomial and reverse the order of the two codes:

$$e_0(B) = z \sum_{j=0}^{K-1} B^j + \mathcal{D}(B), \quad V = \ell_0(Z) + e_0(Z)Z^L. \quad (11.2)$$

Both e_0 and ℓ_0 now have degree at most t_* ; their coefficients are still in $[0, Z)$, with the coefficients of e_0 positive. Choose the fixed power of two H so that

$$H > \max\{2Z^{2L+1}, 2^{t_*+4}Z(1890)^2, 3L, 16\}. \quad (11.3)$$

This includes $H > 4Z(1890)^2$ and all earlier radix bounds. These index choices depend only on the represented r.e. set.

Start from the system of Theorem 10.2, retaining $B = Hb^2$, $\mathcal{C} = 1 + xB + g$, and all its Pell equations. The changed coding equations and packing expressions are

$$Y = \ell + eq, \quad Y + \mathcal{C}^2 + \alpha = q^2, \quad Y = V + t\theta, \quad (11.4)$$

$$S_3 = (2e - Z\lambda)\mathcal{C}^2 + B\lambda(1 + q), \quad \sigma = S_3, \quad (11.5)$$

$$S = g + q^2(Y + q^2 S_3), \quad (11.6)$$

$$T + 1 = q^2(1 + \theta\lambda) - (b - 1)\ell + (B - 2)\ell q^4, \quad (11.7)$$

$$r = S(n^2 - n) + (T + 1)(n^2 - 1). \quad (11.8)$$

Here Y, S_3, S, T remain computed abbreviations. The only new unknown is σ , required to be positive. Its equality to the already computed S_3 is a free equality test. The unchanged coding equations include

$$b = x + \beta, \quad \lambda + q^2 = 1 + \lambda B, \quad \theta + Z = B, \quad n = q^8.$$

11.2 Bounds before decoding and a possible borrow

Equation (11.4) implies

$$\ell < Y < q^2, \quad e < q, \quad \mathcal{C} < q, \quad g < q.$$

In particular $q > B$, since $x > 0$ and $\mathcal{C} > B$. With $N = \theta\lambda$, the geometric equation and the radix bounds give

$$q < \lambda < 2q^2/B, \quad b < N < q^2 - 1.$$

The new unknown supplies $S_3 > 0$. Before any decoding we also have

$$S_3 < 2q^3 + B\lambda(1 + q) < 5q^3 < q^4.$$

The three blocks of S therefore have widths q^2, q^2, q^4 and give $0 < S < n = q^8$.

The first raw mask $q^2 - 1 - (b-1)\ell$ may be negative at this stage. Its complete packed value is positive, because

$$q^2(1+N) - (b-1)\ell > q^2, \quad (B-2)\ell < q^3.$$

In addition $T+1 < q^7 + q^4 + q^2 < q^8$. Thus $0 \leq T < n$ and $r \geq n^2 - 1 \geq n$. The hypotheses used by the unchanged Pell implications hold without assuming the sign of that first raw mask. Those implications give $q = B^L$, the required powers of two, and $\tau_2(S, T) = 0$.

To read the masks, write

$$d = \left\lfloor \frac{(b-1)\ell}{q^2} \right\rfloor, \quad u = (b-1)\ell - dq^2.$$

Then $0 \leq d < b$, $0 \leq u < q^2$, and the actual block decomposition is

$$T = (q^2 - 1 - u) + q^2(N - d) + q^4(B - 2)\ell.$$

All three blocks are in their stated ranges. Every base- B digit of $N = (B - Z)\lambda$ equals $B - Z$, in positions $0, \dots, 2L - 1$. Since $d < b < B - Z$, subtracting d changes only the unit digit. The middle no-carry condition consequently makes every nonunit digit of Y less than Z , and makes its unit digit at most $Z + d - 1$. Hence

$$Y < \frac{Zq^2}{B-1} + b, \quad (b-1)\ell < \frac{(b-1)Zq^2}{B-1} + b(b-1) < q^2.$$

For the last inequality, (11.3) gives $B - 1 > 4(b-1)Z$ and $q > 4b$. Therefore $d = 0$ after all.

All digits of Y are now less than Z . The digit-evaluation lemma applied to (11.2) and the packed congruence gives

$$Y = \ell_0(B) + e_0(B)q, \quad \ell = \ell_0(B) + mq, \quad e = e_0(B) - m \quad (11.9)$$

for an integer m with $0 \leq m < e_0(B)$. Positivity of ℓ and $\ell_0(B) < q$ exclude negative m . The following argument rules out positive m as well.

11.3 The high mask checks the quotient

The fixed support gives $(b-1)\ell_0(B) < q$. The low q block of the first mask is therefore $q - 1 - (b-1)\ell_0(B)$. Since $g < q$, its no-carry condition decodes exactly the earlier low coordinate positions, with every digit less than b . Extra coordinates at positions belonging to mq cannot occur in g , because those positions are at least L . Thus $\mathcal{C}$ has degree at most t_* , constant digit 1, and input digit x . Writing A for the sum of the coefficients of $\mathcal{C}^2$, we obtain

$$A = \mathcal{C}(1)^2 < (1890b)^2, \quad 4 \leq ZA < B/4. \quad (11.10)$$

Here $\mathcal{C}(1)$ denotes evaluation of its digit polynomial, not of the integer expression $1 + xB + g$ with g held constant.

By (11.9), both e and m are smaller than $e_0(B) < B^K$. The decoded $\mathcal{C}$ also satisfies $\mathcal{C} < B^K$. The explicit margin (11.1) consequently gives an integer bound which includes all carries:

$$2e\mathcal{C}^2 < 2B^{3K} < B^L = q. \quad (11.11)$$

Put

$$X = (B - 2)m = \sum_{j=0}^{t_*+1} X_j B^j, \quad 0 \leq X_j < B.$$

Since $(B - 2)\ell_0(B) < q$, the digits of the third mask at positions $L, \dots, L + t_* + 1$ are exactly the X_j . In this window the coefficient of $\lambda\mathcal{C}^2$ is always A : the window is above $2t_*$ and below $2L$. By (11.11), the positive integer $2e\mathcal{C}^2$ has no digit there. The offset $B\lambda(1 + q)$ has raw coefficient 1 at position L and raw coefficient 2 at the subsequent positions of the window.

The incoming carry at L lies in $\{-1, 0, 1\}$. Indeed the raw negative coefficients below L have absolute value at most $ZA < B/4$, while the positive contribution below L is less than $q + q/(B - 1) < 2q$. Thus the normalized digit at L lies between $B - ZA$ and $B - ZA + 2$; because $ZA \geq 4$, its outgoing carry is -1 . The later digits in the window are $B - ZA + 1$ and their outgoing carry remains -1 . In particular every digit in this window is at least $B - ZA$.

No carry with the third mask now implies $X_j \leq ZA - 1$. Consider the ordinary polynomial $F(U) = \sum_j X_j U^j$. Its value $F(B) = (B - 2)m$ is divisible by $B - 2$, so $F(2)$ is divisible by $B - 2$. But (11.3) and (11.10) give

$$0 \leq F(2) < ZA 2^{t_*+2} < Z(1890)^2 b^2 2^{t_*+2} < B/4 < B - 2.$$

Therefore $F(2) = 0$. All coefficients of F are nonnegative, so $X = 0$ and $m = 0$. We have recovered both canonical codes $e = e_0(B)$ and $\ell = \ell_0(B)$.

Every target position is below K . Below that cutoff, $e_0 - z\lambda = \mathcal{D}$, and the offset in $S_3/2$ still supplies the digit $B/2$ at every target. The coefficient bound and the support separation of Section 9 apply unchanged. Carries from overlapping offset blocks above L cannot affect these lower positions. Thus all original quadratic residuals vanish.

11.4 Positive witnesses and the verified count

Conversely, encode a genuine quadratic-system solution, assign one dummy coordinate the value 1, and choose a power of two b larger than x and all coordinate values. Use the canonical $e_0(B)$ and $\ell_0(B)$. Their degrees are at most t_* , and (11.1) gives $Y + \mathcal{C}^2 < q^2$, with a strictly positive slack. It also gives $Z\mathcal{C}^2 < q$, so

$$S_3 \geq \lambda(B(1 + q) - Z\mathcal{C}^2) > 0.$$

Set $\sigma = S_3$. The quotient in the packed congruence is positive because $B > Z$ and its fixed digit polynomial has positive nonconstant coefficients. The three no-carry conditions hold, and the preceding Pell necessity constructions supply all remaining positive unknowns.

Theorem 11.1. *For each r.e. set, an effective admissible index (Z, V, H) defined by (11.2)–(11.3) gives a system in 33 positive unknowns and 21 equations such that, for positive input x , membership is equivalent to a valid certificate with **111 arithmetic instructions**: 61 multiplications and 50 additions. Equality tests and supplied parameters are free. Generating the fixed numerals 2, 4, 5, L from 1 adds seven instructions, giving 118 under that convention.*

Proof. The equivalence, including both positive-witness constructions, was proved above. Relative to Theorem 10.2, the product $\ell\mathcal{C}^2$, addition of e , and addition of the positive slack are replaced by $Y + \mathcal{C}^2$ and addition of that slack, saving one multiplication. The reversed code $\ell + eq$ has the same cost as $e + \ell q$. Computing S still uses two multiplications and two additions. The $T + 1$ calculation uses

$$q^2(\lambda + q^2 - Z\lambda) - (b - 1)\ell + (B - 2)(\ell q^4),$$

which has the same cost as before. Replacing $1 + q^2$ by $1 + q$ also has equal cost, and $\sigma = S_3$ is an equality test on an existing computed register. The exact primitive instructions, equality tests, and triangular polynomial residual corrections are checked by `round7_1980_certificate.py`. The numeral construction is unchanged. $\square$

12 Decoding the unit coordinate

The addition inserting the constant digit in $\mathcal{C} = 1 + xB + g$ can also be removed. We encode a coordinate δ in that position, homogenize the quadratic residuals using δ , and use a slightly less restrictive mask to force $\delta = 1$. The resulting system has 110 arithmetic instructions. The encoded coordinates introduced in this construction remain digits of g ; they do not increase the number of system unknowns.

12.1 Homogeneous rows and the unit test

Take the 1830 integer quadratic residuals $F_i(x, X_1, \dots, X_{58})$ obtained by the basis selection in Section 9. Put $X_0 = x$ and homogenize each residual:

$$F_i^h(\delta, X) = \sum_{|I| \leq 2} a_{i,I} \delta^{2-|I|} X^I.$$

Introduce one more nonnegative encoded coordinate $u = X_{59}$ and one ordinary row $u\delta - x^2$. There are now 1831 ordinary homogeneous quadratic rows. Their target coefficients will be twice their values. Add a final special target whose coefficient is δ^2 . The new mask will permit a target coefficient to be either zero or one. Thus each ordinary target, being even, must be zero, while the special target forces $\delta \in \{0, 1\}$. The guard $u\delta - x^2 = 0$ and the positive input x then force $\delta = 1$.

The distinction in scaling is essential: the special target encodes δ^2 , rather than $2\delta^2$. It is not an additional equation requiring the special value to vanish.

12.2 The uniform support and admissible index

Give δ weight zero. For the input, the 58 original coordinates, and u , use the 60 positive weights

$$v_i = 1 + 3(6963i + i^2) \quad (0 \leq i \leq 59), \quad v_* = v_{59} = 1242895.$$

There are 1891 homogeneous quadratic monomials in these coordinates and δ , and all their weights are distinct. Reduction modulo three distinguishes the number of positive-weight factors. For two such factors, division by $6963 = 2 \cdot 59^2 + 1$ recovers the sum and sum of squares of their indices, hence their unordered pair. This is the same elementary construction used in the preceding sections.

There are $m = 1832$ target positions. Define

$$\begin{aligned} D_* &= 4v_* + 1 = 4971581, \\ T_* &= (m + 1)D_* + 2v_* = 9115393763, \\ t_j &= T_* + jD_* \quad (0 \leq j < m), \\ t_* &= 18218358574, \quad K = t_* + 1 = 18218358575. \end{aligned}$$

The exact checker verifies

$$\begin{aligned} D_* &> 2v_*, \quad T_* - 2v_* > v_*, \quad 2T_* - 2v_* > t_*, \\ L = 5^{16} &= 152587890625 > 3K + 2 = 54655075727. \end{aligned} \tag{12.1}$$

The 1830 original basis rows, 1832 target positions, and 1891 monomial weights count different objects.

For each ordinary homogeneous quadratic monomial, let a be its coefficient in its row and $c \in \{1, 2\}$ its multinomial coefficient in a square. Insert the integer $2a/c$ at the complementary exponent $t_j - w$, where w is the monomial weight. For the special row insert the coefficient 1 at t_{m-1} , complementary

to the weight zero of δ^2 . Let $\mathcal{D}(B)$ be the resulting signed coefficient polynomial. Its row intervals are disjoint by (12.1); within a row, weight uniqueness prevents coefficient collisions.

Choose a power of two $z \geq 2$ larger than the absolute values of all these coefficients, and put $Z = 2z$. Define

$$\begin{aligned}\ell_0(B) &= 1 + \sum_{i=1}^{59} B^{v_i} + \sum_{j=0}^{m-1} B^{t_j}, \\ e_0(B) &= z \sum_{h=0}^{K-1} B^h + \mathcal{D}(B), \\ V &= \ell_0(Z) + e_0(Z)Z^L.\end{aligned}$$

The coefficients of ℓ_0 are zero or one, and those of e_0 lie in $[1, Z - 1]$. Both codes have degree t_* . The mask includes 1892 non-input coordinates: the unit, 59 other genuine coordinates, and 1832 dummy row coordinates. Including the input gives 1893 coordinates; we use the convenient strict bound 1900b for their coefficient sum.

Choose the fixed power of two H so that

$$H > \max\{2Z^{2L+1}, 4^{t_*+3}Z 1900^2, 3L, 16\}. \quad (12.2)$$

This also gives $H > 4Z 1900^2$ and $H > 4Z$. All index construction depends only on the represented r.e. set, independently of the queried input x .

12.3 The changed system and recovery of the codes

Retain the reversed-packing system of Section 11, including its positive unknown $\sigma = S_3$. Make only the replacements

$$\mathcal{C} = xB + g, \quad T_3 = (B - 4)\ell.$$

Thus its coding and packing expressions are

$$\begin{aligned}B &= Hb^2, & Y &= \ell + eq, \\ Y + \mathcal{C}^2 + \alpha &= q^2, & Y &= V + t\theta, \\ S_3 &= (2e - Z\lambda)\mathcal{C}^2 + B\lambda(1 + q), & \sigma &= S_3, \\ S &= g + q^2(Y + q^2S_3), \\ T + 1 &= q^2(1 + \theta\lambda) - (b - 1)\ell + (B - 4)\ell q^4, \\ r &= S(n^2 - n) + (T + 1)(n^2 - 1).\end{aligned}$$

All the preceding geometric and Pell equations are retained.

The bootstrap remains valid before any unit coordinate has been decoded. Indeed $\mathcal{C} = xB + g > B$, since $x, g > 0$. The bound still gives $\mathcal{C} < q$ and $q > B$. Replacing $B - 2$ by the smaller positive number $B - 4$ preserves the complete bounds $0 < S < n$ and $0 \leq T < n$, including the treatment of a possibly negative first raw mask. The preceding borrow argument therefore applies unchanged: it first recovers the Pell conclusions and then proves that the first-block borrow is zero. The middle mask decodes

$$Y = \ell_0(B) + e_0(B)q, \quad \ell = \ell_0(B) + mq, \quad e = e_0(B) - m, \quad 0 \leq m < e_0(B).$$

The low first mask restricts the digits of g to the support of ℓ_0 , each less than b . Its unit digit is some $0 \leq \delta < b$. Position one is excluded from ℓ_0 , so the input digit of $\mathcal{C} = xB + g$ is exactly x . The

digit polynomial $\mathcal{C}(B)$ has degree at most t_* . Write

$$\mathcal{A} = \mathcal{C}(1)^2, \quad 1 \leq \mathcal{A} < 1900^2 b^2, \quad 4 \leq Z\mathcal{A} < B/4.$$

The lower bound follows from $x > 0$ and does not assume $\delta = 1$. Also $e, \mathcal{C} < B^K$, so $2e\mathcal{C}^2 < 2B^{3K} < q$.

To remove the remaining quotient ambiguity, put

$$X = (B - 4)m = \sum_{j=0}^{t_*+1} X_j B^j, \quad 0 \leq X_j < B.$$

The term $(B - 4)\ell_0$ is less than q . Thus the digits X_j occupy exactly the high third-mask positions L through $L + t_* + 1$. The raw-coefficient and carry calculation in Section 11 gives digits of S_3 at least $B - Z\mathcal{A}$ throughout this window. It applies unchanged, because the formula for S_3 is unchanged and its hypotheses were just proved. The third no-carry condition consequently implies $X_j \leq Z\mathcal{A} - 1$.

Consider the digit polynomial $F(U) = \sum_j X_j U^j$. Since $F(B) = (B - 4)m$, the integer $F(4)$ is divisible by $B - 4$. But (12.2) gives

$$0 \leq F(4) < Z\mathcal{A} 4^{t_*+2} < Z 1900^2 b^2 4^{t_*+2} < B/4 < B - 4.$$

Hence $F(4) = 0$. Nonnegativity of its coefficients forces every X_j to be zero, so $m = 0$. We have recovered both canonical codes. The evaluation point four is required here by the changed mask; the earlier evaluation-at-two argument is not used for this step.

12.4 The relaxed mask forces the unit

At every tested position, which is below K , the residual factor $e_0 - z\lambda$ agrees with $\mathcal{D}$. The support separation in (12.1) proves that dummy row coordinates cannot contribute to any target. Distinct rows cannot interfere, and quadratic weight uniqueness isolates the intended monomial within each row. The coefficient of $\mathcal{C}^2(e_0 - z\lambda)$ is therefore zero at every low mask position, twice the homogeneous residual at each ordinary target, and δ^2 at the special target.

Every coefficient of this product has absolute value at most $z\mathcal{A} < B/4$. Below position L , the offset in $S_3/2$ supplies the digit $B/2$ at every position. The resulting digits lie strictly between $B/4$ and $3B/4$, without carries between them. The even third block and mask can both be divided by two. The mask then has digit

$$B/2 - 2$$

at each selected position. Its only zero bits within a base- B digit are the unit bit and the highest bit. A digit in $(B/4, 3B/4)$ with no overlap with this mask must consequently be $B/2$ or $B/2 + 1$. Its centered coefficient is zero or one.

Every ordinary target coefficient is even, so every ordinary row vanishes. At the special target, $\delta^2 \in \{0, 1\}$ gives $\delta \in \{0, 1\}$. The ordinary guard $u\delta - x^2 = 0$ excludes $\delta = 0$ because $x > 0$. Thus $\delta = 1$, and the original quadratic residuals all vanish. This proves soundness without an additional system equation imposing the unit coordinate.

Conversely, given genuine witnesses, set $\delta = 1$ and $u = x^2$, and set all dummy row coordinates to zero. Choose the power of two b greater than x , u , and every original coordinate. The unit digit guarantees $g > 0$. Use the canonical code values. The degree margin and (12.2) give $Y + \mathcal{C}^2 < q^2$ and $Z\mathcal{C}^2 < q$, hence positive α and $\sigma = S_3$. The packed quotient is positive, as before. Ordinary target coefficients are zero and the special coefficient is one, so all masks hold. The preceding Pell necessity constructions furnish the remaining positive unknowns.

Theorem 12.1. *For each r.e. set, the effective admissible index (Z, V, H) above gives a system in 33 positive unknowns and 21 equations equivalent to membership of the positive input x . Its straight-line certificate has **110 arithmetic instructions**: 61 multiplications and 49 additions. Charging the seven fixed-numeral construction instructions gives 117 operations.*

Proof. The equivalence has just been proved. Relative to the 111-instruction schedule, replacing $1 + xB + g$ by $xB + g$ removes one addition. Replacing $B - 2$ by $B - 4$ has equal cost, and the numeral 4 is already used elsewhere. No positive system unknown or equality is added. The complete primitive list and all 21 residual identities, including the existing triangular corrections, are verified by `round8_1980_certificate.py`. The fixed exponent remains $L = 5^{16}$, so numeral construction still requires seven instructions. $\square$

13 Putting the input in the unit position

The supplied input can occupy the constant coefficient of the code. Exchanging its position with that of the encoded unit changes $\mathcal{C} = xB + g$ to $\mathcal{C} = x + g$, removing one multiplication. The main proof obligation is to recover the Pell conclusions without assuming $\mathcal{C} > B$. The geometric equation provides the weaker bound $B \leq q^2$, which suffices.

13.1 Remapping the homogeneous code

Retain the homogeneous rows, guard $u\delta - x^2$, special target δ^2 , and all numerical layout constants of Section 12. Give x weight zero and δ weight $v_0 = 1$. The original coordinates retain weights $v_1, \dots, v_{58}$, and u retains weight v_{59} . The mask becomes

$$\ell_0(B) = \sum_{i=0}^{59} B^{v_i} + \sum_{j=0}^{1831} B^{t_j}.$$

It excludes position zero, which now belongs to the input. There are still 1892 non-input encoded coordinates and 1893 coordinates including the input. The same bound $1900b$ applies to their sum.

Recompute each complementary coefficient position using the new monomial weights. The ordinary coefficient rule is still $2a/c$. The special row has coefficient 1 at $t_{1831} - 2$, since δ^2 now has weight two. The guard's two monomial weights are $v_{59} + 1$ and zero. All 1891 homogeneous monomial weights remain distinct: exchanging which coordinate has weight zero does not affect the number of positive-weight factors or the Sidon argument. All support-separation bounds consequently remain valid.

With the remapped signed polynomial $\mathcal{D}$, define

$$e_0(B) = z \sum_{h=0}^{K-1} B^h + \mathcal{D}(B), \quad V = \ell_0(Z) + e_0(Z)Z^L,$$

and use the same admissibility bound (12.2). These indices are again fixed independently of the input value. Make the sole arithmetic replacement

$$\mathcal{C} = x + g$$

in the 110-operation system. In particular $Y = \ell + eq$, $Y + \mathcal{C}^2 + \alpha = q^2$, $\sigma = S_3 > 0$, the mask $(B - 4)\ell$, and all Pell equations remain unchanged.

13.2 The weaker bounds before the Pell step

The positive bound gives

$$0 < \ell < Y < q^2, \quad e < q, \quad \mathcal{C} < q, \quad g < q.$$

The geometric equation and $\lambda \geq 1$ imply

$$q^2 - 1 = \lambda(B - 1) \geq B - 1, \quad B \leq q^2 \leq n = q^8. \quad (13.1)$$

Since $B = Hb^2$ and $H > 16$, we already have $q > 4b \geq 8$. Put $\mathcal{N} = (B - Z)\lambda$. Then

$$b < B - Z \leq \mathcal{N} < q^2 - 1, \quad B\lambda < 2q^2.$$

No inequality $\lambda > q$ is needed. The imposed positive register and its elementary upper bound give

$$0 < S_3 < 2q^3 + 2q^2(1 + q) < 5q^3 < q^4.$$

Thus $S = g + q^2(Y + q^2 S_3)$ satisfies $0 < S < n$.

For completeness, the possibly negative first raw mask still has an exact normalized packing. Set

$$d_0 = \left\lfloor \frac{(b-1)\ell}{q^2} \right\rfloor, \quad \varepsilon_0 = (b-1)\ell - d_0 q^2.$$

Then $0 \leq d_0 < b$, $0 \leq \varepsilon_0 < q^2$, and

$$T = (q^2 - 1 - \varepsilon_0) + q^2(\mathcal{N} - d_0) + q^4(B - 4)\ell.$$

The first two blocks belong to $[0, q^2)$, since $\mathcal{N} > b > d_0$. By (13.1), the last block satisfies $0 < (B - 4)\ell < q^4$. Hence $0 \leq T < n$, and

$$r = S(n^2 - n) + (T + 1)(n^2 - 1) \geq n^2 - 1 \geq n.$$

This establishes all packing bounds before decoding or any conclusion that $B < q$.

13.3 The Pell bootstrap with $B \leq n$

Here is the dependency check for the relaxed radix comparison. The information now available is

$$3 < 3L \leq B \leq n \leq r, \quad q, b \leq n, \quad n, r \geq 8. \quad (13.2)$$

Moreover B is even because the fixed parameter H is an even power of two, independently of any assertion about b .

Write $N = n$, $R = r$, $U = wN^2$, $Y_P = sN^2$, $M = RY_P$, $a = M(U + 1)$, $X_P = 2UM^2$, $P = X_P + 1$, and $J = 2R + 1$. We have $U, Y_P \geq 64$ and $M \geq 512$. The index congruence gives $k \geq R + 2$, and the positive interval $Y_P < c/k < Y_P + 1$ gives $c > J$. The signed Pell block therefore gives $c = \psi_a(J)$ and $d = \chi_a(J)$. The growth and index-congruence argument of Section 10 gives $k = \psi_P(R + 1)$ and

$$\xi \left(1 - \frac{R}{a} - \frac{R}{X_P} \right) < \frac{c}{k} < \xi, \quad \xi = \frac{(U + 1)^{2R}}{U^R}, \quad \frac{R}{a} + \frac{R}{X_P} < \frac{2}{Y_P(U + 1)} < \frac{1}{2}.$$

These arguments use the numerical lower bounds just stated, not a comparison between B and q . The interval consequently gives $Y_P > U^{R-1}$ and $a > RU^R$.

The base-two exponent argument has the same strict size bounds:

$$2^{3J} = 8 \cdot 8^{2R} \leq 8N^{2R} \leq RU^R < a, \quad (bw)^3 \leq U^R < a.$$

Here $b \leq N$ gives $bw \leq U$. Equation E14 and the exponent lemma yield $bw = 2^J$; thus b is a power of two and $U > 4 \cdot 2^{2R}$.

Next (13.2) gives $0 < L < J < a$. The positive gap $c > \kappa$ and equations E19, E20 give $\kappa = \psi_a(L)$ and $\mu = \chi_a(L)$. The second exponent argument now uses

$$B^{3L} \leq B^B \leq N^N \leq U^R < a, \quad q^3 \leq N^N \leq U^R < a. \quad (13.3)$$

These follow from $3L \leq B \leq N$, $q \leq N$, and $R \geq N$. Also $a > N \geq B$, so the modulus in E18 has its required positive sign. Equation E18 therefore gives $q = B^L$. Only now do we infer $B < q$.

As B was already even, q and N are even. To make the final rounding dependency explicit, the exact-index estimates give

$$0 < \xi - c/k < 6/(U+1) < 1/4, \quad 0 < \xi - \lfloor \xi \rfloor < 1/4, \quad \lfloor \xi \rfloor \equiv \binom{2R}{R} \pmod{U}.$$

The interval makes $|Y_P - \lfloor \xi \rfloor| < 2$. Both integers are even, so they are equal. Since $Y_P = sN^2$ and $N^2 \mid U$, we obtain $N^2 \mid \binom{2R}{R}$. The binomial criterion and the packing lemma now give $\tau_2(S, T) = 0$. Thus the complete Pell conclusion was recovered using (13.1), without an unproved early inequality $B < q$.

13.4 Decoding and positive necessity

The subsequent first-block borrow argument applies unchanged. It proves $d_0 = 0$ and decodes

$$\ell = \ell_0(B) + mq, \quad e = e_0(B) - m, \quad 0 \leq m < e_0(B).$$

Since $g < q$, the first mask restricts its digits to the positive positions of ℓ_0 . Its unit digit is zero. The inequality $x < b < B$ then ensures that $\mathcal{C} = x + g$ has exactly the supplied input as its unit digit, without a carry; its digit at position one is δ .

The sum of the coefficients of this digit polynomial is positive and less than $1900b$. All degree and coefficient estimates in Section 12 still hold. The same high-digit window bounds the digits of $(B-4)m$ by $ZC(1)^2 - 1$; evaluating its digit polynomial at four forces $m = 0$. The canonical relaxed mask then forces every ordinary homogeneous residual and the guard to vanish, while its special target gives $\delta^2 \in \{0, 1\}$. The guard and $x > 0$ give $\delta = 1$, so all original residuals vanish.

Conversely, choose genuine original coordinates, set $\delta = 1$ and $u = x^2$, and choose a power of two b greater than the input and every encoded coordinate. The digit δ at position one ensures $g \geq B > 0$. The canonical code has the same degree and coefficient bounds as before, giving positive α , σ , and the packed quotient. All masks hold. In this necessity direction the intended code even satisfies $\mathcal{C} > B$, so the preceding Pell necessity constructions retain their stronger original bounds.

Theorem 13.1. *For each r.e. set, the remapped effective admissible index gives a membership-equivalent system in 33 positive unknowns and 21 equations with a straight-line certificate of **109 arithmetic instructions**: 60 multiplications and 49 additions. Generating the fixed numerals from 1 adds seven instructions, giving 116 operations.*

Proof. The preceding arguments establish equivalence with all domains preserved. Replacing $xB + g$ by $x + g$ removes exactly one multiplication from the 110-operation schedule. No other arithmetic instruction, unknown, or equality is added. The complete primitive list and all 21 source residual identities, including the triangular corrections, are verified by `round9_1980_certificate.py`. $\square$

14 A positive coefficient and a triangular Pell equation

Two independent changes remove another addition and another multiplication. The first uses positivity of an already computed coefficient to recover a bound that previously required an explicit addition. The second rewrites the first Pell equation using its odd root and proves that its index modulus can be halved. Each separately gives 108 instructions; together they give 107.

14.1 Removing the square from the explicit bound

Keep the fixed input-unit encoding of Section 13, with $\mathcal{C} = x + g$ and $Y = \ell + eq$. Replace

$$Y + \mathcal{C}^2 + \alpha = q^2$$

by the equations

$$Y + \alpha = q^2, \quad \Omega = Z\lambda - 2e, \quad \Omega > 0. \quad (14.1)$$

Here Ω is a new positive system unknown. Its defining value is the negative of an existing register. Reverse that subtraction and compute

$$P_1 = (Z\lambda - 2e)\mathcal{C}^2, \quad P_2 = B\lambda(1 + q), \quad S_3 = P_2 - P_1.$$

The value of S_3 is unchanged. Reversing these signs has equal arithmetic cost, and Ω is connected to the computed coefficient by a free equality test. Retain the separate positive unknown $\sigma = S_3$.

Proposition 14.1. *The replacements (14.1) give a membership-equivalent system with 108 arithmetic instructions: 60 multiplications and 48 additions. It has 34 positive unknowns and 22 equations.*

Proof. Before decoding, the new bound gives $\ell < Y < q^2$ and $e < q$. The geometric equation still gives $B \leq q^2$ and $q > 4b \geq 8$. Consequently

$$P_2 = B\lambda(1 + q) < 2q^2(1 + q) < 3q^3.$$

Since Ω and σ are positive integers,

$$0 < S_3 = P_2 - \Omega\mathcal{C}^2 < P_2, \quad \mathcal{C}^2 < P_2/\Omega \leq P_2 < 3q^3 < q^4.$$

Thus $g < \mathcal{C} < q^2$. This weaker bound is enough for the first packing block, whose width is q^2 . Together with $Y < q^2$ and $S_3 < q^4$, it gives $0 < S < n = q^8$.

The mask bounds use only $\ell < q^2$, $B \leq q^2$, and $b < (B - Z)\lambda < q^2 - 1$. They are unchanged. Explicitly, if $d_0 = \lfloor (b - 1)\ell/q^2 \rfloor$ and $\varepsilon_0 = (b - 1)\ell - d_0q^2$, then

$$T = (q^2 - 1 - \varepsilon_0) + q^2((B - Z)\lambda - d_0) + q^4(B - 4)\ell$$

has normalized blocks in $[0, q^2)$, $[0, q^2)$, and $[0, q^4)$. Hence $0 \leq T < n$ and $r \geq n^2 - 1 \geq n$. The relaxed Pell bootstrap proved in Section 13 now applies: it uses $3L \leq B \leq n \leq r$, $q, b \leq n$, and the unchanged Pell equations, without a further bound on g or $\mathcal{C}$. It gives $q = B^L$, the required powers of two, and $\tau_2(S, T) = 0$.

We recover the stronger bound before using it in digit decoding. Now $q = B^L > B$, so $\lambda > q$. Since $e < q$ and $Z \geq 4$,

$$2e < 2q < 2\lambda \leq Z\lambda/2, \quad \Omega > Z\lambda/2.$$

It follows that

$$\mathcal{C}^2 < \frac{P_2}{\Omega} < \frac{2B(1 + q)}{Z} \leq \frac{B(1 + q)}{2} < \frac{q(q + 1)}{2} < q^2.$$

Thus $g < \mathcal{C} < q$ holds before the first-mask decoding and high-mask quotient argument require it. Those arguments now apply unchanged, proving canonical code recovery and membership.

Conversely, take the positive witnesses constructed for the 109-operation system. They have $e < q$, $\lambda > q$, and $S_3 > 0$. Set $\Omega = Z\lambda - 2e > 0$ and replace the slack by $\alpha = q^2 - Y > 0$. All other witnesses remain valid.

Only the addition of $\mathcal{C}^2$ to Y disappears. The coefficient subtraction, its product with $\mathcal{C}^2$, and the subtraction defining S_3 have the same costs as before. The positive unknown Ω introduces one free equality and no arithmetic instruction beyond these existing registers. The complete check is `round10_1980_certificate.py`. $\square$

14.2 Factoring the odd Pell root

For this independent change to the 109-operation system, write

$$\begin{aligned} N = n, \quad R = r, \quad U = wN^2, \quad Y_P = sN^2, \quad M = RY_P, \\ a = M(U + 1), \quad Q = UM^2, \quad P = 2Q + 1, \quad J = 2R + 1. \end{aligned}$$

The quantity Q is already computed. The old first Pell norm and index equations were

$$((2Q + 1)^2 - 1)k^2 + 1 = \tau_{\text{old}}^2, \quad k = R + 1 + 2Qh_{\text{old}}.$$

Replace them by

$$\tau(\tau + 1) = Q(Q + 1)k^2, \quad k = R + 1 + hQ, \quad \tau, h > 0. \quad (14.2)$$

The mathematical Pell parameter $P = 2Q + 1$ need not be computed by the new certificate.

Proposition 14.2. *Equations (14.2) give a positive-domain equivalent system with 108 arithmetic instructions: 59 multiplications and 49 additions. This independent variant retains 33 positive unknowns and 21 equations.*

Proof. For necessity, P is odd, and the recurrence for $\chi_P(t)$ shows that every such coordinate is odd. The old positive norm has $\tau_{\text{old}} > 1$. Thus

$$\tau = (\tau_{\text{old}} - 1)/2 \geq 1, \quad h = 2h_{\text{old}} > 0$$

are integers satisfying (14.2). The factored norm is simply the old norm identity divided by four. The certificate verifies the new polynomial equation directly; this witness map does not introduce an uncharged division instruction.

For sufficiency, the initial coding bounds give $R \geq N \geq 8$, so $U, Y_P \geq 64$, $M \geq 512$, and $Q > R + 1$. Put $\hat{\tau} = 2\tau + 1$. The new norm implies

$$\hat{\tau}^2 - (P^2 - 1)k^2 = 1, \quad \hat{\tau} > 1.$$

The new congruence gives $k \geq R + 2$. The unchanged positive interval gives $c > Y_P k \geq 64(R + 2) > J$. The signed Pell block therefore gives $c = \psi_a(J)$ and $d = \chi_a(J)$.

There is a positive index t with $k = \psi_P(t)$ and $\hat{\tau} = \chi_P(t)$. The elementary congruence $\psi_P(t) \equiv t \pmod{P - 1}$ also holds modulo Q . Since $0 < R + 1 < Q$, the new index equation implies

$$t = R + 1 + vQ, \quad v \geq 0.$$

If $v \geq 1$, the elementary Pell growth estimates give

$$\frac{c}{k} \leq \frac{(2a)^{2R}}{(2P - 1)^{R+Q}} \leq \left(\frac{2a}{2P - 1} \right)^{2R} < \frac{1}{2}.$$

Here $Q \geq R$, and $2a/(2P - 1) = 2M(U + 1)/(4UM^2 + 1) < 1/2$ follows from $U \geq 64$ and $M \geq 512$. This contradicts $c/k > Y_P \geq 64$. Therefore $v = 0$ and $k = \psi_P(R + 1)$.

Applying the same Pell congruence now modulo $2Q$ gives $k \equiv R + 1 \pmod{2Q}$. Since $k - R - 1 = hQ$, the new positive integer h is even. Consequently

$$h_{\text{old}} = h/2 > 0, \quad \tau_{\text{old}} = 2\tau + 1 > 0$$

restore both old equations, preserving every other witness. This argument precedes the exponent relations and binomial rounding; it does not assume either of them to prove the required evenness.

After Q is available, the old norm block used seven operations: form $2Q$, add two, multiply these two values, square k , multiply, add one, and square τ_{old} . The new block uses six: form $Q + 1$ and $Q(Q + 1)$, square k , multiply, form $\tau + 1$, and multiply $\tau(\tau + 1)$. The index product hQ has the same cost as the former $2Qh_{\text{old}}$, so $2Q$ is no longer needed anywhere in the schedule. The exact saving is one multiplication, as checked by `round11_1980_pell_certificate.py`. $\square$

14.3 Combining the two savings

Theorem 14.3. *For each r.e. set, the fixed admissible input-unit encoding gives a membership-equivalent system in 34 positive unknowns and 22 equations whose straight-line certificate has **107 arithmetic instructions**: 59 multiplications and 48 additions. Equality tests and supplied parameters are free. Generating the fixed numerals from 1 adds seven instructions, giving 114 operations.*

Proof. Apply both replacements to the 109-operation system. Their necessity maps affect disjoint witnesses: the coefficient-bound change replaces α and supplies Ω , while the Pell change replaces τ and h . Thus both maps preserve the other change's equations and positivity.

For soundness, first use $\Omega, \sigma > 0$ to establish the preliminary bounds $0 < S < n$, $0 \leq T < n$, and $R \geq N \geq 8$, as in Proposition 14.1. The index proof in Proposition 14.2 then restores the old positive Pell witnesses without any exponent or decoding assumption. The positive-coefficient proof can now continue with those witnesses and recover membership. This order establishes the equivalence without a circular appeal between the two changes.

The combined schedule removes one addition and one multiplication from 109 instructions. The coefficient change supplies the sole extra positive unknown and free equality; the Pell change alters neither count. The complete primitive list and all residual identities are checked by `round12_1980_certificate.py`. The numeral set and exponent $L = 5^{16}$ are unchanged. $\square$

15 A smaller fixed support

We use a smaller support for the subsequent constructions. Replace the support positions and exponent of Section 13, keeping its arithmetic structure. The resulting certificate still has 107 instructions. Fixed numerals are free in the complexity measure used for the reductions below.

15.1 Pair recovery over a finite field

For $0 \leq i \leq 59$, let r_i be the least nonnegative residue of i^2 modulo 61, and put

$$a_i = 122i + r_i, \quad v_i = 1 + 3a_i.$$

The sums $a_i + a_j$ distinguish unordered pairs, including repeated indices. Indeed, $0 \leq r_i + r_j \leq 120 < 122$, so division by 122 recovers the integer sum $s = i + j$ and the residue sum $r_i + r_j$. Over the field with

61 elements these determine

$$i + j = s, \quad ij = (s^2 - r_i - r_j)/2.$$

Thus the monic polynomial with roots i, j is determined. Since the indices $0, \dots, 59$ are distinct modulo 61, the integer pair is determined as well. The division by 2 occurs in this proof of the fixed support construction; it is not a certificate instruction.

Every v_i is 1 modulo 3. Hence zero, a single positive weight, and a pair sum are distinguished modulo 3. Within each class the weights are distinct by the preceding argument. There are consequently $1 + 60 + 1830 = 1891$ distinct homogeneous quadratic monomial weights in the 61 coordinates. As before, the input has weight zero, δ has weight $v_0 = 1$, the original witnesses have weights $v_1, \dots, v_{58}$, and u has weight v_{59} .

With $m = 1832$ target rows, use the same formulas for their spacing:

$$\begin{aligned} v_* &= 21607, & D_* &= 4v_* + 1 = 86429, \\ T_* &= (m + 1)D_* + 2v_* = 158467571, & t_j &= T_* + jD_*, \\ t_* &= 316719070, & K &= t_* + 1 = 316719071, & L &= 2^{32}. \end{aligned}$$

The exact margins are

$$\begin{aligned} D_* &> 2v_*, & T_* - 2v_* &> v_*, & 2T_* - 2v_* &> t_*, \\ 3K + 2 &= 950157215 < 2^{30} < L. \end{aligned}$$

Different row intervals $[t_j - 2v_*, t_j]$ are disjoint. A product of a coefficient term and an ordinary quadratic monomial lies within $2v_*$ of its row target, so no other row contributes there. Coefficient terms lie above every low variable position, and products involving a dummy row coordinate lie above every target. These are all the support facts used in the preceding decoding proof.

Retain twice each ordinary homogeneous residual and the special coefficient δ^2 . In particular, its coefficient 1 is placed at $t_{m-1} - 2$. Recompute D, ℓ_0, e_0, V with the new positions, choose the power of two $z \geq 2$ above the absolute coefficients of D , put $Z = 2z$, and choose a power of two

$$H > \max\{2Z^{2L+1}, 4^{t_*+3}Z^{1900^2}, 3L, 16\}.$$

The number of coordinates remains 1893, so their coefficient sum remains less than $1900b$. All bounds in the input-unit and positive-coefficient proofs are therefore preserved. In particular, $e, \mathcal{C} < B^K$ and $L > 3K + 2$ still give $2e\mathcal{C}^2 < q$ after exponent recovery, and evaluation at 4 still rules out the high-mask quotient ambiguity. The fixed indices Z, V, H are computed from the represented set, independently of the queried input.

15.2 The even exponent and the preserved certificate

No preceding Pell argument requires L to be odd. The signed congruence uses the odd index $J = 2r + 1$. The other index argument uses only $0 < L < J < a$ and the congruence $\psi_a(t) \equiv t \pmod{a-1}$, valid for every t . These hypotheses still hold because $3L < B \leq n \leq r$. Necessity gives the positive integer $\Delta = (\psi_a(L) - L)/(a-1)$ since $L > 1$, and the positive gap follows from $J > L$. The exponent comparison $B^{3L} \leq B^B \leq n^n \leq U^r < a$ is unchanged. Thus $q = B^L$ is recovered, and every decoding and necessity argument still applies.

Theorem 15.1. *The modular support gives a membership-equivalent system in 34 positive unknowns and 22 equations with **107 arithmetic instructions**: 59 multiplications and 48 additions.*

Proof. The support and exponent arguments establish the same encoding theorem. The explicit schedule moves the existing register $a_- = a - 1$ before E14 and replace its computations of $4a$ and $4a - 5$ by

$$a_4 = 4a_-, \quad a_{4-5} = a_4 - 1.$$

The latter is identically $4a - 5$, so all 107 arithmetic instructions are preserved. The complete schedule is serialized and checked by `round13_1980_certificate.py`. It checks every primitive and source residual identity and all 1891 monomial weights and support margins. Its 59 multiplications and 48 additions have exactly the same costs as in Appendix H. $\square$

16 A base-four congruence and the main Pell parameter

The third digit mask already computes $B - 4$. This permits a useful joint change: replace the base-two exponent test by a base-four test, and store the main Pell parameter minus four. The two expressions $A - 4$ and $A - B$ then share available values, eliminating one subtraction. The altered Pell approximation lies strictly above its binomial expression; this also makes the final rounding argument independent of parity.

Use the 107-operation system of Theorem 14.3, with the fixed modular-Sidon encoding and $L = 2^{32}$ from Section 15. In this section Y_P denotes the Pell quantity sn^2 , to distinguish it from the packed code. Write

$$\begin{aligned} N &= n, & R &= r, & U &= wN^2, & Y_P &= sN^2, & M &= RY_P, \\ a_0 &= M(U + 1), & A &= a_0 + 4, & Q &= UM^2, & P &= 2Q + 1, \\ J &= 2R + 1, & W &= bw, & \xi &= \frac{(U + 1)^{2R}}{UR}. \end{aligned}$$

The supplied positive unknown named a in the certificate now means a_0 . Equation E12 still defines $a_0 = M(U + 1)$. Every main Pell equation uses the mathematical parameter $A = a_0 + 4$, including $G = 1 + (A + 1)(f^2 - 1)$ in the signed auxiliary block and the modulus $A - 1$ in E20. The first Pell norm and index equations retain the forms

$$\tau(\tau + 1) = Q(Q + 1)k^2, \quad k = R + 1 + hQ. \quad (16.1)$$

The positive interval remains

$$c = kY_P + \eta, \quad k = \eta + \zeta, \quad \eta, \zeta > 0, \quad Y_P < \frac{c}{k} < Y_P + 1. \quad (16.2)$$

Replace E14 by

$$d = W + c(A - 4) + \gamma(8A - 17). \quad (16.3)$$

This is the source's χ exponent congruence for base four. The second exponent congruence retains the form

$$\mu = q + \kappa(A - B) + \rho(2AB - B^2 - 1). \quad (16.4)$$

16.1 Exact indices and the shifted approximation

The pre-Pell part of Proposition 14.1 uses only the encoding equations and gives

$$N = q^8, \quad q > 4b \geq 8, \quad 3L \leq B \leq N \leq R, \quad b, q \leq N.$$

In particular $N, R \geq 64$, $U, Y_P \geq 4096$, $M \geq 262144$, $Q > R + 1$, and $A > J > 1$. These estimates precede both exponent relations; the stronger $N \geq 64$ will be needed for base four.

Since $h > 0$, equation (16.1) gives $k \geq R + 2$. The interval gives $c > Y_P(R + 2) > J$. The generic signed Pell argument of Proposition 7.2 therefore applies with parameter A and gives

$$c = \psi_A(J), \quad d = \chi_A(J). \quad (16.5)$$

The triangular norm gives the ordinary Pell equation for $\hat{\tau} = 2\tau + 1$ and $P = 2Q + 1$, so $k = \psi_P(t)$ for some positive index t . The congruence $\psi_P(t) \equiv t \pmod{2Q}$, reduced modulo Q , implies

$$t = R + 1 + vQ, \quad v \geq 0.$$

If $v \geq 1$, the elementary Pell growth bounds give

$$\frac{c}{k} \leq \frac{(2A)^{2R}}{(2P-1)^{R+Q}} \leq \left(\frac{2A}{2P-1} \right)^{2R} < \frac{1}{2}.$$

Here $Q \geq R$, and $2A/(2P-1) < 1/2$ follows, for example, from

$$a_0 < 2UM, \quad 4A < 8UM + 16 < 4UM^2 + 1 = 2P - 1.$$

This contradicts (16.2). Consequently

$$k = \psi_P(R + 1), \quad \hat{\tau} = \chi_P(R + 1). \quad (16.6)$$

No exponent or rounding conclusion has been used to recover either Pell index.

At these indices, the lower growth estimate gives

$$\begin{aligned} \frac{c}{k} &\geq \frac{(2A-1)^{2R}}{(2P)^R} \\ &= \xi \left(1 + \frac{7}{2a_0} \right)^{2R} \left(1 + \frac{1}{2Q} \right)^{-R} > \xi. \end{aligned} \quad (16.7)$$

Indeed $14Q > a_0$, so $(1 + 7/(2a_0))^2 > 1 + 7/a_0 > 1 + 1/(2Q)$. The upper growth estimate gives

$$\begin{aligned} \frac{c}{k} &\leq \frac{(2A)^{2R}}{(2P-1)^R} < \xi \left(1 + \frac{4}{a_0} \right)^{2R} \\ &< \xi \left(1 + \frac{16R}{a_0} \right) = \xi \left(1 + \frac{16}{Y_P(U+1)} \right). \end{aligned} \quad (16.8)$$

For the last step, binomial coefficients give $(1+t)^m \leq \sum_{j \geq 0} (mt)^j = (1-mt)^{-1} < 1 + 2mt$ when $0 < mt < 1/2$. Here $mt = 8R/a_0 = 8/[Y_P(U+1)] < 1/2$.

The lower bound and the interval imply $\xi < Y_P + 1$. Since $\xi > U^R$, integrality gives $Y_P \geq U^R$ and hence

$$A > a_0 = RY_P(U+1) > RU^R. \quad (16.9)$$

This size bound is established before invoking an exponent lemma. Also $\xi < Y_P + 1 < 2Y_P$, so the upper bound yields

$$0 < \frac{c}{k} - \xi < \frac{32}{U+1} < \frac{1}{2}. \quad (16.10)$$

16.2 The exponent relations and rounding

Since $W = bw \leq U$ and $N, R \geq 64$, we have

$$4^{3J} = 64 \cdot 64^{2R} \leq 64N^{2R} \leq RU^R < A, \quad W^3 \leq U^R < A.$$

The χ version of source Lemma 2.22, explicitly stated after that lemma, applies to (16.3) and (16.5). It proves

$$bw = 4^J. \quad (16.11)$$

Thus b is a power of two, and

$$U = N^2w \geq Nbw = N4^J > 4 \cdot 2^{2R}.$$

E13, E19 and E20 give $0 < \kappa < c$, its Pell norm, and $\kappa \equiv L \pmod{A-1}$. Since $0 < L < J < A$, source Lemma 2.23 gives $\kappa = \psi_A(L)$ and $\mu = \chi_A(L)$. Moreover

$$B^{3L} \leq B^B \leq N^N \leq U^R < A, \quad q^3 \leq N^N \leq U^R < A.$$

The same exponent lemma, applied now to (16.4), proves $q = B^L$.

Let $F = \lfloor \xi \rfloor$. Source Lemma 2.24, or its direct binomial expansion, gives

$$0 < \xi - F < \frac{2^{2R}}{U} < \frac{1}{4}, \quad F \equiv \binom{2R}{R} \pmod{U}. \quad (16.12)$$

Combining the interval with (16.7) and (16.10) gives

$$F < \xi < \frac{c}{k} < Y_P + 1, \quad Y_P < \frac{c}{k} < \xi + \frac{1}{2} < F + \frac{3}{4}.$$

Both F and Y_P are integers. The first chain gives $F \leq Y_P$; the second gives $Y_P \leq F$. Thus

$$Y_P = F, \quad N^2 \mid \binom{2R}{R}. \quad (16.13)$$

This argument requires no parity assertion. The power-of-two facts for b and q , the relation $q = B^L$, and the divisibility in (16.13) are precisely the conclusions needed by the unchanged no-carry and modular-Sidon decoding proof. They therefore imply membership in the represented r.e. set.

16.3 Constructing positive witnesses

For the converse, start from a solution of the represented system and take its canonical modular-Sidon coding witnesses. Choose the usual sufficiently large power of two b , put $B = Hb^2$, $q = B^L$, $N = q^8$, and construct the unchanged packed S, T, R . Their no-carry property gives $N^2 \mid \binom{2R}{R}$. These choices precede the Pell witnesses and supply $b \leq R$ and $R \geq N \geq 64$.

Put $J = 2R + 1$ and choose

$$w = 4^J/b, \quad U = N^2w, \quad Y_P = \left\lfloor \frac{(U+1)^{2R}}{U^R} \right\rfloor, \quad s = Y_P/N^2.$$

Writing $b = 2^t$, the inequality $t \leq b \leq R < 2J$ shows that w is a positive integer. The binomial expansion and central-binomial divisibility make s a positive integer. Define $M = RY_P$, $a_0 = M(U+1)$, $A = a_0 + 4$, $Q = UM^2$, $P = 2Q + 1$, and

$$c = \psi_A(J), \quad d = \chi_A(J), \quad k = \psi_P(R+1).$$

The ratio estimates above hold at these chosen indices. Here $\xi < Y_P + 1$ is immediate from the definition of Y_P , so

$$Y_P < \xi < \frac{c}{k} < \xi + \frac{1}{2} < Y_P + \frac{3}{4}.$$

Consequently

$$\eta = c - kY_P > 0, \quad \zeta = k - \eta > 0, \quad \tau = \frac{\chi_P(R+1) - 1}{2}, \quad h = \frac{k - R - 1}{Q}$$

are positive integers. Positivity and integrality of τ follow from the oddness of the χ sequence for odd P ; those of h follow from the Pell congruence modulo $2Q$ and $\psi_P(R+1) > R+1$. Thus the triangular norm, index equation, positive interval, E12, and E15 all hold.

Choose positive f, i with $f^2 - (A^2 - 1)(ic^2)^2 = 1$ using the same generic Pell divisibility construction as in Proposition 7.2. Put

$$G = 1 + (A+1)(f^2 - 1), \quad I = \chi_G(J), \quad H_J = \psi_G(J), \quad o = (I + d)/f, \quad j = (H_J - J)/c.$$

Because $G \equiv -A \pmod{f}$, $G \equiv 1 \pmod{c}$, and J is odd, o and j are integers. They are positive since $I + d > 0$ and $\psi_G(J) > J$. These values satisfy the signed E17 with $of - d = I$. Next choose

$$\kappa = \psi_A(L), \quad \mu = \chi_A(L), \quad \Delta = \frac{\kappa - L}{A - 1}, \quad \phi = c - \kappa.$$

Since $J > L \geq 2$, these are positive integers with the required Pell norm, congruence and gap.

The exponent lemma supplies integer quotients γ, ρ in (16.3) and (16.4). To check their positivity, use $\chi_A(t) = A\psi_A(t) - \psi_A(t-1)$ to obtain

$$d - (A-4)c > 3c > W, \quad \mu - (A-B)\kappa > (B-1)\kappa > q.$$

For the last inequalities, $A > W^3, q^3$ and $J, L \geq 2$ give $c, \kappa \geq A$. The moduli are positive because $A > B \geq 2$ and $A > 4$. Hence $\gamma, \rho > 0$. All coding witnesses, including Ω and σ , remain the canonical ones. This constructs every required positive witness. In particular the construction re-chooses w, s, a_0 and their dependent Pell coordinates; it does not preserve the predecessor's base-two value of w .

16.4 The complete arithmetic count

Theorem 16.1. *For each r.e. set, the fixed modular-Sidon encoding gives a membership-equivalent system in 34 positive unknowns and 22 equations with a straight-line certificate of **106 arithmetic instructions**: 59 multiplications and 47 additions. Equality tests and supplied parameters are free, as are fixed numerals.*

Proof. The preceding arguments prove both directions over positive integers. The arithmetic expressions are evaluated as

$$\begin{aligned} A - 4 &= a_0, & A - 1 &= a_0 + 3, & A + 1 &= (A - 1) + 2, \\ A^2 - 1 &= (A - 1)(A + 1), & 8A - 17 &= 8a_0 + 15, & A - B &= a_0 - (B - 4). \end{aligned}$$

The previous subtraction computing the base-two coefficient $a - 2$ disappears. The other expressions cost the same numbers of operations as their predecessors; in particular $B - 4$ is already computed by the third mask, and the defining equation for a_0 retains its existing factorization. No register for A itself is required.

The checker `round14_1980_base_four_certificate.py` verifies all source residuals, both triangular corrections, every shifted-register identity, and the full primitive instruction list. It counts 59 multiplications and 47 additions. The positive input and equality counts are unchanged. Its generated schedule records the exponent $L = 2^{32}$. $\square$

17 Forcing a multiple of the Pell index

The first Pell index can be isolated with a single product in place of an affine congruence. Arrange that its parameter is congruent to one modulo the intended index. Then require its denominator to be a positive multiple of that index. Pell growth excludes every multiple after the first.

Retain the fixed encoding and the shifted main parameter from Section 16, but change the intermediate scale to

$$N = n, \quad R = r, \quad U = wN^2, \quad Y_P = sN^2, \quad D = R + 1, \\ M = DY_P, \quad a_0 = M(U + 1), \quad A = a_0 + 4, \quad Q = UM^2, \quad P = 2Q + 1, \quad J = 2R + 1.$$

The supplied unknown a still denotes a_0 . Replace the old equation $k = R + 1 + hQ$ by

$$k = h(R + 1) = hD, \quad h > 0. \quad (17.1)$$

Retain the triangular norm and positive interval

$$\tau(\tau + 1) = Q(Q + 1)k^2, \quad c = kY_P + \eta, \quad k = \eta + \zeta, \quad \tau, \eta, \zeta > 0. \quad (17.2)$$

The scale M is an intermediate value, so the changed scale affects the expanded source polynomials E9 and E12; equation (17.1) changes E11. All other source polynomials in the supplied variables are unchanged.

17.1 Isolating the first positive multiple

The unchanged pre-Pell coding proof supplies

$$N = q^8, \quad q > 4b \geq 8, \quad 3L \leq B \leq N \leq R, \quad b, q \leq N.$$

Thus $N, R \geq 64$, $U, Y_P \geq 4096$, and $M \geq 262144$. Equation (17.1) gives $k \geq R + 1$, and the interval in (17.2) gives

$$c > Y_P k \geq Y_P(R + 1) > 2R + 1 = J.$$

The generic signed Pell argument therefore gives $c = \psi_A(J)$ and $d = \chi_A(J)$.

Put $\hat{\tau} = 2\tau + 1$. The norm in (17.2) is precisely

$$\hat{\tau}^2 - (P^2 - 1)k^2 = 1, \quad \hat{\tau} > 1.$$

Hence $k = \psi_P(t)$ for a positive integer t . The new scale makes

$$P - 1 = 2Q = 2U(R + 1)^2 Y_P^2$$

divisible by D . The elementary Pell congruence $\psi_P(t) \equiv t \pmod{P - 1}$ therefore gives $k \equiv t \pmod{D}$. Equation (17.1) yields

$$t = mD = m(R + 1), \quad m \geq 1. \quad (17.3)$$

If $m \geq 2$, then $t - 1 \geq 2R + 1$, and Pell growth gives

$$\frac{c}{k} \leq \frac{(2A)^{2R}}{(2P - 1)^{2R+1}} < \left(\frac{2A}{2P - 1} \right)^{2R} < \frac{1}{2}.$$

The last inequality follows from $2A/(2P - 1) < 1/2$; with the shifted parameter this is equivalent to

$$4UM(M - 1) - 4M - 15 = 4M(U(M - 1) - 1) - 15 > 0.$$

The initial bounds already imply it. This contradicts $c/k > Y_P$. Consequently

$$k = \psi_P(R + 1), \quad \hat{\tau} = \chi_P(R + 1). \quad (17.4)$$

The conclusion uses no exponent relation or binomial rounding. No parity condition on the positive quotient h is required.

17.2 Preserving the approximation and the encoding

At the recovered indices, the principal ratio still cancels the scale:

$$\frac{(2M(U+1))^{2R}}{(4UM^2)^R} = \xi := \frac{(U+1)^{2R}}{U^R}.$$

The lower estimate of Section 16 uses only $14Q > a_0$, which remains true. Its upper estimate uses

$$\frac{8R}{a_0} = \frac{8R}{(R+1)Y_P(U+1)} < \frac{8}{Y_P(U+1)} < \frac{1}{2}.$$

Thus the same growth and binomial estimates give

$$\xi < \frac{c}{k} < \xi \left(1 + \frac{16R}{a_0}\right) < \xi \left(1 + \frac{16}{Y_P(U+1)}\right). \quad (17.5)$$

In particular $\xi < Y_P + 1$, so $Y_P \geq U^R$ and

$$A > a_0 = (R+1)Y_P(U+1) > RU^R, \quad 0 < \frac{c}{k} - \xi < \frac{32}{U+1} < \frac{1}{2}.$$

These are exactly the size and error bounds needed after index recovery in Section 16; they have been established here for the changed scale.

Explicitly, $N, R \geq 64$ and $bw \leq U$ give

$$4^{3J} \leq 64N^{2R} \leq RU^R < A, \quad (bw)^3 \leq U^R < A.$$

The unchanged base-four congruence yields $bw = 4^J$, so b is a power of two and $U \geq N4^J > 4 \cdot 2^{2R}$. The unchanged second Pell norm, gap and index congruence give $\kappa = \psi_A(L)$ and $\mu = \chi_A(L)$. The inequalities

$$B^{3L} \leq B^B \leq N^N \leq U^R < A, \quad q^3 \leq N^N \leq U^R < A$$

then give $q = B^L$ through the second exponent congruence. For $F = \lfloor \xi \rfloor$, the binomial expansion gives

$$0 < \xi - F < \frac{1}{4}, \quad F \equiv \binom{2R}{R} \pmod{U}.$$

Now $F < \xi < c/k < Y_P + 1$ and $Y_P < c/k < \xi + 1/2 < F + 3/4$ imply $Y_P = F$ by integrality. Thus $N^2 \mid \binom{2R}{R}$. The unchanged no-carry and modular-Sidon decoding proves membership in the represented system.

17.3 Positive witnesses and the saved addition

Conversely, start with the canonical coding witnesses for a solution of the represented system. They determine b, B, q, N, R before any Pell coordinates are chosen. As in Section 16, put

$$J = 2R + 1, \quad w = 4^J/b, \quad U = N^2w, \quad Y_P = \lfloor \xi \rfloor, \quad s = Y_P/N^2.$$

The positive integrality of w and s depends only on these coding values and the binomial expansion, so it is unaffected by the scale change. Now form $M = (R+1)Y_P$, $a_0 = M(U+1)$, $A = a_0 + 4$, $Q = UM^2$, $P = 2Q + 1$, and choose

$$c = \psi_A(J), \quad d = \chi_A(J), \quad k = \psi_P(R+1).$$

Estimate (17.5) at these chosen indices gives $Y_P < c/k < Y_P + 3/4$, so $\eta = c - kY_P$ and $\zeta = k - \eta$ are positive integers. Choose

$$\tau = \frac{\chi_P(R+1) - 1}{2}, \quad h = \frac{k}{R+1}.$$

The oddness of P makes τ positive integral. Since $R+1 \mid P-1$ and $\psi_P(R+1) \equiv R+1 \pmod{P-1}$, h is a positive integer and satisfies the new product equation. It is a supplied positive unknown; no division operation is omitted from the certificate.

Choose the remaining dependent Pell witnesses using the generic constructions at this new A from Section 16. In particular, after choosing positive f, i for E16, take

$$G = 1 + (A+1)(f^2 - 1), \quad o = \frac{\chi_G(J) + d}{f}, \quad j = \frac{\psi_G(J) - J}{c},$$

and take $\kappa = \psi_A(L)$, $\mu = \chi_A(L)$, $\Delta = (\kappa - L)/(A - 1)$ and $\phi = c - \kappa$. The same generic congruences and strict Pell growth make them positive integers. The exponent quotients are positive because

$$d - (A - 4)c > 3c > 4^J, \quad \mu - (A - B)\kappa > (B - 1)\kappa > B^L.$$

Their moduli are positive, and the required inequalities $A > (4^J)^3, (B^L)^3$ were preserved above. Thus every witness is constructed with its required positive domain. All coding witnesses remain valid, while M and its dependent Pell coordinates are recomputed; the old and new values of h are not identified.

Theorem 17.1. *For each r.e. set, the fixed modular-Sidon encoding gives a membership-equivalent system in 34 positive unknowns and 22 equations with a straight-line certificate of **105 arithmetic instructions**: 59 multiplications and 46 additions. Fixed numerals, supplied parameters and equality tests are free.*

Proof. The preceding arguments prove soundness and construct all positive witnesses for necessity. For the count, the existing instruction $r_1 = r + 1$ is moved before the computation of M . Replacing $M = rY_P$ by $M = r_1Y_P$ retains one multiplication. Replacing hQ by hr_1 also retains one multiplication, but the old addition $r_1 + hQ$ disappears: the free equality now compares k directly with hr_1 .

The other instructions retain their counts. Hence the 106-operation certificate loses one addition and has 59 multiplications and 46 additions. The positive unknown and equation counts stay unchanged. The checker `round16_1980_index_multiple_certificate.py` verifies the three changed source residuals, every unchanged residual, both triangular corrections, all modified register identities, and the full serialized primitive certificate. Its fixed exponent is $L = 2^{32}$. $\square$

18 Reusing the interval scale

There is another way to remove an operation from the base-four construction. The scale $M = RY_P$ can be replaced by $M = Y_P$. This saves its multiplication by R . The larger approximation error is still small enough, provided it is bounded only after the base-four exponent relation has been proved. A previously unused upper bound on the packed index supplies the earlier estimates.

Start from Section 16, with the same fixed modular-Sidon encoding and $L = 2^{32}$. Set

$$N = n, \quad R = r, \quad U = wN^2, \quad Y_P = sN^2, \quad M = Y_P, \\ a_0 = Y_P(U + 1), \quad A = a_0 + 4, \quad Q = UY_P^2, \quad P = 2Q + 1, \quad J = 2R + 1.$$

The supplied positive unknown a continues to mean a_0 . Retain the first norm, affine index equation and positive interval:

$$\begin{aligned} \tau(\tau + 1) &= Q(Q + 1)k^2, & k &= R + 1 + hQ, \\ c &= kY_P + \eta, & k &= \eta + \zeta, & \tau, h, \eta, \zeta &> 0. \end{aligned} \tag{18.1}$$

Only the expanded source polynomials E9, E11 and E12 change, through the definitions of Q and a_0 . The main parameter $A = a_0 + 4$, signed auxiliary block and both exponent congruences retain their forms in the supplied coordinates.

18.1 A bound from the packing and the exact indices

The pre-Pell coding proof gives

$$N = q^8, \quad q > 4b \geq 8, \quad 3L \leq B \leq N \leq R, \quad b, q \leq N, \quad 0 < S < N, \quad 0 \leq T < N.$$

The packed-index formula and integrality of the block values give the useful additional bound

$$\begin{aligned} R &= S(N^2 - N) + (T + 1)(N^2 - 1) \\ &\leq (N - 1)(N^2 - N) + N(N^2 - 1) = 2N^3 - 2N^2 < 2N^3. \end{aligned} \quad (18.2)$$

Thus $N, R \geq 64$ and $U, Y_P \geq N^2 \geq 4096$. In particular

$$a_0 > N^4 > 4N^3 + 1 > J, \quad Q \geq N^6 > 2N^3 + 1 > R + 1. \quad (18.3)$$

These conclusions use only the coding bounds and positivity, before either exponent relation is known.

The index equation gives $k \geq R + 2$, and the interval gives $c > Y_P k > J$. The signed Pell block therefore implies $c = \psi_A(J)$ and $d = \chi_A(J)$. The first norm gives the ordinary Pell equation with root $\hat{\tau} = 2\tau + 1$ and parameter $P = 2Q + 1$, so $k = \psi_P(t)$ for a positive index t . The Pell congruence modulo $P - 1 = 2Q$, reduced modulo Q , and (18.3) give

$$t = R + 1 + vQ, \quad v \geq 0.$$

If $v \geq 1$, then Pell growth yields

$$\frac{c}{k} \leq \frac{(2A)^{2R}}{(2P - 1)^{R+Q}} \leq \left(\frac{2A}{2P - 1} \right)^{2R} < \frac{1}{2}.$$

Here $Q \geq R$, and the last inequality follows from

$$(2P - 1) - 4A = 4Y_P(U(Y_P - 1) - 1) - 15 > 0.$$

This contradicts $c/k > Y_P$. Consequently

$$c = \psi_A(J), \quad d = \chi_A(J), \quad k = \psi_P(R + 1), \quad \hat{\tau} = \chi_P(R + 1). \quad (18.4)$$

18.2 The larger error is bounded after exponent recovery

Put $\xi = (U + 1)^{2R}/U^R$. At the exact indices, the same cancellation of M in the principal ratio gives

$$\frac{c}{k} \geq \xi \left(1 + \frac{7}{2a_0} \right)^{2R} \left(1 + \frac{1}{2Q} \right)^{-R} > \xi, \quad (18.5)$$

because $14Q > a_0$. The upper growth bound gives

$$\frac{c}{k} < \xi \left(1 + \frac{4}{a_0} \right)^{2R}.$$

The packed-index bound replaces the old factor R in the scale:

$$\frac{8R}{a_0} < \frac{16N^3}{N^4} = \frac{16}{N} \leq \frac{1}{4} < \frac{1}{2}.$$

The elementary estimate $(1+t)^m \leq (1-mt)^{-1} < 1+2mt$ therefore gives

$$\frac{c}{k} < \xi \left(1 + \frac{16R}{a_0}\right). \quad (18.6)$$

The interval and the strict lower bound give $\xi < Y_P + 1$. Since Y_P is integral and $\xi > U^R$, this implies

$$Y_P \geq U^R, \quad A > a_0 = Y_P(U+1) > U^{R+1}. \quad (18.7)$$

Also $\xi < Y_P + 1 < 2Y_P$, so

$$0 < \frac{c}{k} - \xi < \frac{32R}{U+1}. \quad (18.8)$$

At this point the right-hand side is not assumed to be less than one.

Write $W = bw$. Since $W \leq U$, $N \geq 64$ and $U \geq 4096$, (18.7) gives

$$4^{3J} \leq 64N^{2R} \leq 64U^R < U^{R+1} < A, \quad W^3 \leq U^3 \leq U^R < A.$$

The base-four exponent congruence now proves $W = 4^J$ by source Lemma 2.22. Hence b is a power of two and

$$U = N^2w \geq Nbw = N4^J \geq 64 \cdot 4^J > 64R.$$

Only now does (18.8) yield

$$0 < \frac{c}{k} - \xi < \frac{1}{2}. \quad (18.9)$$

The gap, second Pell norm and index congruence give $\kappa = \psi_A(L)$ and $\mu = \chi_A(L)$, since $0 < L < J < A$. Furthermore

$$B^{3L} \leq B^B \leq N^N \leq U^R < A, \quad q^3 \leq N^N \leq U^R < A.$$

The second exponent congruence therefore gives $q = B^L$. Since also $U \geq N4^J > 4 \cdot 2^{2R}$, the binomial expansion has integer part $F = \lfloor \xi \rfloor$ with

$$0 < \xi - F < \frac{1}{4}, \quad F \equiv \binom{2R}{R} \pmod{U}.$$

The two inequalities $F < \xi < c/k < Y_P + 1$ and $Y_P < c/k < \xi + 1/2 < F + 3/4$ force $Y_P = F$. Thus $N^2 \mid \binom{2R}{R}$, and the unchanged no-carry and modular-Sidon decoding proves membership.

18.3 Positive witnesses and the independent count

For necessity, take the canonical coding witnesses, which determine b, B, q, N, R and satisfy (18.2). Choose

$$J = 2R + 1, \quad w = 4^J/b, \quad U = N^2w, \quad Y_P = \lfloor \xi \rfloor, \quad s = Y_P/N^2,$$

and then form $a_0 = Y_P(U+1)$, $A = a_0 + 4$, $Q = UY_P^2$, $P = 2Q + 1$, $c = \psi_A(J)$, $d = \chi_A(J)$ and $k = \psi_P(R+1)$. The positive integrality of w, s follows from the same power-of-two and binomial

arguments as in Section 16. The bounds (18.5)–(18.6) apply at these indices, and $U \geq N4^J$ holds directly from the chosen w . Hence

$$Y_P < \xi < \frac{c}{k} < \xi + \frac{1}{2} < Y_P + \frac{3}{4}.$$

Choose the positive integers

$$\eta = c - kY_P, \quad \zeta = k - \eta, \quad \tau = \frac{\chi_P(R+1) - 1}{2}, \quad h = \frac{k - R - 1}{Q}.$$

The oddness of P gives the integrality of τ , and the Pell congruence modulo $2Q$ gives that of h . Strict Pell growth gives their positivity.

All remaining dependent Pell witnesses are obtained by the generic constructions of Section 16 at this new A . Explicitly, choose positive f, i for E16 and put

$$G = 1 + (A+1)(f^2 - 1), \quad o = \frac{\chi_G(J) + d}{f}, \quad j = \frac{\psi_G(J) - J}{c}.$$

The congruences $G \equiv -A \pmod{f}$, $G \equiv 1 \pmod{c}$ and oddness of J make o, j positive integers. Choose $\kappa = \psi_A(L)$, $\mu = \chi_A(L)$, $\Delta = (\kappa - L)/(A - 1)$ and $\phi = c - \kappa$; these are positive integers because $2 \leq L < J$. The exponent quotients are positive since

$$d - (A - 4)c > 3c > 4^J, \quad \mu - (A - B)\kappa > (B - 1)\kappa > B^L,$$

using $A > (4^J)^3, (B^L)^3$ and $J, L \geq 2$. Their moduli are positive. All coding witnesses remain unchanged. This constructs every required positive coordinate at the new scale.

Theorem 18.1. *For each r.e. set, reusing Y_P as the Pell scale gives a membership-equivalent system in 34 positive unknowns and 22 equations with a straight-line certificate of **105 arithmetic instructions**: 58 multiplications and 47 additions. Fixed numerals, supplied parameters and equality tests are free.*

Proof. The preceding arguments establish soundness and positive necessity. The predecessor already computes $Y_P = sN^2$. Delete the multiplication $M = RY_P$ and reuse Y_P wherever M was used. The three instructions

$$UY_P = U \cdot Y_P, \quad a_0 = UY_P + Y_P, \quad Q = (UY_P)Y_P$$

retain their previous costs. The affine index equation retains its multiplication and addition. Thus exactly one multiplication disappears from the 106-operation certificate. The complete residual and primitive checks are supplied by `round18_1980_unit_scale_certificate.py`. $\square$

This is independent of Theorem 17.1: its saving is a multiplication rather than an addition. The index-multiple argument there cannot be applied automatically here, since $P - 1 = 2UY_P^2$ is not known to be divisible by $R + 1$.

18.4 Sharing the interval product with the norm

The smaller scale permits another multiplication to be removed. Let

$$D = UY_P, \quad Q = DY_P = UY_P^2, \quad H_P = Y_P k.$$

Both D and H_P are already computed: D is needed for $a_0 = D + Y_P$, and H_P is needed for $c = H_P + \eta$. The polynomial identity

$$Q(Q + 1)k^2 = (UY_P^2)(UY_P^2 + 1)k^2 = ((UY_P)^2 + U)(Y_P k)^2 = (D^2 + U)H_P^2 \quad (18.10)$$

allows the norm to share these two products. To remove the remaining use of the computed value Q , replace the index equation in (18.1) by

$$k = R + 1 + hD, \quad h > 0. \quad (18.11)$$

The norm and every other source polynomial are unchanged.

We first prove that the smaller modulus still gives the exact index. Before exponent decoding, the bounds above give

$$D = UY_P \geq N^4 > 2N^3 + 1 > R + 1, \quad A > J.$$

The new equation gives $k \geq R + 2$, so the interval gives $c > J$. The signed Pell block gives $c = \psi_A(J)$ and $d = \chi_A(J)$. The unchanged triangular norm gives $k = \psi_P(t)$ for some positive index t , where $P = 2Q + 1$. Since $P - 1 = 2Q = 2DY_P$ is divisible by D , the Pell congruence and (18.11) imply

$$t = R + 1 + vD, \quad v \geq 0.$$

If $v \geq 1$, then $D > R$ and the same growth estimate gives

$$\frac{c}{k} \leq \frac{(2A)^{2R}}{(2P - 1)^{R+D}} \leq \left(\frac{2A}{2P - 1} \right)^{2R} < \frac{1}{2},$$

contradicting $c/k > Y_P$. Therefore $t = R + 1$. This recovery uses only the preliminary coding bounds, the interval and the two Pell blocks.

There are direct maps between the positive witnesses of this system and those of Theorem 18.1. From an old solution, put $h = Y_P h_{\text{old}}$ and preserve every other witness. Then $hD = h_{\text{old}}Q$, so the new index equation holds. Conversely, the recovered exact index and the stronger Pell congruence modulo $2Q$ show that

$$h_{\text{old}} = \frac{k - R - 1}{Q}$$

is an integer. It is positive because $\psi_P(R + 1) > R + 1$. It restores the old index equation, preserving every other witness. Equivalently, the new h is a positive multiple of Y_P and the reverse map is $h_{\text{old}} = h/Y_P$. This proves equivalence over the positive domains before invoking the unit-scale approximation, exponent or rounding conclusions. Those conclusions and the complete decoding proof therefore transfer.

Theorem 18.2. *For each r.e. set, the fixed modular-Sidon encoding gives a membership-equivalent system in 34 positive unknowns and 22 equations whose straight-line certificate has **104 arithmetic instructions**: 57 multiplications and 47 additions. Fixed numerals, supplied parameters and equality tests are free.*

Proof. Positive-domain equivalence follows from the preceding witness maps and Theorem 18.1. For the count, move the existing computation of $H_P = Y_P k$ before the norm. The former five operations

$$Q = DY_P, \quad Q_+ = Q + 1, \quad F_Q = QQ_+, \quad k_2 = k^2, \quad L_9 = F_Q k_2$$

are replaced by four:

$$D_2 = D^2, \quad F_D = D_2 + U, \quad H_2 = H_P^2, \quad L_9 = F_D H_2.$$

Identity (18.10) proves that they give the same norm residual. The computation of H_P is moved rather than copied, and the root-side instructions $\tau + 1$ and $\tau(\tau + 1)$ are unchanged. Both displayed lists contain one addition; the second contains three multiplications rather than four.

The product $h_{\text{old}}Q$ is replaced by hD at the same cost. Thus the calculated register Q is no longer required anywhere in the certificate. Its mathematical definition in the proof introduces no additional supplied witness or uncharged computed register. The total is 57 multiplications and 47 additions. The checker `round19_1980_shared_ratio_product_certificate.py` verifies the identity, the forward witness-map polynomial, all source residuals and primitive instructions, and the absence of the discarded registers. It checks explicitly that only E11 changes as a source polynomial. $\square$

19 Allowing a borrow in the coefficient tests

The low part of the coefficient-testing offset can be removed. The resulting digits may receive a borrow from the preceding position, but the residuals can be encoded so that both possible digits give the same answer. Starting from Theorem 18.2, replace

$$S_3 = B\lambda(1 + q) - (Z\lambda - 2e)\mathcal{C}^2$$

by

$$S_3 = B\lambda q - (Z\lambda - 2e)\mathcal{C}^2, \quad \sigma = S_3 > 0. \quad (19.1)$$

Here $\mathcal{C} = x + g$ is the digit code, distinguished from the main Pell denominator c . The supplied positive witness σ remains part of the system. The change removes the instruction computing $q + 1$. It requires a new fixed coefficient index, constructed below.

19.1 Coefficient targets and the admissible index

Keep the modular support of Section 15, including the input at weight zero, δ at $v_0 = 1$, the original 58 coordinates at $v_1, \dots, v_{58}$ and the guard coordinate u at v_{59} . In particular,

$$\begin{aligned} v_i &= 1 + 3(122i + (i^2 \bmod 61)), & 0 \leq i \leq 59, \\ v_* &= 21607, & t_j = 158467571 + 86429j \quad (0 \leq j < 1832), \\ t_* &= 316719070, & K = t_* + 1 = 316719071, \quad L = 2^{32} > 3K + 2. \end{aligned}$$

The 1830 original homogeneous quadratic residuals come from the integer row basis and padding of Section 9. Adjoin the guard residual $u\delta - x^2$ as the 1831st ordinary row. For each ordinary row replace its coefficient target by

$$G_j = 4F_j^h + \delta^2 \quad (0 \leq j < 1831), \quad G_{1831} = \delta^2. \quad (19.2)$$

Thus $G_{1830} = 4(u\delta - x^2) + \delta^2$, and the last row remains the special square test.

For a homogeneous quadratic monomial I , let $c_I \in \{1, 2\}$ be its coefficient in $\mathcal{C}^2$: it is 1 for a square and 2 for a product of distinct coordinates. If $a_{j,I}$ is its coefficient in G_j , define

$$\mathcal{D}(B) = \sum_{j,I} \frac{a_{j,I}}{c_I} B^{t_j - w(I)}.$$

Every displayed coefficient is integral. The coefficients from $4F_j^h$ are divisible by both possible denominators, and the added δ^2 is a square with denominator 1. In the special row the coefficient 1 is placed at $t_{1831} - 2$.

The support separation proved in Section 15 gives the exact identity

$$[B^{t_j}] \mathcal{D}(B) \mathcal{C}(B)^2 = G_j. \quad (19.3)$$

Indeed, distinct monomials have distinct weights, different row intervals are disjoint, and a product involving a dummy row coordinate lies above every row target. The least position of $\mathcal{D}$ is above v_* , so all low coordinate tests remain automatic.

Choose a power of two $z \geq 2$ strictly above the absolute values of the new coefficients of $\mathcal{D}$, and put $Z = 2z$. Define

$$\begin{aligned}\ell_0(B) &= \sum_{i=0}^{59} B^{v_i} + \sum_{j=0}^{1831} B^{t_j}, \\ e_0(B) &= z \sum_{j=0}^{K-1} B^j + \mathcal{D}(B), \\ V &= \ell_0(Z) + e_0(Z)Z^L.\end{aligned}$$

All digits of e_0 below K lie in $[1, Z-1]$; higher digits vanish. Choose a power of two

$$H > \max\{2Z^{2L+1}, 4^{t_*+3}Z^{1900^2}, 3L, 16\}. \quad (19.4)$$

The positions, coordinate count and size formula for H are unchanged. The indices Z, V, H are rebuilt for the new coefficient polynomial. They depend effectively on the represented set and are independent of the queried positive input x .

19.2 Bounds before the coefficients are decoded

Retain the coding equations and positive conditions

$$\begin{aligned}B &= Hb^2, & b &= x + \beta, & \lambda(B-1) &= q^2 - 1, & \theta &= B - Z, \\ S_2 &= \ell + eq = V + t\theta, & S_2 + \alpha &= q^2, & \Omega &= Z\lambda - 2e > 0, & \mathcal{C} &= x + g.\end{aligned}$$

The packing remains

$$\begin{aligned}n &= q^8, & S &= g + q^2(S_2 + q^2S_3), \\ T+1 &= q^2(1 + \theta\lambda) - (b-1)\ell + (B-4)\ell q^4, \\ r &= S(n^2 - n) + (T+1)(n^2 - 1).\end{aligned}$$

The Pell interval scale $Y_P = sn^2$ is distinct from the packed code S_2 .

All preliminary upper bounds from Sections 14 and 18 still apply, since the new positive offset $B\lambda q$ is smaller than $B\lambda(1+q)$. More explicitly, $S_2 < q^2$ gives $e < q$ and $\ell < q^2$, while the geometric equation gives $B \leq q^2$. Positivity of Ω and σ gives

$$\mathcal{C}^2 < B\lambda q < 3q^3 < q^4, \quad 0 < S_3 < q^4.$$

Thus $g < q^2$ and the same normalized masks give

$$0 < S < n, \quad 0 \leq T < n, \quad n \leq r \leq 2n^3 - 2n^2 < 2n^3.$$

The Pell proof of Theorem 18.2 uses these bounds and the positive interval, independently of the coefficient values in the encoding. It therefore recovers b as a power of two, $q = B^L$, and $n^2 \mid \binom{2r}{r}$.

After $q = B^L$ is known, $\lambda > q$ and $e < q$ imply $\Omega > Z\lambda/2$. Equation (19.1) then gives

$$\mathcal{C}^2 < \frac{2Bq}{Z} \leq \frac{Bq}{2} < q^2. \quad (19.5)$$

Hence $g < \mathcal{C} < q$ before either the input code or the quotient of the packed polynomial is decoded.

The first two masks and the packed congruence have not changed. The borrow-removal and base- Z uniqueness arguments of Sections 11 and 13 require only the bounds just proved, the support of ℓ_0 , the digit bounds for e_0 , and (19.4). They therefore give

$$S_2 = \ell_0(B) + e_0(B)q, \quad \ell = \ell_0(B) + mq, \quad e = e_0(B) - m, \quad 0 \leq m < e_0(B). \quad (19.6)$$

The first mask, using $g < q$, decodes the input digit exactly as x . All other coordinates are nonnegative integers less than b ; dummy coordinates can occur only at the row positions. Consequently

$$\mathcal{C} < B^K, \quad A_C := \mathcal{C}(1)^2 < (1900b)^2, \quad 0 < e < B^K, \quad 2e\mathcal{C}^2 < q. \quad (19.7)$$

The last estimate uses $L > 3K + 2$ and is valid before m is excluded.

19.3 The high digits still exclude the quotient ambiguity

Lemma 19.1. *Under the preceding hypotheses, the third no-carry condition forces $m = 0$ in (19.6).*

Proof. Put $J_C = ZA_C < B/4$ and

$$W_C = Z\lambda\mathcal{C}^2 = qW_1 + W_0, \quad 0 \leq W_0 < q.$$

The coefficients of the polynomial $Z\lambda\mathcal{C}^2$ are nonnegative and at most J_C . They are already its base- B digits, so no internal carries occur in this product. Since $\deg \mathcal{C}^2 \leq 2K - 2$, these coefficients equal J_C throughout the interval $[2K - 2, 2L - 1]$. In particular they equal J_C at every position $L, \dots, L + K$, because $L > 3K + 2$.

By (19.7), exact division by q gives

$$\left\lfloor \frac{S_3}{q} \right\rfloor = B\lambda - W_1 + \varepsilon, \quad \varepsilon = \left\lfloor \frac{2e\mathcal{C}^2 - W_0}{q} \right\rfloor \in \{-1, 0\}.$$

The unit digit of $B\lambda$ is zero, and its digits at positions $1, \dots, 2L$ are one. The first $K + 1$ digits of W_1 are all J_C . At the unit position the subtraction therefore gives digit $B - J_C - 1$ or $B - J_C$ and an outgoing borrow of one. At each following position through K , the incoming borrow changes $1 - J_C$ into $-J_C$, giving digit $B - J_C$ and again an outgoing borrow of one. Thus the digits of S_3 at positions $L, \dots, L + K$ are all at least $B - J_C - 1$.

The product $(B - 4)\ell_0(B)$ has digits $B - 4$ at its support positions and no internal carries, and it is less than q . Accordingly the part of the third mask at and above position L is

$$X = (B - 4)m < B^{K+1}.$$

If a digit d has no binary overlap with a digit at least $B - J_C - 1$, then $d \leq B - 1 - (B - J_C - 1) = J_C$. Hence every digit of X is at most J_C .

Let F be the digit polynomial of X , so $F(B) = X$, its coefficients are nonnegative, and $\deg F \leq K$. Since $B - 4 \mid F(B)$, reduction modulo $B - 4$ gives $B - 4 \mid F(4)$. On the other hand,

$$0 \leq F(4) \leq J_C \frac{4^{K+1} - 1}{3} < \frac{B}{12} < B - 4.$$

Here (19.4) and $J_C < Z(1900b)^2$ give $4^{K+2}J_C < B$, because $K = t_* + 1$. It follows that $F(4) = 0$. Nonnegativity of its coefficients then gives $F = 0$, so $X = 0$ and $m = 0$. $\square$

19.4 Low carries determine the intended equations

We now have $\ell = \ell_0(B)$ and $e = e_0(B)$. At positions below K , the offset $qB\lambda$ makes no contribution, and the negative tail of $e_0 - z\lambda$ begins too high to contribute. The raw coefficients there are precisely those of $2\mathcal{D}(B)\mathcal{C}(B)^2$. Each has absolute value less than $2zA_C = J_C < B/4$.

Normalize these signed coefficients starting at the unit position. The initial carry is zero. If an incoming carry is in $\{-1, 0\}$, adding a raw coefficient of absolute value less than $B/4$ gives a number strictly between $-B$ and B . Its Euclidean quotient by B is therefore again -1 or 0. Induction proves that these are the only incoming carries at every target position.

At a target, let v be the raw coefficient and let $\varepsilon \in \{-1, 0\}$ be the incoming carry. The mask digit $B - 4$ permits exactly the digits 0, 1, 2, 3. Since $|v| < B/4$, a negative $v + \varepsilon$ would normalize to a digit greater than 3, and a nonnegative one cannot wrap around B . Thus the target test is exactly

$$0 \leq v + \varepsilon \leq 3. \quad (19.8)$$

At the special row, $v = 2\delta^2$. The test implies $2\delta^2 \leq 4$, so the nonnegative integer δ is 0 or 1. At the guard, the raw value is

$$v = 8(u\delta - x^2) + 2\delta^2.$$

If $\delta = 0$, this is $-8x^2 < 0$, and a carry of -1 or 0 cannot make it satisfy (19.8). Hence $\delta = 1$.

Every ordinary row now has $v = 8F_j^h + 2$. Depending on the carry, (19.8) says

$$0 \leq 8F_j^h + 1 \leq 3 \quad \text{or} \quad 0 \leq 8F_j^h + 2 \leq 3.$$

Both imply $F_j^h = 0$ by integrality. Since $\delta = 1$, these are exactly the original equations. At each low variable position the raw coefficient and incoming carry are zero, since the least position of $\mathcal{D}$ lies above all such positions. Those tests impose no additional condition. This proves soundness of the new encoding.

19.5 Positive witnesses and the resulting certificate

Conversely, start with a solution of the represented system. Set $\delta = 1$, $u = x^2$, and every dummy coordinate to zero. Choose a sufficiently large power of two b , and take the canonical $B, q, \lambda, \ell_0, e_0, \mathcal{C}$ defined above. The positive digit at weight one makes $g > 0$. All previous fixed-index, packed-congruence, bound and Ω witnesses remain positive, because the support and digit bounds have the same form.

The changed witness $\sigma = S_3$ is positive as well. Indeed, $\mathcal{C} < B^K$, $Z < B$ and $L > 2K$ give

$$Z\mathcal{C}^2 < B^{2K+1} < qB, \quad S_3 = \lambda(qB - Z\mathcal{C}^2) + 2e\mathcal{C}^2 > 0.$$

At every ordinary row and the special row the raw coefficient is exactly 2. Its actual digit is consequently 1 or 2, both allowed by the mask. The low coordinate tests have digit zero, and ℓ_0 has no other support. The first two masks work as before. Thus the packed integers have no binary overlap, yielding $n^2 \mid \binom{2r}{r}$ for their newly calculated r .

All dependent positive Pell witnesses are chosen by the constructions of Section 18, using this r . Those constructions require only the packed ranges, the binomial divisibility, and the fixed exponent bounds, which have just been established. This proves necessity without identifying the old and new values of r .

Theorem 19.2. *For each r.e. set, the modified coefficient targets (19.2) and admissible index above give a membership-equivalent system in 34 positive unknowns and 22 equations with a straight-line certificate of **103 arithmetic instructions**: 57 multiplications and 46 additions. Fixed numerals, supplied parameters and equality tests are free.*

Proof. The preceding arguments prove soundness and positive necessity. The product $Lam = B\lambda$ is already required by the geometric equation. Replace the instruction $P_2 = Lam(q + 1)$ by $P_2 = Lam\ q$ and delete the single addition computing $q + 1$. All other instruction counts remain unchanged. The coefficient adjustment changes only the supplied fixed index; it introduces no system unknown or arithmetic operation. The exact primitive and source-residual identities, including the two inherited triangular corrections, are checked by `round20_1980_borrow_mask_certificate.py`. The full high-window argument is Lemma 19.1; it needs no stronger bound on H than the preceding construction. $\square$

20 A common witness for the binomial scale and exponent

The first exponent test can use the binomial scale U itself as its target. That value is already computed, so the separate product bw disappears. This change preserves the unit-scale norm and its index modulus. Combined with the coefficient encoding of Section 19, it gives 102 operations.

Use the fixed coefficient index of Theorem 19.2, including its ordinary targets $4F_i^h + \delta^2$, special target δ^2 , and exponent $L = 2^{32}$. Retain the notation

$$\begin{aligned} N = n, \quad R = r, \quad U = wN^2, \quad Y_P = sN^2, \\ a_0 = Y_P(U + 1), \quad A = a_0 + 4, \quad D_P = UY_P, \quad Q = UY_P^2, \quad P = 2Q + 1, \quad J = 2R + 1. \end{aligned}$$

The positive supplied unknown a still means a_0 . Keep the shared norm, first index equation and interval

$$\begin{aligned} \tau(\tau + 1) &= (D_P^2 + U)(Y_P k)^2, & k &= R + 1 + hD_P, \\ c &= kY_P + \eta, & k &= \eta + \zeta, & \tau, h, \eta, \zeta &> 0. \end{aligned} \tag{20.1}$$

Change only E14 among the supplied-coordinate Pell polynomials, replacing its target bw by U :

$$d = U + c(A - 4) + \gamma(8A - 17). \tag{20.2}$$

No parameter or witness is added. The defining identity $U = wN^2$ and the index modulus $D_P = UY_P$ are unchanged.

20.1 Exact indices before the exponent equations

The smaller positive offset in Section 19 retains all preliminary coding bounds:

$$\begin{aligned} N &= q^8, & q &> 4b \geq 8, & 3L &\leq B \leq N \leq R, \\ 0 < S < N, & 0 \leq T < N, & R &\leq 2N^3 - 2N^2 < 2N^3. \end{aligned}$$

These are established without using any Pell equation. Hence $N, R \geq 64$, $U, Y_P \geq N^2$, $a_0 > N^4 > J$, and $D_P \geq N^4 > R + 1$.

The first index equation gives $k \geq R + 2$, and the interval gives $c > Y_P k > J$. The signed block therefore gives $c = \psi_A(J)$ and $d = \chi_A(J)$. The norm in (20.1) is the triangular form of the ordinary Pell equation with parameter $P = 2Q + 1$, so $k = \psi_P(t)$ for some positive index t . Since $P - 1 = 2D_P Y_P$ is divisible by D_P , the Pell congruence and $D_P > R + 1$ imply

$$t = R + 1 + vD_P, \quad v \geq 0.$$

If $v \geq 1$, then $D_P > R$ and Pell growth gives

$$\frac{c}{k} \leq \frac{(2A)^{2R}}{(2P - 1)^{R + D_P}} \leq \left(\frac{2A}{2P - 1} \right)^{2R} < \frac{1}{2}.$$

The last inequality follows from $(2P - 1) - 4A = 4Y_P[U(Y_P - 1) - 1] - 15 > 0$. This contradicts $c/k > Y_P$. Thus the exact indices are

$$c = \psi_A(J), \quad d = \chi_A(J), \quad k = \psi_P(R + 1), \quad 2\tau + 1 = \chi_P(R + 1). \quad (20.3)$$

The change to E14 has not entered this argument.

20.2 Recovering the radices in the required order

Put $\xi = (U + 1)^{2R}/U^R$. The unit-scale lower estimate at (20.3) gives

$$\frac{c}{k} \geq \xi \left(1 + \frac{7}{2a_0}\right)^{2R} \left(1 + \frac{1}{2Q}\right)^{-R} > \xi,$$

using $14Q > a_0$. The interval gives $\xi < Y_P + 1$, so

$$Y_P \geq U^R, \quad A > a_0 = Y_P(U + 1) > U^{R+1}. \quad (20.4)$$

As in Section 18, the independent bound $R < 2N^3$ gives $8R/a_0 < 16/N < 1/2$. The upper growth estimate and $(1 + t)^m \leq (1 - mt)^{-1} < 1 + 2mt$ therefore yield

$$0 < \frac{c}{k} - \xi < \frac{32R}{U + 1}. \quad (20.5)$$

This error need not be bounded by an absolute constant yet.

The changed exponent target satisfies

$$4^{3J} = 64 \cdot 4096^R \leq 64N^{2R} \leq 64U^R < U^{R+1} < A, \quad U^3 \leq U^R < A.$$

The source χ exponent criterion applied to (20.2) now gives

$$U = 4^J. \quad (20.6)$$

In particular $U > 64R$, so (20.5) is strictly less than $1/2$.

The new first exponent conclusion determines the radices in a different order. Since N^2 divides the power of two $U = wN^2$, N is a power of two. Since $N = q^8$, so is q . No conclusion about b is needed yet.

The unchanged second norm, gap and index congruence identify $\kappa = \psi_A(L)$ and $\mu = \chi_A(L)$, since $0 < L < J < A$. Their exponent test has the size bounds

$$B^{3L} \leq B^B \leq N^N \leq U^R < A, \quad q^3 \leq N \leq R < U^R < A.$$

It therefore gives $q = B^L$. Now B is a power of two, because q is a power of two and $L > 0$. Finally $B = Hb^2$ implies that b is a power of two. Thus all binary radix facts are known before the first-mask decoding uses them.

The binomial expansion has integer part $F = \lfloor \xi \rfloor$ with

$$0 < \xi - F < \frac{2^{2R}}{U} < \frac{1}{4}, \quad F \equiv \binom{2R}{R} \pmod{U}.$$

The interval and the ratio bounds give

$$F < \xi < \frac{c}{k} < Y_P + 1, \quad Y_P < \frac{c}{k} < \xi + \frac{1}{2} < F + \frac{3}{4}.$$

Integrality forces $Y_P = F$. The unchanged definitions $U = wN^2$ and $Y_P = sN^2$ now imply

$$N^2 \mid \binom{2R}{R}. \quad (20.7)$$

20.3 Composition with the new coefficient encoding

The conclusions just obtained are exactly those needed by Section 19: b, q, N are powers of two, $q = B^L$, and (20.7) supplies binary nonoverlap of the packed blocks. Positivity of the new offset then gives

$$\Omega > Z\lambda/2, \quad \mathcal{C}^2 < 2Bq/Z < q^2, \quad \mathcal{C} = x + g.$$

The first two masks recover the possible quotient ambiguity $\ell = \ell_0 + mq$, $e = e_0 - m$ and the bounds required by Lemma 19.1. That lemma forces $m = 0$. The low coefficients have incoming carries in $\{-1, 0\}$; the special row and guard force $\delta = 1$, and every ordinary row then forces $F_i^h = 0$. These decoding arguments do not use E14 internally. They apply unchanged once its replacement has supplied the required exponent and binomial conclusions, proving sufficiency.

For necessity, first use the new canonical coding construction of Section 19. It has $\delta = 1$, $u = x^2$, zero dummy coordinates and a sufficiently large power of two b . Its changed σ is positive because

$$S_3 = \lambda(Bq - Z\mathcal{C}^2) + 2e\mathcal{C}^2 > 0.$$

The ordinary and special target digits are one or two, so the masks pass. Form its resulting S, T, R . This supplies the packed bounds and (20.7) before choosing Pell witnesses.

Now set

$$\begin{aligned} J &= 2R + 1, \quad U = 4^J, \quad w = U/N^2, \quad Y_P = \lfloor \xi \rfloor, \quad s = Y_P/N^2, \\ a_0 &= Y_P(U + 1), \quad A = a_0 + 4, \quad Q = UY_P^2, \quad P = 2Q + 1, \\ c &= \psi_A(J), \quad d = \chi_A(J), \quad k = \psi_P(R + 1). \end{aligned}$$

The quotient w is positive integral: canonical N is a power of two and $N^2 \leq R^2 \leq 4^R < 4^J = U$. The binomial expansion and divisibility make s positive integral. The same estimates at these indices give $Y_P < c/k < Y_P + 3/4$, so choose the positive integers

$$\eta = c - kY_P, \quad \zeta = k - \eta, \quad \tau = \frac{\chi_P(R + 1) - 1}{2}, \quad h = \frac{k - R - 1}{UY_P}.$$

Odd P gives the positive integrality of τ . The congruence modulo $P - 1 = 2UY_P^2$ and strict Pell growth give that of h . This construction needs no parity condition on R .

Choose the remaining Pell witnesses at the new A by the generic constructions already established in Section 16. After choosing positive f, i for E16, put

$$G = 1 + (A + 1)(f^2 - 1), \quad o = \frac{\chi_G(J) + d}{f}, \quad j = \frac{\psi_G(J) - J}{c},$$

and take $\kappa = \psi_A(L)$, $\mu = \chi_A(L)$, $\Delta = (\kappa - L)/(A - 1)$ and $\phi = c - \kappa$. Their congruences and strict growth make every required witness positive integral. The exponent quotients are positive because

$$d - (A - 4)c > 3c > U, \quad \mu - (A - B)\kappa > (B - 1)\kappa > q,$$

using $A > U^3, q^3$ and $J, L \geq 2$; their moduli are positive. Thus the new E14 and every other equation hold. The coding witnesses are chosen first, and all dependent Pell coordinates are then chosen for their new packed R .

Theorem 20.1. *For each r.e. set, the fixed coefficient encoding of Section 19 gives a membership-equivalent system in 34 positive unknowns and 22 equations whose straight-line certificate has **102 arithmetic instructions**: 56 multiplications and 46 additions. Fixed numerals, supplied parameters and equality tests are free.*

Proof. The preceding arguments establish the complete positive-domain equivalence and the order of the two constructions. Starting from the 104-operation schedule, the offset change deletes the addition $q + 1$ and uses $(B\lambda)q$ in its place. The common-witness change deletes the product bw and uses the existing register $U = wN^2$ directly in E14. These edits concern disjoint register definitions, so they remove one addition and one multiplication without introducing an instruction, unknown or equation.

The resulting count is 56 multiplications and 46 additions. The rebuilt coefficient index affects only supplied constants. The checker `round22_1980_composed_certificate.py` verifies the complete primitive list, all source residuals and both triangular corrections. It also checks that the two schedule edits commute and that both discarded registers are absent. $\square$

21 Fixing the coefficient digit base at four

The coefficient digit base can be fixed at $Z = 4$ by expanding the represented equations into elementary arithmetic gates. Their number affects the fixed index, but not the number of unknowns in the final system: the additional coordinates are digits of the existing code. The exponent L becomes a fixed index parameter in place of Z . This identifies the already supplied $\theta = B - 4$ with a separately computed register of Theorem 20.1, saving one addition.

21.1 Primitive rows with a fixed coefficient alphabet

Start with a finite polynomial system representing the desired set over nonnegative witnesses and a positive input x . Split each equation into an equality of polynomials with nonnegative coefficients, and evaluate those polynomials by finite circuits of additions, multiplications, copies, and constants zero and one. Every gate has a fresh nonnegative output coordinate. Arbitrary integer coefficients are implemented by finite arithmetic from one. Thus this construction places no bound on the coefficients of the original system.

Replace repeated operands of an addition or multiplication by distinct copied coordinates. With a separate homogenizing coordinate δ , the quadratic residuals have the following forms:

gate or constraint	homogeneous residual F
multiplication	$XY - W\delta$
addition	$(X + Y - W)\delta$
copy or equality	$(X - Y)\delta$
zero	$X\delta$
unit	$X\delta - \delta^2$.

Here X, Y may include the input x , but are distinct from δ ; outputs are fresh, and trivial identical equalities are omitted. At $\delta = 1$, these rows express the original circuit exactly. Every solution of the original system extends by its nonnegative gate values. Adjoin a fresh coordinate u and the guard $F_g = u\delta - x^2$.

For every ordinary row, including the guard, use the target

$$G_j = 2F_j + \delta^2, \quad G_{\text{special}} = \delta^2. \quad (21.1)$$

For a homogeneous quadratic monomial I , let $c_I = 1$ for a square and $c_I = 2$ for a product of distinct coordinates. Every coefficient of every G_j , divided by c_I , belongs to

$$\{-2, -1, 0, 1\}. \quad (21.2)$$

Indeed, ordinary cross terms give ± 1 , and the added δ^2 gives 1. The unit row gives $2X\delta - \delta^2$, with divided coefficients 1, -1 . The guard gives $2u\delta - 2x^2 + \delta^2$, with divided coefficients 1, -2 , 1. The special target has coefficient 1. Distinct copied operands prevent a repeated positive square or coincident cross terms from enlarging this alphabet. The displayed orientation of the unit row is part of the construction.

21.2 Support and admissible indices of arbitrary finite size

Let m be the number of coordinates other than the input, before adding dummy row coordinates. It includes δ , all circuit coordinates, and u . Give x weight zero, and give the remaining coordinates weights

$$v_i = 3^i \quad (0 \leq i < m), \quad v_0 = 1 \text{ for } \delta, \quad M = 3^{m-1}.$$

All quadratic monomial weights are distinct. The weights are zero, a single 3^i , or a sum of two such powers; their base-three digits recover the unordered pair, including repeated coordinates and the weight-zero input.

Let s be the total number of targets, including the guard and the special target. Define

$$\begin{aligned} d_0 &= 4M + 1, & t_0 &= (s + 1)d_0 + 2M, \\ t_j &= t_0 + jd_0 \quad (0 \leq j < s), & t_* &= 2sd_0 + 2M, & K &= t_* + 1. \end{aligned}$$

Choose a power of two $L > 3K + 2$. The coefficient intervals $[t_j - 2M, t_j]$ are disjoint and lie above M . Moreover

$$2t_0 - 2M > t_*.$$

Include a dummy coordinate at each row position t_j . This last inequality makes every contribution involving a dummy lie above all targets, so the dummies cannot alter a tested residual.

Write $a_{j,I}$ for the coefficient of monomial I in G_j , and set

$$\begin{aligned} \mathcal{D}(B) &= \sum_{j,I} \frac{a_{j,I}}{c_I} B^{t_j - w(I)}, \\ \ell_0(B) &= \sum_{i=0}^{m-1} B^{v_i} + \sum_{j=0}^{s-1} B^{t_j}, \\ e_0(B) &= 2 \sum_{j=0}^{K-1} B^j + \mathcal{D}(B), & V &= \ell_0(4) + e_0(4)4^L. \end{aligned}$$

Monomial uniqueness, disjoint row intervals, and the dummy inequality give

$$[B^{t_j}] \mathcal{D}(B) \mathcal{C}(B)^2 = G_j. \quad (21.3)$$

They also show that the coefficients of $\mathcal{D}$ lie in (21.2). Consequently all digits of e_0 belong to $\{0, 1, 2, 3\}$. Zero digits are permitted. Since $\mathcal{D}$ has no constant term, the unit digit of e_0 is exactly 2; in particular e_0 and V are positive.

Put $c_* = m + s + 1$, the number of code coordinates including the input. Choose a power of two

$$H > \max\{2 \cdot 4^{2L+1}, 4^{t_*+3} 4c_*^2, 3L, 16\}. \quad (21.4)$$

An admissible index is a triple (V, H, L) obtained by this construction from a finite representing system. All choices, including the circuit size, are made independently of the queried input x . The supplied positive parameters are exactly (x, V, H, L) . The variable number of decoded coordinates introduces no new unknown in the final system.

21.3 The fixed-four system and its preliminary bounds

In the system of Theorem 20.1, set $Z = 4$ and replace the former fixed exponent by the index parameter L . With the same charged abbreviations and positive witnesses, the coding equations become

$$\begin{aligned} B &= Hb^2, & b &= x + \beta, & \lambda(B - 1) &= q^2 - 1, & \theta + 4 &= B, \\ S_2 &= \ell + eq = V + t\theta, & S_2 + \alpha &= q^2, & \Omega &= 4\lambda - 2e > 0, & \mathcal{C} &= x + g, \\ S_3 &= B\lambda q - \Omega\mathcal{C}^2, & \sigma &= S_3 > 0, & n &= q^8. \end{aligned}$$

The packing is

$$\begin{aligned} S &= g + q^2(S_2 + q^2 S_3), \\ T + 1 &= q^2(1 + \theta\lambda) - (b - 1)\ell + \theta\ell q^4, \\ r &= S(n^2 - n) + (T + 1)(n^2 - 1). \end{aligned}$$

All common-witness Pell equations of Section 20 are retained. In the second exponential congruence the expression $a - (B - 4)$ is now $a - \theta$.

Before decoding, $S_2 < q^2$ gives $e < q$ and $\ell < q^2$. The geometric equation and $H > 16$ give $B \leq q^2$, $q > 4b$, and $3L \leq B \leq n$. Since $\Omega \geq 1$ and $S_3 > 0$,

$$\mathcal{C}^2 < B\lambda q < 3q^3 < q^4.$$

The first block borrows fewer than b units from the middle block; the latter remains positive because $\theta = B - 4 > b$. The width-(2, 2, 4) argument of Section 19 therefore gives

$$0 < S < n, \quad 0 < T + 1 \leq n, \quad n \leq r \leq 2n^3 - 2n^2.$$

These are the hypotheses of the common-witness Pell argument. Its exponent comparisons use $L \geq 2$ and $3L \leq B \leq n \leq r$, which hold for every admissible index here. It follows that b, B, q are powers of two, $q = B^L$, and the required central-binomial divisibility holds. No bound on the number of compiled rows is used in that argument.

21.4 Canonical codes, zero digits, and the target tests

Now $\lambda > q$, so $\Omega > 2\lambda$. Positivity of S_3 gives $\mathcal{C}^2 < Bq/2 < q^2$, hence $g < \mathcal{C} < q$. Let $d = \lfloor (b-1)\ell/q^2 \rfloor < b$. Subtracting d from $(B-4)\lambda$ changes only its unit digit. The middle binary mask therefore bounds all nonunit digits of S_2 by three and its unit digit by $3 + d$. Thus

$$\ell \leq S_2 < \frac{4q^2}{B-1} + b.$$

Using $B = Hb^2$ and $H > 16$ shows $(b-1)\ell < q^2$, so $d = 0$. The canonical evaluation lemma applies to $\ell_0(B) + e_0(B)B^L$, whose digits are below four and whose degree is below $2L$. The first term of (21.4) supplies the required evaluation bound. It yields

$$S_2 = \ell_0(B) + e_0(B)q, \quad \ell = \ell_0(B) + a_{\text{al}}q, \quad e = e_0(B) - a_{\text{al}}, \quad 0 \leq a_{\text{al}} < e_0(B).$$

The quantity a_{al} is the temporary quotient alias.

The first mask, with $g < q$, decodes the low positions specified by ℓ_0 . Every coordinate is below b , and the unit digit of $\mathcal{C}$ is x . Put

$$A_{\mathcal{C}} = \mathcal{C}(1)^2, \quad J_{\mathcal{C}} = 4A_{\mathcal{C}}.$$

There are c_* coordinates, so $\mathcal{C}(1) < c_*b$ and $J_{\mathcal{C}} < B/4$. Also $e, \mathcal{C} < B^K$ and $L > 3K + 2$ give $2e\mathcal{C}^2 < q$.

Lemma 19.1 applies with $Z = 4$. Its coefficient plateau is $J_{\mathcal{C}}$: the digit of S_3 at L is $B - J_{\mathcal{C}} - 1$ or $B - J_{\mathcal{C}}$, and the digits at $L + 1, \dots, L + K$ are $B - J_{\mathcal{C}}$. The high third-mask part is $X = (B - 4)a_{\text{al}}$, of degree at most K . No overlap bounds every digit of X by $J_{\mathcal{C}}$. For its digit polynomial F_X , evaluation at four gives

$$B - 4 \mid F_X(4), \quad 0 \leq F_X(4) \leq J_{\mathcal{C}} \frac{4^{K+1} - 1}{3} < \frac{B}{12} < B - 4.$$

The second bound on H proves the strict inequality. Hence $X = 0$ and $a_{\text{al}} = 0$: both codes are canonical. Zero digits of e_0 have caused no change, since the evaluation lemma requires digits in $[0, 4)$ and the high-window argument requires $e_0 > 0$, not positivity of each digit.

Below K , the raw coefficients of S_3 are those of $2\mathcal{DC}^2$. The non-strict coefficient bound $|\mathcal{D}_j| \leq 2$ is sufficient because

$$|[B^j] 2\mathcal{DC}^2| \leq 4\mathcal{C}(1)^2 = J_{\mathcal{C}} < B/4.$$

All incoming carries in this range are zero or minus one. The mask $B - 4$ permits just the digits $0, 1, 2, 3$; a negative raw value plus its incoming carry normalizes to a digit greater than three. The special target has raw coefficient $2\delta^2$, forcing $\delta \in \{0, 1\}$. If $\delta = 0$, the guard has raw coefficient $-4x^2$ and fails. Thus $\delta = 1$. Each ordinary target now has raw coefficient $4F_j + 2$. With either possible incoming carry, it is permitted exactly when $F_j = 0$. The original system therefore holds at the actual queried input x .

Conversely, extend an original solution by its gate values, take $\delta = 1$ and $u = x^2$, and choose a power-of-two b larger than all coordinates. Use the canonical codes. The low coordinate tests vanish by support separation, and every ordinary or special target has raw coefficient 2, hence normalized digit 1 or 2. The inequalities $e_0, \ell_0 < q$ give $S_2 < q^2$ and $\Omega > 2\lambda$. Since $2K < L$, one has $4\mathcal{C}^2 < Bq$, which gives $S_3 > 0$. All required positive slack witnesses therefore exist. The positive Pell witnesses from Section 20 complete the final system. This proves both directions.

Theorem 21.1. *For every recursively enumerable subset of the positive integers, there is an admissible fixed index (V, H, L) such that its membership relation is represented by a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 101 operations: 56 multiplications and 45 additions. The supplied parameters are (x, V, H, L) , with three fixed index parameters. Fixed numerals are free.*

Proof. The preceding construction proves the representation and both positive-domain implications. In the 102-operation certificate, replace Z by 4, replace the old exponent numeral by L , and replace all uses of the separately computed $B - 4$ by θ . The already present equation $\theta + 4 = B$ justifies that substitution. It removes one subtraction, with no added arithmetic, unknown, or equation. The exact primitive and polynomial-residual check is recorded in `round23_1980_fixed_four_certificate.py` and its JSON receipt. Thus the count is $56 + 45 = 101$; no optimality is asserted. $\square$

22 A doubled-index congruence in the Pell block

Two changes allow the main norm coefficient to reuse the first exponential modulus. The auxiliary signed norm can compare doubled Pell indices, and the final index equation can use a fixed Pell value in place of the index itself. Together they remove both affine parameters $A - 1$ and $A + 1$ from the arithmetic certificate, reducing Theorem 21.1 by one addition.

Retain the fixed-four encoding of Section 21 and the common exponential witness of Section 20. Write

$$A = a + 4, \quad D = A^2 - 1, \quad J = 2r + 1, \quad H_0 = J + jc,$$

where the supplied positive unknown a still satisfies

$$a = Y_P(U + 1), \quad U = wn^2, \quad Y_P = sn^2.$$

The first exponential equation and its modulus remain

$$M = 8a + 15 = 8A - 17, \quad d = U + ac + \gamma M. \quad (22.1)$$

Replace the signed auxiliary norm by

$$\begin{aligned} v &= of - d^2, & K &= D(f^2 - 1), & G &= 2K - 1, \\ v(v + 1) &= K(K - 1)H_0^2. \end{aligned} \quad (22.2)$$

The quantities v, K, G are abbreviations; v is a computed integer and need not be positive. No unknown is added. If L is the underlying power-of-two exponent used to construct the coefficient index, replace the supplied component L by

$$T_L = \psi_4(L)$$

and replace the final index equation by

$$\kappa = T_L + \Delta a. \quad (22.3)$$

Thus the four supplied inputs are (x, V, H, T_L) . The admissible index is constructed by choosing the same finite circuit layout and exponent L as before, constructing V, H from them, and supplying $\psi_4(L)$. These choices are independent of x . Every other source equation is unchanged.

22.1 Recovering the main index from doubled coordinates

Lemma 22.1. *Let $A > 1$, let $J > 1$ be odd, and suppose $c > J$. Over positive integer unknowns, the two norms*

$$d^2 - (A^2 - 1)c^2 = 1, \quad f^2 - (A^2 - 1)(ic^2)^2 = 1$$

together with (22.2) imply $c = \psi_A(J)$ and $d = \chi_A(J)$. Conversely, when those two identities hold, positive f, i, j, o satisfying the auxiliary equations exist.

Proof. Pell classification gives positive indices p, m such that

$$d = \chi_A(p), \quad c = \psi_A(p), \quad f = \chi_A(m), \quad ic^2 = \psi_A(m).$$

The divisibility fact used in Proposition 7.2 gives $c \mid m$ from $c^2 \mid \psi_A(m)$. The case $p = 1$ is excluded by $c > J > 1$. For $p = 2$ one has $\psi_A(2) = 2A \geq 4$; for $p \geq 3$,

$$\psi_A(p) \geq (2A - 1)^{p-1} \geq 3^{p-1} \geq 2p.$$

Consequently

$$0 < 2p \leq c \leq m. \quad (22.4)$$

Here $K > 1$ and $G > 1$. The new norm is exactly

$$(2v + 1)^2 - (G^2 - 1)H_0^2 = 1.$$

Its right-hand product $K(K - 1)H_0^2$ is positive, so $v \geq 1$ or $v \leq -2$. Taking $I = |2v + 1|$, another application of Pell classification gives

$$I = \chi_G(t), \quad H_0 = \psi_G(t)$$

for a positive index t . Reduction modulo f gives

$$\begin{aligned} G &= 2D(f^2 - 1) - 1 \equiv -\chi_A(2), \\ 2v + 1 &= 2of - 2d^2 + 1 \equiv -\chi_A(2p). \end{aligned}$$

The identities $\chi_{-z}(t) = (-1)^t \chi_z(t)$ and $\chi_{\chi_A(2)}(t) = \chi_A(2t)$ therefore imply

$$\chi_A(2t) \equiv \pm \chi_A(2p) \pmod{\chi_A(m)}.$$

By Lemma 7.1, whose comparison hypothesis is (22.4),

$$2t \equiv \pm 2p \pmod{2m}, \quad t \equiv \pm p \pmod{m}, \quad t \equiv \pm p \pmod{c}.$$

The middle implication divides an integer equality by two; it does not assume that two is invertible modulo m .

Since $c \mid f^2 - 1$, one has $G \equiv -1 \pmod{c}$. The recurrence for ψ then gives

$$H_0 \equiv \psi_{-1}(t) = (-1)^{t-1} t \pmod{c}.$$

Together with $H_0 = J + jc$, this proves $J \equiv \pm p \pmod{c}$. Both J and p lie in $(0, c)$, so $J = p$ or $J + p = c$. The latter contradicts $\psi_A(p) \equiv p \pmod{2}$ and the oddness of J . Thus $p = J$. No oddness condition on f was used in this direction.

For necessity, start with $c = \psi_A(J)$ and $d = \chi_A(J)$. Choose

$$m = 2cJ, \quad f = \chi_A(m), \quad i = \frac{\psi_A(m)}{c^2}.$$

The quotient is a positive integer: in $(d + c\sqrt{D})^{2c}$ the term with one square root contains $(2c)c = 2c^2$, and every remaining square-root term contains at least c^3 . Moreover m is even, so $f = 2\chi_A(m/2)^2 - 1$ is odd. Put

$$K = D(f^2 - 1), \quad G = 2K - 1, \quad I = \chi_G(J), \quad H_0 = \psi_G(J).$$

The parameter G is odd and exceeds one, so I is odd and greater than one. Because J is odd,

$$I \equiv -\chi_A(2J) = 1 - 2d^2 \pmod{f}.$$

Thus $I + 2d^2 - 1$ is divisible by both 2 and the odd number f . The choices

$$v = \frac{I - 1}{2} > 0, \quad o = \frac{I + 2d^2 - 1}{2f} > 0$$

are integers and satisfy $v = of - d^2$. Also $G \equiv -1 \pmod{c}$ and J is odd, whence $H_0 \equiv J \pmod{c}$. Strict Pell growth makes $j = (H_0 - J)/c$ a positive integer. The Pell norm for I, H_0 gives (22.2). $\square$

The required hypotheses $A > 1$ and $1 < J < c$ are established by the unchanged preliminary packing and interval arguments. Thus this lemma is available at the same point as the preceding signed-block lemma, before either exponential relation or binary decoding is used.

22.2 Encoding the fixed index by a Pell value

Lemma 22.2. *In the new system, (22.3) and the unchanged second norm and gap force $\kappa = \psi_A(L)$ and $\mu = \chi_A(L)$. Conversely, those coordinates give a positive integer Δ satisfying (22.3).*

Proof. After Lemma 22.1, the common-witness argument is unchanged through the first exponential equation. The ratio lower bound first gives $Y_P \geq U^r$ and $a = Y_P(U + 1) > U^{r+1}$. Equation (22.1) then gives $U = 4^J$. Since $r + 1 > 3$, the modulus in the new index equation satisfies the stronger bound

$$a > U^{r+1} > 4^{3J}. \quad (22.5)$$

This is a bound for $a = A - 4$, not merely for A .

The unchanged second norm and $c = \kappa + \phi$ give

$$\kappa = \psi_A(t_0), \quad \mu = \chi_A(t_0), \quad 0 < t_0 < J.$$

Since $A \equiv 4 \pmod{a}$, the integer polynomial recurrence for ψ and (22.3) yield

$$\psi_4(t_0) \equiv \psi_4(L) \pmod{a}.$$

The underlying encoding still satisfies $3L \leq B \leq n \leq r$, so $L < J$. For $1 \leq s < J$,

$$0 < \psi_4(s) \leq 8^{s-1} < 4^{3J} < a.$$

The two residues therefore agree as integers. Strict growth of ψ_4 gives $t_0 = L$.

Conversely, for the intended coordinates the quotient

$$\Delta = \frac{\psi_A(L) - \psi_4(L)}{a}$$

is integral by the same recurrence and strictly positive because $A > 4$ and $L \geq 2$. Strict growth in the base follows, for example, from the positive binomial expansion of the Pell solution. $\square$

Once this index is recovered, the second exponent test proves $q = B^L$ with the same bounds as before. The common-witness proof has already supplied the power-of-two radix and the requisite binomial divisibility. All fixed-four coefficient decoding is consequently unchanged. In necessity, the new fixed component T_L is chosen with V, H before querying x ; the remaining Pell witnesses are chosen by the common-witness construction and the two lemmas above.

22.3 The complete count

The earlier certificate computed the norm coefficient by

$$a_- = a + 3, \quad a_+ = a_- + 2, \quad D = a_- a_+.$$

Neither affine value is required after the two changes. Since $M = 8a + 15$ is already computed for the first exponent test, use instead

$$a_2 = a^2, \quad D = a_2 + M.$$

This replaces three instructions by two. The auxiliary block has the same length as before: use the existing d^2 in $v = of - d^2$; replace the former root square and final addition of one by $v + 1$ and $v(v + 1)$; and use the three coefficient instructions $K = DAE$, $K_- = K - 1$, and KK_- , where the existing E16 register is $AE = D(ic^2)^2$. The final index equation still takes one product and one addition.

For exact comparison with the source equations, let

$$F_{16} = f^2 - AE - 1, \quad K_s = D(f^2 - 1), \quad K_c = DAE.$$

The changed norm has the triangular identity

$$F_{17}^{\text{calc}} = F_{17}^{\text{source}} + DF_{16}(K_s + K_c - 1)H_0^2.$$

The executable certificate checks this identity, the inherited E7 and E18 corrections from $\theta = B - 4$, every other residual, and every primitive instruction. The fixed-index and positive-domain conclusions are supplied by the proofs above, independently of these symbolic checks.

Theorem 22.3. *For every represented set, an admissible fixed index (V, H, T_L) can be chosen such that the system described above represents it over 34 positive integer unknowns with 22 equality tests. Its straight-line certificate has*

$$\boxed{100 = 56 \text{ multiplications} + 44 \text{ additions}}.$$

Fixed numerals and supplied index parameters are free. Relative to Theorem 21.1, the saving is one addition.

The full source equations, primitive schedule, domains and residual identities are recorded by `round24_1980_doubled_pell_certificate.py` and its JSON receipt.

23 Halving the coefficient and sharing the geometric factor

The fixed-four construction admits another saving, independent of the Pell change in the preceding section. Starting from Theorem 21.1, replace its positive coefficient and third block by

$$\Omega = 2\lambda - e > 0, \quad S_3 = \theta\lambda q - \Omega\mathcal{C}^2 = \sigma > 0, \quad \mathcal{C} = x + g, \quad \theta = B - 4. \quad (23.1)$$

The factor $\theta\lambda$ can now be shared with the geometric equation. To accommodate the smaller coefficient, rebuild the fixed index by pairing ordinary equation tests and placing the special target first. No packed bound or Pell equation is removed.

23.1 Paired rows and a first special target

Use the finite primitive circuit construction of Section 21, with distinct copied operands and fresh outputs, all separate from the homogenizing coordinate δ . For every multiplication, addition, copy, final equality, or zero residual

$$XY - W\delta, \quad (X + Y - W)\delta, \quad (X - Y)\delta, \quad X\delta,$$

include both F and $-F$ among the ordinary rows. Omit trivial identical equalities. Replace each unit gate $X = 1$ by two unpaired rows

$$F_{X,1} = X\delta - \delta^2, \quad F_{X,2} = X\delta - X^2. \quad (23.2)$$

Include the unpaired guard $F_g = u\delta - x^2$, with u fresh. Every ordinary row has target

$$G_j = 2F_j + \delta^2. \quad (23.3)$$

The special target $G_0 = \delta^2$ is placed *first*, before all ordinary targets.

After division by the monomial coefficient in $\mathcal{C}(T)^2$, the coefficient alphabet is still $\{-2, -1, 0, 1\}$. Paired primitive rows have cross-term quotients ± 1 and the added square δ^2 has quotient 1. The two unit targets are

$$2X\delta - \delta^2, \quad 2X\delta - 2X^2 + \delta^2,$$

and the guard has the latter coefficient pattern. The special target has its single coefficient 1. Thus the pairing does not enlarge the digit alphabet. At a genuine circuit solution with $\delta = 1$, every ordinary F_j is zero, including both unit rows; choose $u = x^2$ for the guard.

23.2 The rebuilt fixed index and the absence of an initial carry

Apply the dynamic support construction of Section 21 to this finite list. With m positive-weight coordinates and s targets, write

$$\begin{aligned} v_i &= 3^i \quad (0 \leq i < m), & M &= 3^{m-1}, & d_0 &= 4M + 1, \\ t_0 &= (s + 1)d_0 + 2M, & t_j &= t_0 + jd_0, & t_* &= 2sd_0 + 2M, \\ K &= t_* + 1, & L &> 3K + 2, \end{aligned}$$

where L is a power of two, x has weight zero, and δ has weight $v_0 = 1$. The special target is at t_0 . Include the usual dummy coordinate at each t_j . Monomial uniqueness, separation of the intervals $[t_j - 2M, t_j]$, and $2t_0 - 2M > t_*$ give

$$[T^{t_j}] \mathcal{D}(T)\mathcal{C}(T)^2 = G_j. \tag{23.4}$$

In particular, no term involving a dummy contributes to any target.

Define the fixed polynomials and parameters by

$$\begin{aligned} \ell_0(T) &= \sum_{i=0}^{m-1} T^{v_i} + \sum_{j=0}^{s-1} T^{t_j}, \\ e_0(T) &= 2 \sum_{j=0}^{K-1} T^j + \mathcal{D}(T), & V &= \ell_0(4) + e_0(4)4^L. \end{aligned}$$

The coefficients of e_0 are in $\{0, 1, 2, 3\}$, with unit digit 2. Put $c_* = m + s + 1$, counting all code coordinates including the input, and choose a power of two

$$H > \max\{2 \cdot 4^{2L+1}, 4^{t_*+3} 4c_*^2, 3L, 16\}. \tag{23.5}$$

The resulting (V, H, L) is an admissible index. Its construction is independent of the queried input and introduces no additional unknown in the final system.

The first target has a stronger support property than the others. Its sole coefficient in $\mathcal{D}$ is $+1$ at $t_0 - 2$. Every later row has its entire coefficient support above t_0 , because

$$t_1 - 2M = t_0 + d_0 - 2M > t_0.$$

Consequently every raw coefficient of $\mathcal{D}(T)\mathcal{C}(T)^2$ through position t_0 is nonnegative. After coordinate decoding these coefficients are below B , as proved below. Hence the incoming carry at the special target is exactly zero. This argument assumes neither $\delta = 1$ nor any ordinary equation.

23.3 Bounds before the Pell argument

Retain $B = Hb^2$, $b = x + \beta$, $n = q^8$, and

$$\lambda(B - 1) = q^2 - 1, \quad \theta + 4 = B, \quad S_2 = \ell + eq = V + t\theta, \quad S_2 + \alpha = q^2. \quad (23.6)$$

All supplied unknowns, including Ω, σ, α , remain positive integers. Keep the packing

$$\begin{aligned} S &= g + q^2(S_2 + q^2 S_3), \\ T + 1 &= q^2(1 + \theta\lambda) - (b - 1)\ell + \theta\ell q^4, \\ r &= S(n^2 - n) + (T + 1)(n^2 - 1). \end{aligned} \quad (23.7)$$

The packed bound gives $e < q$ and $\ell < q^2$. Geometry and (23.5) give

$$B \leq q^2, \quad q > 4b \geq 8, \quad 3L \leq B \leq n.$$

Since $\Omega \geq 1$ and $S_3 > 0$, the new offset gives

$$\mathcal{C}^2 < \theta\lambda q < B\lambda q < 3q^3 < q^4. \quad (23.8)$$

Thus $g < \mathcal{C} < q^2$ and $0 < S_3 < q^4$. The mask $T + 1$ is unchanged; its first block borrows fewer than b units from the middle block, which remains positive since $\theta = B - 4 > b$. The same width-(2, 2, 4) bounds therefore give

$$0 < S < n, \quad 0 < T + 1 \leq n, \quad n \leq r \leq 2n^3 - 2n^2 < 2n^3. \quad (23.9)$$

These bounds have been established before any Pell equation is used. In particular $n, r \geq 64$, $U = wn^2 \geq n^2$ and $Y_P = sn^2 \geq n^2$. The main parameter satisfies $a = Y_P(U + 1) > n^4 > 2r + 1$, and the first index modulus satisfies $UY_P \geq n^4 > r + 1$. Moreover $L \geq 2$ and $3L \leq B \leq n \leq r$. Thus every preliminary hypothesis of the common-witness Pell proof in Section 20 is available. All its equations are retained. That proof yields powers of two b, B, q , the exact equality $q = B^L$, and

$$n^2 \mid \binom{2r}{r}. \quad (23.10)$$

No internal target equation has entered this argument.

Now $\lambda > q$ and $e < q$, so $\Omega = 2\lambda - e > \lambda$. Positivity in (23.1) improves the code bound to

$$\mathcal{C}^2 < \theta q < Bq < q^2, \quad g < \mathcal{C} < q. \quad (23.11)$$

The original factor one-half in the bound for $\mathcal{C}^2$ is no longer asserted; the strict inequality $\mathcal{C} < q$ is what the subsequent decoding needs.

23.4 Canonical decoding and the revised high window

By (23.9) and (23.10), the packed binary no-carry tests hold. Let $d = \lfloor (b - 1)\ell/q^2 \rfloor < b$. Subtracting d from $(B - 4)\lambda$ changes only its unit digit. The middle mask therefore bounds all nonunit digits of S_2 by three and its unit digit by $3 + d$. It follows that

$$\ell \leq S_2 < \frac{4q^2}{B - 1} + b.$$

Together with $B = Hb^2$ and $H > 16$, this forces $(b-1)\ell < q^2$, hence $d = 0$. The base-four evaluation lemma now applies to the fixed polynomial defining V : its degree is below $2L$, all digits are below four, and the first term of (23.5) supplies the evaluation bound. Consequently

$$\begin{aligned} S_2 &= \ell_0(B) + e_0(B)q, \\ \ell &= \ell_0(B) + a_{\text{al}}q, \quad e = e_0(B) - a_{\text{al}}, \quad 0 \leq a_{\text{al}} < e_0(B). \end{aligned} \tag{23.12}$$

The first mask and $g < q$ decode the low positions specified by ℓ_0 . The unit digit of g is zero; every coordinate is a nonnegative integer below b , and the unit digit of $\mathcal{C}$ is x . Dummy digits occur only at row positions. Write

$$A_{\mathcal{C}} = \mathcal{C}(1)^2, \quad J_{\mathcal{C}} = 2A_{\mathcal{C}}.$$

Then $\mathcal{C}(1) < c_*b$, and (23.5) gives $J_{\mathcal{C}} < B/4$. Also $e, \mathcal{C} < B^K$ and $L > 3K + 2$ give $e\mathcal{C}^2 < q$. Since $g > 0$ and $x > 0$, the decoded coefficient sum is at least two, so

$$A_{\mathcal{C}} \geq 4. \tag{23.13}$$

This holds before identifying δ , even if the nonzero digit of g occupies a dummy position.

Write

$$2\lambda\mathcal{C}^2 = qV_1 + V_0, \quad 0 \leq V_0 < q.$$

Its polynomial coefficients are at most $J_{\mathcal{C}} < B$, so they are its base- B digits. The stable convolution interval $L, \dots, L + K$ has all digits equal to $J_{\mathcal{C}}$, because $L > 3K + 2$. Exact division of (23.1) by q therefore gives

$$\left\lfloor \frac{S_3}{q} \right\rfloor = \theta\lambda - V_1 + \epsilon, \quad \epsilon = \left\lfloor \frac{e\mathcal{C}^2 - V_0}{q} \right\rfloor \in \{-1, 0\}.$$

Every digit of $\theta\lambda$ through position $2L - 1$ is $B - 4$. Subtracting V_1 creates no borrow in the window: its first digit $B - 4 - J_{\mathcal{C}} + \epsilon$ is positive, as are all subsequent digits. Thus

$$(S_3)_L = B - 4 - J_{\mathcal{C}} + \epsilon, \quad (S_3)_{L+j} = B - 4 - J_{\mathcal{C}} \quad (1 \leq j \leq K). \tag{23.14}$$

The high part of the third mask is $X = (B - 4)a_{\text{al}} < B^{K+1}$. Binary nonoverlap bounds each digit of X by

$$J_{\mathcal{C}} + 4 = 2A_{\mathcal{C}} + 4 \leq 4A_{\mathcal{C}},$$

using (23.13). If F_X is its digit polynomial, then $B - 4 \mid F_X(4)$, whereas

$$0 \leq F_X(4) \leq 4A_{\mathcal{C}} \frac{4^{K+1} - 1}{3} < \frac{B}{12} < B - 4.$$

The strict bound follows from (23.5), without enlarging H . Therefore $F_X(4) = 0$, $X = 0$, and $a_{\text{al}} = 0$. Both supplied codes are canonical.

23.5 The low target tests and positive necessity

With the canonical codes, the raw coefficients of S_3 below K are those of $\mathcal{D}(T)\mathcal{C}(T)^2$. The positive offset starts at $L > K$, and the missing baseline tail starts at K . Every raw coefficient has absolute value at most $2A_{\mathcal{C}} < B/4$, so every incoming carry is zero or minus one. At an ordinary target put $v = 2F + \delta^2$. The mask $B - 4$ permits exactly the digits 0, 1, 2, 3 and is equivalent to

$$0 \leq v + \epsilon \leq 3, \quad \epsilon \in \{-1, 0\}. \tag{23.15}$$

A negative value would normalize to a digit greater than three.

At the first special target there is no incoming carry, by its support property. The target is δ^2 , so $0 \leq \delta^2 \leq 3$, forcing $\delta \in \{0, 1\}$. If $\delta = 0$, the guard has $v = -2x^2 < 0$ and cannot pass. Thus $\delta = 1$. For either possible carry, (23.15) now says

$$0 \leq 2F + 1 + \epsilon \leq 3 \iff F \in \{0, 1\}.$$

Paired rows F and $-F$ force $F = 0$. The unit rows give

$$X - 1 \in \{0, 1\}, \quad X - X^2 \in \{0, 1\}.$$

The first restricts X to $\{1, 2\}$, and the second restricts it to $\{0, 1\}$, so $X = 1$. The guard need not vanish in sufficiency: it excludes $\delta = 0$, and its coordinate u is absent from the original circuit. Every original gate and equality has now been recovered exactly at the input x . The low coordinate tests are automatic because $\mathcal{D}$ starts above their positions.

Conversely, take a solution of the represented system, extend it by its circuit values, and choose $\delta = 1$, $u = x^2$, and zero dummy coordinates. Choose a power-of-two b larger than every coordinate and use the canonical codes and $q = B^L$. Every ordinary F is zero. The special target has raw coefficient one and carry zero; each ordinary target has raw coefficient one and actual digit zero or one. These digits are allowed, and the low variable tests vanish. The first two masks hold as in Section 21.

All positive slack witnesses exist. In particular, the unit digit of e_0 is two, the positive δ digit makes $g > 0$, and $S_2 < q^2$. The packed-congruence quotient is positive because $P(B) > P(4)$ for the nonconstant polynomial $P(T) = \ell_0(T) + e_0(T)T^L$ with nonnegative coefficients. Also $\Omega = 2\lambda - e > \lambda > 0$. Since $\mathcal{C} < B^K$ and $L > 3K + 2$,

$$2\mathcal{C}^2 < q \leq \theta q, \quad S_3 = \lambda(\theta q - 2\mathcal{C}^2) + e\mathcal{C}^2 > 0.$$

The three masks give (23.10), and their packing satisfies (23.9). The complete positive Pell witness construction of Section 20 therefore applies to the newly calculated r . This proves necessity for the rebuilt fixed index.

23.6 Seven shared instructions in place of eight

The predecessor uses eight instructions for the relevant quantities:

$$\begin{aligned} L_2 &= \lambda + q^2, & \Lambda &= B\lambda, & R_2 &= \Lambda + 1, \\ z_\lambda &= 4\lambda, & e_2 &= 2e, & t_2 &= z_\lambda - e_2, \\ T_{\text{coef}} &= L_2 - z_\lambda, & P_2 &= \Lambda q. \end{aligned}$$

Use instead the seven instructions

$$\begin{aligned} U_\lambda &= \theta\lambda, & \lambda_2 &= 2\lambda, & T_{\text{coef}} &= U_\lambda + 1, \\ \lambda_3 &= \lambda_2 + \lambda, & R_2 &= T_{\text{coef}} + \lambda_3, \\ t_2 &= \lambda_2 - e, & P_2 &= U_\lambda q. \end{aligned} \tag{23.16}$$

Test $q^2 = R_2$ and $t_2 = \Omega$. In conjunction with $\theta + 4 = B$, the first test is exactly $\lambda(B - 1) = q^2 - 1$. All other arithmetic instructions remain, including the packed upper bound and $S_3 = P_2 - t_2\mathcal{C}^2$.

Theorem 23.1. *Every recursively enumerable subset of the positive integers has an admissible fixed index (V, H, L) for a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 100 operations: 55 multiplications and 45 additions. The supplied parameters are (x, V, H, L) ; fixed numerals are free.*

Proof. The construction above proves both positive-domain implications. The replacement (23.16) has three multiplications and four additions, instead of four multiplications and four additions, and every other instruction is retained. Hence the 101-operation certificate loses one multiplication.

The exact verification is `round25_1980_half_center_certificate.py` and its JSON receipt. It checks the full 100-instruction primitive list and all 22 polynomial residuals. If F_3 is the original geometric residual and $F_4 = \theta + 4 - B$, the new geometric test has residual $F_3 - \lambda F_4$. The new packing and positive- S_3 source equations use θ explicitly; they need no geometric correction. The existing signed Pell and second-exponent corrections are also verified. Thus the count concerns an explicit certificate for the proved system; no optimality is claimed. $\square$

24 A certificate of 99 operations

The reductions in Sections 23 and 22 can be applied together. Their arithmetic changes occupy disjoint parts of the certificate. The composition also preserves the order of the mathematical proof: the new packing first supplies the bounds for the new Pell block; that block then recovers the exponent and binary radix needed by the new digit tests.

Theorem 24.1. *For every recursively enumerable subset of the positive integers, there is an admissible fixed index (V, H, T_L) such that its membership relation is represented by a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 99 operations: 55 multiplications and 44 additions. Here $T_L = \psi_4(L)$ for the underlying exponent of the index construction. The supplied parameters are (x, V, H, T_L) , with three fixed index parameters, and fixed numerals are free.*

Proof. Construct the coefficient index using the entire paired-row construction of Section 23. In particular, the special target δ^2 comes first; multiplication, addition, copy, equality, and zero rows are paired with their negatives; each unit gate uses the two unpaired rows $X\delta - \delta^2$ and $X\delta - X^2$; and the guard is $u\delta - x^2$. Every ordinary target is $2F + \delta^2$.

If there are m positive-weight coordinates and s targets, use

$$\begin{aligned} v_i &= 3^i \quad (0 \leq i < m), & M &= 3^{m-1}, & d_0 &= 4M + 1, \\ t_j &= (s + 1)d_0 + 2M + jd_0 \quad (0 \leq j < s), \\ t_* &= 2sd_0 + 2M, & K_0 &= t_* + 1, & c_* &= m + s + 1. \end{aligned}$$

Choose a power of two $L > 3K_0 + 2$, form the corresponding ℓ_0, e_0 , and set

$$V = \ell_0(4) + e_0(4)4^L, \quad H > \max\{2 \cdot 4^{2L+1}, 4^{t_*+3} 4c_*^2, 3L, 16\},$$

with H a power of two. Finally supply $T_L = \psi_4(L)$ instead of L . Thus V, H come from the new paired layout throughout; they are not reused from an earlier unpaired index. These choices are independent of the queried input x .

Keep the coding equations and packing (23.1)–(23.7), with

$$\Omega = 2\lambda - e > 0, \quad \sigma = \theta\lambda q - \Omega\mathcal{C}^2 > 0, \quad \theta = B - 4,$$

and apply both Pell changes (22.2) and (22.3). All other common-witness Pell equations remain unchanged.

Before either code is decoded, the positive bound $S_2 + \alpha = q^2$ gives $e < q$ and $\ell < q^2$. The geometric equation gives $B \leq q^2$, and the index bound gives $q > 4b \geq 8$ and $3L \leq B \leq n$. Positivity of Ω and σ gives

$$\mathcal{C}^2 < \theta\lambda q < 3q^3 < q^4.$$

The preliminary borrow argument of Section 23 therefore proves

$$0 < S < n, \quad 0 < T + 1 \leq n, \quad n \leq r \leq 2n^3 - 2n^2 < 2n^3. \quad (24.1)$$

No coefficient target has been tested at this stage.

Put $U = wn^2$, $Y_P = sn^2$, $a = Y_P(U + 1)$, $A = a + 4$, and $J = 2r + 1$. Then $U, Y_P \geq n^2$, $UY_P \geq n^4 > r + 1$, and the first-index congruence and interval give $c > J$. Also $A > n^4 > J$. Thus every hypothesis of Lemma 22.1 holds: $A > 1$, $1 < J < c$, and J is odd. That lemma proves

$$c = \psi_A(J), \quad d = \chi_A(J).$$

In particular, its signed comparison does not depend on either coefficient decoding or an already known value of q .

The unchanged first Pell index and lower ratio argument now give, with $\xi = (U + 1)^{2r}/U^r$,

$$k = \psi_{2UY_P^2+1}(r + 1), \quad \xi < c/k < Y_P + 1, \quad Y_P \geq U^r, \quad a > U^{r+1}.$$

The first exponential congruence yields $U = 4^J$. Therefore n and q are powers of two, and $a > 4^{3J}$. The positive gap and second norm give $\kappa = \psi_A(t)$ for some $0 < t < J$. Admissibility gives $L < J$. Lemma 22.2 applies to $\kappa = T_L + \Delta a$: reduction modulo a gives $\psi_4(t) \equiv \psi_4(L)$, while both values lie strictly between zero and a . It follows that $t = L$.

The remaining exponent comparison uses

$$B^{3L} \leq B^B \leq n^n \leq U^r < A, \quad q^3 \leq n^n \leq U^r < A,$$

so it yields $q = B^L$, and hence powers of two B and b . The unchanged upper ratio and binomial arguments give

$$n^2 \mid \binom{2r}{r}. \quad (24.2)$$

These conclusions precede all internal circuit tests.

Now $\lambda > q$ and $e < q$, so $\Omega > \lambda$ and $\mathcal{C}^2 < \theta q < Bq < q^2$. Thus $g < \mathcal{C} < q$. All post-Pell hypotheses of Section 23 have been restored. Its first two masks remove the preliminary borrow and identify the packed coefficient polynomial, up to the quotient alias. The first mask bounds every coordinate by b . Positivity of x, g gives $\mathcal{C}(1) \geq 2$ before δ is identified. Accordingly, with $A_{\mathcal{C}} = \mathcal{C}(1)^2$, the modified high-window bound $2A_{\mathcal{C}} + 4 \leq 4A_{\mathcal{C}}$ and the displayed bound on H remove the alias. The two codes are canonical.

The first special target has incoming carry zero and forces $\delta \in \{0, 1\}$. The guard excludes zero. At $\delta = 1$, each ordinary row tests $F \in \{0, 1\}$; its paired negative forces $F = 0$, and the two unit rows force the unit coordinate to equal one. This recovers the represented system at the actual input x .

For the converse, take a solution of that system, its circuit values, $\delta = 1$, $u = x^2$, and zero dummy digits. Choose a power-of-two b larger than every coordinate and use the canonical codes. Then $\beta, \theta, \lambda, \alpha$ and the packed-congruence quotient are positive integers. The half-center necessity proof gives positive Ω, σ and all three no-carry conditions. Define r by the packing; (24.1) and (24.2) follow.

Choose $J = 2r + 1$, $U = 4^J$, and $w = U/n^2$. This quotient is a positive integer since n is a power of two and $n^2 \leq r^2 < 4^J$. Take $Y_P = \lfloor \xi \rfloor$ and $s = Y_P/n^2$, integral by the binomial congruence, and define $a = Y_P(U + 1)$. The common-witness construction supplies positive $k, c, d, \tau, h, \eta, \zeta, \gamma, \rho$. Choose

$$\kappa = \psi_A(L), \quad \mu = \chi_A(L), \quad \phi = c - \kappa, \quad \Delta = \frac{\psi_A(L) - \psi_4(L)}{a}.$$

Here $L < J$ makes $\phi > 0$, and $A = a + 4 > 4$, $L \geq 2$ make Δ a positive integer.

Finally choose the auxiliary witnesses exactly as in Lemma 22.1: set

$$\begin{aligned} m_0 &= 2cJ, & f &= \chi_A(m_0), & i &= \psi_A(m_0)/c^2, \\ K &= (A^2 - 1)(f^2 - 1), & G &= 2K - 1, & I &= \chi_G(J), & H_0 &= \psi_G(J), \\ o &= \frac{I + 2d^2 - 1}{2f}, & j &= \frac{H_0 - J}{c}. \end{aligned}$$

The proved divisibility, oddness of f , and Pell congruences make all four auxiliary unknowns positive integers and establish the doubled norm. This last choice changes no coding, interval, gap, or exponential witness. Thus every required positive unknown exists.

For the arithmetic count, the half-center substitution saves one multiplication from Theorem 21.1; the doubled-index substitution saves one addition. They change disjoint source rows and disjoint instruction groups. The exact checker `round26_1980_composed_certificate.py` verifies that both orders give the same complete schedule and source system, together with all primitive instructions, 22 equality tests, and three triangular residual corrections. The resulting count is $(56 - 1) + (45 - 1) = 55 + 44 = 99$. $\square$

25 A coefficient baseline of one

The coefficient baseline in Theorem 24.1 can be reduced from two to one. Copies remove the negative square coefficient in the unit test, while a direct guard excludes a zero homogenizing coordinate before any copy relation is used. The resulting coefficient is

$$\Omega = \lambda - e > 0, \quad S_3 = \theta \lambda q - \Omega \mathcal{C}^2 = \sigma > 0, \quad \mathcal{C} = x + g, \quad \theta = B - 4. \quad (25.1)$$

The offset, geometric equation, packed upper bound and all Pell equations are retained. Computing 3λ directly now saves one addition, giving 98 operations.

25.1 Copied unit tests and a direct guard

Keep the paired primitive rows of Section 23: each multiplication, addition, copy, final equality and zero residual F occurs together with $-F$, with targets $2F + \delta^2$. Operands are distinct copied coordinates, outputs are fresh, and circuit coordinates are distinct from δ .

For each unit gate $X = 1$, introduce a fresh nonnegative coordinate X' and the paired copy rows $(X - X')\delta$ and $(X' - X)\delta$. Replace its two unpaired unit rows by

$$F_{X,1} = X\delta - \delta^2, \quad F_{X,2} = X\delta - XX'. \quad (25.2)$$

All four rows still use targets $2F + \delta^2$. The negative product XX' has distinct factors, so its divided target coefficient is -1 , in place of the -2 of the old square X^2 .

Give the fresh guard coordinate u the direct target

$$G_g = 2u\delta - x^2 + \delta^2. \quad (25.3)$$

This target is not required to be $2F + \delta^2$ for an integer polynomial F . It will exclude $\delta = 0$ directly. Replacing x^2 by a copied product would not do this: its homogenized copy equation would impose no condition when $\delta = 0$. Keep the special target $G_0 = \delta^2$ first.

Every target coefficient, divided by its monomial coefficient in $\mathcal{C}(T)^2$, now belongs to

$$\{-1, 0, 1\}. \quad (25.4)$$

Paired primitive rows have cross-term quotients ± 1 and added δ^2 coefficient 1. The first unit target has coefficients 1, -1 after division; the second has cross-term quotients 1, -1 and square coefficient 1. The direct guard has cross-term quotient 1, coefficient -1 on x^2 , and coefficient 1 on δ^2 . Thus all these divisions are integral.

25.2 The explicit admissible index

Apply the support construction to the entire new finite list, including all unit-copy coordinates and paired copy rows. Let m be the number of positive-weight coordinates and s the number of targets. The input x has weight zero and δ has weight one. Set

$$\begin{aligned} v_i &= 3^i \quad (0 \leq i < m), & M &= 3^{m-1}, & d_0 &= 4M + 1, \\ t_j &= (s+1)d_0 + 2M + jd_0 \quad (0 \leq j < s), \\ t_* &= 2sd_0 + 2M, & K_0 &= t_* + 1, & c_* &= m + s + 1. \end{aligned}$$

The special target occupies t_0 . Include a dummy coordinate at every target position and choose a power of two $L > 3K_0 + 2$. The complementary coefficient polynomial $\mathcal{D}$ has coefficients in (25.4) and satisfies

$$[T^{t_j}] \mathcal{D}(T) \mathcal{C}(T)^2 = G_j.$$

The base-three monomial uniqueness and disjoint row intervals prove this identity exactly; $2t_0 - 2M > t_*$ excludes every dummy contribution. The support of $\mathcal{D}$ lies above every low coordinate and below K_0 . Through t_0 its only term is $+T^{t_0-2}$, since every later row starts above t_0 .

Define

$$\begin{aligned} \ell_0(T) &= \sum_{i=0}^{m-1} T^{v_i} + \sum_{j=0}^{s-1} T^{t_j}, \\ e_0(T) &= \sum_{j=0}^{K_0-1} T^j + \mathcal{D}(T), & V &= \ell_0(4) + e_0(4)4^L. \end{aligned} \tag{25.5}$$

The digits of e_0 are in $\{0, 1, 2\}$ and its unit digit is one. In particular, $e_0 > 0$, while the packed digit polynomial has all digits below four and degree below $2L$. Choose a power of two

$$H > \max\{2 \cdot 4^{2L+1}, 4^{t_*+3} 4c_*^2, 3L, 16\}. \tag{25.6}$$

Finally supply $T_L = \psi_4(L)$. The admissible index is (V, H, T_L) ; L is the underlying exponent defining it, not an additional system input. Both V and H are rebuilt from the new layout and baseline. All choices depend only on the represented system, independently of the queried positive input x .

25.3 Establish the Pell hypotheses before decoding

Besides (25.1), retain

$$\begin{aligned} B &= Hb^2, & b &= x + \beta, & n &= q^8, \\ \lambda(B-1) &= q^2 - 1, & \theta + 4 &= B, \\ S_2 &= \ell + eq = V + t\theta, & S_2 + \alpha &= q^2, \\ S &= g + q^2(S_2 + q^2 S_3), \\ T + 1 &= q^2(1 + \theta\lambda) - (b-1)\ell + \theta\ell q^4, \\ r &= S(n^2 - n) + (T+1)(n^2 - 1). \end{aligned}$$

Every supplied unknown remains positive. The packed bound gives $e < q$ and $\ell < q^2$, and geometry gives $B \leq q^2$. The bound on H gives $q > 4b \geq 8$ and $3L \leq B \leq n$. Since $\Omega \geq 1$, positivity of the new block gives directly

$$\mathcal{C}^2 < \theta\lambda q < B\lambda q < 3q^3 < q^4.$$

Thus $g < \mathcal{C} < q^2$ and $0 < S_3 < q^4$. The first-block borrow is fewer than b , while $\theta\lambda > b$, so the unchanged width-(2, 2, 4) estimates give

$$0 < S < n, \quad 0 < T + 1 \leq n, \quad n \leq r \leq 2n^3 - 2n^2 < 2n^3. \quad (25.7)$$

These are direct bounds for the new coefficient; no comparison with an old value of S_3 is assumed.

Write $U = wn^2$, $Y_P = sn^2$, $a = Y_P(U + 1)$, $A = a + 4$ and $J = 2r + 1$. Then $n, r \geq 64$, $U, Y_P \geq n^2$, $UY_P \geq n^4 > r + 1$, and $a > n^4 > J$. The first index congruence and positive interval give $c > J$. Thus $A > 1$, $1 < J < c$ and odd J are available, as are $L \geq 2$ and $3L \leq B \leq n \leq r$.

All Pell source equations of Theorem 24.1 are unchanged. Their proof therefore applies in its original order. First the doubled-index auxiliary norm identifies $c = \psi_A(J)$, $d = \chi_A(J)$. The first norm and index give $k = \psi_{2UY_P^2+1}(r + 1)$. The lower ratio estimate gives $Y_P \geq U^r$ and $a > U^{r+1}$, before the first exponent equation yields $U = 4^J$. In particular $a > 4^{3J}$. The positive gap and the equation $\kappa = T_L + \Delta a$ now identify the second Pell index as L : its two residues $\psi_4(t)$ and $\psi_4(L)$ lie strictly between zero and a . The second exponent comparison uses

$$B^{3L} \leq B^B \leq n^n \leq U^r < A, \quad q^3 \leq n^n \leq U^r < A,$$

and yields $q = B^L$. The upper ratio and binomial rounding arguments then give

$$b, B, q \text{ are powers of two}, \quad q = B^L, \quad n^2 \mid \binom{2r}{r}. \quad (25.8)$$

Every internal coefficient test still awaits decoding.

The weaker coefficient requires a revised post-Pell estimate. Since $L \geq 2$ and $B > 16$,

$$\lambda \geq B^{2L-1} \geq Bq > 2q, \quad q = B^L \geq B^2 > 2B.$$

Therefore $e < q$ implies $\Omega = \lambda - e > \lambda/2$, and (25.1) yields

$$\mathcal{C}^2 < 2\theta q < 2Bq < q^2, \quad g < \mathcal{C} < q. \quad (25.9)$$

25.4 Canonical codes and the smaller coefficient plateau

The first two masks, congruence and bound are unchanged. Their borrow-removal proof gives $d = \lfloor (b-1)\ell/q^2 \rfloor = 0$: the middle mask first implies $\ell \leq S_2 < 4q^2/(B-1) + b$, which forces $(b-1)\ell < q^2$. The base-four evaluation lemma applies to (25.5); it permits digits in $\{0, 1, 2\}$ and does not require the digit three. Consequently

$$S_2 = \ell_0(B) + e_0(B)q, \quad \ell = \ell_0(B) + a_{\text{al}}q, \quad e = e_0(B) - a_{\text{al}}, \quad 0 \leq a_{\text{al}} < e_0(B).$$

Using (25.9), the first mask decodes every coordinate below b , with input x at the unit position and dummy digits only at row positions. Thus

$$e, \mathcal{C} < B^{K_0}, \quad e\mathcal{C}^2 < q, \quad A_{\mathcal{C}} = \mathcal{C}(1)^2 < (c_*b)^2.$$

Since $x, g > 0$, one has $\mathcal{C}(1) \geq 2$ and $A_{\mathcal{C}} \geq 4$ before recovering δ .

Now the polynomial $\lambda\mathcal{C}^2$ has coefficients at most $J_C = A_C < B/4$, and its digits from L through $L + K_0$ all equal J_C . Write

$$\lambda\mathcal{C}^2 = qV_1 + V_0, \quad 0 \leq V_0 < q.$$

Exact division gives

$$\left\lfloor \frac{S_3}{q} \right\rfloor = \theta\lambda - V_1 + \epsilon, \quad \epsilon = \left\lfloor \frac{e\mathcal{C}^2 - V_0}{q} \right\rfloor \in \{-1, 0\}.$$

There is no borrow, since the digits of $\theta\lambda$ are $B - 4$ throughout this interval. The first digit of S_3 in the window is $B - 4 - J_C + \epsilon$, and every subsequent one is $B - 4 - J_C$. Binary nonoverlap therefore bounds every digit of $X = (B - 4)a_{\text{al}}$ by

$$J_C + 4 = A_C + 4 \leq 4A_C.$$

For its digit polynomial F_X ,

$$B - 4 \mid F_X(4), \quad 0 \leq F_X(4) \leq 4A_C \frac{4^{K_0+1} - 1}{3} < \frac{B}{12} < B - 4.$$

The unchanged bound (25.6) supplies the strict inequality. Hence $X = 0$ and $a_{\text{al}} = 0$, so both codes are canonical before any circuit row is tested.

25.5 Recover the special coordinate before the copies

Below K_0 , the raw coefficients of S_3 are now those of $\mathcal{D}(T)\mathcal{C}(T)^2$. Each has absolute value at most $A_C < B/4$, so incoming carries are zero or minus one. At a target raw value v , the mask $B - 4$ is equivalent to

$$0 \leq v + \epsilon \leq 3, \quad \epsilon \in \{-1, 0\}. \quad (25.10)$$

The first special target has incoming carry zero: its single positive coefficient precedes all later row intervals, making every raw coefficient through that target nonnegative and below B . Thus $0 \leq \delta^2 \leq 3$, so $\delta \in \{0, 1\}$.

At $\delta = 0$, the direct guard (25.3) has raw value $-x^2 < 0$, which fails (25.10). Hence $\delta = 1$ without any copy assumption. Ordinary targets now test $F \in \{0, 1\}$, and paired rows force $F = 0$. In particular, the paired unit copies give $X' = X$. The two rows (25.2) then impose

$$X - 1 \in \{0, 1\}, \quad X - X^2 \in \{0, 1\},$$

forcing $X = 1$. Every original circuit gate and equality is recovered at the actual input. The guard is auxiliary; it is not asserted to be an original equation or to have zero target value.

25.6 Positive necessity and the exact saving

Given a solution of the represented system, take its circuit values, $\delta = 1$, all unit copies $X' = X = 1$, zero dummy coordinates, and

$$u = \left\lfloor \frac{x^2}{2} \right\rfloor.$$

The direct guard then has raw value $2u - x^2 + 1 = 1 + (x^2 \bmod 2)$, which is one or two and survives either incoming carry. All ordinary targets have raw value one, and the first special target has value one with carry zero. The low coordinate tests vanish by support separation.

Choose a power-of-two b larger than every coordinate and use the canonical codes, $B = Hb^2$, $q = B^L$ and $n = q^8$. Their positive slack witnesses exist as before. In particular, e_0 has unit digit one,

the δ digit makes $g > 0$, and the nonconstant polynomial $\ell_0(T) + e_0(T)T^L$ has nonnegative coefficients, so its value at B exceeds its value at four and the congruence quotient t is positive. Also $\Omega = \lambda - e > 0$ and

$$\mathcal{C}^2 < q \leq \theta q, \quad S_3 = \lambda(\theta q - \mathcal{C}^2) + e\mathcal{C}^2 > 0.$$

All three masks hold. Their newly calculated r satisfies (25.7) and the binomial divisibility in (25.8).

The positive Pell construction of Section 24 now applies with these new coding values. Choose $U = 4^{2r+1}$, $w = U/n^2$, $Y_P = \lfloor (U+1)^{2r}/U^r \rfloor$, and $s = Y_P/n^2$. Its binomial congruence makes the latter quotient integral. Choose the main and first Pell coordinates and interval witnesses, then the second coordinates at L , the gap and positive exponent quotients. In particular $\Delta = (\psi_{a+4}(L) - \psi_4(L))/a$ is a positive integer. Finally choose the doubled-index auxiliary witnesses at the new main parameter. This changes no coding value. Every required positive unknown is therefore supplied in the same proved order.

The former seven-instruction block computed 2λ , then $3\lambda = 2\lambda + \lambda$. Replace it by the six instructions

$$\begin{aligned} U_\lambda &= \theta\lambda, & T_{\text{coef}} &= U_\lambda + 1, & \lambda_3 &= 3\lambda, \\ R_2 &= T_{\text{coef}} + \lambda_3, & t_2 &= \lambda - e, & P_2 &= U_\lambda q. \end{aligned} \tag{25.11}$$

The tests $q^2 = R_2$ and $\theta + 4 = B$ still give exactly $\lambda(B-1) = q^2 - 1$. The new block has three multiplications and three additions; the old block had three multiplications and four additions. Every other arithmetic instruction is retained.

Theorem 25.1. *Every recursively enumerable subset of the positive integers has an admissible fixed index (V, H, T_L) for a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 98 operations: 55 multiplications and 43 additions. Here $T_L = \psi_4(L)$ for the underlying exponent of the rebuilt index. The supplied parameters are (x, V, H, T_L) , and fixed numerals are free.*

Proof. The preceding construction proves the representation in both directions and gives every required positive witness. Formula (25.11) saves one addition from Theorem 24.1, giving $55 + 43 = 98$. The exact checker `round27_1980_unit_center_certificate.py` verifies all 98 primitive instructions, all 22 source residuals, and the same three residual corrections as its predecessor. Only the packed r equation and the positive- σ and positive- Ω equations change; geometry and every Pell source equation are unchanged. Its finite regressions are distinct from the universal proof above. No optimality is asserted. $\square$

26 Recovering the integral auxiliary Pell sequence

The auxiliary norm can be weakened while preserving the index information used by the doubled signed block. Starting with Theorem 24.1, replace only E16 by

$$(ic^2)^2 = D(f^2 - 1), \quad D = A^2 - 1, \quad A = a + 4. \tag{26.1}$$

The previous norm was $f^2 - 1 = D(ic^2)^2$. The two individual equations are not equivalent: the new equation can admit additional rational Pell solutions. The existing size bounds exclude the unwanted solutions and recover the divisibility needed by the signed-index argument. At the arithmetic level, the square $(ic^2)^2$ is now exactly the coefficient $K = D(f^2 - 1)$ used by that argument, saving one multiplication.

26.1 A preliminary bound on the main index

Keep the definitions

$$U = wn^2, \quad Y_P = sn^2, \quad a = Y_P(U + 1), \quad A = a + 4, \quad P = 2UY_P^2 + 1, \quad J = 2r + 1.$$

Before any auxiliary norm is used, the coding bootstrap supplies

$$n \geq 64, \quad n \leq r < 2n^3, \quad U, Y_P \geq n^2, \quad UY_P > r + 1.$$

The unchanged main norm and first triangular norm give positive indices p, t with

$$d = \chi_A(p), \quad c = \psi_A(p), \quad 2\tau + 1 = \chi_P(t), \quad k = \psi_P(t).$$

Since $P \equiv 1 \pmod{UY_P}$ and $k = r + 1 + hUY_P$, the congruence for ψ_P gives $t \equiv r + 1 \pmod{UY_P}$. Thus $t \geq r + 1$. Also

$$P - A = UY_P(2Y_P - 1) - Y_P - 3 > 0.$$

If $p \leq r + 1$, Pell growth would give

$$c \leq (2A)^{p-1} \leq (2A)^r < (2P - 1)^r \leq k,$$

contrary to $c = kY_P + \eta > k$. Consequently

$$p \geq r + 2 \geq 66, \quad c > A^6 > AD^2. \quad (26.2)$$

For the second assertion, use $\psi_A(p) \geq (2A - 1)^{p-1}$ and $D < A^2$. This argument uses neither the auxiliary norm, the signed index equation, an exponential identity, nor internal coefficient decoding.

26.2 The rank argument

Lemma 26.1. *Let $A > 1$, $D = A^2 - 1$, and let $c = \psi_A(p)$, $d = \chi_A(p)$ for a positive index p . Suppose $c > AD^2$ and positive integers f, i satisfy (26.1). Then there is a positive index m such that*

$$f = \chi_A(m), \quad p \mid m, \quad c \mid m, \quad ic^2 = D\psi_A(m).$$

Proof. Write $D = d_0 s_0^2$ with $d_0 > 1$ squarefree. We first justify the use of a fundamental unit within the integer-coefficient ring $\mathbb{Z}[\sqrt{d_0}]$. The known unit $A + s_0\sqrt{d_0}$ ensures that positive solutions to $x^2 - d_0 y^2 = 1$ exist. Choose the least positive unit

$$\varepsilon = F + y_0\sqrt{d_0} > 1, \quad F, y_0 \in \mathbb{Z}_{>0}, \quad F^2 - d_0 y_0^2 = 1.$$

Such a choice exists by minimizing its first coordinate. Every other positive integer-coefficient unit $\beta > 1$ is a power of ε . Indeed, choose $j \geq 0$ with $\varepsilon^j \leq \beta < \varepsilon^{j+1}$. The quotient $\beta\varepsilon^{-j}$ has integer coefficients and norm one, because $\varepsilon^{-1} = F - y_0\sqrt{d_0}$. If it belonged to $(1, \varepsilon)$, both coordinates would be positive integers, contradicting minimality. It is therefore one. No assertion about the full ring of algebraic integers is needed.

There is accordingly an $e \geq 1$ such that

$$A = \chi_F(e), \quad s_0 = y_0 S, \quad S = \psi_F(e), \quad D = (F^2 - 1)S^2.$$

Composition of the Pell solutions gives

$$\psi_F(ep) = Sc. \quad (26.3)$$

Put $R_0 = ic^2$. The relaxed equation gives $R_0^2 = d_0 s_0^2 (f^2 - 1)$. Thus $s_0 \mid R_0$, and squarefreeness implies $d_0 \mid R_0/s_0$. The positive integer $y = R_0/(d_0 s_0)$ satisfies $f^2 - d_0 y^2 = 1$. Hence for some $n_0 > 0$,

$$f = \chi_F(n_0), \quad R_0 = (F^2 - 1)S\psi_F(n_0). \quad (26.4)$$

We use the strong divisibility identity

$$\gcd(\psi_F(u), \psi_F(v)) = \psi_F(\gcd(u, v)). \quad (26.5)$$

For completeness, the Pell norm gives $\gcd(\chi_F(j), \psi_F(j)) = 1$. The addition formula therefore reduces $\gcd(\psi_F(u), \psi_F(u+v))$ to $\gcd(\psi_F(u), \psi_F(v))$, and the Euclidean algorithm proves (26.5).

Let $T = (F^2 - 1)S$, $g = \gcd(n_0, ep)$, and

$$h = \gcd(\psi_F(n_0), \psi_F(ep)) = \psi_F(g).$$

Because $c^2 \mid R_0 = T\psi_F(n_0)$,

$$\begin{aligned} \gcd(c^2, T\psi_F(ep)) &= \gcd(c^2, Dc) \\ &= c \gcd(c, D) \geq c \end{aligned}$$

divides Th . Since $T = D/S \leq D$,

$$h \geq c/T \geq c/D. \quad (26.6)$$

If $g < ep$, then $2g \leq ep$, since g is a proper divisor. The duplication identity gives

$$2h^2 < 2\chi_F(g)\psi_F(g) = \psi_F(2g) \leq \psi_F(ep) = Sc.$$

Here $S < A$, by $D = (F^2 - 1)S^2$ and $F \geq 2$. Thus $h^2 < Ac/2$, whereas (26.6) and $c > AD^2$ give $h^2 \geq c^2/D^2 > Ac$, a contradiction. Therefore $ep \mid n_0$. Writing $n_0 = em$ proves $p \mid m$ and, by composition,

$$f = \chi_A(m), \quad R_0 = D\psi_A(m). \quad (26.7)$$

To obtain $c \mid m$, write $m = pk$. Expansion of $(d + c\sqrt{D})^k$, followed by division of the square-root coefficient by c , gives

$$\frac{\psi_A(pk)}{c} \equiv kd^{k-1} \pmod{c}.$$

As $c^2 \mid D\psi_A(m)$ and $\gcd(d, c) = 1$, this implies $c \mid Dk$. The main Pell expansion modulo D also gives

$$c = \psi_A(p) \equiv pA^{p-1} \pmod{D}, \quad \gcd(c, D) = \gcd(p, D),$$

using $\gcd(A, D) = 1$. Put $g_0 = \gcd(c, D)$. Then $c/g_0 \mid k$ and $g_0 \mid p$, so $c \mid pk = m$. $\square$

The lemma recovers precisely the integral sequence and auxiliary index divisibility needed below. It does not assert $c^2 \mid \psi_A(m)$, which is stronger than necessary and is not an automatic consequence of the relaxed norm.

26.3 The signed index and all positive witnesses

By (26.2), Lemma 26.1 applies before the signed auxiliary norm. Its source polynomial remains

$$v = of - d^2, \quad K = D(f^2 - 1) = (ic^2)^2, \quad v(v+1) = K(K-1)(J+jc)^2.$$

For $G = 2K - 1$, reduction modulo f still gives

$$G \equiv -\chi_A(2), \quad 2v+1 \equiv -\chi_A(2p) \pmod{f}.$$

The conclusions $f = \chi_A(m)$ and $c \mid m$ give the comparison bound $0 < 2p \leq c \leq m$. Also $K = (ic^2)^2$ gives $G \equiv -1 \pmod{c}$ directly. The signed chi step-down and parity argument in the proof of Lemma 22.1 therefore apply unchanged: they yield $J \equiv \pm p \pmod{c}$, exclude $J + p = c$, and conclude $p = J$. The old stronger divisibility of the second Pell coordinate is not used at this step.

The exact first index, ratio estimates, common exponential witness, fixed-index recovery through $T_L = \psi_4(L)$, and coefficient decoding now proceed in the order proved for Theorem 24.1. All their hypotheses have been recovered before use. This proves sufficiency of the modified system.

For necessity, start with that theorem's positive witness construction. Its auxiliary index is the even number $m = 2cJ$, and

$$f = \chi_A(m), \quad i_{\text{old}} = \psi_A(m)/c^2 > 0.$$

Keep f and set $i_{\text{new}} = Di_{\text{old}} > 0$. Then

$$(i_{\text{new}}c^2)^2 = D^2\psi_A(m)^2 = D(f^2 - 1).$$

The source value $K = D(f^2 - 1)$ has not changed. Thus the same odd f , the same G , and the positive o, j constructed in Lemma 22.1 still satisfy the signed norm. All other witnesses and fixed index components are preserved.

26.4 One multiplication is removed

Previously E16 and the shared coefficient K required six instructions:

$$\begin{aligned} z &= ic^2, & z_2 &= z^2, & AE &= Dz_2, \\ f_2 &= f^2, & R_{16} &= AE + 1, & K &= DAE. \end{aligned}$$

The relaxed equation needs five:

$$z = ic^2, \quad K = z^2, \quad f_2 = f^2, \quad f_- = f_2 - 1, \quad R_{16} = Df_-.$$

Test $K = R_{16}$ and use the already computed square K in the signed norm. All other instructions remain unchanged.

The exact triangular correction also becomes simpler. If

$$F_{16} = (ic^2)^2 - D(f^2 - 1), \quad K_s = D(f^2 - 1), \quad K_c = (ic^2)^2,$$

then

$$F_{17}^{\text{calc}} = F_{17}^{\text{source}} - F_{16}(K_s + K_c - 1)(J + jc)^2.$$

The complete verifier checks this correction, the retained geometric and second-exponent corrections, all source residuals and primitive instructions, and the input domains. The Pell integrality and index claims are proved above, independently of the symbolic checks.

Theorem 26.2. *Replacing E16 by (26.1) in the system of Theorem 24.1 preserves its existential representation over 34 positive integer unknowns and 22 equality tests. Its straight-line certificate has*

$$\boxed{98 = 54 \text{ multiplications} + 44 \text{ additions}}.$$

Fixed numerals and supplied index parameters are free. The saving relative to Theorem 24.1 is one multiplication.

The full schedule, source polynomials and exact residual identities are recorded in the standalone checker

`round28_1980_relaxed_auxiliary_certificate.py`

and its JSON receipt.

27 A certificate of 97 operations

The unit-centered encoding of Section 25 and the relaxed auxiliary norm of Section 26 compose. The arithmetic substitutions are disjoint, but their mathematical compatibility also requires an ordered proof: the new coding bounds must force a sufficiently large main Pell index before the relaxed norm is used, and the Pell conclusions must precede digit decoding.

Theorem 27.1. *Every recursively enumerable subset of the positive integers has an admissible fixed index (V, H, T_L) for a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 97 operations: 54 multiplications and 43 additions. Here $T_L = \psi_4(L)$ for the underlying exponent of the index construction. The supplied parameters are (x, V, H, T_L) , with three fixed index parameters besides the positive input x . Fixed numerals and equality tests are free.*

Proof. Use the entire construction of Section 25 for the represented system. Compile it into nonnegative primitive circuit coordinates with distinct operands and fresh outputs. Pair multiplication, addition, copy, zero and final equality rows F with their negatives, using targets $2F + \delta^2$. For each unit coordinate X , introduce X' , include the paired copy rows $(X - X')\delta$ and $(X' - X)\delta$, and the two unpaired unit rows

$$X\delta - \delta^2, \quad X\delta - XX'.$$

Place the special target δ^2 first, and include the direct guard $2u\delta - x^2 + \delta^2$. The actual input square is used in the guard, independently of any copy. Every divided target coefficient belongs to $\{-1, 0, 1\}$.

With m positive-weight coordinates and s targets, use

$$\begin{aligned} v_i &= 3^i \quad (0 \leq i < m), & v_0 &= 1 \text{ for } \delta, & M &= 3^{m-1}, & d_0 &= 4M + 1, \\ t_j &= (s+1)d_0 + 2M + jd_0 \quad (0 \leq j < s), \\ t_* &= 2sd_0 + 2M, & K_0 &= t_* + 1, & c_* &= m + s + 1. \end{aligned}$$

The input has weight zero. Include a dummy coordinate at each t_j . The complementary polynomial $\mathcal{D}$ satisfies $[T^{t_j}]\mathcal{D}(T)\mathcal{C}(T)^2 = G_j$ exactly, by the proved monomial and row separation. Form

$$\ell_0(T) = \sum_i T^{v_i} + \sum_j T^{t_j}, \quad e_0(T) = \sum_{j=0}^{K_0-1} T^j + \mathcal{D}(T).$$

Choose a power of two $L > 3K_0 + 2$ and a power of two

$$\begin{aligned} H &> \max\{2 \cdot 4^{2L+1}, 4^{t_*+3} 4c_*^2, 3L, 16\}, \\ V &= \ell_0(4) + e_0(4)4^L, & T_L &= \psi_4(L). \end{aligned} \tag{27.1}$$

These three index values use the complete new coordinate and target list. They are fixed independently of x ; the underlying L is not an additional supplied parameter. The digits of e_0 are in $\{0, 1, 2\}$, with unit digit one. The special target has no incoming carry by the support proof in Section 25.

Retain the coding equations and packing

$$\begin{aligned} B &= Hb^2, & \mathcal{C} &= x + g, & b &= x + \beta, \\ \lambda(B-1) &= q^2 - 1, & \theta + 4 &= B, & n &= q^8, \\ S_2 &= \ell + eq = V + t\theta, & S_2 + \alpha &= q^2, \\ \Omega &= \lambda - e > 0, & S_3 &= \theta\lambda q - \Omega\mathcal{C}^2 = \sigma > 0, \\ S &= g + q^2(S_2 + q^2S_3), \\ T + 1 &= q^2(1 + \theta\lambda) - (b-1)\ell + \theta\ell q^4, \\ r &= S(n^2 - n) + (T+1)(n^2 - 1). \end{aligned}$$

In the Pell block write $A = a + 4$ and $D = A^2 - 1$. Replace only its auxiliary norm by

$$(ic^2)^2 = D(f^2 - 1). \quad (27.2)$$

The doubled signed norm still uses

$$\begin{aligned} v &= of - d^2, & K_{\text{aux}} &= D(f^2 - 1), & J &= 2r + 1, \\ v(v + 1) &= K_{\text{aux}}(K_{\text{aux}} - 1)(J + jc)^2. \end{aligned} \quad (27.3)$$

The fixed index equation remains $\kappa = T_L + \Delta a$, and every other Pell source equation is unchanged. The integer register v may be signed; every supplied unknown, including o , is positive.

Before using any auxiliary Pell equation or internal target, the positive packed bound and geometry give

$$e < q, \quad \ell < q^2, \quad B \leq q^2, \quad q > 4b \geq 8, \quad 3L \leq B \leq n.$$

Also $\Omega \geq 1$ implies $\mathcal{C}^2 < \theta\lambda q < 3q^3 < q^4$ and $0 < S_3 < q^4$. The unit-center preliminary borrow argument consequently proves

$$0 < S < n, \quad 0 < T + 1 \leq n, \quad n \leq r \leq 2n^3 - 2n^2 < 2n^3. \quad (27.4)$$

Put $U = wn^2$, $Y_P = sn^2$, $a = Y_P(U + 1)$ and $P = 2UY_P^2 + 1$. Then $n \geq 64$, $U, Y_P \geq n^2$, $UY_P > r + 1$ and $a > n^4 > J$. The positive interval and first-index equation give $c > J$.

The unchanged main norm and triangular first norm give positive indices p, t such that

$$c = \psi_A(p), \quad d = \chi_A(p), \quad k = \psi_P(t), \quad 2\tau + 1 = \chi_P(t).$$

Since $P \equiv 1 \pmod{UY_P}$ and $k = r + 1 + hUY_P$, $t \equiv r + 1 \pmod{UY_P}$. The inequality $0 < r + 1 < UY_P$ therefore gives $t \geq r + 1$, without knowing the main index. Moreover

$$P - A = UY_P(2Y_P - 1) - Y_P - 3 > 0.$$

If $p \leq r + 1$, then

$$c \leq (2A)^{p-1} \leq (2A)^r < (2P - 1)^r \leq \psi_P(t) = k,$$

contradicting $c = kY_P + \eta > k$. Thus

$$p \geq r + 2 \geq 66, \quad c = \psi_A(p) > A^6 > AD^2. \quad (27.5)$$

The growth bound $\psi_A(p) \geq (2A - 1)^{p-1}$ proves the second inequality. Neither the relaxed norm, the signed norm nor an exponential equation has been used here.

Lemma 26.1 now applies to (27.2) and (27.5). Its integer-coefficient fundamental-unit and strong-divisibility argument supplies an index m_0 with

$$f = \chi_A(m_0), \quad p \mid m_0, \quad c \mid m_0, \quad ic^2 = D\psi_A(m_0). \quad (27.6)$$

These are exactly the needed conclusions; the stronger old divisibility $c^2 \mid \psi_A(m_0)$ is not asserted.

Equation (27.2) identifies $K_{\text{aux}} = (ic^2)^2$, so the signed auxiliary parameter $G = 2K_{\text{aux}} - 1$ is -1 modulo c . In conjunction with (27.6), the doubled signed-index argument has all its hypotheses, including $0 < 2p \leq c \leq m_0$. It gives $J \equiv \pm p \pmod{c}$. Both indices lie strictly between zero and c . The possibility $J + p = c$ contradicts odd J and $\psi_A(p) \equiv p \pmod{2}$. Hence $p = J$, and

$$c = \psi_A(J), \quad d = \chi_A(J).$$

The subsequent common-witness proof proceeds in its established order. It recovers the exact first index, proves $Y_P \geq U^r$ and $a > U^{r+1}$, and then obtains $U = 4^J$ from the first exponent equation. The fixed value $T_L = \psi_4(L)$, reduced modulo a , determines the second Pell index as L : both possible residues lie in $(0, a)$ because $a > 4^{3J}$. Its exponent comparison gives $q = B^L$. The necessary bounds follow from (27.4) and $3L \leq B \leq n \leq r$. Finally the unchanged upper ratio and binomial fractional-part estimates give

$$b, B, q \text{ are powers of two,} \quad q = B^L, \quad n^2 \mid \binom{2r}{r}. \quad (27.7)$$

These conclusions precede coefficient decoding.

Now $\lambda \geq B^{2L-1} \geq Bq > 2q$ and $q \geq B^2 > 2B$. Thus $\Omega = \lambda - e > \lambda/2$ and

$$\mathcal{C}^2 < 2\theta q < 2Bq < q^2, \quad g < \mathcal{C} < q.$$

The first two masks remove their preliminary borrow and recover the combined coefficient polynomial up to its quotient alias. They also bound each decoded coordinate by b . Before identifying δ , positivity of x, g gives $\mathcal{C}(1) \geq 2$. With $A_{\mathcal{C}} = \mathcal{C}(1)^2$, the high plateau of $\lambda\mathcal{C}^2$ has height $A_{\mathcal{C}}$. The high-mask digits of $(B-4)a_{\text{al}}$ are consequently bounded by $A_{\mathcal{C}} + 4 \leq 4A_{\mathcal{C}}$. The bound on H in (27.1) and evaluation at four force the alias to vanish, by the complete argument in Section 25. Both codes are canonical.

The first special target has incoming carry zero and forces $\delta \in \{0, 1\}$. The direct guard excludes zero before any copy equation is used. At $\delta = 1$, paired rows force $F = 0$; unit copies agree and the two unit rows force $X = 1$. All original circuit equations therefore hold at the actual input x . This proves sufficiency of the composition.

For necessity, take a solution of the represented system, its circuit values, $\delta = 1$, unit copies $X' = X = 1$, and zero dummy digits. Put $u = \lceil x^2/2 \rceil$, so the guard target is one or two. Choose a power-of-two b larger than every coordinate and use the canonical codes, $B = Hb^2$, $q = B^L$ and $n = q^8$. Section 25 supplies all positive coding witnesses, the three masks, (27.4) and (27.7) for the newly calculated r .

Construct the full doubled-index Pell witnesses as in Section 24. Choose $U = 4^J$, $Y_P = \lfloor (U+1)^{2r}/U^r \rfloor$, $a = Y_P(U+1)$, the main and first Pell coordinates, positive interval witnesses, the second coordinates at L , and the positive exponent quotients. All these choices use the new fixed value $T_L = \psi_4(L)$. In particular $w = U/n^2$, $s = Y_P/n^2$ and $\Delta = (\psi_{a+4}(L) - \psi_4(L))/a$ are positive integers. Its auxiliary construction chooses

$$m_0 = 2cJ, \quad f = \chi_A(m_0), \quad i_{\text{old}} = \frac{\psi_A(m_0)}{c^2} > 0,$$

and supplies positive o, j for the doubled signed norm. Replace only the auxiliary witness

$$i = Di_{\text{old}} > 0.$$

Then

$$(ic^2)^2 = D^2\psi_A(m_0)^2 = D(f^2 - 1),$$

while the mathematical value $K_{\text{aux}} = D(f^2 - 1)$ is unchanged. Consequently the same f, o, j still satisfy the signed equation. The witness i occurs nowhere else. Every one of the 34 unknowns has therefore received a positive integer value.

For the count, the unit-centered encoding saves one addition from the 99-operation certificate, and the relaxed auxiliary norm saves one multiplication from that same certificate. The full checker `round29_1980_composed_certificate.py` verifies that both orders give the same 97 primitive instructions and the same 22 source residuals. Relative to 99, only the packed equation, auxiliary norm, and positive- σ and positive- Ω equations change. Hence the final count is $54 + 43 = 97$.

The three residual corrections are acyclic: the geometric and second-exponent corrections use the independently exact equation $\theta + 4 = B$, while the signed norm correction is

$$-F_{16}(K_{\text{source}} + K_{\text{calculated}} - 1)(J + jc)^2, \quad F_{16} = (ic^2)^2 - D(f^2 - 1),$$

whose auxiliary equality is checked exactly. Each subtraction is one reversed addition equation with a free integer intermediate. Thus 97 is the full addition/multiplication certificate length for the proved positive-domain system. No optimality is claimed. $\square$

28 Linear radix scaling and a certificate of 96 operations

The quadratic radix scale can be replaced by a linear one when the coefficient tests are allowed two base- B digits. An explicit bound on the indicator code removes its quotient alias. Positive padding then resets the incoming carry at every ordinary target, and a second unit test excludes a scale of order $\sqrt{B}$.

Starting from Section 27, replace the radix and third packed block by

$$B = Hb, \quad C = x + g, \quad \Omega = \lambda - e > 0, \quad \sigma = \Omega(q - C^2) > 0, \quad (28.1)$$

and add the positive gap equation

$$\ell + \alpha_2 = q. \quad (28.2)$$

Every Pell equation is retained, with $B = Hb$ wherever the radix occurs. The coefficient index is rebuilt below.

Theorem 28.1. *Every recursively enumerable subset of the positive integers has an admissible fixed index (V, H, T_L) for a system with 35 positive unknowns and 23 equations admitting a straight-line certificate of 96 operations: 52 multiplications and 44 additions. Here $T_L = \psi_4(L)$ for the underlying exponent in the index construction. The supplied parameters are (x, V, H, T_L) , with three fixed index parameters besides the positive input x . Fixed numerals and equality tests are free.*

The following subsections prove the theorem. The complete arithmetic table is in Appendix U. The exact source and primitive checker is located in `../verification/`:

`round30_1980_linear_radix_certificate.py.`

28.1 The homogeneous circuit and its normalization rows

Compile the represented finite Diophantine system into an arithmetic circuit over nonnegative integers, using the actual positive input x , addition, multiplication, zero and one. Introduce a homogeneous coordinate δ . Repeated operands are replaced by fresh copies, and every gate output is fresh. All circuit coordinates other than x have positive weights in the code.

For a multiplication, addition or zero equation use, respectively,

$$2XY - 2W\delta = 0, \quad 2(X + Y - W)\delta = 0, \quad 2X\delta = 0. \quad (28.3)$$

All products in these expressions have distinct factors. A copy $X' = X$ can instead be imposed by

$$X'^2 - X^2 = 0. \quad (28.4)$$

Nonnegativity makes this an exact copy equation. It also supplies arbitrarily many distinct copies of δ without introducing a diagonal coefficient of magnitude two. A circuit occurrence of the constant

one is represented by δ or a copy of δ . Whenever using δ directly would create $\delta\delta$ in (28.3), use a fresh square-difference copy instead. Likewise give repeated gate operands distinct copies. Thus every product in (28.3) has distinct factors, including when a gate operand is the constant one. Final equalities can use either distinct-coordinate products with δ or square differences. Include both F and $-F$ for every zero equation $F = 0$ in this construction.

For normalization, introduce distinct coordinates X, Y, δ', u and include the following equations and their negatives:

$$\begin{aligned} X^2 - x^2 &= 0, & Y^2 - x^2 &= 0, & \delta'^2 - \delta^2 &= 0, \\ 2xX - 2\delta\delta' - 2u\delta &= 0. \end{aligned} \tag{28.5}$$

Once their zero tests have been decoded, these equations give

$$X = Y = x, \quad \delta' = \delta, \quad x^2 = \delta^2 + u\delta. \tag{28.6}$$

Thus $\delta > 0$ because $x > 0$, and $\delta \leq x$. Conversely, $\delta = \delta' = 1$, $X = Y = x$ and $u = x^2 - 1$ are nonnegative witnesses for every positive x , including $x = 1$.

Finally include two unpaired targets with raw polynomial δ^2 . The second will receive extra positive padding. These two tests force $\delta = 1$ below; the argument does not presume an incoming carry at an unprotected special target.

Divide every square coefficient of a target polynomial by one and every distinct-coordinate product coefficient by two. All results lie in $\{-1, 0, 1\}$. This includes (28.3), (28.4), both signs of (28.5), and the two unit targets. The number of rows and encoded coordinates may depend on the represented system; they are digits of the integer codes, not additional unknowns of the final system.

28.2 Two-digit support and reset positions

Let m count all non-input circuit and normalization coordinates. Give them weights

$$v_i = 4 \cdot 3^i \quad (0 \leq i < m), \quad M = 4 \cdot 3^{m-1}.$$

The input x has weight zero. Quadratic monomial weights determine their unordered coordinate pair, including squares and products with x : divide by four and read the base-three digits. All these weights are multiples of four and are at most $2M$.

Let s be the number of targets, counting both signs of each zero equation and the two final unit tests. Put the unit tests last, and distinguish the last target t_* as the additional-carry test. Define

$$\begin{aligned} d_0 &= 4M + 4, & t_j &= (s + 1)d_0 + 2M + jd_0 \quad (0 \leq j < s), \\ t_* &= t_{s-1} = 2sd_0 + 2M, & K &= t_* + 2. \end{aligned} \tag{28.7}$$

All t_j are multiples of four. For a target G_j and a monomial of weight w with divided coefficient a , put aT^{t_j-w} in $\mathcal{D}_{\text{main}}(T)$. The intervals $[t_j - 2M, t_j]$ are disjoint, so these instructions do not collide. Add the reset polynomial and the last-target padding:

$$\begin{aligned} \mathcal{D}_{\text{reset}}(T) &= \sum_j T^{t_j-2}, \\ \mathcal{D}_{\text{extra}}(T) &= T^{t_*-1} + T^{t_*-1-v_X} + T^{t_*-1-v_Y}, \\ \mathcal{D}(T) &= \mathcal{D}_{\text{main}}(T) + \mathcal{D}_{\text{reset}}(T) + \mathcal{D}_{\text{extra}}(T). \end{aligned} \tag{28.8}$$

The three parts have exponents congruent to 0, 2 and 3 modulo four, respectively. The extra exponents are distinct. All $\mathcal{D}$ coefficients therefore still lie in $\{-1, 0, 1\}$. Every exponent is positive, greater than M , and less than K .

Include a dummy coordinate at each t_j and $t_j + 1$, and set

$$\begin{aligned}\ell_0(T) &= \sum_i T^{v_i} + \sum_j (T^{t_j} + T^{t_j+1}), \\ e_0(T) &= \sum_{h=0}^{K-1} T^h + \mathcal{D}(T).\end{aligned}\tag{28.9}$$

The digits of ℓ_0 are zero or one, and those of e_0 are zero, one or two. The unit digit of e_0 is one. Let c_* count the input, all m true coordinates and the $2s$ dummies, and put

$$D_1 = \sum_h |[T^h]\mathcal{D}(T)|.$$

Choose a power of two $L > 3K + 2$ and a power of two H such that

$$\begin{aligned}H &> \max\{2 \cdot 4^{2L+1}, 64 \max(1, D_1)c_*^2, 3L, 64\}, \\ V &= \ell_0(4) + e_0(4)4^L, \quad T_L = \psi_4(L).\end{aligned}\tag{28.10}$$

These choices depend only on the represented system and precede the queried input x .

No arbitrary decoded dummy value contributes to $\mathcal{D}(T)\mathcal{C}(T)^2$ through position $t_* + 1$. Indeed, the least $\mathcal{D}$ exponent is at least $t_0 - 2M$, the least dummy weight is t_0 , and

$$2t_0 - 2M > t_* + 1.\tag{28.11}$$

This matters because the dummies at $t_j + 1$ are not in the zero residue class modulo four. Up to the last tested digit, however, they have no effect on the product.

28.3 Preliminary bounds and the Pell conclusions

Retain the coding equations and packing

$$\begin{aligned}\mathcal{C} &= x + g, & b &= x + \beta, \quad B = Hb, \quad \theta = B - 4, \\ \lambda(B - 1) &= q^2 - 1, & n &= q^8, \\ S_2 &= \ell + eq = V + t\theta, & S_2 + \alpha &= q^2, \\ S &= g + q^2(S_2 + q^2\sigma), \\ T + 1 &= q^2(1 + \theta\lambda) - (b - 1)\ell + \theta\ell q^4, \\ r &= S(n^2 - n) + (T + 1)(n^2 - 1).\end{aligned}\tag{28.12}$$

All supplied unknowns, including α , α_2 , Ω and σ , are positive integers. Before any circuit row or Pell conclusion is used, positivity gives

$$\ell < q, \quad e < q, \quad S_2 < q^2, \quad \mathcal{C}^2 < q, \quad g < q, \quad 0 < \sigma < \lambda q < q^3.\tag{28.13}$$

The last inequality follows from $\lambda = (q^2 - 1)/(B - 1) < q^2$. Geometry gives $B \leq q^2$. Also $b \geq 2$ and (28.10) imply $q \geq 6$, $b < B < n$ and $3L \leq B \leq n$. The former estimate $q > 4b$ is not used or asserted for the linear radix. Moreover,

$$\theta\lambda \geq B - 4 = Hb - 4 > b.$$

The integer bounds in (28.13) give $0 < S < q^7 < n$. For the other packed number,

$$\begin{aligned}T + 1 &> q^2\theta\lambda - b\ell > bq^2 - bq > 0, \\ T + 1 &< q^4(1 + \theta\ell) < Bq^5 \leq q^7 < n.\end{aligned}\tag{28.14}$$

The upper bound uses $1 + \theta\lambda < q^2$ and $1 + \theta\ell \leq 1 + (B - 4)(q - 1) < Bq$. Consequently

$$n \leq n^2 - 1 \leq r < 2n^3. \quad (28.15)$$

These are all the packing-size inputs to the main Pell and binomial arguments. Write

$$\begin{aligned} U &= wn^2, & Y_p &= s_p n^2, & a &= Y_p(U + 1), & A &= a + 4, \\ D_p &= A^2 - 1, & P &= 2UY_p^2 + 1, & J &= 2r + 1. \end{aligned}$$

The notation Y_p, s_p distinguishes the Pell values from the encoded input copy Y . We have $U, Y_p \geq n^2$, $UY_p \geq n^4 > r + 1$ and $a > n^4 > J$. The first index equation and positive interval give $c > J$. The norm classifications and $P > A$ imply that the preliminary main index is at least $r + 2$, hence $c > AD_p^2$, exactly as in Section 27.

The retained relaxed auxiliary norm

$$(ic^2)^2 = D_p(f^2 - 1)$$

therefore has the same fundamental-unit and strong-divisibility hypotheses. It recovers the required auxiliary multiple of c . The doubled signed-index argument identifies $c = \psi_A(J)$ and $d = \chi_A(J)$. The first-index and ratio argument gives $Y_p \geq U^r$ and $a > U^{r+1}$ before either exponent relation is used. The common-witness exponent equation then gives $U = 4^J$. The retained fixed congruence $\kappa = T_L + \Delta a$ determines the second index as L , and the second exponent equation gives $q = B^L$.

Every exponent-size hypothesis follows from (28.15) and $3L \leq B \leq n \leq r$ as in the preceding proof. None requires $B = Hb^2$. Since n^2 divides $U = 4^J$, q is a power of two; $q = B^L$ makes B a power of two; and the fixed power of two H divides $B = Hb$, so b is a power of two. The unchanged upper ratio and binomial fractional-part estimates give

$$n^2 \mid \binom{2r}{r}. \quad (28.16)$$

This applies the full Pell argument before any coefficient equation or unit value is assumed. Its necessity construction will be used only after the new coding witnesses have been chosen.

28.4 Canonical codes without a high-mask estimate

After the Pell conclusions, $b < B < q$ and $\ell < q$. Therefore

$$q^2 - 1 - (b - 1)\ell > 0. \quad (28.17)$$

The packed number T now has the exact three-block expansion

$$T = (q^2 - 1 - (b - 1)\ell) + q^2\theta\lambda + q^4\theta\ell.$$

The blocks have widths $(2, 2, 4)$ in radix q . The binary no-carry packing lemma, using (28.12), (28.15) and (28.16), gives

$$\begin{aligned} g \& (q^2 - 1 - (b - 1)\ell) &= 0, \\ S_2 \& \theta\lambda &= 0, & \sigma \& \theta\ell &= 0. \end{aligned} \quad (28.18)$$

Here $\&$ denotes bitwise intersection only in the proof of the polynomial system's meaning.

Since $q = B^L$ and $\theta = B - 4$, the middle mask restricts the base- B digits of S_2 to $\{0, 1, 2, 3\}$ in fewer than $2L$ positions. If $F(T)$ is its digit polynomial, then $F(B) = S_2$, and the congruence in (28.12)

gives $F(4) \equiv V \pmod{B-4}$. Both $F(4)$ and V are less than 4^{2L} , while (28.10) makes $B-4$ larger than their possible difference. Hence $F(4) = V$. Uniqueness of the base-four expansion gives

$$S_2 = \ell_0(B) + e_0(B)q.$$

Both ℓ, e and $\ell_0(B), e_0(B)$ are less than q . Uniqueness of quotient and remainder on division by q yields

$$\ell = \ell_0(B), \quad e = e_0(B). \quad (28.19)$$

No high-mask estimate proportional to the square of the digit bound is required.

The first mask, together with b and B being powers of two, now bounds every coordinate digit of g by $b-1$ and permits nonzero digits only at the indicator positions. Its unit digit is zero, so the unit coordinate of $\mathcal{C}$ is the queried x . Write

$$\mathcal{C}(T) = x + \sum_i z_i T^{v_i} + \sum_j (d_j T^{t_j} + d'_j T^{t_j+1}).$$

All decoded coordinates are nonnegative and below b . Put $A_{\mathcal{C}} = \mathcal{C}(1)^2$. Then

$$A_{\mathcal{C}} < (c_* b)^2 < B^2/64. \quad (28.20)$$

Also $e, \mathcal{C} < B^K$, and $L > 3K + 2$ gives $e\mathcal{C}^2 < q$. Through position $K-1$, the raw signed coefficients of σ are those of $\mathcal{D}(T)\mathcal{C}(T)^2$: the q terms in (28.1) start at L , and the difference between the finite baseline and λ starts at K . Each raw coefficient has absolute value at most $A_{\mathcal{C}}$, because the coefficients of $\mathcal{D}$ have absolute value at most one.

28.5 Reset padding and exact paired zero tests

Consider the signed base- B carry recursion

$$\begin{aligned} h_0 &= 0, & a_j &= [T^j]\mathcal{D}(T)\mathcal{C}(T)^2, \\ h_{j+1} &= \left\lfloor \frac{a_j + h_j}{B} \right\rfloor, & \text{digit}_j &= a_j + h_j - Bh_{j+1}. \end{aligned} \quad (28.21)$$

It computes the actual low digits of the positive integer σ . Bound (28.20) gives $|h_j| < B/8$ by induction. Indeed,

$$\left\| \left\lfloor \frac{a_j + h_j}{B} \right\rfloor \right\| \leq \frac{B}{64} + \frac{1}{8} + 1 < \frac{B}{8}$$

when $|a_j| < B^2/64$, $|h_j| < B/8$ and $B > 64$.

By (28.11), dummies do not affect the tested range. All true non-input weights are zero modulo four. The raw coefficients can therefore be nonzero only in residue classes 0, 2 and 3 modulo four; class 1 is empty. The carry into $t_j - 2$, following an empty class-1 position, is consequently either -1 or zero.

At $t_j - 2$, the raw coefficient is nonnegative and at least x^2 : the row's own reset contributes x^2 , and every other reset contribution is nonnegative. It is less than $B^2/64$. The carry into $t_j - 1$ is therefore a nonnegative integer $\alpha_j < B/64$.

At every target except t_* , the raw coefficient at $t_j - 1$ is zero. Only $\mathcal{D}_{\text{extra}}$ contributes in this residue class, and its contributions are within distance $2M$ of $t_* - 1$; different targets are separated by $4M + 4$. Dividing $\alpha_j < B$ at this empty position leaves exactly zero incoming carry at t_j .

There is no raw coefficient at $t_j + 1$. Thus the two tested digits at $t_j, t_j + 1$ form the residue modulo B^2 of the raw target polynomial G_j . The third mask permits both digits to lie only in $\{0, 1, 2, 3\}$.

For a paired zero equation, $|G_j| < B^2/64$. A negative nonzero value has a residue modulo B^2 whose high digit exceeds three, so it fails the mask. Both G_j and $-G_j$ have their own reset and zero incoming carry. They can both pass exactly when $G_j = 0$. This proves every circuit and normalization equation exactly, before δ is known to be one. In particular (28.6) holds.

The first unpaired δ^2 target also receives zero incoming carry. The sole exception to this rule is the deliberately modified last target.

28.6 The last carry fixes the unit

The first unit test has raw value δ^2 and zero incoming carry. Its two allowed digits give

$$\delta^2 = kB + s, \quad k, s \in \{0, 1, 2, 3\}. \quad (28.22)$$

The guard gives $0 < \delta \leq x$, and the coordinate bound gives $\delta < b = B/H < B/4$. Since B is a sufficiently large power of two, $s = 2$ or 3 is impossible modulo four. If $s = 1$, the square roots of one modulo B are congruent to $1, B/2 - 1, B/2 + 1, B - 1$. The bound $\delta < B/4$ forces $\delta = 1$. The only remaining case for $\delta > 1$ is $s = 0$, and therefore

$$\delta^2 \geq B, \quad x^2 \geq B. \quad (28.23)$$

At $t_* - 1$, monomial uniqueness and the input-copy equations make the raw extra-padding coefficient exactly

$$x^2 + 2xX + 2xY = 5x^2.$$

The reset at $t_* - 2$ remains present and supplies a nonnegative carry $\alpha_* < B/64$. Thus the incoming carry at the last target is

$$k_{\text{pad}} = \left\lfloor \frac{5x^2 + \alpha_*}{B} \right\rfloor \geq 0. \quad (28.24)$$

The global coefficient bound also gives $k_{\text{pad}} < B/8 < B$. If (28.23) held, then $k_{\text{pad}} \geq 5$. Since δ^2 is a multiple of B , the low digit at the last target would be k_{pad} , between five and $B - 1$, contradicting its mask. Therefore $\delta = 1$.

The circuit equations now recover the represented system at the actual positive input x . This proves sufficiency.

28.7 Necessity and all positive supplied witnesses

Given a solution of the represented system, take its circuit values, $\delta = \delta' = 1$, $X = Y = x$, $u = x^2 - 1$, all fresh copies equal, and every dummy zero. Each paired row has value zero, and both unpaired unit targets have value one.

The fixed index was already chosen in (28.7)–(28.10). Now choose a power of two b arbitrarily large relative to the finitely many coordinate values. Besides exceeding every coordinate, choose it so that every raw coefficient of $\mathcal{D}(T)\mathcal{C}(T)^2$ has absolute value less than $B/4$ for $B = Hb$, and $5x^2 < B/4$. This is possible because the coordinates, $\mathcal{D}$ and their raw coefficients are fixed before b is chosen, whereas Hb grows without bound.

Take the canonical codes, $q = B^L$ and $n = q^8$. The support margin gives $\mathcal{C}^2 < q$, and $\lambda = (q^2 - 1)/(B - 1) > e$. Hence Ω and σ are positive. The values

$$\begin{aligned} \alpha_2 &= q - \ell, & \alpha &= q^2 - (\ell + eq), & \beta &= b - x, \\ t &= \frac{\ell + eq - V}{B - 4} \end{aligned}$$

are positive integers. Positivity of t follows by evaluating the nonconstant nonnegative polynomial $\ell_0(T) + e_0(T)T^L$ at $B > 4$. The coordinate $\delta = 1$ ensures $g > 0$.

The first two masks hold by the coordinate and digit bounds. For the third, all variable positions and their preceding positions lie below the least $\mathcal{D}$ exponent, so their raw coefficients and carries are zero. Every paired target has zero value and receives zero incoming carry from its reset. The first unit target has value one. At the last target, the chosen small-coefficient bounds give $\alpha_* = 0$ and $k_{\text{pad}} = 0$, so its tested digits are also one and zero. Thus all masks hold. The packing lemma gives (28.16) for the newly determined r , with the required bounds already established.

Construct the positive Pell witnesses at this new r as in Section 27: choose $U = 4^J$,

$$Y_p = \left\lfloor \frac{(U+1)^{2r}}{U^r} \right\rfloor, \quad w = \frac{U}{n^2}, \quad s_p = \frac{Y_p}{n^2}, \quad a = Y_p(U+1).$$

Take the main and first Pell coordinates at J and $r+1$, their positive interval and first-index quotients, and the second coordinates at L . The second index uses the newly fixed $T_L = \psi_4(L)$. The exponent and rounding bounds give every required positive quotient as in that proof.

Choose the doubled-index auxiliary exponent last as $m_{\text{aux}} = 2cJ$. Set $f = \chi_A(m_{\text{aux}})$ and $i_{\text{old}} = \psi_A(m_{\text{aux}})/c^2 > 0$; the doubled-index construction supplies positive o, j . Use $i = D_p i_{\text{old}}$. This gives the relaxed norm $(ic^2)^2 = D_p(f^2 - 1)$ and preserves the signed auxiliary value. No coding witness changes. Together with the new positive α_2 , all 35 supplied unknowns are positive integers. This proves necessity and completes the mathematical proof of Theorem 28.1.

28.8 The complete arithmetic count

Relative to the 97-operation certificate, replacing $B = Hb^2$ by Hb removes one multiplication. Replacing

$$\sigma = \theta\lambda q - (\lambda - e)\mathcal{C}^2 \quad \text{by} \quad \sigma = (\lambda - e)(q - \mathcal{C}^2)$$

replaces two multiplications and one subtraction by one multiplication and one subtraction, removing another multiplication. Equation (28.2) costs one addition. All normalization, padding and two-digit support changes belong to the fixed index rather than to the arithmetic of the final system. Therefore

$$97 - 1 - 1 + 1 = 96, \quad (54 \text{ multiplications, } 43 \text{ additions}) \longrightarrow (52 \text{ multiplications, } 44 \text{ additions}).$$

The standalone checker verifies all 96 primitive instructions, the complete 23 source residuals and their acyclic corrections. The geometric and second-exponent corrections depend only on the exact equality $\theta + 4 = B$. The signed auxiliary correction depends only on the exact relaxed auxiliary norm. The sparse coefficient/carry regression supplies additional finite evidence; it does not replace the support, carry, index or positive-witness arguments above. The result is an upper bound, not an optimality claim.

29 Affine radix scaling and a certificate of 95 operations

An affine radix saves one further operation. Its first mask forbids one fixed bit in each coordinate digit. Three coordinates with that bit clear represent an arbitrary nonnegative circuit value. The larger possible digits require three-digit target windows; reset positions make paired zero equations exact, and two different padding carries exclude every nonunit homogeneous scaling.

Start with the system of Section 28, replacing its two positive gaps by the combined positive gap

$$\ell + e + \alpha = q. \tag{29.1}$$

This preserves the two additions used for the gaps and removes one positive unknown and one equation. Indeed, it implies $\ell, e < q$ and

$$S_2 = \ell + eq \leq (e + 1)(q - 1) \leq (q - 1)^2 < q^2.$$

Conversely, the canonical codes below satisfy $\ell + e < q$. Now replace the radix and the first packed mask by

$$H = H_0 + 1, \quad B = H + b, \quad q^2 - 1 - b\ell, \quad (29.2)$$

where H_0 is a sufficiently large fixed power of two. The supplied fixed index remains (V, H, T_L) ; $H_0 = H - 1$ is not an additional parameter. Every Pell equation is retained.

Theorem 29.1. *Every recursively enumerable subset of the positive integers has an admissible fixed index (V, H, T_L) for a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 95 operations: 51 multiplications and 44 additions. Here $T_L = \psi_4(L)$ for the underlying exponent in the index construction. The supplied parameters are (x, V, H, T_L) , with three fixed index parameters besides the positive input x . Fixed numerals and equality tests are free.*

The following proof includes the rebuilt coefficient index and all positive witnesses. Appendix V gives the complete arithmetic table. The exact source and primitive checker is in `../verification/`:

`round32_1980_affine_radix_certificate.py.`

29.1 Logical values and physical coordinates

Compile the represented finite Diophantine system into a circuit over nonnegative integers with actual positive input x , addition, multiplication, zero and one. Signed integer witnesses can first be represented by differences of nonnegative witnesses. Introduce a homogeneous coordinate δ , retained as one physical coordinate. Every other logical non-input coordinate is the sum of three distinct nonnegative physical coordinates. Different logical coordinates have disjoint groups. The input x is unsplit and has weight zero.

Every nonnegative integer z has such a representation with the H_0 bit clear in each entry. If that bit of z is zero, take $(z, 0, 0)$. If it is one, take

$$(z - H_0, H_0/2, H_0/2). \quad (29.3)$$

All entries are nonnegative and have the required bit clear. They become base- B digits once B exceeds these finitely many values. Their choice depends on z, H_0 , not on how much larger B is subsequently chosen. In the reverse direction, the sum of three decoded physical digits is a nonnegative logical value; no further bound on that sum is needed.

Use fresh outputs and fresh square-difference copies whenever a product would repeat a logical factor. For multiplication, addition and zero equations use respectively

$$2XY - 2W\delta = 0, \quad 2(X + Y - W)\delta = 0, \quad 2X\delta = 0. \quad (29.4)$$

A copy $X' = X$ can be imposed by

$$X'^2 - X^2 = 0. \quad (29.5)$$

Nonnegativity makes this exact. Constant-one occurrences use δ or a fresh logical copy of δ . Whenever a use in (29.4) would create $\delta\delta$, use a fresh copy instead. Repeated operands are copied too. Every displayed product thus has distinct logical factors and expands into distinct physical factors. Final equalities may use square differences or distinct-factor products with δ . Include both F and $-F$ for every zero equation $F = 0$.

Introduce fresh logical coordinates X, Y, Z, δ', u with disjoint physical groups and include both signs of

$$\begin{aligned} X^2 - x^2 &= 0, & Y^2 - x^2 &= 0, & Z^2 - x^2 &= 0, & \delta'^2 - \delta^2 &= 0, \\ 2xX - 2\delta\delta' - 2u\delta &= 0. \end{aligned} \quad (29.6)$$

Their exact zero tests will give

$$X = Y = Z = x, \quad \delta' = \delta, \quad x^2 = \delta^2 + u\delta. \quad (29.7)$$

Consequently $\delta > 0$ and $\delta \leq x$. Conversely, every positive input has the nonnegative normalization witnesses

$$\delta = \delta' = 1, \quad X = Y = Z = x, \quad u = x^2 - 1. \quad (29.8)$$

Add three unpaired targets whose raw polynomial is δ^2 , and put them last. The first is ordinary; the second and third will have different padding at their preceding positions.

Expand each logical row in the physical coordinates and x . Divide square coefficients by one and distinct-coordinate product coefficients by two. Every divided coefficient lies in $\{-1, 0, 1\}$. A square of a three-coordinate sum has diagonal coefficients one and cross coefficients two. Disjoint groups and fresh copies prevent two logical monomials from contributing to the same physical monomial in (29.4)–(29.6). The three unit targets have divided coefficient one. These rows and physical coordinates affect the fixed index, not the number of unknowns of the final system.

29.2 Support, reset positions, and the fixed index

Let m count all true non-input physical coordinates, including the unsplit δ . Assign weights

$$v_i = 6 \cdot 3^i \quad (0 \leq i < m), \quad M = 6 \cdot 3^{m-1}.$$

The input has weight zero. Quadratic monomial weights are multiples of six, at most $2M$, and determine their unordered coordinate pair, including squares and products with x . Divide by six and read the base-three digits to see this.

Let s count all targets, including both signs of every zero equation and the three final unit targets. Define

$$\begin{aligned} d_0 &= 4M + 6, & t_j &= (s + 1)d_0 + 2M + jd_0 \quad (0 \leq j < s), \\ t_{\max} &= t_{s-1} = 2sd_0 + 2M, & K &= t_{\max} + 3. \end{aligned} \quad (29.9)$$

All targets are multiples of six. Write $\tau_5 = t_{s-2}$ and $\tau_7 = t_{s-1}$ for the two modified unit targets. For a target G_j , place divided coefficient a of monomial weight w at exponent $t_j - w$ in $\mathcal{D}_{\text{main}}(T)$. The intervals $[t_j - 2M, t_j]$ are disjoint. Multiplication by the physical coordinate polynomial squared recovers exactly G_j at t_j .

Add the resets

$$\mathcal{D}_{\text{reset}}(T) = \sum_j T^{t_j-3}. \quad (29.10)$$

Let $\mathcal{P}(X)$ be the three physical indices belonging to logical coordinate X , and set

$$\begin{aligned} \mathcal{D}_5(T) &= T^{\tau_5-1} + \sum_{p \in \mathcal{P}(X) \cup \mathcal{P}(Y)} T^{\tau_5-1-v_p}, \\ \mathcal{D}_7(T) &= T^{\tau_7-1} + \sum_{p \in \mathcal{P}(X) \cup \mathcal{P}(Y) \cup \mathcal{P}(Z)} T^{\tau_7-1-v_p}, \\ \mathcal{D}(T) &= \mathcal{D}_{\text{main}}(T) + \mathcal{D}_{\text{reset}}(T) + \mathcal{D}_5(T) + \mathcal{D}_7(T). \end{aligned} \quad (29.11)$$

The three types have exponent residues zero, three, and five modulo six. Each padding polynomial has distinct exponents, and their intervals are disjoint because $\tau_7 - \tau_5 = d_0 > M$. All $\mathcal{D}$ coefficients remain in $\{-1, 0, 1\}$. Every exponent is positive, greater than M , and below K .

Allow dummy physical digits at all three tested positions $t_j, t_j + 1, t_j + 2$, and define

$$\begin{aligned}\ell_0(T) &= \sum_i T^{v_i} + \sum_j (T^{t_j} + T^{t_j+1} + T^{t_j+2}), \\ e_0(T) &= \sum_{h=0}^{K-1} T^h + \mathcal{D}(T).\end{aligned}\tag{29.12}$$

The digits of ℓ_0 are zero or one, those of e_0 are zero, one or two, and the unit digit of e_0 is one. Put

$$c_* = 1 + m + 3s, \quad D_1 = \sum_h |[T^h]\mathcal{D}(T)|.$$

Choose a power of two $L > 3K + 2$. After the complete layout is fixed, choose a power of two H_0 satisfying

$$\begin{aligned}H_0 &> \max\{2 \cdot 4^{2L+1}, 128 \max(1, D_1)c_*^2, 3L, 1024\}, \\ H &= H_0 + 1, \quad V = \ell_0(4) + e_0(4)4^L, \quad T_L = \psi_4(L).\end{aligned}\tag{29.13}$$

These choices precede the queried input. The fixed integer H need not be a power of two.

No decoded dummy contributes to $\mathcal{D}(T)\mathcal{C}(T)^2$ through $t_{\max} + 2$. The least $\mathcal{D}$ exponent is at least $t_0 - 2M$, the least dummy weight is t_0 , and

$$2t_0 - 2M > t_{\max} + 2.\tag{29.14}$$

Thus the residue analysis below involves only true coordinates of weight zero modulo six. This exclusion is needed because some dummy weights are not zero modulo six.

The spacing also makes the relevant padding coefficients exact. At $t_j - 3$ the own reset contributes x^2 . Another reset would require a monomial weight of absolute value at least $d_0 > 2M$, so cannot contribute. At $\tau_5 - 1$, the extra terms give exactly $x^2 + 2xX + 2xY$; at $\tau_7 - 1$, they give exactly $x^2 + 2xX + 2xY + 2xZ$. Other targets' preceding positions receive no extra term. Monomial uniqueness identifies these terms, the different bands are separated by more than $4M$, and the other parts of $\mathcal{D}$ have different residues modulo six.

29.3 Preliminary bounds and the Pell conclusions

The source definitions and coding equations are

$$\begin{aligned}\mathcal{C} &= x + g, \quad b = x + \beta, \quad B = H + b, \quad \theta = B - 4, \\ \lambda(B - 1) &= q^2 - 1, \quad S_2 = \ell + eq = V + t\theta, \quad \ell + e + \alpha = q, \\ \Omega &= \lambda - e > 0, \quad \sigma = \Omega(q - \mathcal{C}^2) > 0, \quad n = q^8, \\ S &= g + q^2(S_2 + q^2\sigma), \\ T + 1 &= q^2(1 + \theta\lambda) - b\ell + \theta\ell q^4, \\ r &= S(n^2 - n) + (T + 1)(n^2 - 1).\end{aligned}\tag{29.15}$$

All supplied unknowns, including $\alpha, \beta, \Omega, \sigma$, are positive integers. Before using any Pell conclusion or decoded coefficient row, positivity gives

$$\ell, e < q, \quad S_2 < q^2, \quad \mathcal{C}^2 < q, \quad g < q, \quad 0 < \sigma < \lambda q < q^3.\tag{29.16}$$

The combined gap even gives $S_2 \leq (q-1)^2$. Geometry gives $B \leq q^2$. Since $b \geq 2$ and $H > 1024$, we have $q \geq 6$, $b < B < n$, $3L \leq B \leq n$, and

$$\theta\lambda \geq B - 4 = H + b - 4 > b.$$

Integer gaps give

$$0 < S \leq q - 1 + q^2(q^2 - 1) + q^4(q^3 - 1) < q^7 < n.$$

For the other packed number,

$$\begin{aligned} T + 1 &> q^2\theta\lambda - b\ell > bq^2 - bq > 0, \\ T + 1 &< q^4(1 + \theta\ell) < Bq^5 \leq q^7 < n. \end{aligned} \tag{29.17}$$

The upper bound uses $1 + \theta\lambda < q^2$ and $1 + \theta\ell \leq 1 + (B - 4)(q - 1) < Bq$. Consequently

$$n \leq n^2 - 1 \leq r < 2n^3. \tag{29.18}$$

These are the packing and growth hypotheses of the retained Pell block. To specify the proof order, write

$$\begin{aligned} U &= wn^2, \quad Y_p = s_p n^2, \quad a = Y_p(U + 1), \quad A = a + 4, \\ D_p &= A^2 - 1, \quad P = 2UY_p^2 + 1, \quad J = 2r + 1. \end{aligned}$$

The subscripts distinguish Pell values from encoded logical coordinates. We have $U, Y_p \geq n^2$, $UY_p \geq n^4 > r + 1$, and $a > n^4 > J$. The positive interval and first index equation imply $c > J$. Classification of the main and first norms, together with $P > A$, gives preliminary main index at least $r + 2$. Thus $c > AD_p^2$, as in Sections 27 and 26.

The retained relaxed norm

$$(ic^2)^2 = D_p(f^2 - 1)$$

has the required fundamental-unit and divisibility hypotheses. The doubled signed-index argument gives $c = \psi_A(J)$ and $d = \chi_A(J)$. The first-index and ratio argument gives $Y_p \geq U^r$ and $a > U^{r+1}$ before either exponent relation is used. The common-witness exponent equation gives $U = 4^J$. The fixed congruence $\kappa = T_L + \Delta a$ determines the second index as L , and its exponent equation gives $q = B^L$. All exponent-size inequalities follow from (29.18) and $3L \leq B \leq n \leq r$ as in those proofs.

Since $n^2 \mid U = 4^J$, q is a power of two; $q = B^L$ then makes B a power of two. From $B > H_0 + 1$ we get $B \geq 2H_0$. Neither b nor H is asserted to be a power of two, and no Pell step requires it. The retained upper ratio and binomial fractional-part argument gives

$$n^2 \mid \binom{2r}{r}. \tag{29.19}$$

All these conclusions precede canonical decoding, internal equations, and the proof that $\delta = 1$.

29.4 Canonical codes and the forbidden bit

After the Pell conclusions, $b < B < q$ and $\ell < q$, so

$$q^2 - 1 - b\ell > 0. \tag{29.20}$$

The exact block expansion is

$$T = (q^2 - 1 - b\ell) + q^2\theta\lambda + q^4\theta\ell,$$

with widths $(2, 2, 4)$ in radix q . The binary no-carry packing lemma, using (29.15), (29.18), and (29.19), gives

$$g \& (q^2 - 1 - b\ell) = 0, \quad S_2 \& \theta\lambda = 0, \quad \sigma \& \theta\ell = 0. \quad (29.21)$$

The ampersand is bitwise intersection, used only to interpret the polynomial system.

Since $q = B^L$ and $\theta = B - 4$, the middle mask bounds each base- B digit of S_2 by three. Its digit polynomial $F(T)$ has degree less than $2L$ and $F(B) = S_2$. The source congruence gives $F(4) \equiv V \pmod{B-4}$. Both $F(4), V < 4^{2L}$, whereas (29.13) makes $B - 4$ larger than their possible difference. Hence $F(4) = V$. Uniqueness of base-four expansion yields $S_2 = \ell_0(B) + e_0(B)q$. Both $\ell, e < q$, and both fixed codes are below $B^K < q$; quotient and remainder uniqueness gives

$$\ell = \ell_0(B), \quad e = e_0(B). \quad (29.22)$$

This removes the quotient alias without a quadratic digit bound.

The integer $q^2 - 1$ has every base- B digit equal to $B - 1$. Subtracting $b\ell_0(B)$ causes no borrow. At indicator positions the resulting digit is

$$B - 1 - b = H - 1 = H_0,$$

and elsewhere it is $B - 1$. Since B, H_0 are powers of two with $H_0 < B$, the first mask says exactly that each physical digit of g has its H_0 bit clear and that digits outside the indicator support vanish. Every digit is nonnegative and below B . The unit position is outside the support, so g has unit digit zero. Since $x < b < B$, the unit coordinate of $\mathcal{C} = x + g$ is exactly the queried input. Write

$$\mathcal{C}(T) = x + \sum_i z_i T^{v_i} + \sum_j (d_j T^{t_j} + d'_j T^{t_j+1} + d''_j T^{t_j+2}).$$

Every physical digit, including the unsplit δ , is below B . The input is below B too. Therefore

$$A_{\mathcal{C}} = \mathcal{C}(1)^2 < (c_* B)^2 < B^3/64. \quad (29.23)$$

The last inequality follows from the fixed bound and $B \geq 2H_0$. Every raw coefficient of $\mathcal{D}(T)\mathcal{C}(T)^2$ has absolute value at most $A_{\mathcal{C}}$: distinct exponents of $\mathcal{D}$ select distinct nonnegative coefficients of $\mathcal{C}(T)^2$, and each multiplier has absolute value at most one.

Also $e, \mathcal{C} < B^K$, hence $e\mathcal{C}^2 < q$ because $L > 3K + 2$. Through position $K - 1$, the raw signed coefficients of σ are those of $\mathcal{D}(T)\mathcal{C}(T)^2$. Indeed,

$$\sigma = (\lambda - e)q + (e - \lambda)\mathcal{C}^2.$$

The first summand starts at L , and the coefficient polynomial of $e - \lambda$ agrees with $\mathcal{D}(T)$ below K . Terms starting at K or L cannot affect lower digits. This identifies the actual low digits even when some raw coefficients are negative.

29.5 Reset padding and exact three-digit zero tests

Use the signed carry recursion

$$\begin{aligned} h_0 &= 0, & a_j &= [T^j]\mathcal{D}(T)\mathcal{C}(T)^2, \\ h_{j+1} &= \left\lfloor \frac{a_j + h_j}{B} \right\rfloor, & \text{digit}_j &= a_j + h_j - Bh_{j+1}. \end{aligned} \quad (29.24)$$

It computes the low base- B digits of the positive integer σ . The coefficient bound gives $|h_j| < B^2/8$ by induction throughout the tested range: the induction step has upper bound $B^2/64 + B/8 + 1 < B^2/8$ for $B > 64$.

By (29.14), dummies do not contribute. Only residue classes zero, three and five modulo six can have nonzero raw coefficients; classes one, two and four are empty. Two consecutive empty positions of classes one and two reduce a carry of magnitude below $B^2/8$ to -1 or zero, since their combined operation is division by B^2 followed by the floor. Thus the incoming carry at every reset $t_j - 3$ is in $\{-1, 0\}$.

Its raw coefficient is exactly x^2 . Since $1 \leq x < B$, the sum with the incoming carry is nonnegative and below B^2 . The carry to $t_j - 2$ is a nonnegative integer below B . That position is empty, so the carry to $t_j - 1$ is exactly zero. Except at τ_5, τ_7 , the raw coefficient at $t_j - 1$ is also zero. Every ordinary target therefore receives zero incoming carry. Its next two raw coefficients are zero as well, so its three tested digits form the residue of G_j modulo B^3 . The third mask requires each digit to be in $\{0, 1, 2, 3\}$.

For a paired zero row, $|G_j| < B^3/64$. A negative nonzero value has residue modulo B^3 greater than $63B^3/64$; its top digit exceeds three, so it fails the mask. Both signs have separate resets and zero incoming carry. They can both pass only at zero, which passes both. Every circuit equation and every normalization equation is now exact, without assuming a unit value. In particular (29.7) holds.

The first unpaired unit target also receives zero carry. At $\tau_5 - 1$ and $\tau_7 - 1$, the exact padding coefficients and the input-copy equations give $5x^2$ and $7x^2$, with zero incoming carry. Thus the target carries are exactly

$$h_5 = \left\lfloor \frac{5x^2}{B} \right\rfloor, \quad h_7 = \left\lfloor \frac{7x^2}{B} \right\rfloor. \quad (29.25)$$

Both targets still have raw value δ^2 and two empty following raw positions. Their tested digits are the residues of $\delta^2 + h_5$ and $\delta^2 + h_7$ modulo B^3 .

29.6 Two different padding carries fix the unit

The guard gives $0 < \delta \leq x < B$; the unsplit physical digit bound also gives $\delta < B$ directly. Hence $\delta^2 < B^2$. The first unit test implies

$$\delta^2 = kB + s, \quad k, s \in \{0, 1, 2, 3\}. \quad (29.26)$$

Since $4 \mid B$, a square cannot have $s = 2$ or 3 . If $s = 1$, then $\delta^2 \leq 3B + 3 < B^2/16$ for $B \geq 64$, so $\delta < B/4$. The roots of one modulo a power of two $B \geq 8$ are congruent to $1, B/2 - 1, B/2 + 1, B - 1$; only the first lies in $(0, B/4)$. Thus $\delta = 1$.

For $\delta > 1$ the only remaining case is

$$\delta^2 = kB, \quad 1 \leq k \leq 3, \quad x^2 \geq kB. \quad (29.27)$$

For $c = 5, 7$, write

$$h_c = m_c B + r_c, \quad 0 \leq r_c < B.$$

Since $x < B$, we have $h_c < cB$. The tested integer is $(k + m_c)B + r_c < (3 + c)B < B^2$. There is no reduction modulo B^3 and no overflow into its third digit. Acceptance forces

$$0 \leq r_5, r_7 \leq 3, \quad 0 \leq m_5, m_7 \leq 3 - k \leq 2. \quad (29.28)$$

The two floors are related. For $z = x^2/B$, using braces for fractional part,

$$7h_5 - 5h_7 = 5\{7z\} - 7\{5z\}, \quad -7 < 7h_5 - 5h_7 < 5. \quad (29.29)$$

Thus its absolute value is less than seven. Substituting the quotients and remainders and using $|7r_5 - 5r_7| \leq 21$ gives

$$|B(7m_5 - 5m_7)| < 28.$$

Since $B > 32$, we obtain $7m_5 = 5m_7$. Its nonnegative solutions are $(m_5, m_7) = (5h, 7h)$. Bound (29.28) forces both quotients to be zero. Then $h_5 = r_5 \leq 3$, whereas (29.27) gives $h_5 \geq 5k \geq 5$, a contradiction. Therefore $\delta = 1$.

The paired circuit equations now recover the represented system at its actual positive input. This proves sufficiency.

29.7 Necessity and all positive supplied witnesses

Given a solution of the represented system, assign its logical circuit values and fresh copies, and use (29.8) for normalization. Split each non-input logical value other than δ into three physical coordinates as above; retain $\delta = 1$ unsplit. All physical values have their H_0 bit clear, including δ since $H_0 > 1$. Set every dummy to zero. Each paired target has value zero and the three unit targets have value one.

The fixed index was chosen before this input. With the physical assignment fixed, choose B an arbitrarily large power of two. Require $B > H + x$ and B larger than every physical coordinate. Also require every raw coefficient of $\mathcal{D}(T)\mathcal{C}(T)^2$ to have absolute value below $B/4$, and require $7x^2 < B/4$. These conditions are possible because all those finitely many values are fixed while B grows without bound. Set

$$\begin{aligned} b &= B - H, & \beta &= b - x > 0, & \ell &= \ell_0(B), & e &= e_0(B), \\ q &= B^L, & n &= q^8. \end{aligned}$$

The sum of the two code polynomials has digits at most three, so $\ell + e < B^K < q$. The support margin gives $\mathcal{C}^2 < q$ and $\lambda = (q^2 - 1)/(B - 1) > e$. Consequently

$$\begin{aligned} \alpha &= q - \ell - e, & \Omega &= \lambda - e, & \sigma &= \Omega(q - \mathcal{C}^2), \\ t &= \frac{\ell + eq - V}{B - 4} \end{aligned}$$

are positive integers. For t , evaluate the nonconstant nonnegative polynomial $\ell_0(T) + e_0(T)T^L$ at $B > 4$; its difference from its value at four is positive and divisible by $B - 4$. The encoded coordinate $\delta = 1$ ensures $g > 0$.

The first mask holds because the physical digits have the specified bit clear; the second holds because the combined code digits are at most three. For the third mask, each true variable position lies below the least $\mathcal{D}$ exponent and has zero raw coefficient and zero carry. Every paired target receives zero carry from its reset and has value zero. The first unit target has value one. Since $7x^2 < B/4$, both special carries vanish, so those targets also have digits $(1, 0, 0)$. All masks hold. The packing lemma supplies (29.19) for the newly determined r ; the preliminary bounds give its required size estimates.

Construct every positive Pell witness for this r as in Section 27. Choose $U = 4^J$ and

$$Y_p = \left\lfloor \frac{(U + 1)^{2r}}{U^r} \right\rfloor, \quad w = \frac{U}{n^2}, \quad s_p = \frac{Y_p}{n^2}, \quad a = Y_p(U + 1).$$

Take the main and first Pell coordinates at J and $r + 1$, their positive interval and first-index quotients, and the second Pell coordinates at L . That index uses the newly fixed $T_L = \psi_4(L)$. The retained exponent and rounding bounds give the positive exponential quotients. They require B, q to be powers of two, as they are, but place no such requirement on b or H .

Choose the doubled-index auxiliary exponent last as $m_{\text{aux}} = 2cJ$. Set $f = \chi_A(m_{\text{aux}})$, $i_{\text{old}} = \psi_A(m_{\text{aux}})/c^2 > 0$ and $i = D_p i_{\text{old}}$. The doubled-index construction supplies positive o, j . The relaxed auxiliary norm holds and its signed auxiliary value is unchanged. Together with the new positive β and combined-bound α , all 34 supplied unknowns are positive integers. This proves necessity and completes the proof of Theorem 29.1.

29.8 The complete arithmetic count

The combined-bound 96-operation schedule uses a multiplication for $B = Hb$ and a subtraction for $b - 1$. The new definition $B = H + b$ uses one addition instead of one multiplication. Replacing $(b - 1)\ell$ by $b\ell$ preserves that multiplication and removes the subtraction. Therefore

$$(52 \text{ multiplications, } 44 \text{ additions}) \longrightarrow (51 \text{ multiplications, } 44 \text{ additions}), \quad 51 + 44 = 95.$$

There are 34 positive unknowns and 22 source equations. The logical coordinates, physical triples, three-digit windows, resets, and both padding patterns change only the fixed integer index. No numeral construction is charged. Each subtraction can be realized by one reversed addition equation with a free integer intermediate.

The complete arithmetic checker compares the changed radix and first-mask expressions with all source residuals, including the retained acyclic corrections for geometry, the second exponent, and the relaxed signed norm. The separate sparse encoding regression gives finite evidence for coefficient isolation and carries. It does not replace the universal support, unit, Pell, or positive-witness arguments above. The result is an upper bound, with no optimality assertion.

30 Translating the fixed index: 94 operations

The affine-radix construction permits one more saving by translating a fixed index parameter. This changes neither its coefficient encoding nor any positive witness. Let $H_{\text{old}} = H_0 + 1$ be the fixed parameter of Section 29, and supply instead

$$H = H_{\text{old}} - 4 = H_0 - 3. \quad (30.1)$$

The actual radix is still

$$B = H + b + 4 = H_{\text{old}} + b, \quad (30.2)$$

so the fixed-bit mask and every part of the encoding are unchanged. In particular, $B - 1 - b = H_0$ as before. The positive source equation for θ becomes

$$\theta = H + b. \quad (30.3)$$

All other source equations are obtained by replacing the former fixed parameter H_{old} by $H + 4$. Every occurrence of the mathematical radix therefore has the value in (30.2).

Theorem 30.1. *Every recursively enumerable subset of the positive integers has an admissible fixed index (V, H, T_L) for a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 94 operations: 51 multiplications and 43 additions. Here $H = H_0 - 3$, with H_0 chosen by (29.13), and $T_L = \psi_4(L)$. The supplied parameters are (x, V, H, T_L) ; fixed numerals and equality tests are free.*

Proof. Fix an index (V, H_{old}, T_L) furnished by Theorem 29.1. The translated index is $(V, H_{\text{old}} - 4, T_L)$, with a positive second component because $H_0 > 1024$. For every positive input x and every tuple $\mathbf{z}$ of the 34 positive supplied unknowns, define the new source residuals by

$$F_i^{94}(x, V, H, T_L; \mathbf{z}) = F_i^{95}(x, V, H + 4, T_L; \mathbf{z}), \quad 1 \leq i \leq 22.$$

This is an exact polynomial substitution. The old radix equation $\theta + 4 - (H_{\text{old}} + b) = 0$ simplifies to (30.3); the remaining equations are exactly their old values at the translated fixed index. Thus

the identity map on $\mathbf{z}$ is a bijection between the positive solutions of the two source systems, in both directions.

For sufficiency, a solution of the translated system is a solution of the 95-operation system at $H_{\text{old}} = H + 4$. Its complete Pell, canonical-code, carry and unit arguments recover the represented set at the actual input x . For necessity, use every positive witness of that construction unchanged. In particular, choosing a sufficiently large power of two B gives

$$b = B - H - 4 = B - H_{\text{old}}, \quad \beta = b - x > 0,$$

and preserves the same α , codes, packing, and positive Pell witnesses. The value $H_0 = H + 3$ is the same fixed power of two, so its forbidden bit and all support bounds are identical. This proves the full source-system equivalence without imposing any new restriction on the witnesses.

It remains to realize the translated source equations with one fewer primitive operation. In the 95-operation schedule the two additions

$$B_{\text{reg}} = H_{\text{old}} + b, \quad L_3 = \theta + 4$$

are followed by the free equality test $L_3 = B_{\text{reg}}$. The register B_{reg} has no other use in that schedule. All other radix-dependent expressions already use the existing θ registers, together with the exact source relation $\theta + 4 = B$.

In the translated schedule retain just

$$\theta_{\text{sum}} = H + b$$

and replace the equality test by $\theta = \theta_{\text{sum}}$. Delete the instruction for L_3 . Renaming the retained register has no arithmetic cost. The new register is the value of θ ; it is not the mathematical radix, which remains $B = H + b + 4$ and is never constructed as a separate register. No instruction depending on it needs correction, since its sole use is this equality. The other primitive instructions and all other equality tests are retained, giving

$$51 \text{ multiplications} + 43 \text{ additions} = 94$$

with 34 positive unknowns and 22 source equations. Translating the fixed integer H_{old} has no numeral-construction cost. $\square$

Appendix W gives the complete 94-instruction table.

The standalone checker is in `../verification/`:

```
round33_1980_shifted_affine_certificate.py.
```

It checks the entire primitive schedule and all 22 translated source residuals, including the retained acyclic corrections. The unchanged coefficient construction retains the support, carry, unit, and positive-witness proof of Section 29. The number 94 is a proved upper bound, with no assertion of optimality.

31 Factoring the packed mask: 93 operations

The system and fixed indices of Section 30 admit a shorter arithmetic certificate by factoring the indicator code out of two terms of the packed mask. No source equation, encoding, admissibility condition, or positive unknown changes.

Theorem 31.1. *The universal system of Theorem 30.1, with exactly the same admissible fixed indices and positive witnesses, admits a straight-line certificate of 93 operations: 50 multiplications and 43 additions. There remain 34 positive unknowns and 22 equations. Its supplied parameters are (x, V, H, T_L) , with three fixed index parameters besides the positive input x . The literal numerals are $\{1, 3, 8, 15\}$; fixed numerals and equality tests are free.*

Proof. The existing register $T_{\text{coef}} = 1 + \theta\lambda$ gives the packed mask in the form

$$T + 1 = q^2 T_{\text{coef}} - \ell b + \theta \ell q^4.$$

The 94-operation schedule evaluates this expression using four multiplications and two additions or subtractions:

$$\begin{aligned} T_{q^4} &= T_{\text{coef}} q^2, & L_b &= \ell b, & T_2 &= T_{q^4} - L_b, \\ L_{q^4} &= \ell q^4, & M &= \theta L_{q^4}, & T + 1 &= T_2 + M. \end{aligned}$$

Replace these six instructions by the following five:

$$\begin{aligned} T_{q^4} &= T_{\text{coef}} q^2, & Q_\theta &= \theta q^4, & G &= Q_\theta - b, \\ I &= \ell G, & T + 1 &= T_{q^4} + I. \end{aligned} \tag{31.1}$$

These use three multiplications and two additions or subtractions. Distributivity gives the unconditional integer identity

$$q^2 T_{\text{coef}} + \ell(\theta q^4 - b) = q^2 T_{\text{coef}} - \ell b + \theta \ell q^4. \tag{31.2}$$

Thus the retained packed output, named **T** in the certificate, has its old value for every integer assignment, before any source equation is assumed. The deleted temporary registers have no consumers outside this block and occur in none of the free equality tests.

Every other instruction and equality test is retained. The source residuals are therefore exactly those of the 94-operation system. The identity map on the input parameters and all 34 positive supplied unknowns is the positive-witness map in both directions. Each schedule determines its own temporary registers from that same tuple. Consequently the full universality, canonical-code, carry, unit, and Pell proof transfers without any changed hypothesis.

The new temporary registers introduce no supplied unknown or domain condition. They are in fact positive on the existing domain: $\theta = H + b > b$, $q \geq 1$, and hence $\theta q^4 - b > 0$. Each subtraction is certified by its reversed addition, as before. The complete count is

$$94 - 1 = 93 = 50 \text{ multiplications} + 43 \text{ additions.}$$

No numeral is introduced or changed. □

Appendix X gives all 93 instructions and the 22 unchanged free equality tests. The complete checker is in `../verification/`:

`round34_1980_factored_mask_certificate.py.`

It verifies the exact six-to-five replacement, the absence of outside uses for deleted temporaries, equality of every retained register, all primitive checks, and all source residuals with their unchanged acyclic corrections. The shorter schedule is a certificate for the same proved system; 93 is an upper bound, with no optimality assertion.

32 A smaller auxiliary Pell parameter: 92 operations

The square $K = (ic^2)^2$ already present in the relaxed auxiliary norm allows a smaller Pell parameter. Replacing the ten-operation signed block by a nine-operation block saves one multiplication. The new block uses one additional positive unknown and one additional equation. Its positive witnesses are constructed afresh; the coefficient encoding and fixed index do not change.

Theorem 32.1. *Every recursively enumerable subset of the positive integers has an admissible fixed index (V, H, T_L) for a system with 35 positive unknowns and 23 equations admitting a straight-line certificate of 92 operations: 49 multiplications and 43 additions. The fixed index and coefficient encoding are those of Theorem 31.1. The supplied inputs are (x, V, H, T_L) , with three fixed components besides the positive input x . Fixed numerals and equality tests are free.*

32.1 The new equations

Retain the definitions

$$J = 2r + 1, \quad A = a + 4, \quad D = A^2 - 1, \quad R = ic^2, \quad K = R^2. \quad (32.1)$$

In particular, the relaxed auxiliary equation is retained:

$$R^2 = D(f^2 - 1). \quad (32.2)$$

The former signed equation was

$$v(v + 1) = K(K - 1)(J + jc)^2, \quad v = of - d^2.$$

Replace it by

$$u = J + jc, \quad u = c + of, \quad (32.3)$$

and

$$K(u^2 - y_{\text{aux}}^2) = 1 - y_{\text{aux}}^2. \quad (32.4)$$

The quantity u is an arithmetic register, not another supplied unknown. The old variables o, j and the new variable y_{aux} are positive integers. The last equation is equivalently

$$(Ru)^2 - (R^2 - 1)y_{\text{aux}}^2 = 1. \quad (32.5)$$

Its ordinary Pell parameter is R , whereas the preceding signed construction used $2R^2 - 1$. The certificate does not have to construct either Ru or that larger parameter.

32.2 Independent bounds before the new block

The preliminary proof of Section 29, retained by the subsequent arithmetic improvements, gives

$$\begin{aligned} n &= q^8, & n &\geq 64, & n &\leq r < 2n^3, & 3L &\leq B \leq n, \\ U &= wn^2, & Y_P &= spn^2, & a &= Y_P(U + 1), & P &= 2UY_P^2 + 1, \\ U, Y_P &\geq n^2, & UY_P &> r + 1, & a &> J, & c &> J > 1. \end{aligned} \quad (32.6)$$

The subscripts distinguish these Pell quantities from the encoded logical coordinates. These bounds use positivity, the combined gap, geometry, packing, and the positive interval. They precede coefficient decoding and any use of either auxiliary signed norm.

Classification of the unchanged main and first norms gives positive indices p, t with

$$c = \psi_A(p), \quad d = \chi_A(p), \quad k = \psi_P(t).$$

The first-index equation implies $t \geq r + 1$. Since $P > A$ and $c > Y_P k > k$, the growth comparison in Section 26 forces

$$p \geq r + 2 \geq 66, \quad c > AD^2. \quad (32.7)$$

The retained relaxed-norm argument therefore applies to (32.2). Its integer-coefficient fundamental-unit and divisibility proof supplies a positive index m such that

$$f = \chi_A(m), \quad p \mid m, \quad c \mid m, \quad R = D\psi_A(m). \quad (32.8)$$

That proof uses neither the former nor the new signed-index equation. Elementary Pell growth now gives

$$0 < 2p \leq c \leq m, \quad R = ic^2 \geq c^2 > A > 1. \quad (32.9)$$

In particular, R is an integer Pell parameter greater than one. No exact value of p , no parity of r , no conclusion $q = B^L$, and no decoded circuit equation has been assumed here.

32.3 The odd-index polynomial identity

For the standard Pell sequences, write

$$\begin{aligned} \chi_X(0) = 1, \quad \chi_X(1) = X, \quad \psi_X(0) = 0, \quad \psi_X(1) = 1, \\ z(n+2) = 2Xz(n+1) - z(n). \end{aligned}$$

For every $h \geq 0$ there is an integer polynomial $Q_h(T)$ with

$$\chi_X(2h+1) = XQ_h(X^2). \quad (32.10)$$

Its initial values and doubled-step recurrence are

$$\begin{aligned} Q_0(T) = 1, \quad Q_1(T) = 4T - 3, \\ Q_{h+2}(T) = (4T - 2)Q_{h+1}(T) - Q_h(T). \end{aligned} \quad (32.11)$$

The doubled recurrence follows by applying the ordinary recurrence twice, or from $\chi_X(2) = 2X^2 - 1$ and the addition identities. Thus these formulas prove divisibility by X as an integer-polynomial identity; they do not require division modulo an integer.

The odd subsequence $\psi_A(2h+1)$ has recurrence coefficient $4A^2 - 2$ and initial values $1, 4A^2 - 1$. Consequently $(-1)^h \psi_A(2h+1)$ has coefficient $2 - 4A^2$ and initial values $1, 1 - 4A^2$. Substituting $T = 1 - A^2$ in (32.11) gives exactly these data. Induction proves

$$Q_h(1 - A^2) = (-1)^h \psi_A(2h+1). \quad (32.12)$$

Similarly, substitution of $T = 0$ gives

$$Q_h(0) = (-1)^h (2h+1). \quad (32.13)$$

Both identities hold for every nonnegative h . Their two uses below have different moduli.

32.4 Sufficiency: the exact main index

Suppose the new source equations have positive integer witnesses. By (32.5), positivity, and $R > 1$, ordinary Pell classification gives a positive index s such that

$$Ru = \chi_R(s), \quad y_{\text{aux}} = \psi_R(s). \quad (32.14)$$

The index s is odd. Indeed, the chi recurrence at $R = 0$ gives $\chi_R(2h) \equiv (-1)^h \pmod{R}$. An even s would make $R > 1$ divide one, since R divides the first coordinate in (32.14). Write $s = 2h + 1$. Equation (32.10) gives the exact equality

$$u = Q_h(R^2). \quad (32.15)$$

Reduction of (32.2) modulo f gives

$$R^2 = D(f^2 - 1) \equiv 1 - A^2 \pmod{f}.$$

Equations (32.12), (32.15), and the second equality in (32.3) therefore imply

$$(-1)^h \psi_A(s) \equiv u \equiv c = \psi_A(p) \pmod{f}.$$

Squaring and using the doubling identity

$$\chi_A(2z) = 1 + 2D\psi_A(z)^2$$

gives

$$\chi_A(2s) \equiv \chi_A(2p) \pmod{\chi_A(m)}. \quad (32.16)$$

Apply Lemma 7.1: for $A > 1$ and $0 < k \leq m$, the congruence $\chi_A(n) \equiv \pm \chi_A(k) \pmod{\chi_A(m)}$ implies $n \equiv \pm k \pmod{2m}$. The positive-sign case also follows from the ordinary chi step-down statement with its stronger modulus $4m$. Here its comparison-index hypothesis is exactly $0 < 2p \leq m$ from (32.9). Hence

$$2s \equiv \pm 2p \pmod{2m}, \quad s \equiv \pm p \pmod{m}, \quad s \equiv \pm p \pmod{c}. \quad (32.17)$$

The middle implication divides an integer equality $2s = \pm 2p + 2mz$ by two. It does not assume that two is invertible in an even residue ring.

Independently, $c \mid R$ because $R = ic^2$. Equations (32.13) and (32.15) give

$$u \equiv (-1)^h s \pmod{c}.$$

The first equality in (32.3), together with (32.17), now yields

$$J \equiv \pm p \pmod{c}. \quad (32.18)$$

Both J and p lie strictly between zero and c , by (32.6) and Pell growth. Thus either $J = p$ or $J + p = c$. The latter is impossible: the psi recurrence gives $\psi_A(p) \equiv p \pmod{2}$ for every integer A , whereas $J = 2r + 1$ is odd. Therefore $J + p$ has parity opposite to $c = \psi_A(p)$. We have proved

$$p = J, \quad c = \psi_A(J), \quad d = \chi_A(J). \quad (32.19)$$

This deduction did not assume that r is even. In particular, it does not use the subsequent binary decoding circularly.

32.5 The remaining sufficiency proof

Once (32.19) is available, the retained first-index and ratio arguments recover the first index $r + 1$, prove $Y_P \geq U^r$ and $a > U^{r+1}$, and use the common-witness exponent equation to obtain $U = 4^J$. The unchanged fixed value $T_L = \psi_4(L)$, gap, and second Pell equations recover the second index L and $q = B^L$. The same bounds give $n^2 \mid \binom{2r}{r}$.

These are exactly the hypotheses and conclusions of Section 29. The replaced auxiliary variables o, j occur nowhere in these later equations. Since $n^2 \mid U$, the integers q and then B are powers of two. Neither b nor H is required to be a power of two. The unchanged masks recover the canonical codes, the three-way physical coordinates, all paired rows, and the unit $\delta = 1$. They therefore recover the represented system at its actual positive input x .

The supplied fixed index remains

$$H = H_0 - 3, \quad V = \ell_0(4) + e_0(4)4^L, \quad T_L = \psi_4(L), \quad B = H + b + 4.$$

The new block adds no index parameter, compiler convention, or support requirement.

32.6 Necessity: parity and positive auxiliary witnesses

Start with a solution of the represented Diophantine system. Choose its canonical physical encoding and all positive coding and Pell witnesses as in Section 29, translating the fixed index and retaining the factored mask as in Sections 30 and 31. All those choices remain available.

The canonical indicator has zero unit digit, so $B \mid \ell$. The integer g also has zero unit digit, so $B \mid g$. Since B and $q = B^L$ are even, both packed integers

$$\begin{aligned} S &= g + q^2(\ell + eq + q^2\sigma), \\ T + 1 &= q^2(1 + \theta\lambda) + \ell(\theta q^4 - b) \end{aligned}$$

are even. Hence $r = S(n^2 - n) + (T + 1)(n^2 - 1)$ is even, and

$$J = 2r + 1 \equiv 1 \pmod{4}. \quad (32.20)$$

Keep the canonical relaxed auxiliary witnesses

$$\begin{aligned} m &= 2cJ, & f &= \chi_A(m), & i_{\text{old}} &= \frac{\psi_A(m)}{c^2} > 0, \\ i &= Di_{\text{old}}, & R &= ic^2 = D\psi_A(m), & K &= R^2. \end{aligned} \quad (32.21)$$

The quotient is integral: expanding $(d + c\sqrt{D})^{2c}$, its term with one square root contains $2c^2$, while every later square-root term contains at least c^3 . Positivity follows from $m > 0$. The Pell norm gives (32.2). Thus the same i, f can be retained. Keep every other canonical witness except o, j , and define

$$\begin{aligned} y_{\text{aux}} &= \psi_R(J), & u &= \frac{\chi_R(J)}{R} = Q_{(J-1)/2}(R^2), \\ o &= \frac{u - c}{f}, & j &= \frac{u - J}{c}. \end{aligned} \quad (32.22)$$

The division defining u is exact by (32.10). Equation (32.20) makes $(J - 1)/2$ even, so (32.12) and (32.13) give

$$u \equiv \psi_A(J) = c \pmod{f}, \quad u \equiv J \pmod{c}.$$

Both quotients in (32.22) are therefore integers.

They are strictly positive. For $R > 1$, the positive chi sequence is strictly increasing, and its recurrence gives

$$\chi_R(k+1) > (2R-1)\chi_R(k) \quad (k \geq 1).$$

Since $\chi_R(1) = R$ and $J \geq 3$, we obtain

$$\frac{\chi_R(J)}{R} > (2R-1)^{J-1} > (2A)^{J-1} > \psi_A(J) = c > J. \quad (32.23)$$

The middle inequality uses the integer bound $R \geq c^2 > A$, which implies $2R-1 > 2A$; the next is the usual strict psi upper bound at index $J \geq 3$. Thus $o, j > 0$, and $y_{\text{aux}} > 0$ too.

The Pell equation for $(\chi_R(J), \psi_R(J))$ is precisely (32.5), hence (32.4). The definitions give both equalities in (32.3). Every other source equation is retained and uses none of the changed auxiliary values. All 35 supplied unknowns are therefore positive integers. This proves necessity without asserting an identity map for the auxiliary coordinates, and shows exactly why $J \equiv 1 \pmod{4}$ was needed only in this direction.

32.7 The complete arithmetic count

The former signed block used ten operations:

$$\begin{aligned} O &= of, & v &= O - d^2, & v_1 &= v + 1, & L &= vv_1, \\ K_1 &= K - 1, & K_2 &= KK_1, \\ J_c &= jc, & u &= J + J_c, & U_2 &= u^2, & P &= K_2U_2. \end{aligned}$$

The quantities d^2, K, c, J already exist outside this block. Replace it by nine operations:

$$\begin{aligned} J_c &= jc, & u &= J + J_c, & O &= of, \\ u_{\text{rhs}} &= c + O, & U_2 &= u^2, & Y_2 &= y_{\text{aux}}^2, \\ G &= U_2 - Y_2, & L &= KG, & P &= 1 - Y_2. \end{aligned} \quad (32.24)$$

The free tests are $u = u_{\text{rhs}}$ and $L = P$. The old block has six multiplications and four additions or subtractions; the new block has five multiplications and four additions or subtractions. Every other primitive remains unchanged. The complete count is

$$93 - 1 = 92 = 49 \text{ multiplications} + 43 \text{ additions.}$$

The variable d and its squared register remain in the main norm; their absence from the new auxiliary block gives no further saving.

The new intermediates are ordinary integer registers. In fact G and P in (32.24) are negative on solutions. Since $u = J + jc > 1$, Pell index one is excluded in (32.14); thus $y_{\text{aux}} > 1$ and

$$u^2 - y_{\text{aux}}^2 = \frac{1 - y_{\text{aux}}^2}{K} < 0.$$

These subtractions are represented by reversed addition equations. No positivity condition is imposed on the intermediate gaps; all 35 supplied unknowns, including y_{aux} , are positive.

The checker writes the source norm with $K_{\text{source}} = D(f^2 - 1)$ and uses the already computed $K = (ic^2)^2$ in the primitive product. The difference of its residuals is exactly

$$(K - K_{\text{source}})(u^2 - y_{\text{aux}}^2).$$

Its first factor is the independently exact retained equation (32.2), so this is an acyclic correction. It replaces the preceding signed norm's quadratic-factor correction. The new linear congruence check is exact, and the geometric and second-exponent corrections remain as before.

Appendix Y gives the complete table of 92 instructions and all 23 equality tests. The checker is in `../verification/`:

`round35_1980_half_parameter_pell_certificate.py.`

Its JSON receipt records the full primitive schedule, domains, source residuals, and remaining corrections. Those arithmetic checks and the positive-witness proof above establish Theorem 32.1. The number 92 is an upper bound; no optimality assertion or new compiled Lean theorem is claimed.

33 A positive product bound and two copied inputs: 91 operations

The coefficient polynomial can be encoded as the difference of the two already supplied codes. This removes the subtraction $q - C^2$ from the product used in the third binary mask. Every negative coefficient then creates an additional tested position. Two encoded copies of the input, and positive complementary terms, make those extra tests exact without adding arithmetic operations to the final system.

Theorem 33.1. *Every recursively enumerable subset of the positive integers has an admissible fixed index (V, H, T_L) for a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 91 operations: 49 multiplications and 42 additions. The supplied inputs are (x, V, H, T_L) , with three fixed components besides the positive queried input x . Fixed numerals and equality tests are free.*

33.1 The changed source and its preliminary bounds

Start from the system of Theorem 32.1, retaining its half-parameter Pell block and factored packed mask. Replace the coding bound and product by

$$\ell + \sigma + \alpha = q, \quad \sigma = (e - \ell)C^2, \quad C = x + g. \quad (33.1)$$

All supplied unknowns in these equations are positive integers. The previous equations were

$$\ell + e + \alpha = q, \quad \Omega = \lambda - e > 0, \quad \sigma = \Omega(q - C^2).$$

The new system does not supply Ω . Indeed, $C \geq 2$ and $\sigma > 0$ imply $e - \ell > 0$. If desired, a supplied positive $\Omega = e - \ell$ and its free equality can be restored uniquely. Their elimination therefore preserves the positive solution set.

Write $\Omega = e - \ell$ only as notation for this computed positive integer. Equations (33.1) give, before any Pell or coefficient decoding,

$$\ell < q, \quad \sigma < q, \quad C^2 \leq \sigma < q, \quad e = \ell + \Omega \leq \ell + \sigma < q. \quad (33.2)$$

The remaining definitions are

$$\begin{aligned} H &= H_0 - 3, \quad B = H + b + 4 = H_0 + 1 + b, \quad \theta = B - 4, \quad b = x + \beta, \\ \lambda(B - 1) &= q^2 - 1, \quad S_2 = \ell + eq = V + t\theta, \quad n = q^8, \\ S &= g + q^2(S_2 + q^2\sigma), \\ T_+ &= q^2(1 + \theta\lambda) + \ell(\theta q^4 - b), \\ r &= S(n^2 - n) + T_+(n^2 - 1). \end{aligned} \quad (33.3)$$

Here H_0 is a sufficiently large fixed power of two, specified below. In particular $S_2 < q^2$ and $g < C < q$. Geometry gives $B \leq q^2$; the fixed threshold gives $3L \leq B \leq n$ and $\theta\lambda > b$. The estimates used in Section 29 consequently give

$$0 < S, T_+ < q^7 < n, \quad n^2 - 1 \leq r < 2n^3. \quad (33.4)$$

For example,

$$T_+ > q^2\theta\lambda - b\ell > bq^2 - bq > 0, \quad T_+ < q^4(1 + \theta\ell) < Bq^5 \leq q^7.$$

The upper estimate uses $1 + \theta\lambda < q^2$ and $1 + \theta\ell < Bq$. The bound on S is stronger than its predecessor because $\sigma < q$.

These are the same independent hypotheses used by the half-parameter Pell proof in Section 32. That proof now applies in its stated order, before any coefficient test. It yields the required central-binomial divisibility, $U = 4^{2r+1}$, $q = B^L$, and that B, q, n are powers of two. Its sufficiency proof does not assume even r . No discarded formula for σ enters that Pell argument.

33.2 Nonnegative coordinates and two helpers

Compile the represented finite Diophantine system into a circuit over nonnegative integers, using the fresh-copy conventions of Section 29. The actual positive input x has weight zero and is not split. Keep the homogeneous unit δ as one physical coordinate. Every other logical value is a sum of three distinct nonnegative physical coordinates. Different logical values have disjoint physical groups.

For a nonnegative value z , the fixed forbidden-bit decomposition is

$$(z, 0, 0) \quad \text{if its } H_0 \text{ bit is zero,} \quad (z - H_0, H_0/2, H_0/2) \quad \text{otherwise.} \quad (33.5)$$

All pieces have that bit zero and are nonnegative. They can be fixed before choosing an arbitrarily large power-of-two radix B .

Introduce two reserved logical helpers V_0, V_1 , each the sum of its own three physical coordinates. Every ordinary circuit row uses V_0 in place of the input and mentions neither x nor V_1 . Every monomial of such a row therefore has two non-input physical factors. Fresh copies ensure that each nonzero physical square coefficient is ± 1 and each nonzero physical cross coefficient is ± 2 . Dividing by the multiplicity in $C(T)^2$ gives coefficients in $\{-1, 1\}$.

The first four main rows are helper rows. For each $i = 0, 1$, use a seed with base polynomial $-x^2$ and a reverse row with base polynomial $x^2 - V_i^2$. The seed will be augmented so that its actual tested polynomial is $V_i^2 - x^2$. The reverse row keeps its base polynomial. This is the only intentional change to a main row caused by the augmentation.

Include both signs of every ordinary zero row. In addition to the circuit rows, include both signs of

$$\begin{aligned} X^2 - V_0^2, \quad Y^2 - V_0^2, \quad Z^2 - V_0^2, \quad \delta'^2 - \delta^2, \\ 2V_0X - 2\delta\delta' - 2u\delta. \end{aligned} \quad (33.6)$$

All these logical groups are distinct. Finish with three unpaired δ^2 rows. The last two receive the fivefold and sevenfold padding described below. They have no negative coefficients.

33.3 Weights and the positive compensation terms

Let m count the true non-input physical coordinates, including δ . Give them distinct weights

$$v_i = 8 \cdot 7^i \quad (0 \leq i < m), \quad M = 8 \cdot 7^{m-1}. \quad (33.7)$$

After division by eight, a sum of at most six such weights has base-seven digits below seven. Equality of any two such sums is therefore equality of their multisets. In particular, a sum with formal degree four cannot equal a sum with formal degree at most two. The actual input has weight zero; its seed exception is treated separately.

Let s be the number of main rows, including the four helper rows and the three final unit rows. Put

$$\begin{aligned} d_0 &= 12M + 8, & t_j &= (s+2)d_0 + 8M + jd_0 \quad (0 \leq j < s), \\ t_* &= t_{s-1} = (2s+1)d_0 + 8M, & K &= t_* + 3. \end{aligned} \quad (33.8)$$

For every divided base coefficient a at a physical monomial of weight w in row j , put aT^{t_j-w} in a complementary polynomial.

For each negative divided coefficient at $r = t_j - w$, choose a helper: V_1 for an ordinary row, its own V_i for seed i , and the other V_{1-i} for reverse row i . If its physical group is P , add the six positive terms

$$\sum_{h \leq h', \ h, h' \in P} T^{r-v_h-v_{h'}}. \quad (33.9)$$

The cross multiplicities in $C(T)^2$ recover the square of the sum of the three helper coordinates. Mark every main target and every negative base exponent as a tested start. Denote their set by $\mathcal{R}$. In a seed, the negative exponent and main target are the same start.

There are four relevant coefficient identities. First, an ordinary or reverse augmentation has complementary weight $w_{\text{negative}} + w_{\text{helper}}$ of degree four. It cannot affect the main target, because a term of C^2 has degree at most two. Second, at a negative position $r = t_j - w_{\text{negative}}$, the negative base term contributes $-x^2$. Another non-input base monomial could contribute only if

$$w_{\text{base}} = w_{\text{negative}} + w_C.$$

Degree and multiset uniqueness force $w_C = 0$ and the identical base monomial. The positive input square in a reverse row has weight zero and cannot satisfy this equality.

Third, an augmentation contributes at that position only if

$$w_{\text{negative},1} + w_{\text{helper}} = w_{\text{negative},2} + w_C. \quad (33.10)$$

Every factor of a negative monomial in this row is disjoint from the helper group. Both helper factors in (33.10) must consequently belong to the monomial on the right. Uniqueness forces the identical negative pair and the identical helper pair. Thus the extra tested polynomial is exactly

$$V_{\text{helper}}^2 - x^2. \quad (33.11)$$

Fourth, a seed has only $-x^2$ at its main target and its degree-two helper complements. Its actual target is exactly $V_i^2 - x^2$.

These arguments also exclude coefficient accumulation. Within a non-seed augmentation, the partition into a negative pair and a helper pair is unique by their disjoint groups. Seed helper pairs have distinct weights. Base and augmented complementary exponents have different degrees. Different row bands are separated by $d_0 > 12M$. Thus all coefficients constructed so far remain ± 1 .

33.4 Resets, unit padding, and the two code polynomials

Add a positive reset at every distinct tested start:

$$D_{\text{reset}}(T) = \sum_{r \in \mathcal{R}} T^{r-4}. \quad (33.12)$$

Write $t_5 = t_{s-2}$ and $t_7 = t_{s-1}$ for the last two unit targets. If $P(X)$ denotes the three physical indices of X , add

$$\begin{aligned} D_5(T) &= T^{t_5-1} + \sum_{h \in P(X) \cup P(Y)} T^{t_5-1-v_h}, \\ D_7(T) &= T^{t_7-1} + \sum_{h \in P(X) \cup P(Y) \cup P(Z)} T^{t_7-1-v_h}. \end{aligned} \quad (33.13)$$

Let $D(T)$ be the full complementary polynomial. Its base and augmentation exponents are zero modulo eight, its resets are four modulo eight, and its padding exponents are seven modulo eight. These parts do not collide. Resets have distinct exponents, and the padding terms are distinct within their disjoint final bands. Consequently every nonzero coefficient of D is still ± 1 .

Define

$$\ell_0(T) = \sum_{i=0}^{m-1} T^{v_i} + \sum_{r \in \mathcal{R}} (T^r + T^{r+1} + T^{r+2}), \quad e_0(T) = \ell_0(T) + D(T). \quad (33.14)$$

All indicator positions are distinct. Every negative coefficient of D occurs at a tested start and is canceled in e_0 by an indicator digit one. Hence the digits of ℓ_0 are zero or one and those of e_0 are zero, one, or two. Both polynomials have zero constant coefficient. The highest nonzero D coefficient is the positive padding term at $t_* - 1$. Its leading $+1$ and remaining digits ± 1 imply $D(B) > 0$ for every integer $B \geq 2$.

The indicator also permits dummy coordinate digits at the tested positions. None can affect the low coefficient tests: the least D exponent is at least $t_0 - 4M$, the least dummy weight is at least $t_0 - 2M$, and

$$2t_0 - 6M > t_* + 2. \quad (33.15)$$

Every true-coordinate weight lies below the least D exponent. Its raw coefficient and incoming carry are therefore zero.

Let c_* count the input, all true physical coordinates, and all permitted dummies, and put $D_1 = \sum_h |[T^h]D(T)|$. Choose a power of two $L > 3K + 2$ and then a power of two H_0 such that

$$H_0 > \max(2 \cdot 4^{2L+1}, 128 \max(1, D_1) c_*^2, 3L, 1024). \quad (33.16)$$

The supplied fixed index is

$$H = H_0 - 3, \quad V = \ell_0(4) + 4^L e_0(4), \quad T_L = \psi_4(L). \quad (33.17)$$

These choices precede the queried input and its witnesses.

33.5 Canonical decoding after the Pell conclusions

The bounds (33.2) and the conclusion $q = B^L$ give $b\ell < Bq < q^2$. Thus

$$T_+ - 1 = (q^2 - 1 - b\ell) + q^2\theta\lambda + q^4\theta\ell \quad (33.18)$$

is the genuine block decomposition: its first two blocks are below q^2 , and the last is below q^4 . The packed binomial condition therefore gives the three independent binary masks for g, S_2, σ .

The middle mask is $\theta\lambda = (B - 4) \sum_{j < 2L} B^j$. It requires every base- B digit of S_2 to be at most three. Write its digit polynomial as $F(T)$, with degree below $2L$. Since $S_2 \equiv V \pmod{B - 4}$, one has

$$F(4) \equiv V \pmod{B - 4}.$$

Both sides are bounded by $4^{2L} - 1$, while (33.16) makes $B - 4$ larger than their possible difference. Hence $F(4) = V$. Uniqueness of the base-four digit expansion gives

$$F(T) = \ell_0(T) + T^L e_0(T).$$

Because $\ell, e < q$, the low and high q -blocks are their actual values. Consequently $\ell = \ell_0(B)$ and $e = e_0(B)$.

The first mask now has digit $H_0 = B - 1 - b$ at an indicator position and digit $B - 1$ elsewhere. It forces all non-indicator digits of g to vanish and clears the H_0 bit of each permitted digit. The indicator has zero unit digit, so g has zero unit digit. Since $x < B$, adding x introduces no carry, and $C = x + g$ is exactly the physical coordinate polynomial evaluated at B .

Finally $\theta\ell_0(B)$ has digit $B - 4$ at every indicator position and zero elsewhere. The third mask says that each corresponding digit of

$$\sigma = (e - \ell)C^2 = D(B)C(B)^2$$

is at most three. This includes every extra negative-coefficient position and all three digits of each tested window.

33.6 Signed carries and exact zero equations

Every decoded coordinate digit is less than B , and $B \geq 2H_0$. Thus every raw coefficient of $D(T)C(T)^2$ has absolute value at most

$$D_1 C(1)^2 < D_1 c_*^2 B^2 < B^3/256. \quad (33.19)$$

The signed carry recurrence keeps every incoming carry in absolute value below $B^2/128$. By (33.15), the only raw residue classes through $t_* + 2$ are zero, four, and seven modulo eight.

At a tested start r , the positions $r - 6, r - 5$ are empty. Two successive divisions by B reduce the carry entering $r - 4$ to -1 or zero. Its raw reset polynomial is nonnegative and contains x^2 : the reset belonging to r contributes x^2 , and all other reset contributions are positive. The other parts of D have different residue classes. Since $x \geq 1$, the outgoing carry is nonnegative and below B^2 . The two empty positions $r - 3, r - 2$ therefore reduce it to zero.

At every start other than t_5, t_7 , the position $r - 1$ is also empty. The following positions $r + 1, r + 2$ are always empty. Such a test therefore sees its exact raw value modulo B^3 . A negative nonzero value satisfying (33.19) has a top digit greater than three and fails. The seed and reverse tests thus give $V_i^2 \geq x^2$ and $V_i^2 \leq x^2$, respectively. Since these values are nonnegative,

$$V_0 = V_1 = x. \quad (33.20)$$

All extra target values (33.11) now vanish. Both signs of every ordinary row must pass, so every ordinary zero equation is exact. In particular,

$$X = Y = Z = x, \quad \delta' = \delta, \quad x^2 = \delta^2 + u\delta, \quad 0 < \delta \leq x < B. \quad (33.21)$$

This establishes the input and guard constraints before using any unit test.

At the two remaining preceding positions, the exact raw values from (33.13) are $x^2 + 2xX + 2xY$ and $x^2 + 2xX + 2xY + 2xZ$. Their incoming carries are zero. By (33.21), the three unit tests are therefore the residues modulo B^3 of

$$\delta^2, \quad \delta^2 + \lfloor 5x^2/B \rfloor, \quad \delta^2 + \lfloor 7x^2/B \rfloor. \quad (33.22)$$

There is no padding at an extra negative start: the two padded rows contain no negative term, and other row bands are disjoint.

33.7 The unit is one

For completeness, the three-unit argument can be applied directly to (33.22). Since $\delta < B$, the first test gives

$$\delta^2 = kB + s, \quad k, s \in \{0, 1, 2, 3\}.$$

A square modulo four cannot have $s = 2$ or $s = 3$. If $s = 1$, then $\delta^2 \leq 3B + 3 < B^2/16$, so $0 < \delta < B/4$. The four square roots of one modulo a power of two $B \geq 8$ are represented by $1, B/2 - 1, B/2 + 1, B - 1$. Only the first lies in this range, giving $\delta = 1$.

Otherwise a nonunit solution would have $\delta^2 = kB$ with $1 \leq k \leq 3$ and $x^2 \geq kB$. Write

$$h_c = \lfloor cx^2/B \rfloor = m_c B + r_c \quad (c = 5, 7), \quad 0 \leq r_c < B.$$

Because $x < B$, one has $h_c < cB$. The padded tests then force $r_5, r_7 \leq 3$ and $m_5, m_7 \leq 3 - k \leq 2$. On the other hand,

$$-7 < 7h_5 - 5h_7 < 5, \quad |7r_5 - 5r_7| \leq 21.$$

It follows that $|B(7m_5 - 5m_7)| < 28$. Since $B > 32$, $7m_5 = 5m_7$, and their bounds imply $m_5 = m_7 = 0$. This contradicts $h_5 \geq 5k \geq 5$ and $h_5 = r_5 \leq 3$. Thus $\delta = 1$ in every solution. The decoded circuit consequently represents the original system at its actual input x , proving sufficiency.

33.8 Necessity, including the auxiliary parity condition

Given a represented solution, assign both helpers the value x , set $\delta = \delta' = 1$, $X = Y = Z = x$, and $u = x^2 - 1$, and supply all circuit values and fresh copies. Split the non-input logical values by (33.5); keep δ unsplit and set all dummies to zero. Every ordinary, helper, and extra target has raw value zero, and all three unit targets have raw value one.

After these finitely many physical values are fixed, choose B to be an arbitrarily large power of two. Require $B > H_0 + 1 + x$, every raw coefficient in absolute value below $B/4$, and $7x^2 < B/4$. Set

$$\begin{aligned} b &= B - H_0 - 1, & \beta &= b - x, & g &= C(B) - x, \\ \ell &= \ell_0(B), & e &= e_0(B), & q &= B^L, & n &= q^8, \\ \sigma &= D(B)C(B)^2, & \alpha &= q - \ell - \sigma. \end{aligned} \tag{33.23}$$

The leading coefficient of D makes $\sigma > 0$. The encoded unit makes $g > 0$. Since $\deg(DC^2) < 3K < L$, choosing B still larger makes $\alpha > 0$. All forbidden-bit tests pass, and all middle-code digits are at most two. Resets clear every signed carry. The two padded unit carries are zero because $7x^2 < B/4$. All third-mask windows are therefore $(0, 0, 0)$ or $(1, 0, 0)$.

Geometry supplies a positive integer λ , and

$$t = \frac{\ell + eq - V}{B - 4} > 0$$

is an integer: the numerator is the difference at B and at four of the nonconstant polynomial $\ell_0(T) + T^L e_0(T)$, which has nonnegative coefficients. The three masks give the required binomial divisibility for the newly determined r , whose growth bounds were proved in (33.4).

Construct all non-auxiliary Pell witnesses as in Section 29, with $U = 4^{2r+1}$, the binomial ratio value Y_P , the main and first Pell coordinates, and second index L with fixed $T_L = \psi_4(L)$. The half-parameter auxiliary construction has one additional necessity hypothesis. Here B divides both g and ℓ , since their unit digits vanish. Thus both S and T_+ in (33.3) are even, and so is r . Consequently $J = 2r + 1$ is one modulo four.

Write $A = a + 4$ and $D_P = A^2 - 1$, distinguishing the Pell coefficient from the encoding polynomial $D(T)$. The positive construction in Section 32 now applies with

$$\begin{aligned} m &= 2cJ, & f &= \chi_A(m), & i &= D_P \psi_A(m)/c^2, & R &= ic^2, \\ y_{\text{aux}} &= \psi_R(J), & u_{\text{aux}} &= \chi_R(J)/R, \\ o &= (u_{\text{aux}} - c)/f, & j &= (u_{\text{aux}} - J)/c. \end{aligned} \tag{33.24}$$

That proof establishes integrality and strict positivity using $J \equiv 1 \pmod{4}$, and gives exactly the retained auxiliary norm and congruence. All changed positive coding witnesses were supplied in (33.23). The difference $e - \ell$ is positive but is a computed register, not another supplied unknown. This proves necessity for all 34 positive unknowns.

33.9 Complete arithmetic certificate

Relative to the 92-operation schedule, replace the three instructions

$$t_2 = \lambda - e, \quad b_\Sigma = \ell + e, \quad S_3 = t_2(q - C^2)$$

by $t_2 = e - \ell$, $b_\Sigma = \ell + \sigma$, and $S_3 = t_2 C^2$, and delete the subtraction $q - C^2$. The supplied Ω and its free equality are also deleted. Exactly one arithmetic operation disappears, giving 49 multiplications and 42 additions.

The complete checker states all 22 source residuals afresh and compares them with every certificate equality. If $P_{\text{old}} = (\lambda - e)(q - C^2)$ and $P_{\text{new}} = (e - \ell)C^2$, the aligned predecessor residuals change only by

$$\sigma - e, \quad -q^4(n^2 - n)(P_{\text{new}} - P_{\text{old}}), \quad P_{\text{new}} - P_{\text{old}}, \tag{33.25}$$

in the bound, packed- r , and product equations respectively. The old positive-factor equation is omitted; the auxiliary congruence is unchanged. To state the inherited residual corrections precisely, let F_j denote source residual j in the checker's zero-based order, and put $u_* = 2r + 1 + jc$. The nonzero corrections are

$$\begin{aligned} \mathcal{E}_2 &= -\lambda F_3, \\ \mathcal{E}_{16} &= F_{15}(u_*^2 - y_{\text{aux}}^2), \\ \mathcal{E}_{17} &= F_3(\kappa + \rho(F_3 - 2(A - B))). \end{aligned}$$

Residuals F_3 and F_{15} are independently exact geometry and relaxed-norm equations. These corrections are therefore triangular and vanish without circular reasoning. The generated schedule gives all 91 primitive instructions and 22 free equality tests.

The general compiler and positive-domain proof received two independent reviews. A supplementary sparse regression checks exact symbolic target and padding identities in a sample with 35 true coordinates, 25 main rows, 128 tested starts, 418 indicator digits, and 976 complementary coefficients. Six canonical assignments and four wide signed-carry assignments pass. These finite checks support the general proof and do not assert exhaustive coverage of circuits or integer witnesses. The construction proves an upper bound; it does not claim that 91 is optimal.

34 Binary coefficient codes and a matching Pell shift: 90 operations

Changing the helper reverse rows makes every positive complementary coefficient lie outside the indicator. Both fixed codes then have binary digits. The geometric identity uses λ directly in place of 3λ , saving one multiplication. A simultaneous change of the main Pell base from $a + 4$ to $a + 2$ preserves all shared expressions.

Theorem 34.1. *Every recursively enumerable subset of the positive integers has an admissible fixed index (V, H, T_L) for a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 90 operations: 48 multiplications and 42 additions. The supplied inputs are (x, V, H, T_L) , with three fixed components besides the positive queried input x . Fixed numerals and equality tests are free.*

34.1 The binary helper compiler

Retain the physical representation of Section 33. The actual positive input x has weight zero and is not split; the homogeneous unit δ is one physical coordinate. Every other logical value is a sum of three distinct nonnegative physical coordinates, with disjoint groups for distinct logical values. The fixed forbidden-bit decomposition of a value z is $(z, 0, 0)$ if its H_0 bit is zero, and $(z - H_0, H_0/2, H_0/2)$ otherwise.

Use the same two reserved helper triples V_0, V_1 . Every ordinary circuit row uses V_0 as input and mentions neither x nor V_1 . Fresh square-difference copies handle repeated operands and repeated unit factors. Expanded physical square coefficients are ± 1 and cross coefficients are ± 2 , so their divided coefficients are ± 1 .

For each $i = 0, 1$, retain the seed base row $-x^2$, augmented by its own helper so that its tested value is $V_i^2 - x^2$. Replace the reverse row by

$$2x\delta - 2V_i\delta. \quad (34.1)$$

Its positive monomial has one non-input physical factor, while all its negative monomials have two. Its augmenting helper is the other V_{1-i} . The implication $V_i = x$ will be proved only after $\delta > 0$ has been recovered.

Include both signs of each ordinary zero equation, including

$$\begin{aligned} X^2 - V_0^2, \quad Y^2 - V_0^2, \quad Z^2 - V_0^2, \quad \delta'^2 - \delta^2, \\ 2V_0X - 2\delta\delta' - 2u\delta. \end{aligned} \quad (34.2)$$

The logical groups are distinct. Keeping the extra copies Y, Z is convenient and has no effect on the final operation count. Finish with two unpaired δ^2 rows. Only the last is padded, by a preceding row coefficient $2xX$.

For m true non-input physical coordinates, use

$$v_i = 8 \cdot 7^i \quad (0 \leq i < m), \quad M = 8 \cdot 7^{m-1}.$$

Sums of at most six such weights have unique multisets. If s counts the main rows, including the four helper rows and two unit rows, put

$$\begin{aligned} d_0 = 12M + 8, \quad t_j = (s + 2)d_0 + 8M + jd_0 \quad (0 \leq j < s), \\ t_* = t_{s-1} = (2s + 1)d_0 + 8M, \quad K = t_* + 3. \end{aligned} \quad (34.3)$$

Place each divided base coefficient a of monomial weight w at $t_j - w$. For a negative coefficient at $r = t_j - w$, add the six positive helper complements

$$\sum_{h \leq h', \, h, h' \in P} T^{r-v_h-v_{h'}}, \quad (34.4)$$

where P is the own helper group for a seed, the other group for a reverse row, and the V_1 group for an ordinary row. Let $\mathcal{R}$ contain every main target and every negative base exponent. The seed target

coincides with its negative exponent. Add one reset T^{r-4} for each distinct $r \in \mathcal{R}$, and add the final padding

$$D_{\text{pad}}(T) = \sum_{h \in P(X)} T^{t_* - 1 - v_h}. \quad (34.5)$$

Call the resulting polynomial $D(T)$.

The coefficient-isolation proof from Section 33 has one new degree case. Helper complements in non-seed rows have degree four, so cannot affect a main target after multiplication by $C(T)^2$. At a negative position, another base monomial could contribute only if $w_{\text{base}} = w_{\text{negative}} + w_C$. Ordinary base monomials have degree two; the new positive reverse monomial has degree one and cannot satisfy this equality with a degree-two negative monomial. At an extra negative-position target, an augmentation contributes only if

$$w_{\text{negative},1} + w_{\text{helper}} = w_{\text{negative},2} + w_C.$$

The negative factors and helper group are disjoint. Multiset uniqueness forces the same negative pair and helper pair, giving exactly $V_{\text{helper}}^2 - x^2$. A seed separately gives $V_i^2 - x^2$ at its coincident target. Reverse main targets remain (34.1).

The same uniqueness and row spacing prevent coefficient accumulation. Base and augmentation terms have residue zero modulo eight; resets have residue four and padding has residue seven. All nonzero D coefficients therefore remain ± 1 . Define

$$\ell_0(T) = \sum_{i=0}^{m-1} T^{v_i} + \sum_{r \in \mathcal{R}} (T^r + T^{r+1} + T^{r+2}), \quad e_0(T) = \ell_0(T) + D(T). \quad (34.6)$$

Every negative D coefficient is at an indicator start. Every positive coefficient is now outside the indicator. Ordinary positive base complements are distinct degree-two monomials; reverse positive complements have degree one; non-seed helper complements have degree four; and seed helper complements differ from their sole main start. Reset and padding residues are outside the three tested residues. Other row bands and the low true-coordinate region are disjoint. In particular, the previous positive x^2 term at the reverse main target has disappeared. Thus both ℓ_0 and e_0 have only digits zero and one.

Both code polynomials have zero constant coefficient and support below K . The leading D term is the last reset T^{t_*-4} with coefficient $+1$; all other terms in the final positive-only row have smaller degree. Consequently $D(B) > 0$ for every integer $B \geq 2$.

Permit dummy coordinate digits at the indicator windows. Their least weight is at least $t_0 - 2M$, while the least D exponent is at least $t_0 - 4M$. Their sum exceeds $t_* + 2$, so no dummy affects any tested coefficient or reset. The least D exponent also exceeds M , so every true-coordinate indicator position has raw coefficient and incoming carry zero.

34.2 Fixed constants and bounds before Pell decoding

Let c_* count the input, true coordinates, and permitted dummies, and let $D_1 = \sum_h |[T^h]D(T)|$. Choose a power of two $L > 3K + 2$, and then a power of two H_0 such that

$$H_0 > \max(2 \cdot 2^{2L+1}, 128 \max(1, D_1) c_*^2, 3L, 1024). \quad (34.7)$$

The fixed index is

$$H = H_0 - 1, \quad V = \ell_0(2) + 2^L e_0(2), \quad T_L = \psi_2(L). \quad (34.8)$$

It is chosen before the queried input.

The source retains the positive product bound and uses the new alphabet:

$$\begin{aligned} C &= x + g, & \sigma &= (e - \ell)C^2, & \ell + \sigma + \alpha &= q, \\ b &= x + \beta, & \theta &= H + b = B - 2, & B &= H_0 + 1 + b, & \lambda(B - 1) &= q^2 - 1. \end{aligned} \quad (34.9)$$

There is no supplied Ω . Positivity gives $C \geq 2$, $e - \ell \geq 1$, and therefore

$$\ell < q, \quad C^2 \leq \sigma < q, \quad e \leq \ell + \sigma < q, \quad g < C < q. \quad (34.10)$$

Set

$$\begin{aligned} S_2 &= \ell + eq = V + t\theta, & n &= q^8, \\ S &= g + q^2(S_2 + q^2\sigma), \\ T_+ &= q^2(1 + \theta\lambda) + \ell(\theta q^4 - b), \\ r &= S(n^2 - n) + T_+(n^2 - 1). \end{aligned} \quad (34.11)$$

Geometry gives $B \leq q^2$, $\theta\lambda > b$, and $1 + \theta\lambda = q^2 - \lambda < q^2$. Hence

$$T_+ > bq^2 - bq > 0, \quad T_+ < q^4(1 + \theta\ell) < Bq^5 \leq q^7, \quad 0 < S < q^7 < n.$$

Together with the fixed threshold, these imply

$$n \geq 64, \quad n \leq n^2 - 1 \leq r < 2n^3, \quad 2 \leq L, \quad 3L \leq B \leq n. \quad (34.12)$$

No exponent or coefficient decoding has been used.

34.3 The main Pell base is now $a + 2$

Distinguish the Pell discriminant from the coefficient polynomial:

$$\begin{aligned} U &= wn^2, & Y_P &= s_P n^2, & E &= UY_P, & Q &= UY_P^2, & P &= 2Q + 1, \\ a &= Y_P(U + 1), & A &= a + 2, & D_P &= A^2 - 1, & J &= 2r + 1. \end{aligned} \quad (34.13)$$

Retain the first norm, positive interval, and index congruence:

$$\begin{aligned} \tau(\tau + 1) &= (E^2 + U)(Y_P k)^2, \\ c &= Y_P k + \eta, & k &= \eta + \zeta, & k &= r + 1 + hE. \end{aligned} \quad (34.14)$$

The main norm, relaxed auxiliary norm, and half-parameter block are

$$\begin{aligned} d^2 &= 1 + D_P c^2, & R &= ic^2, & K_P &= R^2 = D_P(f^2 - 1), \\ u_* &= J + jc = c + of, & K_P(u_*^2 - y_{\text{aux}}^2) &= 1 - y_{\text{aux}}^2. \end{aligned} \quad (34.15)$$

Replace the first exponent congruence by

$$d = U + ac + \gamma(4a + 3). \quad (34.16)$$

The second norm, positive gap, fixed index, and exponent congruence are

$$\begin{aligned} c &= \kappa + \phi, & \mu^2 &= 1 + D_P \kappa^2, & \kappa &= T_L + \Delta a, \\ \mu &= q + \kappa(A - B) + \rho(D_P - (A - B)^2). \end{aligned} \quad (34.17)$$

Because $A = a + 2$ and $B = \theta + 2$, the shared difference is still $A - B = a - \theta$.

34.4 Exact indices before either exponential

The preliminary bounds give

$$U, Y_P \geq n^2 \geq 4096, \quad E \geq n^4 > r + 1, \quad a > n^4 > J.$$

Classification of the first norm yields $k = \psi_P(t)$ at a positive index. Its congruence modulo E gives $t = r + 1 + vE$ with $v \geq 0$. Thus $t \geq r + 1$. Since $P > A$ and $c > Y_P k > k$, comparison with the main norm $c = \psi_A(p)$, $d = \chi_A(p)$ gives

$$p \geq t + 1 \geq r + 2 \geq 66, \quad c > AD_P^2, \quad c > J.$$

These are the generic hypotheses of the relaxed auxiliary and half-parameter proofs above. They use only $A > 1$, not its former shift by four or evenness of r . Applying them to (34.15) recovers

$$p = J, \quad c = \psi_A(J), \quad d = \chi_A(J).$$

If $v \geq 1$, ordinary Pell growth and $E > r$ give

$$\frac{c}{k} \leq \frac{(2A)^{2r}}{(2P-1)^{r+E}} \leq \left(\frac{2A}{2P-1} \right)^{2r} < \frac{1}{2},$$

because the changed exact difference is

$$(2P-1) - 4A = 4Y_P(U(Y_P-1) - 1) - 7 > 0. \quad (34.18)$$

This contradicts $c/k > Y_P$. Consequently $k = \psi_P(r+1)$, still without either exponential identity or a decoded circuit equation.

34.5 Ratio growth, exponent recovery, and exact rounding

Put $\xi = (U+1)^{2r}/U^r$. Standard lower and upper Pell bounds give

$$\begin{aligned} \frac{c}{k} &\geq \xi \frac{(1 + 3/(2a))^{2r}}{(1 + 1/(2Q))^r} > \xi, \\ \frac{c}{k} &< \xi(1 + 2/a)^{2r}. \end{aligned} \quad (34.19)$$

The strict lower bound uses $(1 + 3/(2a))^2 > 1 + 3/a > 1 + 1/(2Q)$, since $6Q > a$. Also $4r/a < 8/n \leq 1/8 < 1/2$. The elementary estimate $(1+t)^m \leq (1+mt)^{-1} < 1+2mt$ therefore gives $c/k < \xi(1 + 8r/a)$.

The positive interval and the lower ratio bound imply $\xi < Y_P + 1$. Since $\xi > U^r$ and Y_P is an integer,

$$Y_P \geq U^r, \quad a = Y_P(U+1) > U^{r+1}. \quad (34.20)$$

Using $\xi < Y_P + 1 < 2Y_P$ in the upper estimate gives

$$0 < c/k - \xi < 16r/(U+1). \quad (34.21)$$

The right side is not yet asserted small.

The chi-congruence exponent criterion used above applies to (34.16) with base two, since $A-2 = a$ and $4A-5 = 4a+3$. Its growth assumptions hold before its use:

$$2^{3J} = 8 \cdot 64^r < U^{r+1} < a < A, \quad U^3 \leq U^r < a < A.$$

It follows that $U = 2^J$. Since $U = wn^2$ and $n = q^8$, both n and q are powers of two.

For the second index, the norm and gap in (34.17) give $\kappa = \psi_A(t_0)$, $\mu = \chi_A(t_0)$ with $0 < t_0 < J$. Moreover, $0 < L < J$ follows from $3L \leq B \leq n \leq r$. Reduction of the fixed index congruence modulo a , using $A \equiv 2 \pmod{a}$, gives $\psi_2(t_0) \equiv \psi_2(L) \pmod{a}$. Both quantities are strictly smaller than a : for $0 < t < J$,

$$0 < \psi_2(t) \leq 4^{t-1} < 2^{3J} < a.$$

Thus they are equal as integers, and strict growth implies $t_0 = L$. The second exponent criterion has the required bounds

$$B^{3L} \leq B^B \leq n^n \leq U^r < a < A, \quad q^3 \leq n^n \leq U^r < a < A.$$

Its congruence in (34.17) therefore gives $q = B^L$. As q is a power of two, so is B .

Now $U = 2^J = 2 \cdot 4^r > 32r$, so (34.21) is less than $1/2$. Split the binomial expansion as $\xi = F + \varepsilon$, where

$$F = \sum_{j=r}^{2r} \binom{2r}{j} U^{j-r}, \quad \varepsilon = \sum_{j=1}^r \frac{\binom{2r}{r-j}}{U^j}. \quad (34.22)$$

Exact symmetry gives the sharper bound

$$0 < \varepsilon \leq \frac{4^r - \binom{2r}{r}}{2U} < \frac{1}{4}. \quad (34.23)$$

Hence F is the integer part of ξ and $F \equiv \binom{2r}{r} \pmod{U}$. The interval and ratio bounds yield

$$F < \xi < c/k < Y_P + 1, \quad Y_P < c/k < \xi + \frac{1}{2} < F + \frac{3}{4}.$$

Integrality forces $Y_P = F$. Since n^2 divides Y_P and U , it divides $\binom{2r}{r}$. This establishes every hypothesis for the packed binary masks without relying on a decoded coefficient row.

34.6 Canonical codes, carry clearing, and the unit

From $b < B < q$ and $\ell < q$, one has $b\ell < q^2$. Thus the exact normalized mask is

$$T_+ - 1 = (q^2 - 1 - b\ell) + q^2\theta\lambda + q^4\theta\ell,$$

with block widths q^2, q^2, q^4 . The three independent masks follow from central-binomial divisibility. The middle mask is $(B - 2) \sum_{j < 2L} B^j$, so all digits of S_2 are binary. If $F_0(T)$ is its digit polynomial, the fixed congruence gives $F_0(2) \equiv V \pmod{B - 2}$. Both nonnegative quantities are below 2^{2L} , and (34.7) makes their difference smaller than $B - 2$. Binary uniqueness therefore gives

$$F_0(T) = \ell_0(T) + T^L e_0(T), \quad \ell = \ell_0(B), \quad e = e_0(B).$$

There is no ambiguity between the two codes because $\ell, e < q$.

The first mask has digit $H_0 = B - 1 - b$ at each indicator position and $B - 1$ elsewhere. It confines g to permitted digits and clears their fixed bit. Its unit digit is zero; $x < B$ then makes $C = x + g$ exactly the physical polynomial evaluated at B . The third mask requires binary digits of $\sigma = D(B)C(B)^2$ at every indicator.

Since $B \geq 2H_0$, all raw coefficients satisfy

$$|[T^h]D(T)C(T)^2| \leq D_1 C(1)^2 < D_1 c_*^2 B^2 < B^3/256.$$

The signed carry recurrence keeps incoming carries below $B^2/128$ in absolute value. Dummy exclusion makes the low raw residues 0, 4, 7 modulo eight. At a start r , empty positions $r - 6, r - 5$ reduce the carry entering the reset at $r - 4$ to -1 or zero. Its nonnegative raw polynomial contains $x^2 \geq 1$, so its outgoing carry is nonnegative and below B^2 . The empty positions $r - 3, r - 2$ reduce it to zero. Every $r - 1$ is empty except the last unit pad, and every $r + 1, r + 2$ is empty.

Every ordinary, helper, and extra target therefore tests its exact raw value modulo B^3 . Any negative nonzero value in the proved range has top digit greater than one and fails. Use these facts in the following order. The seeds give $V_i \geq x > 0$. Both signs of every ordinary row force it to vanish independently of the helper values. In particular (34.2) gives

$$X = Y = Z = V_0, \quad \delta' = \delta, \quad V_0^2 = \delta^2 + u\delta,$$

so $\delta > 0$. The reverse target $2\delta(x - V_i)$ cannot be negative, proving $V_i \leq x$ and hence $V_0 = V_1 = x$. All negative targets now vanish, and the guard gives $0 < \delta \leq x < B$.

The final preceding coefficient from (34.5) is exactly $2xX = 2x^2$. Its incoming carry is zero. The two unit tests are consequently

$$\delta^2, \quad \delta^2 + \lfloor 2x^2/B \rfloor$$

with two empty following positions. The first gives $\delta^2 = kB + s$, $k, s \in \{0, 1\}$. If $s = 1$, then $\delta^2 \leq B + 1 < B^2/16$, and the four roots of one modulo a power of two leave only $\delta = 1$ in $(0, B/4)$. Otherwise a nonunit solution has $\delta^2 = B$ and $x^2 \geq B$. Writing $h = \lfloor 2x^2/B \rfloor$ gives $2 \leq h < 2B$. Binary digits of $B + h < 3B < B^2$ would force $1 + \lfloor h/B \rfloor \leq 1$ and then $h \leq 1$, a contradiction. Thus $\delta = 1$, and the exact homogeneous circuit recovers the represented system at its actual input.

34.7 Positive necessity at the new Pell base

Given a represented solution, choose $V_0 = V_1 = x$, $\delta = \delta' = 1$, $X = Y = Z = x$, $u = x^2 - 1$, all circuit values and fresh copies, and their fixed forbidden-bit splits. Set all dummy digits to zero. Every ordinary, seed, reverse, and extra target is zero, and both unit targets are one.

Choose an arbitrarily large power of two B after these physical values have been fixed. Require $B > H_0 + 1 + x$, every raw coefficient in absolute value below $B/4$, and $2x^2 < B/4$. Set

$$\begin{aligned} b &= B - H_0 - 1, & \beta &= b - x, & g &= C(B) - x, \\ \ell &= \ell_0(B), & e &= e_0(B), & q &= B^L, & n &= q^8, \\ \sigma &= D(B)C(B)^2, & \alpha &= q - \ell - \sigma. \end{aligned} \tag{34.24}$$

The positive leading coefficient of D and the unit coordinate give $\sigma, g > 0$. The degree bound $\deg(DC^2) < 3K < L$ allows B large enough that $\alpha > 0$. The first-mask bits clear, the middle digits are binary, and every third window is $(0, 0, 0)$ or $(1, 0, 0)$; the last pad has zero carry. Geometry supplies positive λ , and the congruence quotient $(\ell + eq - V)/(B - 2)$ is positive integral by evaluation of the nonconstant nonnegative polynomial $\ell_0(T) + T^L e_0(T)$ at B and at two.

The masks give central-binomial divisibility for the resulting r , with bounds (34.12). Also B divides both g and ℓ , so (34.11) makes S, T_+, r even. Put $J = 2r + 1$ and construct the new Pell witnesses:

$$\begin{aligned} U &= 2^J, & w &= U/n^2, & Y_P &= \lfloor (U + 1)^{2r}/U^r \rfloor, & s_P &= Y_P/n^2, \\ a &= Y_P(U + 1), & A &= a + 2, & E &= UY_P, & Q &= UY_P^2, & P &= 2Q + 1, \\ c &= \psi_A(J), & d &= \chi_A(J), & k &= \psi_P(r + 1). \end{aligned} \tag{34.25}$$

The quotient w is positive integral because n^2 and U are powers of two and $n^2 \leq r^2 < 2^{2r+1}$. Binomial divisibility and (34.23) make s_P positive integral. The same ratio bounds at these chosen indices give

$Y_P < \xi < c/k < Y_P + 3/4$. Thus

$$\eta = c - kY_P > 0, \quad \zeta = k - \eta > 0.$$

Set $\tau = (\chi_P(r+1) - 1)/2$ and $h = (k - r - 1)/E$. The odd parameter P makes τ integral; congruence modulo $P - 1 = 2Q$ gives integrality of h , and strict Pell growth gives their positivity.

For the relaxed and half-parameter auxiliaries choose

$$m = 2cJ, \quad f = \chi_A(m), \quad i = D_P\psi_A(m)/c^2, \quad R = ic^2,$$

then take

$$y_{\text{aux}} = \psi_R(J), \quad u_* = \chi_R(J)/R, \quad o = (u_* - c)/f, \quad j = (u_* - J)/c.$$

The generic positive construction of Section 32 applies because even r gives $J \equiv 1 \pmod{4}$. It proves every required divisibility and strict positivity at the new value of A .

Finally choose $\kappa = \psi_A(L)$, $\mu = \chi_A(L)$, $\phi = c - \kappa$, and $\Delta = (\psi_A(L) - \psi_2(L))/a$. The inequalities $0 < L < J$, $A > 2$, and $L \geq 2$ give $\phi, \Delta > 0$; polynomial congruence modulo a makes Δ integral. Both exponent congruences have integer quotients. Their positivity follows explicitly from

$$d - ac = 2c - \psi_A(J - 1) > c > U,$$

and

$$\mu - (A - B)\kappa = B\kappa - \psi_A(L - 1) > (B - 1)\kappa > q.$$

Here $c, \kappa \geq A$, and the established growth gives $A > U^3, q^3$. Both moduli are positive. This supplies every positive unknown at the changed base; it does not retain old base-four witnesses without justification.

34.8 Exact operation count and verification

Geometry now reads

$$q^2 = (1 + \theta\lambda) + \lambda.$$

The former instruction 3λ is deleted. The shared exponent modulus and discriminant are now $4a + 3$ and $a^2 + (4a + 3)$ in place of $8a + 15$ and $a^2 + (8a + 15)$, with the same count; $A - B = a - \theta$ still uses one subtraction. Thus exactly one multiplication disappears: 48 multiplications and 42 additions.

The complete checker states all 22 source residuals afresh. If $E_0 = 4a + 12$ is the old discriminant minus the new one, their only changes, in zero-based source order, are

2	2λ
13	γE ₀
14	c ² E ₀
15	(f ² - 1)E ₀
16	-(f ² - 1)(u _* ² - y _{aux} ²)E ₀
17	ρE ₀
18	-κ ² E ₀ .

The three inherited residual corrections remain triangular, using independently exact geometry and relaxed-norm equations. Every primitive, equality, source difference, and supplied input is checked. The generated table lists all 90 instructions and 22 free equalities.

The full compiler proof and exact certificate passed independent review. The sparse regression checks a sample with 35 true coordinates, 24 main rows, 121 tested starts, 397 indicator digits, and 912 complementary coefficients. It verifies binary code digits, every extra target, the exact pad, dummy exclusion, six canonical assignments, four wide signed-carry assignments, and 1,252 local unit cases. A separate exact Pell regression checks the changed ratio, norms, congruences, and positive witnesses through indices $r = 64, 66$. These finite receipts supplement the general proofs and do not assert exhaustive coverage or optimality.

35 Narrow binary packing: 89 operations

The binary coefficient compiler of Section 34 permits a narrower packing. A change in its lowest mask digit removes one multiplication from the power chain. The proof must recover the mask's size after exponent decoding and then exclude a possible carry into the fixed-code field; neither property is assumed in advance.

Theorem 35.1. *Every recursively enumerable subset of the positive integers has the same admissible fixed index (V, H, T_L) as in Theorem 34.1 for a system with 34 positive unknowns and 22 equations admitting a straight-line certificate of 89 operations: 47 multiplications and 42 additions. The supplied inputs are (x, V, H, T_L) , with three fixed components besides the positive queried input x . Fixed numerals and equality tests are free.*

35.1 The changed source and preliminary bounds

Retain the compiler, fixed index, product equation, positive bounds, geometry, and Pell equations of Section 34. In particular,

$$\begin{aligned} C &= x + g, & \sigma &= (e - \ell)C^2, & \ell + \sigma + \alpha &= q, \\ b &= x + \beta, & \theta &= H + b = B - 2, & \lambda(B - 1) &= q^2 - 1, \\ S_2 &= \ell + eq = V + t\theta. \end{aligned} \tag{35.1}$$

All supplied unknowns are strictly positive. Replace only the packing definitions and their index equation by

$$\begin{aligned} n &= q^4, & S &= g + q\sigma + q^2 S_2, \\ T_+ &= q^2 \theta \lambda + \ell(\theta q - b), & r &= S(n^2 - n) + T_+(n^2 - 1). \end{aligned} \tag{35.2}$$

Thus the unchanged bound gives

$$0 < g, \sigma, \ell, e < q, \quad 0 < S < q^4 = n.$$

Before any exponent is decoded, geometry gives $B \leq q^2$ and $\theta\lambda > b$. Therefore

$$T_+ > bq^2 - b\ell > 0, \quad T_+ < q^4 + \theta\ell q < 2q^4 = 2n.$$

Here $\theta\lambda = q^2 - 1 - \lambda < q^2$ and $\theta\ell q < Bq^2 \leq q^4$. The fixed compiler threshold then gives

$$n \geq 64, \quad n \leq n^2 - 1 \leq r < 3n^3, \quad 2 \leq L, \quad 3L \leq B \leq n. \tag{35.3}$$

In particular, the narrower bound $T_+ < n$ has not yet been used.

35.2 Pell recovery with the wider preliminary range

Use the unchanged Pell definitions and equations (34.13)–(34.17) at this new n and r . Their proof extends to (35.3). Indeed,

$$U, Y_P \geq n^2, \quad E \geq n^4 > 3n^3 \geq r + 1, \quad a > n^4 > 6n^3 > 2r + 1.$$

The first Pell index is at least $r + 1$, the main index is at least $r + 2$, and the relaxed auxiliary and half-parameter arguments recover the exact main index $J = 2r + 1$. The same Pell growth comparison then recovers the first index $r + 1$.

For $\xi = (U + 1)^{2r}/U^r$, the lower ratio bound remains $c/k > \xi$. The only changed estimate in the upper bound is

$$\frac{4r}{a} < \frac{12}{n} \leq \frac{3}{16} < \frac{1}{2}.$$

Consequently the same elementary estimate gives

$$\frac{c}{k} < \xi \left(1 + \frac{8r}{a}\right), \quad Y_P \geq U^r, \quad a > U^{r+1}, \quad 0 < \frac{c}{k} - \xi < \frac{16r}{U + 1}.$$

The first exponent criterion proves $U = 2^{2r+1}$, independently of mask decoding. The positive gap $c > \kappa$ and fixed index congruence give the second Pell index L , since $0 < L < J$ and both values $\psi_2(L), \psi_2(t_0)$ are less than a . Its size hypotheses are unchanged:

$$B^{3L} \leq B^B \leq n^n \leq U^r < a, \quad q^3 \leq n^n \leq U^r < a.$$

Thus the second exponent criterion proves $q = B^L$. Both B and q are powers of two. Exact binomial symmetry and the ratio estimate give the same rounding conclusion as before,

$$n^2 \mid \binom{2r}{r}. \tag{35.4}$$

This proof applies to arbitrary positive source solutions. It has used neither canonical fixed codes nor nonintersection of packed fields.

35.3 Recover the packing range after exponent decoding

Now $q = B^L$ with $L \geq 2$, so

$$\lambda = \frac{q^2 - 1}{B - 1} \geq B + 1 > \theta.$$

Using $\ell < q$ in the exact identity

$$T_+ = q^4 - q^2 - \lambda q^2 + \ell(\theta q - b)$$

shows $T_+ < q^4 = n$. Together with $0 < S < n$, the ordinary packed binary lemma therefore applies to (35.4):

$$S \ \& \ (T_+ - 1) = 0. \tag{35.5}$$

There is no appeal to that lemma outside its proved range.

35.4 A spill into the fixed-code mask is impossible

Write

$$b\ell = k_0q + r_0, \quad 0 \leq r_0 < q, \quad \theta\ell - 1 - k_0 = s_0q + u_0, \quad 0 \leq u_0 < q.$$

Since $b < B \leq q$, we have $0 \leq k_0 \leq b - 1$. Also $\theta\ell - 1 - k_0 \geq \theta - b = H > 0$, and it is less than θq . Hence $0 \leq s_0 \leq B - 3$. The normalized mask is

$$T_+ - 1 = (q - 1 - r_0) + qu_0 + q^2(\theta\lambda + s_0). \quad (35.6)$$

The last coefficient is below q^2 , because $\theta\lambda = q^2 - 1 - \lambda$ and $s_0 < \lambda$. The corresponding fields of S are g, σ, S_2 , of widths q, q, q^2 . Their bounds justify separate extraction from (35.5).

The base- B digits of $\theta\lambda$ are all $B - 2$. If $s_0 = 1$, the unit mask digit becomes $B - 1$, while the others remain $B - 2$. Every digit of S_2 , and therefore of ℓ , is then Boolean, so

$$\ell \leq \frac{q-1}{B-1} < \frac{q}{B-1}, \quad \theta\ell < q.$$

This contradicts $s_0 = 1$. If $s_0 \geq 2$, the first two mask digits become $s_0 - 2, B - 1$ and all later digits remain $B - 2$. The unit digit of ℓ is at most $B - 1$, its next digit is zero, and its higher digits are Boolean. Since $L \geq 2$,

$$\ell \leq B - 1 + \sum_{i=2}^{L-1} B^i = \frac{q - 2B + 1}{B - 1} < \frac{q}{B - 1}.$$

Again $\theta\ell < q$, contradicting $s_0 \geq 2$. Thus $s_0 = 0$.

The unperturbed middle mask now proves that S_2 has a Boolean digit polynomial $F(T)$ of degree below $2L$. Reduction of $S_2 = V + t(B - 2)$ modulo $B - 2$ gives $F(2) \equiv V \pmod{B - 2}$. The fixed threshold (34.7) puts both integers below $B - 2$. Binary uniqueness therefore recovers

$$\ell = \ell_0(B), \quad e = e_0(B). \quad (35.7)$$

The unchanged support bound $K + 1 < L$ implies $b\ell < q$ and $\theta\ell < q$. Consequently $k_0 = 0$, and the other two masks are exactly

$$g \& (q - 1 - b\ell) = 0, \quad \sigma \& (\theta\ell - 1) = 0. \quad (35.8)$$

35.5 The altered low mask preserves every circuit test

The first mask in (35.8) is the same coordinate mask as before, with its unused high block removed. At an indicator position its digit is $H_0 = B - 1 - b$, and elsewhere it is $B - 1$. Since $g < q$, the removed high block makes no difference. It recovers the physical coordinate polynomial $C(B) = x + g$ exactly.

The second mask differs from $\theta\ell$ only at and below the least indicator position $v_0 = 8$. Below v_0 its digits are $B - 1$; at v_0 the digit is $B - 3$ instead of $B - 2$. But the unchanged compiler has $\min \text{supp } D > M \geq v_0$. Independently of the values of the coordinate digits,

$$\sigma = D(B)(x + g)^2 \equiv 0 \pmod{B^{v_0+1}}.$$

Every altered digit of σ is therefore zero. The new test is equivalent to the old product mask. All coefficient-isolation, carry-reset, helper-normalization and unit arguments of Section 34 apply unchanged. This proves the soundness direction for the same fixed index and the same raw input.

35.6 Positive converse and exact operation count

Given a represented input, retain the canonical coding witnesses of Section 34 at a sufficiently large power-of-two radix. They satisfy all unchanged bounds and give the three masks just proved, including the modified low product mask. Form the new S, T_+, n by (35.2) and define its actual new r . The packed lemma gives $n^2 \mid \binom{2r}{r}$ and $r \geq n^2 - 1$.

The canonical g and ℓ have zero radix-unit digit. Thus S and T_+ are even, and the new r is even. Apply the positive Pell construction of Section 34 afresh at this n, r : set $U = 2^{2r+1}$ and $Y_P = \lfloor (U+1)^{2r}/U^r \rfloor$, use the main index $2r+1$, the first index $r+1$, and the second index L . The same exact congruences, ratio gaps and even-index half-parameter construction make every quotient and every remaining supplied witness positive. No old packed index or Pell witness is reused.

Both packings still cost the same number of operations. The new power chain computes $q^2, q^4, n^2 = q^8$ in three multiplications, where the old one computed $q^2, q^4, q^8, n^2 = q^{16}$ in four. The baseline term q^2 formerly supplied by $q^2(1 + \theta\lambda)$ is absent, so the reordered mask incurs no extra addition. This gives 47 multiplications and 42 additions, proving Theorem 35.1.

The complete source checker compares all 22 independently written polynomial residuals with all 89 primitive instructions, including the same acyclic norm and geometric corrections. The full table is Appendix AB. The companion files `BINARY_PRODUCT_89_PROOF.md` and `BASE_TWO_PELL_89_PROOF.md` state the outer and Pell proofs separately. Finite regression checks corroborate these arguments; they do not replace either unbounded positive-domain direction.

36 Theorem 5 in a precise form

Let T be an effectively axiomatizable theory: its language is countable with a Gödel numbering under which the sets of axioms and of instances of the inference rules are recursive. Let $\text{Thm}_T \subseteq \mathbb{N}$ be the set of Gödel numbers of theorems of T .

Lemma 36.1. *Thm_T is recursively enumerable, and there is an admissible coding triple $\langle z, u, y \rangle$ with $\text{Thm}_T \cap \mathbb{Z}_{>0} = W_{\langle z, u, y \rangle}$.*

Proof. The first assertion is standard: enumerate all finite sequences of formulas, check recursively whether each is a derivation, and output the Gödel number of its last formula. For the second, every r.e. set is Diophantine [1, 6]; by [4, §5] it has a representation with degree $\delta = 4$ in $\nu = 58$ unknowns, and [4, (4.1)] turns such a representation into an admissible triple $\langle z, u, y \rangle$ with $x \in \text{Thm}_T \Leftrightarrow x \in W_{\langle z, u, y \rangle}$ for $x > 0$. $\square$

Theorem 36.2 (Theorem 5 of [3], certificate form). *Let T be an effectively axiomatizable theory and let $\langle z, u, y \rangle$ be the triple of Lemma 36.1. For every proposition P with Gödel number $x > 0$ the following are equivalent:*

- (a) *P has a proof in T ;*
- (b) *there are 32 positive integers satisfying the twenty equations of Σ with parameters x, z, u, y ;*
- (c) *there is a valid straight-line certificate of the form given in the Appendix A: 32 positive integers, the auxiliary integers, and 129 true statements of the form “integer + integer = integer” or “integer · integer = integer” (139 if the numerals are generated from 1), together with 20 equality tests;*
- (d) *for the packed admissible index (Z, V, H) constructed from $\langle z, u, y \rangle$ in Section 6, there is a valid certificate of Appendix B: 32 positive witnesses, auxiliary integers, 120 true addition or multiplication statements, and 20 equality tests.*

- (e) for the admissible index (Z, V, H) constructed from the quadratic representation of Thm_T in Section 8, there is a valid certificate of Appendix C: 33 positive witnesses, auxiliary integers, 114 true addition or multiplication statements (123 with the fixed numerals generated from 1), and 21 equality tests.
- (f) for the direct quadratic index of Section 9, there is a valid certificate of Appendix D: 32 positive witnesses, auxiliary integers, 112 true addition or multiplication statements (119 with the fixed numerals generated from 1), and 20 equality tests.
- (g) for the reversed quadratic index of Section 11, there is a valid certificate of Appendix E: 33 positive witnesses, auxiliary integers, 111 true addition or multiplication statements (118 with the fixed numerals generated from 1), and 21 equality tests.
- (h) for the homogeneous index of Section 12, there is a valid certificate of Appendix F: 33 positive witnesses, auxiliary integers, 110 true addition or multiplication statements (117 with the fixed numerals generated from 1), and 21 equality tests.
- (i) for the input-unit index of Section 13, there is a valid certificate of Appendix G: 33 positive witnesses, auxiliary integers, 109 true addition or multiplication statements (116 with the fixed numerals generated from 1), and 21 equality tests.
- (j) for the same fixed index, there is a valid certificate of Appendix H: 34 positive witnesses, auxiliary integers, 107 true addition or multiplication statements (114 with the fixed numerals generated from 1), and 22 equality tests.
- (k) for the modular index of Section 15, there is a valid certificate with 34 positive witnesses, auxiliary integers, 107 true addition or multiplication statements, and 22 equality tests.
- (l) for the same modular index and the shifted main Pell parameter of Section 16, there is a valid certificate of Appendix I: 34 positive witnesses, auxiliary integers, 106 true addition or multiplication statements, and 22 equality tests.
- (m) for the same modular index and the positive index-multiple condition of Section 17, there is a valid certificate of Appendix J: 34 positive witnesses, auxiliary integers, 105 true addition or multiplication statements, and 22 equality tests.
- (n) for the same modular index and the shared-product Pell system of Section 18, there is a valid certificate of Appendix K: 34 positive witnesses, auxiliary integers, 104 true addition or multiplication statements, and 22 equality tests.
- (o) for the new coefficient index of Section 19, there is a valid certificate of Appendix L: 34 positive witnesses, auxiliary integers, 103 true addition or multiplication statements, and 22 equality tests.
- (p) for that coefficient index and the common exponent witness of Section 20, there is a valid certificate of Appendix M: 34 positive witnesses, auxiliary integers, 102 true addition or multiplication statements, and 22 equality tests.
- (q) for the fixed-four circuit index (V, H, L) of Section 21, there is a valid certificate of Appendix N: 34 positive witnesses, auxiliary integers, 101 true addition or multiplication statements, and 22 equality tests.
- (r) for the fixed index $(V, H, \psi_4(L))$ and doubled-index Pell system of Section 22, there is a valid certificate of Appendix O: 34 positive witnesses, auxiliary integers, 100 true addition or multiplication statements, and 22 equality tests.
- (s) for the paired circuit index (V, H, L) of Section 23, there is a valid certificate of Appendix P: 34 positive witnesses, auxiliary integers, 100 true addition or multiplication statements, and 22 equality tests.
- (t) for the combined index $(V, H, \psi_4(L))$ of Section 24, there is a valid certificate of Appendix Q: 34 positive witnesses, auxiliary integers, 99 true addition or multiplication statements, and 22 equality tests.
- (u) for the unit-centered index $(V, H, \psi_4(L))$ of Section 25, there is a valid certificate of Appendix R: 34 positive witnesses, auxiliary integers, 98 true addition or multiplication statements, and 22 equality tests.

equality tests.

- (v) for the index of Section 24 with the relaxed auxiliary Pell norm of Section 26, there is a valid certificate of Appendix S: 34 positive witnesses, auxiliary integers, 98 true addition or multiplication statements, and 22 equality tests.
- (w) for the combined index $(V, H, \psi_4(L))$ of Section 27, there is a valid certificate of Appendix T: 34 positive witnesses, auxiliary integers, 97 true addition or multiplication statements, and 22 equality tests.
- (x) for the reset-padded index $(V, H, \psi_4(L))$ of Section 28, there is a valid certificate of Appendix U: 35 positive witnesses, auxiliary integers, 96 true addition or multiplication statements, and 23 equality tests. Fixed numerals are free.
- (y) for the affine-radix index $(V, H, \psi_4(L))$ of Section 29, there is a valid certificate of Appendix V: 34 positive witnesses, auxiliary integers, 95 true addition or multiplication statements, and 22 equality tests. Fixed numerals are free.
- (z) for the translated index $(V, H, \psi_4(L))$ of Section 30, there is a valid certificate of Appendix W: 34 positive witnesses, auxiliary integers, 94 true addition or multiplication statements, and 22 equality tests. Fixed numerals are free.
- (aa) for the same translated index and the same system, there is a valid certificate of Appendix X: 34 positive witnesses, auxiliary integers, 93 true addition or multiplication statements, and 22 equality tests. Fixed numerals are free.
- (ab) for the same translated index and the smaller auxiliary Pell parameter of Section 32, there is a valid certificate of Appendix Y: 35 positive witnesses, auxiliary integers, 92 true addition or multiplication statements, and 23 equality tests. Fixed numerals are free.
- (ac) for the product-bound coefficient index of Section 33 and the retained smaller auxiliary Pell parameter, there is a valid certificate of Appendix Z: 34 positive witnesses, auxiliary integers, 91 true addition or multiplication statements, and 22 equality tests. Fixed numerals are free.
- (ad) for the binary coefficient index $(V, H, \psi_2(L))$ of Section 34 and its matching Pell shift, there is a valid certificate of Appendix AA: 34 positive witnesses, auxiliary integers, 90 true addition or multiplication statements, and 22 equality tests. Fixed numerals are free.
- (ae) for the same binary coefficient index and the narrow packing of Section 35, there is a valid certificate of Appendix AB: 34 positive witnesses, auxiliary integers, 89 true addition or multiplication statements, and 22 equality tests. Fixed numerals are free.

Moreover, the seventeen polynomial equations of Theorem 3 that the certificate verifies contain 100 indicated additions, subtractions and multiplications, which is the number reported in [3].

Proof. (a) $\Leftrightarrow$ (b) is Lemma 36.1 and Theorem 4.3; (b) $\Leftrightarrow$ (c) is Theorem 5.1, and (a) $\Leftrightarrow$ (d) is Theorem 6.5. The quadratic-system construction of [4, §5] and Corollary 8.2 give (a) $\Leftrightarrow$ (e), and Theorem 10.2 gives (a) $\Leftrightarrow$ (f), and Theorem 11.1 gives (a) $\Leftrightarrow$ (g). Theorems 12.1 and 13.1 give the equivalences with (h) and (i), respectively. Theorem 14.3 gives the equivalence with (j). Theorems 15.1 and 16.1 give the equivalences with (k) and (l), and Theorem 17.1 gives the equivalence with (m), and Theorem 18.2 gives the equivalence with (n). Theorems 19.2 and 20.1 give the equivalences with (o) and (p). Theorem 21.1 gives the equivalence with (q). Theorems 22.3, 23.1 and 24.1 give the equivalences with (r), (s) and (t), respectively. Theorems 25.1, 26.2 and 27.1 give the equivalences with (u), (v) and (w), respectively. Theorem 28.1 gives the equivalence with (x). Theorem 29.1 gives the equivalence with (y). Theorem 30.1 gives the equivalence with (z). Theorem 31.1 gives the equivalence with (aa). Theorem 32.1 gives the equivalence with (ab). Theorem 33.1 gives the equivalence with (ac). Theorem 34.1 gives the equivalence with (ad). Theorem 35.1 gives the equivalence with (ae). The last sentence is Proposition 3.1. $\square$

Remark 36.3 (what the “other proof” is). Each certificate in (c)–(ae) is the “proof consisting of additions

and multiplications of integers” of Theorem 5. Its correctness is checked by performing the listed operations and comparing results; no knowledge of T , of logic, or of the meaning of the integers is needed, because the soundness directions are guaranteed by Theorems 4.3 and 6.5, Corollary 8.2, and Theorems 10.2, 11.1, 12.1, 13.1, 14.3, 15.1, 16.1, 17.1, 18.2, 19.2, 20.1, 21.1, 22.3, 23.1, 24.1, 25.1, 26.2, 27.1, 28.1, 29.1, 30.1, 31.1, 32.1, 33.1, 34.1, and 35.1 for the respective fixed indices. Three things are not claimed. The integers are astronomically large: the original construction requires $b > y \geq 2^{559}$, while the packed construction has the huge fixed scale H , with still larger q, n, r, a, κ, μ . The count bounds neither their lengths nor the number of bit operations. Nothing is said about finding the certificate, which is as hard as finding a proof in T . And the certificate does not preserve the length or structure of the proof in T ; a derivation is recoverable from it only through the encoding of Theorem 4.3, as the editorial footnote to Theorem 5 already says.

37 How the count arose: typographical complexity

That an author of Jones’s care should leave the cost of q^{16} out of a count of “additions and multiplications” is at first bewildering. The papers themselves, read in order, make it intelligible. What Jones counted was the typographical size of the system; the proof-theoretic reading was attached to that number afterwards, and the convention travelled from one paper to the next without being re-examined.

The measure was defined as a count of what is written. The 1978 paper introduces o as follows:

A natural measure of the size or complexity of a system of Diophantine equations is the number of operations (additions and multiplications), necessary to write the system. If we regard subtraction as merely a special form of addition, then the system (1.3) will be seen to involve 243 operations. (Combining the 36 equations into one equation would increase this number to 350.) [2, p. 338]

and, a few lines later,

As a measure of complexity, the number of operations in a Diophantine equation is very closely associated with the number of symbols necessary to write down the equation. [2, p. 338]

Operations “necessary to write the system” and “the number of symbols necessary to write down the equation” are counts of signs on the page. An exponent is a superscript, not an operation sign, and under this measure a monomial q^{16} is no longer than q by more than one small numeral. That is precisely Definition 2.1, and it is why the rule reproduces 87, 241 and 100.

The convention was inherited from 1976. The primality paper states its bound in the same words and adds the only comment on its derivation that any of the papers contains:

If p is a composite number, then there is a proof that p is composite consisting of a single multiplication. Julia Robinson remarks in [14] that prior to the solution of Hilbert’s tenth problem in 1970, it was not known that there was a similar proof establishing primality in a bounded number of steps. Yet it follows from [4] [8] that this is so. And our construction permits us to compute a bound on this number.

THEOREM 5. If p is a prime number, then there is a proof that p is prime consisting of only 87 additions and multiplications.

The number is easily calculated from the equations of Theorem 2.12. [5, p. 451]

“Easily calculated” is a count, not a construction. The system of Theorem 2.12 contains almost nothing but squares and a few cubes, so on that system the sign count and an honest straight-line count happen to agree (Proposition 3.2 and [5, §1], editorial verification note). The habit that was harmless there was carried over to Theorem 3 of 1980, where the powers q^{16} , q^8 , q^7 , q^5 , q^3 , b^5 , n^6 , c^4 and a fourth

power of a trinomial make the sign count diverge from the initial certificate count. Section 6 reduces the latter by changing both the calculations and the encoding.

Powers were treated as notation, as numerals were. The 1982 paper restates the measure and immediately draws the proof-theoretic conclusion:

Perhaps therefore a better measure of size and complexity of diophantine equations is to consider the total number of operations, o necessary to evaluate the polynomial. By an arithmetical operation we shall mean an addition or multiplication. We will consider subtraction as merely a signed form of addition. From the equations of Theorem 3, modified to do away with $q = b^{5^{60}}$, it can be seen that $o = 100$. Thus 100 arithmetical operations are universally sufficient to determine membership of a number in any r.e. set. [4, p. 553]

“It can be seen” (1980 has “it can be shown”) again describes reading the number off the page. In the same sentence the exponent 5^{60} and the coefficients 2, 4, 5 are numerals that cost nothing, and a power x^k with a fixed exponent is a fixed-degree monomial, a piece of notation of the same kind. For measuring the size of a polynomial this is a defensible choice; for counting the steps of a calculator that only adds and multiplies it is not; and the paper does not separate the two readings.

The inconsistency was not a considered position. The parenthesis “would increase this number to 350” in the first quotation charges, for each of the 36 equations, one subtraction for transposing and one squaring, plus 35 additions to sum: $243 + 36 + 36 + 35 = 350$. So the squaring of a transposed equation cost one operation while the z^{18} inside (1.3.06) cost nothing. That is the signature of counting the operation signs that become visible when the equations are rewritten, not of applying a definition in which exponentiation is free.

The proof-theoretic gloss was added on top. Both 1978 and 1980 pass from the size measure to a statement about proofs in one step:

Hence from Theorem 3 we obtain an absolute epistemological upper bound on the complexity of mathematical proofs. [2, p. 338]

These solutions faithfully reflect the logical complexity of the original proof, for the entire deduction is effectively recoverable from the solution. Hence we have THEOREM 5. [3, p. 862]

The step from “the theorems form an r.e. set, so provability is solvability of a fixed system” to “a proof consisting of o additions and multiplications” is sound only for the calculator count of Definition 2.2; the number that was actually computed is the sign count. The corrected edition already softened “faithfully reflect” to “effectively recoverable, without preservation of length or complexity”; the present article supplies the number that the proof-theoretic statement needs: an initial 129-operation certificate, improved to 89 by Sections 6–35, alongside the number that was printed.

Nobody checked. The count was announced without a derivation, repeated verbatim in the full paper, and reported since as a curiosity; earlier revisions of this edition also left it as “the author’s reported count”. A number that is never derived in print is never corrected. In sum: Jones counted the typographical complexity of his system, said so in 1978, and did not notice that the corollary about proofs requires the other count.

38 Limits of this verification

Proposition 3.1 and Theorem 5.1 are exact computations on the transcribed systems and are reproduced by the companion program. Theorem 4.3 relies on Lemmas 2.22, 2.25 and 2.26 and the proof of

Theorem 3 of [4], as corrected in this edition; the Lean formalization of those results is in progress and is reported in `Lean/STATUS.md`. The identification of Jones’s convention in Remark 3.3 is an inference from the published numbers, not a statement found in the papers: the two exact matches (87 and 100) and the near match (241 against 243) are the evidence, and the unexplained two signs in (1.3) and the treatment of the squaring in the “350” remark are the residue. The improved results additionally use the recoding, Pell, and interval proofs in Sections 6–35; symbolic residual verification does not establish their positive-domain or decoding arguments. Those arguments are supplied as mathematical proofs and have not been formalized in Lean. All operation bounds in this article are upper bounds; no optimality is claimed.

References

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A The original 129-instruction certificate

Names: the parameters x, z, u, y ; the unknowns of Theorem 3 with $al = \alpha$, $ga = \gamma$, $eta = \eta$, $th = \theta$, $la = \lambda$, $tau = \tau$, $phi = \varphi$; the new unknowns $ka = \kappa$, $mu = \mu$, $rho = \rho$, $Delta = \Delta$. Every other name is an auxiliary integer supplied with the certificate. The column “Eq.” gives the equation of Σ served by the instruction; “pow” marks the shared powers of q and b . The last column is the instruction in addition/multiplication form. The twenty equality tests that follow the instructions are

$$\begin{aligned}
L1 &= R1, \quad L2 = R2, \quad L3 = b5, \quad l = R4, \quad e = R5, \quad n = q16, \quad r = R7, \\
p &= R8, \quad L9 = R9, \quad L10 = k2, \quad k = R11, \quad a = R12, \quad c = R13, \quad d = R14, \\
L15 &= R15, \quad L16 = R16, \quad L17 = R17, \quad mu = R18, \quad L19 = R19, \quad ka = R20.
\end{aligned}$$

No.	Eq.	Assignment	Addition/multiplication check
1	pow	$q^2 = q \cdot q$	$q \cdot q = q^2$
2	pow	$q^3 = q^2 \cdot q$	$q^2 \cdot q = q^3$
3	pow	$q^4 = q^2 \cdot q^2$	$q^2 \cdot q^2 = q^4$
4	pow	$q^8 = q^4 \cdot q^4$	$q^4 \cdot q^4 = q^8$
5	pow	$q16 = q8 \cdot q8$	$q8 \cdot q8 = q16$
6	pow	$b^2 = b \cdot b$	$b \cdot b = b^2$

No.	Eq.	Assignment	Addition/multiplication check
7	pow	$b4 = b2 \cdot b2$	$b2 \cdot b2 = b4$
8	pow	$b5 = b4 \cdot b$	$b4 \cdot b = b5$
9	E1	$g2 = g \cdot g$	$g \cdot g = g2$
10	E1	$el = e \cdot l$	$e \cdot l = el$
11	E1	$elg2 = el \cdot g2$	$el \cdot g2 = elg2$
12	E1	$L1 = elg2 + al$	$elg2 + al = L1$
13	E1	$xy = x \cdot y$	$x \cdot y = xy$
14	E1	$bxy = b - xy$	$bxy + xy = b$
15	E1	$R1 = bxy \cdot q2$	$bxy \cdot q2 = R1$
16	E2	$L2 = la + q4$	$la + q4 = L2$
17	E2	$Lam = la \cdot b5$	$la \cdot b5 = Lam$
18	E2	$R2 = Lam + 1$	$Lam + 1 = R2$
19	E3	$z2 = 2 \cdot z$	$2 \cdot z = z2$
20	E3	$L3 = th + z2$	$th + z2 = L3$
21	E4	$tth = t \cdot th$	$t \cdot th = tth$
22	E4	$R4 = u + tth$	$u + tth = R4$
23	E5	$mth = m \cdot th$	$m \cdot th = mth$
24	E5	$R5 = y + mth$	$y + mth = R5$
25	E7	$lq2 = l \cdot q2$	$l \cdot q2 = lq2$
26	E7	$S2 = e + lq2$	$e + lq2 = S2$
27	E7	$zl = z \cdot la$	$z \cdot la = zl$
28	E7	$ezl = e - zl$	$ezl + zl = e$
29	E7	$t2 = 2 \cdot ezl$	$2 \cdot ezl = t2$
30	E7	$xb5 = x \cdot b5$	$x \cdot b5 = xb5$
31	E7	$c0 = xb5 + 1$	$xb5 + 1 = c0$
32	E7	$C = c0 + g$	$c0 + g = C$
33	E7	$C2 = C \cdot C$	$C \cdot C = C2$
34	E7	$C4 = C2 \cdot C2$	$C2 \cdot C2 = C4$
35	E7	$P1 = t2 \cdot C4$	$t2 \cdot C4 = P1$
36	E7	$q4p1 = q4 + 1$	$q4 + 1 = q4p1$
37	E7	$P2 = Lam \cdot q4p1$	$Lam \cdot q4p1 = P2$
38	E7	$S3 = P1 + P2$	$P1 + P2 = S3$
39	E7	$S3q4 = S3 \cdot q4$	$S3 \cdot q4 = S3q4$
40	E7	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
41	E7	$Sq3 = Sin \cdot q3$	$Sin \cdot q3 = Sq3$
42	E7	$S = g + Sq3$	$g + Sq3 = S$
43	E7	$n2 = n \cdot n$	$n \cdot n = n2$
44	E7	$n2n = n2 - n$	$n2n + n = n2$
45	E7	$SA = S \cdot n2n$	$S \cdot n2n = SA$
46	E7	$bl = b \cdot l$	$b \cdot l = bl$
47	E7	$q3bl = q3 - bl$	$q3bl + bl = q3$
48	E7	$T1 = q3bl + l$	$q3bl + l = T1$
49	E7	$thl = th \cdot la$	$th \cdot la = thl$
50	E7	$thlq3 = thl \cdot q3$	$thl \cdot q3 = thlq3$
51	E7	$T2 = T1 + thlq3$	$T1 + thlq3 = T2$
52	E7	$b5m2 = b5 - 2$	$b5m2 + 2 = b5$
53	E7	$b5m2q8 = b5m2 \cdot q8$	$b5m2 \cdot q8 = b5m2q8$
54	E7	$T = T2 + b5m2q8$	$T2 + b5m2q8 = T$
55	E7	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
56	E7	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
57	E7	$R7 = SA + TB$	$SA + TB = R7$
58	E12	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
59	E12	$wn2p1 = wn2 + 1$	$wn2 + 1 = wn2p1$

No.	Eq.	Assignment	Addition/multiplication check
60	E12	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
61	E12	$rsn2 = r \cdot sn2$	$r \cdot sn2 = rsn2$
62	E12	$R12 = wn2p1 \cdot rsn2$	$wn2p1 \cdot rsn2 = R12$
63	E8	$rsn2sq = rsn2 \cdot rsn2$	$rsn2 \cdot rsn2 = rsn2sq$
64	E8	$wsq = wn2 \cdot rsn2sq$	$wn2 \cdot rsn2sq = wsq$
65	E8	$R8 = 2 \cdot wsq$	$2 \cdot wsq = R8$
66	E9	$p2 = p \cdot p$	$p \cdot p = p2$
67	E9	$p2m1 = p2 - 1$	$p2m1 + 1 = p2$
68	E9	$k2 = k \cdot k$	$k \cdot k = k2$
69	E9	$P9 = p2m1 \cdot k2$	$p2m1 \cdot k2 = P9$
70	E9	$L9 = P9 + 1$	$P9 + 1 = L9$
71	E9	$R9 = tau \cdot tau$	$tau \cdot tau = R9$
72	E10	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
73	E10	$cm = c - ksn2$	$cm + ksn2 = c$
74	E10	$cm2 = cm \cdot cm$	$cm \cdot cm = cm2$
75	E10	$f4 = 4 \cdot cm2$	$4 \cdot cm2 = f4$
76	E10	$L10 = f4 + eta$	$f4 + eta = L10$
77	E11	$r1 = r + 1$	$r + 1 = r1$
78	E11	$pm1 = p - 1$	$pm1 + 1 = p$
79	E11	$hpm1 = h \cdot pm1$	$h \cdot pm1 = hpm1$
80	E11	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
81	E13	$tr1 = r1 + r$	$r1 + r = tr1$
82	E13	$tk = tr1 + ka$	$tr1 + ka = tk$
83	E13	$R13 = tk + phi$	$tk + phi = R13$
84	E14	$bw = b \cdot w$	$b \cdot w = bw$
85	E14	$am2 = a - 2$	$am2 + 2 = a$
86	E14	$cam2 = c \cdot am2$	$c \cdot am2 = cam2$
87	E14	$D1 = bw + cam2$	$bw + cam2 = D1$
88	E14	$a4 = 4 \cdot a$	$4 \cdot a = a4$
89	E14	$a4m5 = a4 - 5$	$a4m5 + 5 = a4$
90	E14	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
91	E14	$R14 = D1 + gam$	$D1 + gam = R14$
92	E15	$a2 = a \cdot a$	$a \cdot a = a2$
93	E15	$A = a2 - 1$	$A + 1 = a2$
94	E15	$c2 = c \cdot c$	$c \cdot c = c2$
95	E15	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
96	E15	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
97	E15	$L15 = d \cdot d$	$d \cdot d = L15$
98	E16	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
99	E16	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
100	E16	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
101	E16	$R16 = AE + 1$	$AE + 1 = R16$
102	E16	$L16 = f \cdot f$	$f \cdot f = L16$
103	E17	$of = o \cdot f$	$o \cdot f = of$
104	E17	$dof = d + of$	$d + of = dof$
105	E17	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
106	E17	$d2ma = L15 - a$	$d2ma + a = L15$
107	E17	$f2d = L16 \cdot d2ma$	$L16 \cdot d2ma = f2d$
108	E17	$G = a + f2d$	$a + f2d = G$
109	E17	$G2 = G \cdot G$	$G \cdot G = G2$
110	E17	$G2m1 = G2 - 1$	$G2m1 + 1 = G2$
111	E17	$jc = j \cdot c$	$j \cdot c = jc$
112	E17	$H = tr1 + jc$	$tr1 + jc = H$

No.	Eq.	Assignment	Addition/multiplication check
113	E17	$H2 = H \cdot H$	$H \cdot H = H2$
114	E17	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
115	E17	$R17 = P17 + 1$	$P17 + 1 = R17$
116	E18	$amb = a - b$	$amb + b = a$
117	E18	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
118	E18	$qk = q + kamb$	$q + kamb = qk$
119	E18	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
120	E18	$Mod = A - amb2$	$Mod + amb2 = A$
121	E18	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
122	E18	$R18 = qk + rM$	$qk + rM = R18$
123	E19	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
124	E19	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
125	E19	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
126	E19	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
127	E20	$am1 = a - 1$	$am1 + 1 = a$
128	E20	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
129	E20	$R20 = Dam1 + 5^{60}$	$Dam1 + 5^{60} = R20$

B The 120-instruction certificate

The four inputs are x, Z, V, H , with (Z, V, H) a fixed admissible packed index. The positive witnesses are those of Appendix A, with m removed and β added. Here $beta = \beta$; the other Greek names use the same transliterations. Every additional name denotes a supplied integer whose value is constrained by the listed checks. In particular, $B = Hb^4$ and $C = 1 + xB + g$ are computed, while c is the distinct Pell witness. The literal 5^{59} is denoted explicitly. All assignments and all 20 final equality tests are generated from `round4_1980_packed_certificate.py`; reversing each subtraction gives the addition in the last column. The E7 test is equivalent to its source equation after the E2 and E3 tests, as proved in Section 6.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q3 = q2 \cdot q$	$q2 \cdot q = q3$
3	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
4	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
5	$q16 = q8 \cdot q8$	$q8 \cdot q8 = q16$
6	$b2 = b \cdot b$	$b \cdot b = b2$
7	$b4 = b2 \cdot b2$	$b2 \cdot b2 = b4$
8	$B = H \cdot b4$	$H \cdot b4 = B$
9	$el = e \cdot l$	$e \cdot l = el$
10	$bx = x + beta$	$x + beta = bx$
11	$L2 = la + q4$	$la + q4 = L2$
12	$Lam = la \cdot B$	$la \cdot B = Lam$
13	$R2 = Lam + 1$	$Lam + 1 = R2$
14	$L3 = th + Z$	$th + Z = L3$
15	$lq2 = l \cdot q2$	$l \cdot q2 = lq2$
16	$S2 = e + lq2$	$e + lq2 = S2$
17	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
18	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
19	$zl = Z \cdot la$	$Z \cdot la = zl$
20	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$

No.	Assignment	Addition/multiplication check
21	$t2 = twice_e - zl$	$t2 + zl = twice_e$
22	$xb5 = x \cdot B$	$x \cdot B = xb5$
23	$c0 = xb5 + 1$	$xb5 + 1 = c0$
24	$C = c0 + g$	$c0 + g = C$
25	$C2 = C \cdot C$	$C \cdot C = C2$
26	$elC2 = el \cdot C2$	$el \cdot C2 = elC2$
27	$L1 = elC2 + al$	$elC2 + al = L1$
28	$C4 = C2 \cdot C2$	$C2 \cdot C2 = C4$
29	$P1 = t2 \cdot C4$	$t2 \cdot C4 = P1$
30	$q4p1 = q4 + 1$	$q4 + 1 = q4p1$
31	$P2 = Lam \cdot q4p1$	$Lam \cdot q4p1 = P2$
32	$S3 = P1 + P2$	$P1 + P2 = S3$
33	$S3q4 = S3 \cdot q4$	$S3 \cdot q4 = S3q4$
34	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
35	$Sq3 = Sin \cdot q3$	$Sin \cdot q3 = Sq3$
36	$S = g + Sq3$	$g + Sq3 = S$
37	$n2 = n \cdot n$	$n \cdot n = n2$
38	$n2n = n2 - n$	$n2n + n = n2$
39	$SA = S \cdot n2n$	$S \cdot n2n = SA$
40	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
41	$Tq3 = Tcoef \cdot q3$	$Tcoef \cdot q3 = Tq3$
42	$bm1 = b - 1$	$bm1 + 1 = b$
43	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
44	$T2 = Tq3 - lbm1$	$T2 + lbm1 = Tq3$
45	$b5m2 = B - 2$	$b5m2 + 2 = B$
46	$b5m2q8 = b5m2 \cdot q8$	$b5m2 \cdot q8 = b5m2q8$
47	$T = T2 + b5m2q8$	$T2 + b5m2q8 = T$
48	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
49	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
50	$R7 = SA + TB$	$SA + TB = R7$
51	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
52	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
53	$rsn2 = r \cdot sn2$	$r \cdot sn2 = rsn2$
54	$UM = wn2 \cdot rsn2$	$wn2 \cdot rsn2 = UM$
55	$R12 = UM + rsn2$	$UM + rsn2 = R12$
56	$wsq = UM \cdot rsn2$	$UM \cdot rsn2 = wsq$
57	$R8 = 2 \cdot wsq$	$2 \cdot wsq = R8$
58	$p2 = p \cdot p$	$p \cdot p = p2$
59	$p2m1 = p2 - 1$	$p2m1 + 1 = p2$
60	$k2 = k \cdot k$	$k \cdot k = k2$
61	$P9 = p2m1 \cdot k2$	$p2m1 \cdot k2 = P9$
62	$L9 = P9 + 1$	$P9 + 1 = L9$
63	$R9 = tau \cdot tau$	$tau \cdot tau = R9$
64	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
65	$cm = c - ksn2$	$cm + ksn2 = c$
66	$cm2 = cm \cdot cm$	$cm \cdot cm = cm2$
67	$f4 = 4 \cdot cm2$	$4 \cdot cm2 = f4$
68	$L10 = f4 + eta$	$f4 + eta = L10$
69	$r1 = r + 1$	$r + 1 = r1$
70	$pm1 = p - 1$	$pm1 + 1 = p$
71	$hpm1 = h \cdot pm1$	$h \cdot pm1 = hpm1$
72	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
73	$tr1 = r1 + r$	$r1 + r = tr1$

No.	Assignment	Addition/multiplication check
74	$R13 = ka + phi$	$ka + phi = R13$
75	$bw = b \cdot w$	$b \cdot w = bw$
76	$am2 = a - 2$	$am2 + 2 = a$
77	$cam2 = c \cdot am2$	$c \cdot am2 = cam2$
78	$D1 = bw + cam2$	$bw + cam2 = D1$
79	$a4 = 4 \cdot a$	$4 \cdot a = a4$
80	$a4m5 = a4 - 5$	$a4m5 + 5 = a4$
81	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
82	$R14 = D1 + gam$	$D1 + gam = R14$
83	$a2 = a \cdot a$	$a \cdot a = a2$
84	$A = a2 - 1$	$A + 1 = a2$
85	$c2 = c \cdot c$	$c \cdot c = c2$
86	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
87	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
88	$L15 = d \cdot d$	$d \cdot d = L15$
89	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
90	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
91	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
92	$R16 = AE + 1$	$AE + 1 = R16$
93	$L16 = f \cdot f$	$f \cdot f = L16$
94	$of = o \cdot f$	$o \cdot f = of$
95	$dof = d + of$	$d + of = dof$
96	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
97	$d2ma = L15 - a$	$d2ma + a = L15$
98	$f2d = L16 \cdot d2ma$	$L16 \cdot d2ma = f2d$
99	$G = a + f2d$	$a + f2d = G$
100	$G2 = G \cdot G$	$G \cdot G = G2$
101	$G2m1 = G2 - 1$	$G2m1 + 1 = G2$
102	$jc = j \cdot c$	$j \cdot c = jc$
103	$H17 = tr1 + jc$	$tr1 + jc = H17$
104	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
105	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
106	$R17 = P17 + 1$	$P17 + 1 = R17$
107	$amb = a - B$	$amb + B = a$
108	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
109	$qk = q + kamb$	$q + kamb = qk$
110	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
111	$Mod = A - amb2$	$Mod + amb2 = A$
112	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
113	$R18 = qk + rM$	$qk + rM = R18$
114	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
115	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
116	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
117	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
118	$am1 = a - 1$	$am1 + 1 = a$
119	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
120	$R20 = Dam1 + 5^{59}$	$Dam1 + 5^{59} = R20$

The twenty free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= \text{pack_rhs}, & n &= q16, \\
r &= R7, & p &= R8, & L9 &= R9, \\
L10 &= k2, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20
\end{aligned}$$

C The 114-instruction certificate

The four inputs are x, Z, V, H , with (Z, V, H) the fixed admissible index of Section 8. The 33 positive witnesses are

$$\begin{aligned}
&a, b, c, d, e, f, g, h, i, j, k, \ell, p, q, r, s, t, w, \\
&\alpha, \beta, \gamma, \eta, \theta, \lambda, \tau, \varphi, \kappa, \mu, \rho, \Delta, n, o, \zeta.
\end{aligned}$$

The names $l, al, beta, ga, eta, th, la, tau, phi, ka, mu, rho, Delta, zeta$ denote the corresponding Greek letters (with $l = \ell$). All other names denote auxiliary integers constrained by the listed arithmetic checks. In particular, the signed value $of - d$ is allowed; positivity is required only of the declared parameters and witnesses. The fixed exponent literal is $L = 5^{64}$.

The table and all 21 equality tests are generated from `round5_1980_certificate.py`. Its exact polynomial checks include the two triangular residual identities for (E7) and (E17). The table contains 63 multiplication checks and 51 addition checks. The optional nine-step numeral construction in Section 8 raises this to 123 while leaving the index parameters supplied. The proofs of Corollary 8.2 establish the positive-domain and decoding equivalences needed to interpret these arithmetic checks.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$b4 = b2 \cdot b2$	$b2 \cdot b2 = b4$
6	$B = H \cdot b4$	$H \cdot b4 = B$
7	$bx = x + beta$	$x + beta = bx$
8	$L2 = la + q2$	$la + q2 = L2$
9	$Lam = la \cdot B$	$la \cdot B = Lam$
10	$R2 = Lam + 1$	$Lam + 1 = R2$
11	$L3 = th + Z$	$th + Z = L3$
12	$lq2 = l \cdot q$	$l \cdot q = lq2$
13	$S2 = e + lq2$	$e + lq2 = S2$
14	$\text{pack_tth} = t \cdot th$	$t \cdot th = \text{pack_tth}$
15	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
16	$zl = Z \cdot la$	$Z \cdot la = zl$
17	$\text{twice_e} = 2 \cdot e$	$2 \cdot e = \text{twice_e}$
18	$t2 = \text{twice_e} - zl$	$t2 + zl = \text{twice_e}$
19	$xb5 = x \cdot B$	$x \cdot B = xb5$
20	$c0 = xb5 + 1$	$xb5 + 1 = c0$
21	$C = c0 + g$	$c0 + g = C$
22	$C2 = C \cdot C$	$C \cdot C = C2$
23	$C4 = C2 \cdot C2$	$C2 \cdot C2 = C4$
24	$lC4 = l \cdot C4$	$l \cdot C4 = lC4$

No.	Assignment	Addition/multiplication check
25	$bound_sum = e + lC4$	$e + lC4 = bound_sum$
26	$L1 = bound_sum + al$	$bound_sum + al = L1$
27	$P1 = t2 \cdot C4$	$t2 \cdot C4 = P1$
28	$q4p1 = q2 + 1$	$q2 + 1 = q4p1$
29	$P2 = Lam \cdot q4p1$	$Lam \cdot q4p1 = P2$
30	$S3 = P1 + P2$	$P1 + P2 = S3$
31	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
32	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
33	$Sq3 = Sin \cdot q$	$Sin \cdot q = Sq3$
34	$S = g + Sq3$	$g + Sq3 = S$
35	$n2 = n \cdot n$	$n \cdot n = n2$
36	$n2n = n2 - n$	$n2n + n = n2$
37	$SA = S \cdot n2n$	$S \cdot n2n = SA$
38	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
39	$Tq3 = Tcoef \cdot q$	$Tcoef \cdot q = Tq3$
40	$bm1 = b - 1$	$bm1 + 1 = b$
41	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
42	$T2 = Tq3 - lbm1$	$T2 + lbm1 = Tq3$
43	$b5m2 = B - 2$	$b5m2 + 2 = B$
44	$b5m2q8 = b5m2 \cdot q4$	$b5m2 \cdot q4 = b5m2q8$
45	$T = T2 + b5m2q8$	$T2 + b5m2q8 = T$
46	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
47	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
48	$R7 = SA + TB$	$SA + TB = R7$
49	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
50	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
51	$rsn2 = r \cdot sn2$	$r \cdot sn2 = rsn2$
52	$UM = wn2 \cdot rsn2$	$wn2 \cdot rsn2 = UM$
53	$R12 = UM + rsn2$	$UM + rsn2 = R12$
54	$wsq = UM \cdot rsn2$	$UM \cdot rsn2 = wsq$
55	$R8 = 2 \cdot wsq$	$2 \cdot wsq = R8$
56	$p2 = p \cdot p$	$p \cdot p = p2$
57	$p2m1 = p2 - 1$	$p2m1 + 1 = p2$
58	$k2 = k \cdot k$	$k \cdot k = k2$
59	$P9 = p2m1 \cdot k2$	$p2m1 \cdot k2 = P9$
60	$L9 = P9 + 1$	$P9 + 1 = L9$
61	$R9 = tau \cdot tau$	$tau \cdot tau = R9$
62	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
63	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
64	$R10b = eta + zeta$	$eta + zeta = R10b$
65	$r1 = r + 1$	$r + 1 = r1$
66	$pm1 = p - 1$	$pm1 + 1 = p$
67	$hpm1 = h \cdot pm1$	$h \cdot pm1 = hpm1$
68	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
69	$tr1 = r1 + r$	$r1 + r = tr1$
70	$R13 = ka + phi$	$ka + phi = R13$
71	$bw = b \cdot w$	$b \cdot w = bw$
72	$am2 = a - 2$	$am2 + 2 = a$
73	$cam2 = c \cdot am2$	$c \cdot am2 = cam2$
74	$D1 = bw + cam2$	$bw + cam2 = D1$
75	$a4 = 4 \cdot a$	$4 \cdot a = a4$
76	$a4m5 = a4 - 5$	$a4m5 + 5 = a4$
77	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$

No.	Assignment	Addition/multiplication check
78	$R14 = D1 + gam$	$D1 + gam = R14$
79	$am1 = a - 1$	$am1 + 1 = a$
80	$ap1 = a + 1$	$a + 1 = ap1$
81	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
82	$c2 = c \cdot c$	$c \cdot c = c2$
83	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
84	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
85	$L15 = d \cdot d$	$d \cdot d = L15$
86	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
87	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
88	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
89	$R16 = AE + 1$	$AE + 1 = R16$
90	$L16 = f \cdot f$	$f \cdot f = L16$
91	$of = o \cdot f$	$o \cdot f = of$
92	$dof = of - d$	$dof + d = of$
93	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
94	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
95	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
96	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
97	$jc = j \cdot c$	$j \cdot c = jc$
98	$H17 = tr1 + jc$	$tr1 + jc = H17$
99	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
100	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
101	$R17 = P17 + 1$	$P17 + 1 = R17$
102	$amb = a - B$	$amb + B = a$
103	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
104	$qk = q + kamb$	$q + kamb = qk$
105	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
106	$Mod = A - amb2$	$Mod + amb2 = A$
107	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
108	$R18 = qk + rM$	$qk + rM = R18$
109	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
110	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
111	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
112	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
113	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
114	$R20 = Dam1 + 5^{64}$	$Dam1 + 5^{64} = R20$

The twenty-one free equality tests are

$$\begin{aligned}
L1 &= q, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & p &= R8, & L9 &= R9, \\
c &= R10a, & k &= R10b, & k &= R11, \\
a &= R12, & c &= R13, & d &= R14, \\
L15 &= R15, & L16 &= R16, & L17 &= R17, \\
mu &= R18, & L19 &= R19, & ka &= R20
\end{aligned}$$

D The 112-instruction certificate

The fixed admissible index (Z, V, H) is now the direct quadratic index of Section 9. The positive system unknowns are the 33 listed in Appendix C with p removed, leaving 32. The notation and auxiliary integer domains otherwise remain the same. In particular, $B = Hb^2$ and the code square C^2 are computed explicitly. The historical name $R8$ now denotes the computed abbreviation $X = 2wn^2(rsn^2)^2$, with no additional equation defining a witness p . The shifted Pell parameter $P = X + 1$ is used only in the proof; the table checks its coefficient as $X(X + 2)$ without computing P .

The table is generated from `round6_1980_certificate.py`, which verifies all 112 primitive statements, the 20 equation residuals, both triangular corrections, and the finite Sidon and support inequalities. There are 62 multiplication checks and 50 addition checks. The literal 5^{16} and the other fixed numerals can be generated in seven additional instructions, giving 119 with that requirement. Theorem 10.2 supplies the mathematical equivalence for this fixed encoding.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + \text{beta}$	$x + \text{beta} = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$lq2 = l \cdot q$	$l \cdot q = lq2$
12	$S2 = e + lq2$	$e + lq2 = S2$
13	$\text{pack_tth} = t \cdot th$	$t \cdot th = \text{pack_tth}$
14	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$\text{twice_e} = 2 \cdot e$	$2 \cdot e = \text{twice_e}$
17	$t2 = \text{twice_e} - zl$	$t2 + zl = \text{twice_e}$
18	$xb5 = x \cdot B$	$x \cdot B = xb5$
19	$c0 = xb5 + 1$	$xb5 + 1 = c0$
20	$C = c0 + g$	$c0 + g = C$
21	$C2 = C \cdot C$	$C \cdot C = C2$
22	$lC4 = l \cdot C2$	$l \cdot C2 = lC4$
23	$\text{bound_sum} = e + lC4$	$e + lC4 = \text{bound_sum}$
24	$L1 = \text{bound_sum} + al$	$\text{bound_sum} + al = L1$
25	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
26	$q4p1 = q2 + 1$	$q2 + 1 = q4p1$
27	$P2 = Lam \cdot q4p1$	$Lam \cdot q4p1 = P2$
28	$S3 = P1 + P2$	$P1 + P2 = S3$
29	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
30	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
31	$Sq3 = Sin \cdot q$	$Sin \cdot q = Sq3$
32	$S = g + Sq3$	$g + Sq3 = S$
33	$n2 = n \cdot n$	$n \cdot n = n2$
34	$n2n = n2 - n$	$n2n + n = n2$
35	$SA = S \cdot n2n$	$S \cdot n2n = SA$
36	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
37	$Tq3 = Tcoef \cdot q$	$Tcoef \cdot q = Tq3$

No.	Assignment	Addition/multiplication check
38	$bm1 = b - 1$	$bm1 + 1 = b$
39	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
40	$T2 = Tq3 - lbm1$	$T2 + lbm1 = Tq3$
41	$b5m2 = B - 2$	$b5m2 + 2 = B$
42	$packed_target_shift = lq2 \cdot q2$	$lq2 \cdot q2 = packed_target_shift$
43	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$
44	$T = T2 + mask_term$	$T2 + mask_term = T$
45	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
46	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
47	$R7 = SA + TB$	$SA + TB = R7$
48	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
49	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
50	$rsn2 = r \cdot sn2$	$r \cdot sn2 = rsn2$
51	$UM = wn2 \cdot rsn2$	$wn2 \cdot rsn2 = UM$
52	$R12 = UM + rsn2$	$UM + rsn2 = R12$
53	$wsq = UM \cdot rsn2$	$UM \cdot rsn2 = wsq$
54	$R8 = 2 \cdot wsq$	$2 \cdot wsq = R8$
55	$Pplus1 = R8 + 2$	$R8 + 2 = Pplus1$
56	$p2m1 = R8 \cdot Pplus1$	$R8 \cdot Pplus1 = p2m1$
57	$k2 = k \cdot k$	$k \cdot k = k2$
58	$P9 = p2m1 \cdot k2$	$p2m1 \cdot k2 = P9$
59	$L9 = P9 + 1$	$P9 + 1 = L9$
60	$R9 = tau \cdot tau$	$tau \cdot tau = R9$
61	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
62	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
63	$R10b = eta + zeta$	$eta + zeta = R10b$
64	$r1 = r + 1$	$r + 1 = r1$
65	$hpm1 = h \cdot R8$	$h \cdot R8 = hpm1$
66	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
67	$tr1 = r1 + r$	$r1 + r = tr1$
68	$R13 = ka + phi$	$ka + phi = R13$
69	$bw = b \cdot w$	$b \cdot w = bw$
70	$am2 = a - 2$	$am2 + 2 = a$
71	$cam2 = c \cdot am2$	$c \cdot am2 = cam2$
72	$D1 = bw + cam2$	$bw + cam2 = D1$
73	$a4 = 4 \cdot a$	$4 \cdot a = a4$
74	$a4m5 = a4 - 5$	$a4m5 + 5 = a4$
75	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
76	$R14 = D1 + gam$	$D1 + gam = R14$
77	$am1 = a - 1$	$am1 + 1 = a$
78	$ap1 = a + 1$	$a + 1 = ap1$
79	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
80	$c2 = c \cdot c$	$c \cdot c = c2$
81	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
82	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
83	$L15 = d \cdot d$	$d \cdot d = L15$
84	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
85	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
86	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
87	$R16 = AE + 1$	$AE + 1 = R16$
88	$L16 = f \cdot f$	$f \cdot f = L16$
89	$of = o \cdot f$	$o \cdot f = of$
90	$dof = of - d$	$dof + d = of$

No.	Assignment	Addition/multiplication check
91	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
92	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
93	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
94	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
95	$jc = j \cdot c$	$j \cdot c = jc$
96	$H17 = tr1 + jc$	$tr1 + jc = H17$
97	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
98	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
99	$R17 = P17 + 1$	$P17 + 1 = R17$
100	$amb = a - B$	$amb + B = a$
101	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
102	$qk = q + kamb$	$q + kamb = qk$
103	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
104	$Mod = A - amb2$	$Mod + amb2 = A$
105	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
106	$R18 = qk + rM$	$qk + rM = R18$
107	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
108	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
109	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
110	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
111	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
112	$R20 = Dam1 + 5^{16}$	$Dam1 + 5^{16} = R20$

The twenty free equality tests are

$$\begin{aligned}
L1 &= q, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20
\end{aligned}$$

E The 111-instruction certificate

The fixed index (Z, V, H) is now that of Section 11. Add the positive unknown σ to the 32 witnesses of Appendix D; the free equality $S3 = \sigma$ enforces the sign of the computed third block. All other auxiliary names retain their integer domains. In particular, positivity is imposed on the 33 declared witnesses, not on every intermediate expression.

The code register is $S2 = \ell + eq$, and the size test is $S2 + C^2 + \alpha = q^2$. The table uses the already available q^2 and q^4 for the new packing shifts. It is generated by `round7_1980_schedule_table.py` from the complete `round7_1980_certificate.py` checker. Every primitive and all 21 source residuals are checked exactly, including the changed triangular identity for (E7). The table contains 61 multiplications and 50 additions; the same seven numeral instructions give 118 when required. Theorem 11.1 supplies the encoding equivalence and positive witness construction. The 112-instruction table remains unchanged above.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = twice_e - zl$	$t2 + zl = twice_e$
18	$xb5 = x \cdot B$	$x \cdot B = xb5$
19	$c0 = xb5 + 1$	$xb5 + 1 = c0$
20	$C = c0 + g$	$c0 + g = C$
21	$C2 = C \cdot C$	$C \cdot C = C2$
22	$bound_sum = S2 + C2$	$S2 + C2 = bound_sum$
23	$L1 = bound_sum + al$	$bound_sum + al = L1$
24	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
25	$qp1 = q + 1$	$q + 1 = qp1$
26	$P2 = Lam \cdot qp1$	$Lam \cdot qp1 = P2$
27	$S3 = P1 + P2$	$P1 + P2 = S3$
28	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
29	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
30	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
31	$S = g + Sq4$	$g + Sq4 = S$
32	$n2 = n \cdot n$	$n \cdot n = n2$
33	$n2n = n2 - n$	$n2n + n = n2$
34	$SA = S \cdot n2n$	$S \cdot n2n = SA$
35	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
36	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
37	$bm1 = b - 1$	$bm1 + 1 = b$

No.	Assignment	Addition/multiplication check
38	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
39	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
40	$b5m2 = B - 2$	$b5m2 + 2 = B$
41	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
42	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$
43	$T = T2 + mask_term$	$T2 + mask_term = T$
44	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
45	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
46	$R7 = SA + TB$	$SA + TB = R7$
47	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
48	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
49	$rsn2 = r \cdot sn2$	$r \cdot sn2 = rsn2$
50	$UM = wn2 \cdot rsn2$	$wn2 \cdot rsn2 = UM$
51	$R12 = UM + rsn2$	$UM + rsn2 = R12$
52	$wsq = UM \cdot rsn2$	$UM \cdot rsn2 = wsq$
53	$R8 = 2 \cdot wsq$	$2 \cdot wsq = R8$
54	$Pplus1 = R8 + 2$	$R8 + 2 = Pplus1$
55	$p2m1 = R8 \cdot Pplus1$	$R8 \cdot Pplus1 = p2m1$
56	$k2 = k \cdot k$	$k \cdot k = k2$
57	$P9 = p2m1 \cdot k2$	$p2m1 \cdot k2 = P9$
58	$L9 = P9 + 1$	$P9 + 1 = L9$
59	$R9 = tau \cdot tau$	$tau \cdot tau = R9$
60	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
61	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
62	$R10b = eta + zeta$	$eta + zeta = R10b$
63	$r1 = r + 1$	$r + 1 = r1$
64	$hpm1 = h \cdot R8$	$h \cdot R8 = hpm1$
65	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
66	$tr1 = r1 + r$	$r1 + r = tr1$
67	$R13 = ka + phi$	$ka + phi = R13$
68	$bw = b \cdot w$	$b \cdot w = bw$
69	$am2 = a - 2$	$am2 + 2 = a$
70	$cam2 = c \cdot am2$	$c \cdot am2 = cam2$
71	$D1 = bw + cam2$	$bw + cam2 = D1$
72	$a4 = 4 \cdot a$	$4 \cdot a = a4$
73	$a4m5 = a4 - 5$	$a4m5 + 5 = a4$
74	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
75	$R14 = D1 + gam$	$D1 + gam = R14$
76	$am1 = a - 1$	$am1 + 1 = a$
77	$ap1 = a + 1$	$a + 1 = ap1$
78	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
79	$c2 = c \cdot c$	$c \cdot c = c2$
80	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
81	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
82	$L15 = d \cdot d$	$d \cdot d = L15$
83	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
84	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
85	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
86	$R16 = AE + 1$	$AE + 1 = R16$
87	$L16 = f \cdot f$	$f \cdot f = L16$
88	$of = o \cdot f$	$o \cdot f = of$
89	$dof = of - d$	$dof + d = of$
90	$L17 = dof \cdot dof$	$dof \cdot dof = L17$

No.	Assignment	Addition/multiplication check
91	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
92	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
93	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
94	$jc = j \cdot c$	$j \cdot c = jc$
95	$H17 = tr1 + jc$	$tr1 + jc = H17$
96	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
97	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
98	$R17 = P17 + 1$	$P17 + 1 = R17$
99	$amb = a - B$	$amb + B = a$
100	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
101	$qk = q + kamb$	$q + kamb = qk$
102	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
103	$Mod = A - amb2$	$Mod + amb2 = A$
104	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
105	$R18 = qk + rM$	$qk + rM = R18$
106	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
107	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
108	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
109	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
110	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
111	$R20 = Dam1 + 5^{16}$	$Dam1 + 5^{16} = R20$

The twenty-one free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma
\end{aligned}$$

F The 110-instruction certificate

Use the homogeneous fixed index of Section 12, with the same 33 positive witnesses and 21 equality tests as in Appendix E. The decoded unit coordinate is part of the integer code, not an additional system unknown. The code register is now $C = xB + g$, and the third mask uses $B - 4$.

The complete checker `round8_1980_certificate.py` verifies every primitive, source residual, and support inequality for this layout. There are 61 multiplication checks and 49 addition checks; the seven numeral instructions give 117 when required. Theorem 12.1 proves the encoding and positive-witness equivalence. The table is generated by `round8_1980_schedule_table.py`.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = twice_e - zl$	$t2 + zl = twice_e$
18	$xb5 = x \cdot B$	$x \cdot B = xb5$
19	$C = xb5 + g$	$xb5 + g = C$
20	$C2 = C \cdot C$	$C \cdot C = C2$
21	$bound_sum = S2 + C2$	$S2 + C2 = bound_sum$
22	$L1 = bound_sum + al$	$bound_sum + al = L1$
23	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
24	$qp1 = q + 1$	$q + 1 = qp1$
25	$P2 = Lam \cdot qp1$	$Lam \cdot qp1 = P2$
26	$S3 = P1 + P2$	$P1 + P2 = S3$
27	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
28	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
29	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
30	$S = g + Sq4$	$g + Sq4 = S$
31	$n2 = n \cdot n$	$n \cdot n = n2$
32	$n2n = n2 - n$	$n2n + n = n2$
33	$SA = S \cdot n2n$	$S \cdot n2n = SA$
34	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
35	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
36	$bm1 = b - 1$	$bm1 + 1 = b$
37	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
38	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
39	$b5m2 = B - 4$	$b5m2 + 4 = B$
40	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
41	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$
42	$T = T2 + mask_term$	$T2 + mask_term = T$

No.	Assignment	Addition/multiplication check
43	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
44	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
45	$R7 = SA + TB$	$SA + TB = R7$
46	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
47	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
48	$rsn2 = r \cdot sn2$	$r \cdot sn2 = rsn2$
49	$UM = wn2 \cdot rsn2$	$wn2 \cdot rsn2 = UM$
50	$R12 = UM + rsn2$	$UM + rsn2 = R12$
51	$wsq = UM \cdot rsn2$	$UM \cdot rsn2 = wsq$
52	$R8 = 2 \cdot wsq$	$2 \cdot wsq = R8$
53	$Pplus1 = R8 + 2$	$R8 + 2 = Pplus1$
54	$p2m1 = R8 \cdot Pplus1$	$R8 \cdot Pplus1 = p2m1$
55	$k2 = k \cdot k$	$k \cdot k = k2$
56	$P9 = p2m1 \cdot k2$	$p2m1 \cdot k2 = P9$
57	$L9 = P9 + 1$	$P9 + 1 = L9$
58	$R9 = tau \cdot tau$	$tau \cdot tau = R9$
59	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
60	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
61	$R10b = eta + zeta$	$eta + zeta = R10b$
62	$r1 = r + 1$	$r + 1 = r1$
63	$hpm1 = h \cdot R8$	$h \cdot R8 = hpm1$
64	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
65	$tr1 = r1 + r$	$r1 + r = tr1$
66	$R13 = ka + phi$	$ka + phi = R13$
67	$bw = b \cdot w$	$b \cdot w = bw$
68	$am2 = a - 2$	$am2 + 2 = a$
69	$cam2 = c \cdot am2$	$c \cdot am2 = cam2$
70	$D1 = bw + cam2$	$bw + cam2 = D1$
71	$a4 = 4 \cdot a$	$4 \cdot a = a4$
72	$a4m5 = a4 - 5$	$a4m5 + 5 = a4$
73	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
74	$R14 = D1 + gam$	$D1 + gam = R14$
75	$am1 = a - 1$	$am1 + 1 = a$
76	$ap1 = a + 1$	$a + 1 = ap1$
77	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
78	$c2 = c \cdot c$	$c \cdot c = c2$
79	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
80	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
81	$L15 = d \cdot d$	$d \cdot d = L15$
82	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
83	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
84	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
85	$R16 = AE + 1$	$AE + 1 = R16$
86	$L16 = f \cdot f$	$f \cdot f = L16$
87	$of = o \cdot f$	$o \cdot f = of$
88	$dof = of - d$	$dof + d = of$
89	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
90	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
91	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
92	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
93	$jc = j \cdot c$	$j \cdot c = jc$
94	$H17 = tr1 + jc$	$tr1 + jc = H17$
95	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$

No.	Assignment	Addition/multiplication check
96	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
97	$R17 = P17 + 1$	$P17 + 1 = R17$
98	$amb = a - B$	$amb + B = a$
99	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
100	$qk = q + kamb$	$q + kamb = qk$
101	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
102	$Mod = A - amb2$	$Mod + amb2 = A$
103	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
104	$R18 = qk + rM$	$qk + rM = R18$
105	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
106	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
107	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
108	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
109	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
110	$R20 = Dam1 + 5^{16}$	$Dam1 + 5^{16} = R20$

The twenty-one free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma
\end{aligned}$$

G The 109-instruction certificate

Use the fixed input-unit index of Section 13, with the same positive witnesses and equality tests. The sole arithmetic change from Appendix F is $C = x + g$, which removes the input multiplication by B . The index assigns the input to weight zero and the decoded unit coordinate to weight one; these are part of its fixed definition and introduce no extra arithmetic checks.

The complete checker `round9_1980_certificate.py` verifies the 109 primitive statements and all 21 source residuals, including both triangular corrections. There are 60 multiplications and 49 additions; the seven numeral instructions give 116 when required. Theorem 13.1 proves the relaxed bootstrap and the resulting encoding equivalence. The table is generated by `round9_1980_schedule_table.py`.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = twice_e - zl$	$t2 + zl = twice_e$
18	$C = x + g$	$x + g = C$
19	$C2 = C \cdot C$	$C \cdot C = C2$
20	$bound_sum = S2 + C2$	$S2 + C2 = bound_sum$
21	$L1 = bound_sum + al$	$bound_sum + al = L1$
22	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
23	$qp1 = q + 1$	$q + 1 = qp1$
24	$P2 = Lam \cdot qp1$	$Lam \cdot qp1 = P2$
25	$S3 = P1 + P2$	$P1 + P2 = S3$
26	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
27	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
28	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
29	$S = g + Sq4$	$g + Sq4 = S$
30	$n2 = n \cdot n$	$n \cdot n = n2$
31	$n2n = n2 - n$	$n2n + n = n2$
32	$SA = S \cdot n2n$	$S \cdot n2n = SA$
33	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
34	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
35	$bm1 = b - 1$	$bm1 + 1 = b$
36	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
37	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
38	$b5m2 = B - 4$	$b5m2 + 4 = B$
39	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
40	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$
41	$T = T2 + mask_term$	$T2 + mask_term = T$

No.	Assignment	Addition/multiplication check
42	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
43	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
44	$R7 = SA + TB$	$SA + TB = R7$
45	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
46	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
47	$rsn2 = r \cdot sn2$	$r \cdot sn2 = rsn2$
48	$UM = wn2 \cdot rsn2$	$wn2 \cdot rsn2 = UM$
49	$R12 = UM + rsn2$	$UM + rsn2 = R12$
50	$wsq = UM \cdot rsn2$	$UM \cdot rsn2 = wsq$
51	$R8 = 2 \cdot wsq$	$2 \cdot wsq = R8$
52	$Pplus1 = R8 + 2$	$R8 + 2 = Pplus1$
53	$p2m1 = R8 \cdot Pplus1$	$R8 \cdot Pplus1 = p2m1$
54	$k2 = k \cdot k$	$k \cdot k = k2$
55	$P9 = p2m1 \cdot k2$	$p2m1 \cdot k2 = P9$
56	$L9 = P9 + 1$	$P9 + 1 = L9$
57	$R9 = tau \cdot tau$	$tau \cdot tau = R9$
58	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
59	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
60	$R10b = eta + zeta$	$eta + zeta = R10b$
61	$r1 = r + 1$	$r + 1 = r1$
62	$hpm1 = h \cdot R8$	$h \cdot R8 = hpm1$
63	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
64	$tr1 = r1 + r$	$r1 + r = tr1$
65	$R13 = ka + phi$	$ka + phi = R13$
66	$bw = b \cdot w$	$b \cdot w = bw$
67	$am2 = a - 2$	$am2 + 2 = a$
68	$cam2 = c \cdot am2$	$c \cdot am2 = cam2$
69	$D1 = bw + cam2$	$bw + cam2 = D1$
70	$a4 = 4 \cdot a$	$4 \cdot a = a4$
71	$a4m5 = a4 - 5$	$a4m5 + 5 = a4$
72	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
73	$R14 = D1 + gam$	$D1 + gam = R14$
74	$am1 = a - 1$	$am1 + 1 = a$
75	$ap1 = a + 1$	$a + 1 = ap1$
76	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
77	$c2 = c \cdot c$	$c \cdot c = c2$
78	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
79	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
80	$L15 = d \cdot d$	$d \cdot d = L15$
81	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
82	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
83	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
84	$R16 = AE + 1$	$AE + 1 = R16$
85	$L16 = f \cdot f$	$f \cdot f = L16$
86	$of = o \cdot f$	$o \cdot f = of$
87	$dof = of - d$	$dof + d = of$
88	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
89	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
90	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
91	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
92	$jc = j \cdot c$	$j \cdot c = jc$
93	$H17 = tr1 + jc$	$tr1 + jc = H17$
94	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$

No.	Assignment	Addition/multiplication check
95	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
96	$R17 = P17 + 1$	$P17 + 1 = R17$
97	$amb = a - B$	$amb + B = a$
98	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
99	$qk = q + kamb$	$q + kamb = qk$
100	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
101	$Mod = A - amb2$	$Mod + amb2 = A$
102	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
103	$R18 = qk + rM$	$qk + rM = R18$
104	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
105	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
106	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
107	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
108	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
109	$R20 = Dam1 + 5^{16}$	$Dam1 + 5^{16} = R20$

The twenty-one free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma
\end{aligned}$$

H The 107-instruction certificate

Use the same fixed input-unit index as in Appendix G. The 34 positive unknowns are the 33 previously listed witnesses, including σ , together with Ω . The name *Omega* denotes Ω , and the free equality to the register $t2 = Z\lambda - 2e$ imposes its positive sign. The size bound is now $S2 + \alpha = q^2$, and the third block is computed as $P2 - P1$.

The first Pell root and index quotient also have new meanings: $\tau_{\text{old}} = 2\tau + 1$ and $h_{\text{old}} = h/2$ in the positive reverse map proved in Section 14. The table directly checks the new equations $\tau(\tau + 1) = Q(Q + 1)k^2$ and $k = r + 1 + hQ$, using the existing register $wsq = Q$. Neither $2Q$ nor the old root is computed.

The complete checker `round12_1980_certificate.py` verifies all 107 primitives, all 22 source residuals, the triangular corrections, and the exact polynomial witness identities. The table is generated by `round12_1980_schedule_table.py`. It has 59 multiplications and 48 additions, or 114 operations including the seven fixed-numeral instructions. Theorem 14.3 supplies the composed positive-domain and encoding proof.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + \text{beta}$	$x + \text{beta} = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$\text{pack_tth} = t \cdot th$	$t \cdot th = \text{pack_tth}$
14	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$\text{twice_e} = 2 \cdot e$	$2 \cdot e = \text{twice_e}$
17	$t2 = zl - \text{twice_e}$	$zl - \text{twice_e} = t2$
18	$C = x + g$	$x + g = C$
19	$C2 = C \cdot C$	$C \cdot C = C2$
20	$L1 = S2 + al$	$S2 + al = L1$
21	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
22	$qp1 = q + 1$	$q + 1 = qp1$
23	$P2 = Lam \cdot qp1$	$Lam \cdot qp1 = P2$
24	$S3 = P2 - P1$	$P2 - P1 = S3$
25	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
26	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
27	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
28	$S = g + Sq4$	$g + Sq4 = S$
29	$n2 = n \cdot n$	$n \cdot n = n2$
30	$n2n = n2 - n$	$n2 - n = n2n$
31	$SA = S \cdot n2n$	$S \cdot n2n = SA$
32	$Tcoef = L2 - zl$	$L2 - zl = Tcoef$
33	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
34	$bm1 = b - 1$	$b - 1 = bm1$
35	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$

No.	Assignment	Addition/multiplication check
36	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
37	$b5m2 = B - 4$	$b5m2 + 4 = B$
38	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
39	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$
40	$T = T2 + mask_term$	$T2 + mask_term = T$
41	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
42	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
43	$R7 = SA + TB$	$SA + TB = R7$
44	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
45	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
46	$rsn2 = r \cdot sn2$	$r \cdot sn2 = rsn2$
47	$UM = wn2 \cdot rsn2$	$wn2 \cdot rsn2 = UM$
48	$R12 = UM + rsn2$	$UM + rsn2 = R12$
49	$wsq = UM \cdot rsn2$	$UM \cdot rsn2 = wsq$
50	$Qplus1 = wsq + 1$	$wsq + 1 = Qplus1$
51	$QQplus1 = wsq \cdot Qplus1$	$wsq \cdot Qplus1 = QQplus1$
52	$k2 = k \cdot k$	$k \cdot k = k2$
53	$L9 = QQplus1 \cdot k2$	$QQplus1 \cdot k2 = L9$
54	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
55	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
56	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
57	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
58	$R10b = eta + zeta$	$eta + zeta = R10b$
59	$r1 = r + 1$	$r + 1 = r1$
60	$hpm1 = h \cdot wsq$	$h \cdot wsq = hpm1$
61	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
62	$tr1 = r1 + r$	$r1 + r = tr1$
63	$R13 = ka + phi$	$ka + phi = R13$
64	$bw = b \cdot w$	$b \cdot w = bw$
65	$am2 = a - 2$	$am2 + 2 = a$
66	$cam2 = c \cdot am2$	$c \cdot am2 = cam2$
67	$D1 = bw + cam2$	$bw + cam2 = D1$
68	$a4 = 4 \cdot a$	$4 \cdot a = a4$
69	$a4m5 = a4 - 5$	$a4m5 + 5 = a4$
70	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
71	$R14 = D1 + gam$	$D1 + gam = R14$
72	$am1 = a - 1$	$am1 + 1 = a$
73	$ap1 = a + 1$	$a + 1 = ap1$
74	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
75	$c2 = c \cdot c$	$c \cdot c = c2$
76	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
77	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
78	$L15 = d \cdot d$	$d \cdot d = L15$
79	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
80	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
81	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
82	$R16 = AE + 1$	$AE + 1 = R16$
83	$L16 = f \cdot f$	$f \cdot f = L16$
84	$of = o \cdot f$	$o \cdot f = of$
85	$dof = of - d$	$dof + d = of$
86	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
87	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
88	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$

No.	Assignment	Addition/multiplication check
89	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
90	$jc = j \cdot c$	$j \cdot c = jc$
91	$H17 = tr1 + jc$	$tr1 + jc = H17$
92	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
93	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
94	$R17 = P17 + 1$	$P17 + 1 = R17$
95	$amb = a - B$	$amb + B = a$
96	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
97	$qk = q + kamb$	$q + kamb = qk$
98	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
99	$Mod = A - amb2$	$Mod + amb2 = A$
100	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
101	$R18 = qk + rM$	$qk + rM = R18$
102	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
103	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
104	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
105	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
106	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
107	$R20 = Dam1 + 5^{16}$	$Dam1 + 5^{16} = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

I The 106-instruction certificate

Use the modular support of Section 15, with $L = 2^{32}$. The positive input named a now denotes $a_0 = RY(U + 1)$; the mathematical main Pell parameter is $A = a_0 + 4$. The base-four congruence forces $bw = 4^{2r+1}$. Theorem 16.1 proves equivalence with the represented set, including the new ratio estimates and the reconstruction of all positive Pell witnesses.

The complete checker `round14_1980_base_four_certificate.py` verifies 59 multiplications and 47 additions, all 22 source residuals, and both triangular corrections. The table is generated by `round14_1980_schedule_table.py`. Fixed numerals are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = zl - twice_e$	$t2 + twice_e = zl$
18	$C = x + g$	$x + g = C$
19	$C2 = C \cdot C$	$C \cdot C = C2$
20	$L1 = S2 + al$	$S2 + al = L1$
21	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
22	$qp1 = q + 1$	$q + 1 = qp1$
23	$P2 = Lam \cdot qp1$	$Lam \cdot qp1 = P2$
24	$S3 = P2 - P1$	$S3 + P1 = P2$
25	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
26	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
27	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
28	$S = g + Sq4$	$g + Sq4 = S$
29	$n2 = n \cdot n$	$n \cdot n = n2$
30	$n2n = n2 - n$	$n2n + n = n2$
31	$SA = S \cdot n2n$	$S \cdot n2n = SA$
32	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
33	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
34	$bm1 = b - 1$	$bm1 + 1 = b$
35	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
36	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
37	$b5m2 = B - 4$	$b5m2 + 4 = B$
38	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
39	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$
40	$T = T2 + mask_term$	$T2 + mask_term = T$
41	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
42	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$

No.	Assignment	Addition/multiplication check
43	$R7 = SA + TB$	$SA + TB = R7$
44	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
45	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
46	$rsn2 = r \cdot sn2$	$r \cdot sn2 = rsn2$
47	$UM = wn2 \cdot rsn2$	$wn2 \cdot rsn2 = UM$
48	$R12 = UM + rsn2$	$UM + rsn2 = R12$
49	$wsq = UM \cdot rsn2$	$UM \cdot rsn2 = wsq$
50	$Qplus1 = wsq + 1$	$wsq + 1 = Qplus1$
51	$QQplus1 = wsq \cdot Qplus1$	$wsq \cdot Qplus1 = QQplus1$
52	$k2 = k \cdot k$	$k \cdot k = k2$
53	$L9 = QQplus1 \cdot k2$	$QQplus1 \cdot k2 = L9$
54	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
55	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
56	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
57	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
58	$R10b = eta + zeta$	$eta + zeta = R10b$
59	$r1 = r + 1$	$r + 1 = r1$
60	$hpm1 = h \cdot wsq$	$h \cdot wsq = hpm1$
61	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
62	$tr1 = r1 + r$	$r1 + r = tr1$
63	$R13 = ka + phi$	$ka + phi = R13$
64	$bw = b \cdot w$	$b \cdot w = bw$
65	$cam2 = c \cdot a$	$c \cdot a = cam2$
66	$D1 = bw + cam2$	$bw + cam2 = D1$
67	$am1 = a + 3$	$a + 3 = am1$
68	$a4 = 8 \cdot a$	$8 \cdot a = a4$
69	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
70	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
71	$R14 = D1 + gam$	$D1 + gam = R14$
72	$ap1 = am1 + 2$	$am1 + 2 = ap1$
73	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
74	$c2 = c \cdot c$	$c \cdot c = c2$
75	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
76	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
77	$L15 = d \cdot d$	$d \cdot d = L15$
78	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
79	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
80	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
81	$R16 = AE + 1$	$AE + 1 = R16$
82	$L16 = f \cdot f$	$f \cdot f = L16$
83	$of = o \cdot f$	$o \cdot f = of$
84	$dof = of - d$	$dof + d = of$
85	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
86	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
87	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
88	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
89	$jc = j \cdot c$	$j \cdot c = jc$
90	$H17 = tr1 + jc$	$tr1 + jc = H17$
91	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
92	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
93	$R17 = P17 + 1$	$P17 + 1 = R17$
94	$amb = a - b5m2$	$amb + b5m2 = a$
95	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$

No.	Assignment	Addition/multiplication check
96	$qk = q + kamb$	$q + kamb = qk$
97	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
98	$Mod = A - amb2$	$Mod + amb2 = A$
99	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
100	$R18 = qk + rM$	$qk + rM = R18$
101	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
102	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
103	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
104	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
105	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
106	$R20 = Dam1 + 2^{32}$	$Dam1 + 2^{32} = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

J The 105-instruction certificate

Retain the same fixed modular index and the main parameter $A = a_0 + 4$. The positive coordinate named a now denotes $a_0 = (r + 1)Y(U + 1)$, and $Q = U((r + 1)Y)^2$. The first index equation is $k = h(r + 1)$. Theorem 17.1 proves that this forces the exact Pell index $r + 1$ and gives the complete positive-domain equivalence.

The schedule moves the existing register $r1 = r + 1$ before the scale product and eliminates the former index-sum register $R11$. The complete checker `round16_1980_index_multiple_certificate.py` verifies all 105 primitives: 59 multiplications and 46 additions. It also verifies all 22 source residuals, including the three changed equations and both triangular corrections. The table is generated by `round16_1980_schedule_table.py`. There are 34 positive unknowns, and fixed numerals are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = zl - twice_e$	$t2 + twice_e = zl$
18	$C = x + g$	$x + g = C$
19	$C2 = C \cdot C$	$C \cdot C = C2$
20	$L1 = S2 + al$	$S2 + al = L1$
21	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
22	$qp1 = q + 1$	$q + 1 = qp1$
23	$P2 = Lam \cdot qp1$	$Lam \cdot qp1 = P2$
24	$S3 = P2 - P1$	$S3 + P1 = P2$
25	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
26	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
27	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
28	$S = g + Sq4$	$g + Sq4 = S$
29	$n2 = n \cdot n$	$n \cdot n = n2$
30	$n2n = n2 - n$	$n2n + n = n2$
31	$SA = S \cdot n2n$	$S \cdot n2n = SA$
32	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
33	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
34	$bm1 = b - 1$	$bm1 + 1 = b$
35	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
36	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
37	$b5m2 = B - 4$	$b5m2 + 4 = B$
38	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
39	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$
40	$T = T2 + mask_term$	$T2 + mask_term = T$

No.	Assignment	Addition/multiplication check
41	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
42	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
43	$R7 = SA + TB$	$SA + TB = R7$
44	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
45	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
46	$r1 = r + 1$	$r + 1 = r1$
47	$rsn2 = r1 \cdot sn2$	$r1 \cdot sn2 = rsn2$
48	$UM = wn2 \cdot rsn2$	$wn2 \cdot rsn2 = UM$
49	$R12 = UM + rsn2$	$UM + rsn2 = R12$
50	$wsq = UM \cdot rsn2$	$UM \cdot rsn2 = wsq$
51	$Qplus1 = wsq + 1$	$wsq + 1 = Qplus1$
52	$QQplus1 = wsq \cdot Qplus1$	$wsq \cdot Qplus1 = QQplus1$
53	$k2 = k \cdot k$	$k \cdot k = k2$
54	$L9 = QQplus1 \cdot k2$	$QQplus1 \cdot k2 = L9$
55	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
56	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
57	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
58	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
59	$R10b = eta + zeta$	$eta + zeta = R10b$
60	$hpm1 = h \cdot r1$	$h \cdot r1 = hpm1$
61	$tr1 = r1 + r$	$r1 + r = tr1$
62	$R13 = ka + phi$	$ka + phi = R13$
63	$bw = b \cdot w$	$b \cdot w = bw$
64	$cam2 = c \cdot a$	$c \cdot a = cam2$
65	$D1 = bw + cam2$	$bw + cam2 = D1$
66	$am1 = a + 3$	$a + 3 = am1$
67	$a4 = 8 \cdot a$	$8 \cdot a = a4$
68	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
69	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
70	$R14 = D1 + gam$	$D1 + gam = R14$
71	$ap1 = am1 + 2$	$am1 + 2 = ap1$
72	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
73	$c2 = c \cdot c$	$c \cdot c = c2$
74	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
75	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
76	$L15 = d \cdot d$	$d \cdot d = L15$
77	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
78	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
79	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
80	$R16 = AE + 1$	$AE + 1 = R16$
81	$L16 = f \cdot f$	$f \cdot f = L16$
82	$of = o \cdot f$	$o \cdot f = of$
83	$dof = of - d$	$dof + d = of$
84	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
85	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
86	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
87	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
88	$jc = j \cdot c$	$j \cdot c = jc$
89	$H17 = tr1 + jc$	$tr1 + jc = H17$
90	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
91	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
92	$R17 = P17 + 1$	$P17 + 1 = R17$
93	$amb = a - b5m2$	$amb + b5m2 = a$

No.	Assignment	Addition/multiplication check
94	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
95	$qk = q + kamb$	$q + kamb = qk$
96	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
97	$Mod = A - amb2$	$Mod + amb2 = A$
98	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
99	$R18 = qk + rM$	$qk + rM = R18$
100	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
101	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
102	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
103	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
104	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
105	$R20 = Dam1 + 2^{32}$	$Dam1 + 2^{32} = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= hpm1, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

K The 104-instruction certificate

Retain the fixed modular index, with $L = 2^{32}$, and put $U = wn^2$, $Y_P = sn^2$, $D = UY_P$. The supplied positive coordinate a denotes $a_0 = D + Y_P$, and the main Pell parameter is $A = a_0 + 4$. The first norm and index equation are

$$\tau(\tau + 1) = (D^2 + U)(Y_P k)^2, \quad k = r + 1 + hD.$$

The products D and $Y_P k$ are shared with the main parameter and interval equations. Theorem 18.2 proves the exact index, positive witness equivalence, and full decoding.

The complete checker `round19_1980_shared_ratio_product_certificate.py` verifies all 104 primitives, all 22 source residuals, both triangular corrections and the forward witness-map identity. The table is generated by `round19_1980_schedule_table.py`. It contains 57 multiplications and 47 additions. There are 34 positive unknowns; fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = zl - twice_e$	$t2 + twice_e = zl$
18	$C = x + g$	$x + g = C$
19	$C2 = C \cdot C$	$C \cdot C = C2$
20	$L1 = S2 + al$	$S2 + al = L1$
21	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
22	$qp1 = q + 1$	$q + 1 = qp1$
23	$P2 = Lam \cdot qp1$	$Lam \cdot qp1 = P2$
24	$S3 = P2 - P1$	$S3 + P1 = P2$
25	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
26	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
27	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
28	$S = g + Sq4$	$g + Sq4 = S$
29	$n2 = n \cdot n$	$n \cdot n = n2$
30	$n2n = n2 - n$	$n2n + n = n2$
31	$SA = S \cdot n2n$	$S \cdot n2n = SA$
32	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
33	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
34	$bm1 = b - 1$	$bm1 + 1 = b$
35	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
36	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
37	$b5m2 = B - 4$	$b5m2 + 4 = B$

No.	Assignment	Addition/multiplication check
38	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
39	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$
40	$T = T2 + mask_term$	$T2 + mask_term = T$
41	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
42	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
43	$R7 = SA + TB$	$SA + TB = R7$
44	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
45	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
46	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
47	$R12 = UM + sn2$	$UM + sn2 = R12$
48	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
49	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
50	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
51	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
52	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
53	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
54	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
55	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
56	$R10b = eta + zeta$	$eta + zeta = R10b$
57	$r1 = r + 1$	$r + 1 = r1$
58	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
59	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
60	$tr1 = r1 + r$	$r1 + r = tr1$
61	$R13 = ka + phi$	$ka + phi = R13$
62	$bw = b \cdot w$	$b \cdot w = bw$
63	$cam2 = c \cdot a$	$c \cdot a = cam2$
64	$D1 = bw + cam2$	$bw + cam2 = D1$
65	$am1 = a + 3$	$a + 3 = am1$
66	$a4 = 8 \cdot a$	$8 \cdot a = a4$
67	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
68	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
69	$R14 = D1 + gam$	$D1 + gam = R14$
70	$ap1 = am1 + 2$	$am1 + 2 = ap1$
71	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
72	$c2 = c \cdot c$	$c \cdot c = c2$
73	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
74	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
75	$L15 = d \cdot d$	$d \cdot d = L15$
76	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
77	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
78	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
79	$R16 = AE + 1$	$AE + 1 = R16$
80	$L16 = f \cdot f$	$f \cdot f = L16$
81	$of = o \cdot f$	$o \cdot f = of$
82	$dof = of - d$	$dof + d = of$
83	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
84	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
85	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
86	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
87	$jc = j \cdot c$	$j \cdot c = jc$
88	$H17 = tr1 + jc$	$tr1 + jc = H17$
89	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
90	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$

No.	Assignment	Addition/multiplication check
91	$R17 = P17 + 1$	$P17 + 1 = R17$
92	$amb = a - b5m2$	$amb + b5m2 = a$
93	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
94	$qk = q + kamb$	$q + kamb = qk$
95	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
96	$Mod = A - amb2$	$Mod + amb2 = A$
97	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
98	$R18 = qk + rM$	$qk + rM = R18$
99	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
100	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
101	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
102	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
103	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
104	$R20 = Dam1 + 2^{32}$	$Dam1 + 2^{32} = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

L The 103-instruction coefficient certificate

Use the rebuilt fixed coefficient index of Section 19, with ordinary targets $4F_i^h + \delta^2$ and special target δ^2 . The code positions and $L = 2^{32}$ are unchanged, but Z, V, H are reconstructed from the new coefficients. The third packed block is

$$S_3 = B\lambda q - (Z\lambda - 2e)(x + g)^2.$$

Theorem 19.2 proves the complete decoding and positive necessity, including the low carries and the elimination of the possible quotient ambiguity. The former $q + 1$ register is absent.

The checker `round20_1980_borrow_mask_certificate.py` verifies all 103 primitives, all 22 source residuals and both triangular corrections.

The table generator is `round20_1980_schedule_table.py`. There are 57 multiplications and 46 additions, with 34 positive unknowns. Fixed numerals are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = zl - twice_e$	$t2 + twice_e = zl$
18	$C = x + g$	$x + g = C$
19	$C2 = C \cdot C$	$C \cdot C = C2$
20	$L1 = S2 + al$	$S2 + al = L1$
21	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
22	$P2 = Lam \cdot q$	$Lam \cdot q = P2$
23	$S3 = P2 - P1$	$S3 + P1 = P2$
24	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
25	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
26	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
27	$S = g + Sq4$	$g + Sq4 = S$
28	$n2 = n \cdot n$	$n \cdot n = n2$
29	$n2n = n2 - n$	$n2n + n = n2$
30	$SA = S \cdot n2n$	$S \cdot n2n = SA$
31	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
32	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
33	$bm1 = b - 1$	$bm1 + 1 = b$
34	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
35	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
36	$b5m2 = B - 4$	$b5m2 + 4 = B$
37	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
38	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$

No.	Assignment	Addition/multiplication check
39	$T = T2 + \text{mask_term}$	$T2 + \text{mask_term} = T$
40	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
41	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
42	$R7 = SA + TB$	$SA + TB = R7$
43	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
44	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
45	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
46	$R12 = UM + sn2$	$UM + sn2 = R12$
47	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
48	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
49	$\text{scaled_norm_coefficient} = UM2 + wn2$	$UM2 + wn2 = \text{scaled_norm_coefficient}$
50	$\text{ratio_product2} = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = \text{ratio_product2}$
51	$L9 = \text{scaled_norm_coefficient} \cdot \text{ratio_product2}$	$\text{scaled_norm_coefficient} \cdot \text{ratio_product2} = L9$
52	$\text{tauplus1} = \text{tau} + 1$	$\text{tau} + 1 = \text{tauplus1}$
53	$R9 = \text{tau} \cdot \text{tauplus1}$	$\text{tau} \cdot \text{tauplus1} = R9$
54	$R10a = ksn2 + \text{eta}$	$ksn2 + \text{eta} = R10a$
55	$R10b = \text{eta} + \text{zeta}$	$\text{eta} + \text{zeta} = R10b$
56	$r1 = r + 1$	$r + 1 = r1$
57	$\text{hpm1} = h \cdot UM$	$h \cdot UM = \text{hpm1}$
58	$R11 = r1 + \text{hpm1}$	$r1 + \text{hpm1} = R11$
59	$\text{tr1} = r1 + r$	$r1 + r = \text{tr1}$
60	$R13 = ka + \text{phi}$	$ka + \text{phi} = R13$
61	$\text{bw} = b \cdot w$	$b \cdot w = \text{bw}$
62	$\text{cam2} = c \cdot a$	$c \cdot a = \text{cam2}$
63	$D1 = \text{bw} + \text{cam2}$	$\text{bw} + \text{cam2} = D1$
64	$\text{am1} = a + 3$	$a + 3 = \text{am1}$
65	$a4 = 8 \cdot a$	$8 \cdot a = a4$
66	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
67	$\text{gam} = ga \cdot a4m5$	$ga \cdot a4m5 = \text{gam}$
68	$R14 = D1 + \text{gam}$	$D1 + \text{gam} = R14$
69	$\text{ap1} = \text{am1} + 2$	$\text{am1} + 2 = \text{ap1}$
70	$A = \text{am1} \cdot \text{ap1}$	$\text{am1} \cdot \text{ap1} = A$
71	$c2 = c \cdot c$	$c \cdot c = c2$
72	$\text{Ac2} = A \cdot c2$	$A \cdot c2 = \text{Ac2}$
73	$R15 = \text{Ac2} + 1$	$\text{Ac2} + 1 = R15$
74	$L15 = d \cdot d$	$d \cdot d = L15$
75	$\text{ic2} = i \cdot c2$	$i \cdot c2 = \text{ic2}$
76	$\text{ic22} = \text{ic2} \cdot \text{ic2}$	$\text{ic2} \cdot \text{ic2} = \text{ic22}$
77	$\text{AE} = A \cdot \text{ic22}$	$A \cdot \text{ic22} = \text{AE}$
78	$R16 = \text{AE} + 1$	$\text{AE} + 1 = R16$
79	$L16 = f \cdot f$	$f \cdot f = L16$
80	$\text{of} = o \cdot f$	$o \cdot f = \text{of}$
81	$\text{dof} = \text{of} - d$	$\text{dof} + d = \text{of}$
82	$L17 = \text{dof} \cdot \text{dof}$	$\text{dof} \cdot \text{dof} = L17$
83	$\text{Gminus1} = \text{ap1} \cdot \text{AE}$	$\text{ap1} \cdot \text{AE} = \text{Gminus1}$
84	$\text{Gplus1} = \text{Gminus1} + 2$	$\text{Gminus1} + 2 = \text{Gplus1}$
85	$\text{G2m1} = \text{Gminus1} \cdot \text{Gplus1}$	$\text{Gminus1} \cdot \text{Gplus1} = \text{G2m1}$
86	$\text{jc} = j \cdot c$	$j \cdot c = \text{jc}$
87	$\text{H17} = \text{tr1} + \text{jc}$	$\text{tr1} + \text{jc} = \text{H17}$
88	$\text{H2} = \text{H17} \cdot \text{H17}$	$\text{H17} \cdot \text{H17} = \text{H2}$
89	$\text{P17} = \text{G2m1} \cdot \text{H2}$	$\text{G2m1} \cdot \text{H2} = \text{P17}$
90	$\text{R17} = \text{P17} + 1$	$\text{P17} + 1 = \text{R17}$
91	$\text{amb} = a - \text{b5m2}$	$\text{amb} + \text{b5m2} = a$

No.	Assignment	Addition/multiplication check
92	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
93	$qk = q + kamb$	$q + kamb = qk$
94	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
95	$Mod = A - amb2$	$Mod + amb2 = A$
96	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
97	$R18 = qk + rM$	$qk + rM = R18$
98	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
99	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
100	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
101	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
102	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
103	$R20 = Dam1 + 2^{32}$	$Dam1 + 2^{32} = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

M The 102-instruction certificate

Use the same coefficient index as Appendix L and retain $U = wn^2$, $Y_P = sn^2$, $a_0 = Y_P(U + 1)$ and $A = a_0 + 4$. Use the already calculated U as the first exponent target:

$$d = U + c(A - 4) + \gamma(8A - 17).$$

Theorem 20.1 proves that this change composes with the coefficient encoding, including the correct order of the radix deductions and the positive witness $w = 4^{2r+1}/n^2$. The separate products bw and the low-offset register $q + 1$ are absent.

The complete checker `round22_1980_composed_certificate.py` verifies all 102 primitives, all 22 source residuals and both triangular corrections. It also verifies that the two transformations commute on the full schedule and every source polynomial.

The table generator is `round22_1980_schedule_table.py`. It contains 56 multiplications and 46 additions, with 34 positive unknowns. Fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + Z$	$th + Z = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = Z \cdot la$	$Z \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = zl - twice_e$	$t2 + twice_e = zl$
18	$C = x + g$	$x + g = C$
19	$C2 = C \cdot C$	$C \cdot C = C2$
20	$L1 = S2 + al$	$S2 + al = L1$
21	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
22	$P2 = Lam \cdot q$	$Lam \cdot q = P2$
23	$S3 = P2 - P1$	$S3 + P1 = P2$
24	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
25	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
26	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
27	$S = g + Sq4$	$g + Sq4 = S$
28	$n2 = n \cdot n$	$n \cdot n = n2$
29	$n2n = n2 - n$	$n2n + n = n2$
30	$SA = S \cdot n2n$	$S \cdot n2n = SA$
31	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
32	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
33	$bm1 = b - 1$	$bm1 + 1 = b$
34	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
35	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$

No.	Assignment	Addition/multiplication check
36	$b5m2 = B - 4$	$b5m2 + 4 = B$
37	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
38	$mask_term = b5m2 \cdot packed_target_shift$	$b5m2 \cdot packed_target_shift = mask_term$
39	$T = T2 + mask_term$	$T2 + mask_term = T$
40	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
41	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
42	$R7 = SA + TB$	$SA + TB = R7$
43	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
44	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
45	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
46	$R12 = UM + sn2$	$UM + sn2 = R12$
47	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
48	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
49	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
50	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
51	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
52	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
53	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
54	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
55	$R10b = eta + zeta$	$eta + zeta = R10b$
56	$r1 = r + 1$	$r + 1 = r1$
57	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
58	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
59	$tr1 = r1 + r$	$r1 + r = tr1$
60	$R13 = ka + phi$	$ka + phi = R13$
61	$cam2 = c \cdot a$	$c \cdot a = cam2$
62	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
63	$am1 = a + 3$	$a + 3 = am1$
64	$a4 = 8 \cdot a$	$8 \cdot a = a4$
65	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
66	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
67	$R14 = D1 + gam$	$D1 + gam = R14$
68	$ap1 = am1 + 2$	$am1 + 2 = ap1$
69	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
70	$c2 = c \cdot c$	$c \cdot c = c2$
71	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
72	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
73	$L15 = d \cdot d$	$d \cdot d = L15$
74	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
75	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
76	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
77	$R16 = AE + 1$	$AE + 1 = R16$
78	$L16 = f \cdot f$	$f \cdot f = L16$
79	$of = o \cdot f$	$o \cdot f = of$
80	$dof = of - d$	$dof + d = of$
81	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
82	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
83	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
84	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
85	$jc = j \cdot c$	$j \cdot c = jc$
86	$H17 = tr1 + jc$	$tr1 + jc = H17$
87	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
88	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$

No.	Assignment	Addition/multiplication check
89	$R17 = P17 + 1$	$P17 + 1 = R17$
90	$amb = a - b5m2$	$amb + b5m2 = a$
91	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
92	$qk = q + kamb$	$q + kamb = qk$
93	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
94	$Mod = A - amb2$	$Mod + amb2 = A$
95	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
96	$R18 = qk + rM$	$qk + rM = R18$
97	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
98	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
99	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
100	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
101	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
102	$R20 = Dam1 + 2^{32}$	$Dam1 + 2^{32} = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

N The 101-instruction certificate

Use the primitive circuit index (V, H, L) of Section 21. Its digit parameter is the universal constant $Z = 4$, and the power-of-two length L is chosen once for the represented set. The supplied parameters are x, V, H, L ; the 34 positive unknowns are unchanged. The existing equation $\theta + 4 = B$ allows the register θ to supply both the third mask and the shift $a - \theta = A - B$ in the second exponent test. Thus the separate subtraction $B - 4$ disappears.

The complete checker `round23_1980_fixed_four_certificate.py` verifies every primitive, all 22 source residuals and all three triangular corrections. The new corrections account for reusing θ in the packed equation and the second exponential congruence. The full positive-domain proof is Theorem 21.1; the arbitrary number of compiled circuit coordinates changes only the index, not this table.

The generator `round23_1980_schedule_table.py` reruns that checker. There are 56 multiplications and 45 additions. Fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + 4$	$th + 4 = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = 4 \cdot la$	$4 \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = zl - twice_e$	$t2 + twice_e = zl$
18	$C = x + g$	$x + g = C$
19	$C2 = C \cdot C$	$C \cdot C = C2$
20	$L1 = S2 + al$	$S2 + al = L1$
21	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
22	$P2 = Lam \cdot q$	$Lam \cdot q = P2$
23	$S3 = P2 - P1$	$S3 + P1 = P2$
24	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
25	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
26	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
27	$S = g + Sq4$	$g + Sq4 = S$
28	$n2 = n \cdot n$	$n \cdot n = n2$
29	$n2n = n2 - n$	$n2n + n = n2$
30	$SA = S \cdot n2n$	$S \cdot n2n = SA$
31	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
32	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
33	$bm1 = b - 1$	$bm1 + 1 = b$
34	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
35	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
36	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$

No.	Assignment	Addition/multiplication check
37	$mask_term = th \cdot packed_target_shift$	$th \cdot packed_target_shift = mask_term$
38	$T = T2 + mask_term$	$T2 + mask_term = T$
39	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
40	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
41	$R7 = SA + TB$	$SA + TB = R7$
42	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
43	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
44	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
45	$R12 = UM + sn2$	$UM + sn2 = R12$
46	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
47	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
48	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
49	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
50	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
51	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
52	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
53	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
54	$R10b = eta + zeta$	$eta + zeta = R10b$
55	$r1 = r + 1$	$r + 1 = r1$
56	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
57	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
58	$tr1 = r1 + r$	$r1 + r = tr1$
59	$R13 = ka + phi$	$ka + phi = R13$
60	$cam2 = c \cdot a$	$c \cdot a = cam2$
61	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
62	$am1 = a + 3$	$a + 3 = am1$
63	$a4 = 8 \cdot a$	$8 \cdot a = a4$
64	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
65	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
66	$R14 = D1 + gam$	$D1 + gam = R14$
67	$ap1 = am1 + 2$	$am1 + 2 = ap1$
68	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
69	$c2 = c \cdot c$	$c \cdot c = c2$
70	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
71	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
72	$L15 = d \cdot d$	$d \cdot d = L15$
73	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
74	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
75	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
76	$R16 = AE + 1$	$AE + 1 = R16$
77	$L16 = f \cdot f$	$f \cdot f = L16$
78	$of = o \cdot f$	$o \cdot f = of$
79	$dof = of - d$	$dof + d = of$
80	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
81	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
82	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
83	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
84	$jc = j \cdot c$	$j \cdot c = jc$
85	$H17 = tr1 + jc$	$tr1 + jc = H17$
86	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
87	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
88	$R17 = P17 + 1$	$P17 + 1 = R17$
89	$amb = a - th$	$amb + th = a$

No.	Assignment	Addition/multiplication check
90	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
91	$qk = q + kamb$	$q + kamb = qk$
92	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
93	$Mod = A - amb2$	$Mod + amb2 = A$
94	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
95	$R18 = qk + rM$	$qk + rM = R18$
96	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
97	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
98	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
99	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
100	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
101	$R20 = Dam1 + L$	$Dam1 + L = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

O The 100-instruction doubled-index Pell certificate

Use the unpaired circuit index of Section 21, replacing its supplied component L by $T_L = \psi_4(L)$. The doubled-index norm and the congruence $\kappa = T_L + \Delta a$ are proved in Section 22. Their joint effect removes the last uses of $A - 1$ and $A + 1$, so $A^2 - 1$ is computed as $a^2 + (8a + 15)$. The register named **A** below holds this norm coefficient; **dof** holds the possibly signed $v = of - d^2$; **Gminus1** and **Gplus1** hold $K = (A^2 - 1)(f^2 - 1)$ and $K - 1$.

The complete checker `round24_1980_doubled_pell_certificate.py` verifies 100 primitives, 22 source residuals and three corrections. There are 56 multiplications and 44 additions, with 34 positive unknowns and supplied parameters x, V, H, T_L .

This and the following two tables are generated by `round24_26_1980_schedule_tables.py`, which reruns each exact checker. Fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$L2 = la + q2$	$la + q2 = L2$
8	$Lam = la \cdot B$	$la \cdot B = Lam$
9	$R2 = Lam + 1$	$Lam + 1 = R2$
10	$L3 = th + 4$	$th + 4 = L3$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
14	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
15	$zl = 4 \cdot la$	$4 \cdot la = zl$
16	$twice_e = 2 \cdot e$	$2 \cdot e = twice_e$
17	$t2 = zl - twice_e$	$zl + twice_e = t2$
18	$C = x + g$	$x + g = C$
19	$C2 = C \cdot C$	$C \cdot C = C2$
20	$L1 = S2 + al$	$S2 + al = L1$
21	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
22	$P2 = Lam \cdot q$	$Lam \cdot q = P2$
23	$S3 = P2 - P1$	$S3 + P1 = P2$
24	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
25	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
26	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
27	$S = g + Sq4$	$g + Sq4 = S$
28	$n2 = n \cdot n$	$n \cdot n = n2$
29	$n2n = n2 - n$	$n2n + n = n2$
30	$SA = S \cdot n2n$	$S \cdot n2n = SA$
31	$Tcoef = L2 - zl$	$Tcoef + zl = L2$
32	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
33	$bm1 = b - 1$	$bm1 + 1 = b$
34	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
35	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
36	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
37	$mask_term = th \cdot packed_target_shift$	$th \cdot packed_target_shift = mask_term$
38	$T = T2 + mask_term$	$T2 + mask_term = T$

No.	Assignment	Addition/multiplication check
39	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
40	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
41	$R7 = SA + TB$	$SA + TB = R7$
42	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
43	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
44	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
45	$R12 = UM + sn2$	$UM + sn2 = R12$
46	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
47	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
48	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
49	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
50	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
51	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
52	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
53	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
54	$R10b = eta + zeta$	$eta + zeta = R10b$
55	$r1 = r + 1$	$r + 1 = r1$
56	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
57	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
58	$tr1 = r1 + r$	$r1 + r = tr1$
59	$R13 = ka + phi$	$ka + phi = R13$
60	$cam2 = c \cdot a$	$c \cdot a = cam2$
61	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
62	$a4 = 8 \cdot a$	$8 \cdot a = a4$
63	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
64	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
65	$R14 = D1 + gam$	$D1 + gam = R14$
66	$a_square = a \cdot a$	$a \cdot a = a_square$
67	$A = a_square + a4m5$	$a_square + a4m5 = A$
68	$c2 = c \cdot c$	$c \cdot c = c2$
69	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
70	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
71	$L15 = d \cdot d$	$d \cdot d = L15$
72	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
73	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
74	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
75	$R16 = AE + 1$	$AE + 1 = R16$
76	$L16 = f \cdot f$	$f \cdot f = L16$
77	$of = o \cdot f$	$o \cdot f = of$
78	$dof = of - L15$	$dof + L15 = of$
79	$vplus1 = dof + 1$	$dof + 1 = vplus1$
80	$L17 = dof \cdot vplus1$	$dof \cdot vplus1 = L17$
81	$Gminus1 = A \cdot AE$	$A \cdot AE = Gminus1$
82	$Gplus1 = Gminus1 - 1$	$Gplus1 + 1 = Gminus1$
83	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
84	$jc = j \cdot c$	$j \cdot c = jc$
85	$H17 = tr1 + jc$	$tr1 + jc = H17$
86	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
87	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
88	$amb = a - th$	$amb + th = a$
89	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
90	$qk = q + kamb$	$q + kamb = qk$
91	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$

No.	Assignment	Addition/multiplication check
92	$Mod = A - amb2$	$Mod + amb2 = A$
93	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
94	$R18 = qk + rM$	$qk + rM = R18$
95	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
96	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
97	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
98	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
99	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
100	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & L2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

P The 100-instruction shared-geometric-factor certificate

Use the paired circuit index (V, H, L) of Section 23, with its first special target and two unit constraints. The coefficient is $\Omega = 2\lambda - e$, and the third block is $S_3 = \theta\lambda q - \Omega\mathcal{C}^2$. The product $\theta\lambda$ also enters the geometric equation and the middle mask. Seven instructions replace eight, saving one multiplication.

The complete checker `round25_1980_half_center_certificate.py` verifies every primitive and source residual, including the changed geometric test. Its correction uses the exact equality $\theta + 4 = B$; the latter is used first in the proof order, regardless of its position in the displayed list. There are 55 multiplications and 45 additions, with 34 positive unknowns and 22 equality tests.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + beta$	$x + beta = bx$
7	$theta_lambda = th \cdot la$	$th \cdot la = theta_lambda$
8	$twice_lambda = 2 \cdot la$	$2 \cdot la = twice_lambda$
9	$Tcoef = theta_lambda + 1$	$theta_lambda + 1 = Tcoef$
10	$three_lambda = twice_lambda + la$	$twice_lambda + la = three_lambda$
11	$R2 = Tcoef + three_lambda$	$Tcoef + three_lambda = R2$
12	$t2 = twice_lambda - e$	$t2 + e = twice_lambda$
13	$P2 = theta_lambda \cdot q$	$theta_lambda \cdot q = P2$
14	$L3 = th + 4$	$th + 4 = L3$
15	$eq = e \cdot q$	$e \cdot q = eq$
16	$S2 = l + eq$	$l + eq = S2$
17	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
18	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
19	$C = x + g$	$x + g = C$
20	$C2 = C \cdot C$	$C \cdot C = C2$
21	$L1 = S2 + al$	$S2 + al = L1$
22	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
23	$S3 = P2 - P1$	$S3 + P1 = P2$
24	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
25	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
26	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
27	$S = g + Sq4$	$g + Sq4 = S$
28	$n2 = n \cdot n$	$n \cdot n = n2$
29	$n2n = n2 - n$	$n2n + n = n2$
30	$SA = S \cdot n2n$	$S \cdot n2n = SA$
31	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
32	$bm1 = b - 1$	$bm1 + 1 = b$
33	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
34	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
35	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
36	$mask_term = th \cdot packed_target_shift$	$th \cdot packed_target_shift = mask_term$
37	$T = T2 + mask_term$	$T2 + mask_term = T$
38	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
39	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
40	$R7 = SA + TB$	$SA + TB = R7$
41	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
42	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$

No.	Assignment	Addition/multiplication check
43	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
44	$R12 = UM + sn2$	$UM + sn2 = R12$
45	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
46	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
47	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
48	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
49	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
50	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
51	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
52	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
53	$R10b = eta + zeta$	$eta + zeta = R10b$
54	$r1 = r + 1$	$r + 1 = r1$
55	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
56	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
57	$tr1 = r1 + r$	$r1 + r = tr1$
58	$R13 = ka + phi$	$ka + phi = R13$
59	$cam2 = c \cdot a$	$c \cdot a = cam2$
60	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
61	$am1 = a + 3$	$a + 3 = am1$
62	$a4 = 8 \cdot a$	$8 \cdot a = a4$
63	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
64	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
65	$R14 = D1 + gam$	$D1 + gam = R14$
66	$ap1 = am1 + 2$	$am1 + 2 = ap1$
67	$A = am1 \cdot ap1$	$am1 \cdot ap1 = A$
68	$c2 = c \cdot c$	$c \cdot c = c2$
69	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
70	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
71	$L15 = d \cdot d$	$d \cdot d = L15$
72	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
73	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
74	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
75	$R16 = AE + 1$	$AE + 1 = R16$
76	$L16 = f \cdot f$	$f \cdot f = L16$
77	$of = o \cdot f$	$o \cdot f = of$
78	$dof = of - d$	$dof + d = of$
79	$L17 = dof \cdot dof$	$dof \cdot dof = L17$
80	$Gminus1 = ap1 \cdot AE$	$ap1 \cdot AE = Gminus1$
81	$Gplus1 = Gminus1 + 2$	$Gminus1 + 2 = Gplus1$
82	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
83	$jc = j \cdot c$	$j \cdot c = jc$
84	$H17 = tr1 + jc$	$tr1 + jc = H17$
85	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
86	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
87	$R17 = P17 + 1$	$P17 + 1 = R17$
88	$amb = a - th$	$amb + th = a$
89	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
90	$qk = q + kamb$	$q + kamb = qk$
91	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
92	$Mod = A - amb2$	$Mod + amb2 = A$
93	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
94	$R18 = qk + rM$	$qk + rM = R18$
95	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$

No.	Assignment	Addition/multiplication check
96	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
97	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
98	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
99	$Dam1 = Delta \cdot am1$	$Delta \cdot am1 = Dam1$
100	$R20 = Dam1 + L$	$Dam1 + L = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & q2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= R17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

Q The 99-instruction composed certificate

Use the paired, special-first coefficient construction and supply its underlying length through $T_L = \psi_4(L)$. The supplied parameters are x, V, H, T_L and the 34 positive unknowns are unchanged. Combine both changes of the preceding appendices; Theorem 24.1 supplies the complete positive-domain composition proof.

The standalone checker `round26_1980_composed_certificate.py` verifies all 99 instructions and 22 source residuals, their three acyclic corrections, and both orders of the full arithmetic and source transformations. Only the packed equation, auxiliary signed norm, fixed index equation, positive S_3 equation and positive Ω equation change after renaming the fixed index. The resulting count is 55 multiplications and 44 additions. Fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q^2 = q \cdot q$	$q \cdot q = q^2$
2	$q^4 = q^2 \cdot q^2$	$q^2 \cdot q^2 = q^4$
3	$q^8 = q^4 \cdot q^4$	$q^4 \cdot q^4 = q^8$
4	$b^2 = b \cdot b$	$b \cdot b = b^2$
5	$B = H \cdot b^2$	$H \cdot b^2 = B$
6	$bx = x + \text{beta}$	$x + \text{beta} = bx$
7	$\text{theta_lambda} = \text{th} \cdot \text{la}$	$\text{th} \cdot \text{la} = \text{theta_lambda}$
8	$\text{twice_lambda} = 2 \cdot \text{la}$	$2 \cdot \text{la} = \text{twice_lambda}$
9	$\text{Tcoef} = \text{theta_lambda} + 1$	$\text{theta_lambda} + 1 = \text{Tcoef}$
10	$\text{three_lambda} = \text{twice_lambda} + \text{la}$	$\text{twice_lambda} + \text{la} = \text{three_lambda}$
11	$R^2 = \text{Tcoef} + \text{three_lambda}$	$\text{Tcoef} + \text{three_lambda} = R^2$
12	$t^2 = \text{twice_lambda} - e$	$t^2 + e = \text{twice_lambda}$
13	$P^2 = \text{theta_lambda} \cdot q$	$\text{theta_lambda} \cdot q = P^2$
14	$L^3 = \text{th} + 4$	$\text{th} + 4 = L^3$
15	$eq = e \cdot q$	$e \cdot q = eq$
16	$S^2 = l + eq$	$l + eq = S^2$
17	$\text{pack_tth} = t \cdot \text{th}$	$t \cdot \text{th} = \text{pack_tth}$
18	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
19	$C = x + g$	$x + g = C$
20	$C^2 = C \cdot C$	$C \cdot C = C^2$
21	$L^1 = S^2 + al$	$S^2 + al = L^1$
22	$P^1 = t^2 \cdot C^2$	$t^2 \cdot C^2 = P^1$
23	$S^3 = P^2 - P^1$	$S^3 + P^1 = P^2$
24	$S^3q^4 = S^3 \cdot q^2$	$S^3 \cdot q^2 = S^3q^4$
25	$\text{Sin} = S^2 + S^3q^4$	$S^2 + S^3q^4 = \text{Sin}$
26	$Sq^4 = \text{Sin} \cdot q^2$	$\text{Sin} \cdot q^2 = Sq^4$
27	$S = g + Sq^4$	$g + Sq^4 = S$
28	$n^2 = n \cdot n$	$n \cdot n = n^2$
29	$n^2n = n^2 - n$	$n^2n + n = n^2$
30	$SA = S \cdot n^2n$	$S \cdot n^2n = SA$
31	$Tq^4 = \text{Tcoef} \cdot q^2$	$\text{Tcoef} \cdot q^2 = Tq^4$
32	$bm^1 = b - 1$	$bm^1 + 1 = b$
33	$lbm^1 = l \cdot bm^1$	$l \cdot bm^1 = lbm^1$
34	$T^2 = Tq^4 - lbm^1$	$T^2 + lbm^1 = Tq^4$
35	$\text{packed_target_shift} = l \cdot q^4$	$l \cdot q^4 = \text{packed_target_shift}$
36	$\text{mask_term} = \text{th} \cdot \text{packed_target_shift}$	$\text{th} \cdot \text{packed_target_shift} = \text{mask_term}$
37	$T = T^2 + \text{mask_term}$	$T^2 + \text{mask_term} = T$
38	$n^2m^1 = n^2 - 1$	$n^2m^1 + 1 = n^2$
39	$TB = T \cdot n^2m^1$	$T \cdot n^2m^1 = TB$
40	$R^7 = SA + TB$	$SA + TB = R^7$

No.	Assignment	Addition/multiplication check
41	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
42	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
43	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
44	$R12 = UM + sn2$	$UM + sn2 = R12$
45	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
46	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
47	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
48	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
49	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
50	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
51	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
52	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
53	$R10b = eta + zeta$	$eta + zeta = R10b$
54	$r1 = r + 1$	$r + 1 = r1$
55	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
56	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
57	$tr1 = r1 + r$	$r1 + r = tr1$
58	$R13 = ka + phi$	$ka + phi = R13$
59	$cam2 = c \cdot a$	$c \cdot a = cam2$
60	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
61	$a4 = 8 \cdot a$	$8 \cdot a = a4$
62	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
63	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
64	$R14 = D1 + gam$	$D1 + gam = R14$
65	$a_square = a \cdot a$	$a \cdot a = a_square$
66	$A = a_square + a4m5$	$a_square + a4m5 = A$
67	$c2 = c \cdot c$	$c \cdot c = c2$
68	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
69	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
70	$L15 = d \cdot d$	$d \cdot d = L15$
71	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
72	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
73	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
74	$R16 = AE + 1$	$AE + 1 = R16$
75	$L16 = f \cdot f$	$f \cdot f = L16$
76	$of = o \cdot f$	$o \cdot f = of$
77	$dof = of - L15$	$dof + L15 = of$
78	$vplus1 = dof + 1$	$dof + 1 = vplus1$
79	$L17 = dof \cdot vplus1$	$dof \cdot vplus1 = L17$
80	$Gminus1 = A \cdot AE$	$A \cdot AE = Gminus1$
81	$Gplus1 = Gminus1 - 1$	$Gplus1 + 1 = Gminus1$
82	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
83	$jc = j \cdot c$	$j \cdot c = jc$
84	$H17 = tr1 + jc$	$tr1 + jc = H17$
85	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
86	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
87	$amb = a - th$	$amb + th = a$
88	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
89	$qk = q + kamb$	$q + kamb = qk$
90	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
91	$Mod = A - amb2$	$Mod + amb2 = A$
92	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
93	$R18 = qk + rM$	$qk + rM = R18$

No.	Assignment	Addition/multiplication check
94	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
95	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
96	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
97	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
98	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
99	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & q2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

R The 98-instruction unit-centered certificate

Use the complete rebuilt index of Section 25, including the copied unit coordinates and the direct guard. The coefficient digits are centered at one, with $\Omega = \lambda - e > 0$ and $S_3 = \theta\lambda q - \Omega\mathcal{C}^2 > 0$. Computing 3λ directly reduces the shared geometric block from seven instructions to six. The result has 55 multiplications and 43 additions, with 34 positive unknowns and 22 equality tests. All Pell source equations are retained.

The standalone checker `round27_1980_unit_center_certificate.py` verifies every primitive and source residual. The positive-domain equivalence and the rebuilt fixed index are proved in Theorem 25.1. This and the following two tables are generated by `round27_29_1980_schedule_tables.py`, which reruns all three exact checkers. Fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q^2 = q \cdot q$	$q \cdot q = q^2$
2	$q^4 = q^2 \cdot q^2$	$q^2 \cdot q^2 = q^4$
3	$q^8 = q^4 \cdot q^4$	$q^4 \cdot q^4 = q^8$
4	$b^2 = b \cdot b$	$b \cdot b = b^2$
5	$B = H \cdot b^2$	$H \cdot b^2 = B$
6	$bx = x + \text{beta}$	$x + \text{beta} = bx$
7	$\text{theta_lambda} = \text{th} \cdot \text{la}$	$\text{th} \cdot \text{la} = \text{theta_lambda}$
8	$\text{Tcoef} = \text{theta_lambda} + 1$	$\text{theta_lambda} + 1 = \text{Tcoef}$
9	$\text{three_lambda} = 3 \cdot \text{la}$	$3 \cdot \text{la} = \text{three_lambda}$
10	$R^2 = \text{Tcoef} + \text{three_lambda}$	$\text{Tcoef} + \text{three_lambda} = R^2$
11	$t^2 = \text{la} - e$	$t^2 + e = \text{la}$
12	$P^2 = \text{theta_lambda} \cdot q$	$\text{theta_lambda} \cdot q = P^2$
13	$L^3 = \text{th} + 4$	$\text{th} + 4 = L^3$
14	$eq = e \cdot q$	$e \cdot q = eq$
15	$S^2 = l + eq$	$l + eq = S^2$
16	$\text{pack_tth} = t \cdot \text{th}$	$t \cdot \text{th} = \text{pack_tth}$
17	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
18	$C = x + g$	$x + g = C$
19	$C^2 = C \cdot C$	$C \cdot C = C^2$
20	$L^1 = S^2 + al$	$S^2 + al = L^1$
21	$P^1 = t^2 \cdot C^2$	$t^2 \cdot C^2 = P^1$
22	$S^3 = P^2 - P^1$	$S^3 + P^1 = P^2$
23	$S^3q^4 = S^3 \cdot q^2$	$S^3 \cdot q^2 = S^3q^4$
24	$\text{Sin} = S^2 + S^3q^4$	$S^2 + S^3q^4 = \text{Sin}$
25	$Sq^4 = \text{Sin} \cdot q^2$	$\text{Sin} \cdot q^2 = Sq^4$
26	$S = g + Sq^4$	$g + Sq^4 = S$
27	$n^2 = n \cdot n$	$n \cdot n = n^2$
28	$n^2n = n^2 - n$	$n^2n + n = n^2$
29	$SA = S \cdot n^2n$	$S \cdot n^2n = SA$
30	$Tq^4 = \text{Tcoef} \cdot q^2$	$\text{Tcoef} \cdot q^2 = Tq^4$
31	$bm^1 = b - 1$	$bm^1 + 1 = b$
32	$lbm^1 = l \cdot bm^1$	$l \cdot bm^1 = lbm^1$
33	$T^2 = Tq^4 - lbm^1$	$T^2 + lbm^1 = Tq^4$
34	$\text{packed_target_shift} = l \cdot q^4$	$l \cdot q^4 = \text{packed_target_shift}$
35	$\text{mask_term} = \text{th} \cdot \text{packed_target_shift}$	$\text{th} \cdot \text{packed_target_shift} = \text{mask_term}$
36	$T = T^2 + \text{mask_term}$	$T^2 + \text{mask_term} = T$
37	$n^2m^1 = n^2 - 1$	$n^2m^1 + 1 = n^2$
38	$TB = T \cdot n^2m^1$	$T \cdot n^2m^1 = TB$
39	$R^7 = SA + TB$	$SA + TB = R^7$
40	$wn^2 = w \cdot n^2$	$w \cdot n^2 = wn^2$

No.	Assignment	Addition/multiplication check
41	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
42	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
43	$R12 = UM + sn2$	$UM + sn2 = R12$
44	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
45	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
46	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
47	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
48	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
49	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
50	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
51	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
52	$R10b = eta + zeta$	$eta + zeta = R10b$
53	$r1 = r + 1$	$r + 1 = r1$
54	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
55	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
56	$tr1 = r1 + r$	$r1 + r = tr1$
57	$R13 = ka + phi$	$ka + phi = R13$
58	$cam2 = c \cdot a$	$c \cdot a = cam2$
59	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
60	$a4 = 8 \cdot a$	$8 \cdot a = a4$
61	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
62	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
63	$R14 = D1 + gam$	$D1 + gam = R14$
64	$a_square = a \cdot a$	$a \cdot a = a_square$
65	$A = a_square + a4m5$	$a_square + a4m5 = A$
66	$c2 = c \cdot c$	$c \cdot c = c2$
67	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
68	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
69	$L15 = d \cdot d$	$d \cdot d = L15$
70	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
71	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
72	$AE = A \cdot ic22$	$A \cdot ic22 = AE$
73	$R16 = AE + 1$	$AE + 1 = R16$
74	$L16 = f \cdot f$	$f \cdot f = L16$
75	$of = o \cdot f$	$o \cdot f = of$
76	$dof = of - L15$	$dof + L15 = of$
77	$vplus1 = dof + 1$	$dof + 1 = vplus1$
78	$L17 = dof \cdot vplus1$	$dof \cdot vplus1 = L17$
79	$Gminus1 = A \cdot AE$	$A \cdot AE = Gminus1$
80	$Gplus1 = Gminus1 - 1$	$Gplus1 + 1 = Gminus1$
81	$G2m1 = Gminus1 \cdot Gplus1$	$Gminus1 \cdot Gplus1 = G2m1$
82	$jc = j \cdot c$	$j \cdot c = jc$
83	$H17 = tr1 + jc$	$tr1 + jc = H17$
84	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
85	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
86	$amb = a - th$	$amb + th = a$
87	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
88	$qk = q + kamb$	$q + kamb = qk$
89	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
90	$Mod = A - amb2$	$Mod + amb2 = A$
91	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
92	$R18 = qk + rM$	$qk + rM = R18$
93	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$

No.	Assignment	Addition/multiplication check
94	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
95	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
96	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
97	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
98	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & q2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
L16 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

S The 98-instruction relaxed-Pell certificate

Retain the index and encoding of Section 24 and replace only the auxiliary norm by $(ic^2)^2 = D(f^2 - 1)$, where $D = (a + 4)^2 - 1$. The existing square $(ic^2)^2$ then supplies the parameter in the signed auxiliary norm directly. Theorem 26.2 proves recovery of the required integer Pell sequence and auxiliary index divisibility from the existing bounds, and constructs all positive witnesses.

The standalone checker `round28_1980_relaxed_auxiliary_certificate.py` verifies all 98 instructions and all 22 source residuals. Its changed signed-norm correction is $-F_{16}(K_{\text{source}} + K_{\text{calc}} - 1)(2r + 1 + jc)^2$; F_{16} is checked exactly first. The result has 54 multiplications and 44 additions. No unknown or equality test is added.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$b2 = b \cdot b$	$b \cdot b = b2$
5	$B = H \cdot b2$	$H \cdot b2 = B$
6	$bx = x + \text{beta}$	$x + \text{beta} = bx$
7	$\text{theta_lambda} = th \cdot la$	$th \cdot la = \text{theta_lambda}$
8	$\text{twice_lambda} = 2 \cdot la$	$2 \cdot la = \text{twice_lambda}$
9	$\text{Tcoef} = \text{theta_lambda} + 1$	$\text{theta_lambda} + 1 = \text{Tcoef}$
10	$\text{three_lambda} = \text{twice_lambda} + la$	$\text{twice_lambda} + la = \text{three_lambda}$
11	$R2 = \text{Tcoef} + \text{three_lambda}$	$\text{Tcoef} + \text{three_lambda} = R2$
12	$t2 = \text{twice_lambda} - e$	$t2 + e = \text{twice_lambda}$
13	$P2 = \text{theta_lambda} \cdot q$	$\text{theta_lambda} \cdot q = P2$
14	$L3 = th + 4$	$th + 4 = L3$
15	$eq = e \cdot q$	$e \cdot q = eq$
16	$S2 = l + eq$	$l + eq = S2$
17	$\text{pack_tth} = t \cdot th$	$t \cdot th = \text{pack_tth}$
18	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
19	$C = x + g$	$x + g = C$
20	$C2 = C \cdot C$	$C \cdot C = C2$
21	$L1 = S2 + al$	$S2 + al = L1$
22	$P1 = t2 \cdot C2$	$t2 \cdot C2 = P1$
23	$S3 = P2 - P1$	$S3 + P1 = P2$
24	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
25	$\text{Sin} = S2 + S3q4$	$S2 + S3q4 = \text{Sin}$
26	$Sq4 = \text{Sin} \cdot q2$	$\text{Sin} \cdot q2 = Sq4$
27	$S = g + Sq4$	$g + Sq4 = S$
28	$n2 = n \cdot n$	$n \cdot n = n2$
29	$n2n = n2 - n$	$n2n + n = n2$
30	$SA = S \cdot n2n$	$S \cdot n2n = SA$
31	$Tq4 = \text{Tcoef} \cdot q2$	$\text{Tcoef} \cdot q2 = Tq4$
32	$bm1 = b - 1$	$bm1 + 1 = b$
33	$lbm1 = l \cdot bm1$	$l \cdot bm1 = lbm1$
34	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$
35	$\text{packed_target_shift} = l \cdot q4$	$l \cdot q4 = \text{packed_target_shift}$
36	$\text{mask_term} = th \cdot \text{packed_target_shift}$	$th \cdot \text{packed_target_shift} = \text{mask_term}$
37	$T = T2 + \text{mask_term}$	$T2 + \text{mask_term} = T$
38	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
39	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
40	$R7 = SA + TB$	$SA + TB = R7$
41	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$

No.	Assignment	Addition/multiplication check
42	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
43	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
44	$R12 = UM + sn2$	$UM + sn2 = R12$
45	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
46	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
47	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
48	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
49	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
50	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
51	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
52	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
53	$R10b = eta + zeta$	$eta + zeta = R10b$
54	$r1 = r + 1$	$r + 1 = r1$
55	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
56	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
57	$tr1 = r1 + r$	$r1 + r = tr1$
58	$R13 = ka + phi$	$ka + phi = R13$
59	$cam2 = c \cdot a$	$c \cdot a = cam2$
60	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
61	$a4 = 8 \cdot a$	$8 \cdot a = a4$
62	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
63	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
64	$R14 = D1 + gam$	$D1 + gam = R14$
65	$a_square = a \cdot a$	$a \cdot a = a_square$
66	$A = a_square + a4m5$	$a_square + a4m5 = A$
67	$c2 = c \cdot c$	$c \cdot c = c2$
68	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
69	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
70	$L15 = d \cdot d$	$d \cdot d = L15$
71	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
72	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
73	$L16 = f \cdot f$	$f \cdot f = L16$
74	$f_square_minus_one = L16 - 1$	$f_square_minus_one + 1 = L16$
75	$R16 = A \cdot f_square_minus_one$	$A \cdot f_square_minus_one = R16$
76	$of = o \cdot f$	$o \cdot f = of$
77	$dof = of - L15$	$dof + L15 = of$
78	$vplus1 = dof + 1$	$dof + 1 = vplus1$
79	$L17 = dof \cdot vplus1$	$dof \cdot vplus1 = L17$
80	$Gplus1 = ic22 - 1$	$Gplus1 + 1 = ic22$
81	$G2m1 = ic22 \cdot Gplus1$	$ic22 \cdot Gplus1 = G2m1$
82	$jc = j \cdot c$	$j \cdot c = jc$
83	$H17 = tr1 + jc$	$tr1 + jc = H17$
84	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
85	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
86	$amb = a - th$	$amb + th = a$
87	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
88	$qk = q + kamb$	$q + kamb = qk$
89	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
90	$Mod = A - amb2$	$Mod + amb2 = A$
91	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
92	$R18 = qk + rM$	$qk + rM = R18$
93	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
94	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$

No.	Assignment	Addition/multiplication check
95	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
96	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
97	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
98	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & q2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

T The 97-instruction composed certificate

Combine the unit-centered index and direct guard with the relaxed auxiliary Pell norm. The supplied parameters are x, V, H, T_L , where $T_L = \psi_4(L)$, and there are 34 positive unknowns and 22 equations. The complete composition proof is Theorem 27.1.

The standalone checker `round29_1980_composed_certificate.py` checks all 97 primitives and 22 source residuals, their three acyclic corrections, and both orders of the full source and schedule changes. Relative to the 99-operation system, only the packed equation, auxiliary norm, and positive S_3 and Ω equations change. The resulting count is 54 multiplications and 43 additions. Fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q^2 = q \cdot q$	$q \cdot q = q^2$
2	$q^4 = q^2 \cdot q^2$	$q^2 \cdot q^2 = q^4$
3	$q^8 = q^4 \cdot q^4$	$q^4 \cdot q^4 = q^8$
4	$b^2 = b \cdot b$	$b \cdot b = b^2$
5	$B = H \cdot b^2$	$H \cdot b^2 = B$
6	$bx = x + \text{beta}$	$x + \text{beta} = bx$
7	$\text{theta_lambda} = \text{th} \cdot \text{la}$	$\text{th} \cdot \text{la} = \text{theta_lambda}$
8	$\text{Tcoef} = \text{theta_lambda} + 1$	$\text{theta_lambda} + 1 = \text{Tcoef}$
9	$\text{three_lambda} = 3 \cdot \text{la}$	$3 \cdot \text{la} = \text{three_lambda}$
10	$R^2 = \text{Tcoef} + \text{three_lambda}$	$\text{Tcoef} + \text{three_lambda} = R^2$
11	$t^2 = \text{la} - e$	$t^2 + e = \text{la}$
12	$P^2 = \text{theta_lambda} \cdot q$	$\text{theta_lambda} \cdot q = P^2$
13	$L^3 = \text{th} + 4$	$\text{th} + 4 = L^3$
14	$eq = e \cdot q$	$e \cdot q = eq$
15	$S^2 = l + eq$	$l + eq = S^2$
16	$\text{pack_tth} = t \cdot \text{th}$	$t \cdot \text{th} = \text{pack_tth}$
17	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
18	$C = x + g$	$x + g = C$
19	$C^2 = C \cdot C$	$C \cdot C = C^2$
20	$L^1 = S^2 + al$	$S^2 + al = L^1$
21	$P^1 = t^2 \cdot C^2$	$t^2 \cdot C^2 = P^1$
22	$S^3 = P^2 - P^1$	$S^3 + P^1 = P^2$
23	$S^3q^4 = S^3 \cdot q^2$	$S^3 \cdot q^2 = S^3q^4$
24	$\text{Sin} = S^2 + S^3q^4$	$S^2 + S^3q^4 = \text{Sin}$
25	$Sq^4 = \text{Sin} \cdot q^2$	$\text{Sin} \cdot q^2 = Sq^4$
26	$S = g + Sq^4$	$g + Sq^4 = S$
27	$n^2 = n \cdot n$	$n \cdot n = n^2$
28	$n^2n = n^2 - n$	$n^2n + n = n^2$
29	$SA = S \cdot n^2n$	$S \cdot n^2n = SA$
30	$Tq^4 = \text{Tcoef} \cdot q^2$	$\text{Tcoef} \cdot q^2 = Tq^4$
31	$bm^1 = b - 1$	$bm^1 + 1 = b$
32	$lbm^1 = l \cdot bm^1$	$l \cdot bm^1 = lbm^1$
33	$T^2 = Tq^4 - lbm^1$	$T^2 + lbm^1 = Tq^4$
34	$\text{packed_target_shift} = l \cdot q^4$	$l \cdot q^4 = \text{packed_target_shift}$
35	$\text{mask_term} = \text{th} \cdot \text{packed_target_shift}$	$\text{th} \cdot \text{packed_target_shift} = \text{mask_term}$
36	$T = T^2 + \text{mask_term}$	$T^2 + \text{mask_term} = T$
37	$n^2m^1 = n^2 - 1$	$n^2m^1 + 1 = n^2$
38	$TB = T \cdot n^2m^1$	$T \cdot n^2m^1 = TB$
39	$R^7 = SA + TB$	$SA + TB = R^7$
40	$wn^2 = w \cdot n^2$	$w \cdot n^2 = wn^2$
41	$sn^2 = s \cdot n^2$	$s \cdot n^2 = sn^2$

No.	Assignment	Addition/multiplication check
42	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
43	$R12 = UM + sn2$	$UM + sn2 = R12$
44	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
45	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
46	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
47	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
48	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
49	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
50	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
51	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
52	$R10b = eta + zeta$	$eta + zeta = R10b$
53	$r1 = r + 1$	$r + 1 = r1$
54	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
55	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
56	$tr1 = r1 + r$	$r1 + r = tr1$
57	$R13 = ka + phi$	$ka + phi = R13$
58	$cam2 = c \cdot a$	$c \cdot a = cam2$
59	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
60	$a4 = 8 \cdot a$	$8 \cdot a = a4$
61	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
62	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
63	$R14 = D1 + gam$	$D1 + gam = R14$
64	$a_square = a \cdot a$	$a \cdot a = a_square$
65	$A = a_square + a4m5$	$a_square + a4m5 = A$
66	$c2 = c \cdot c$	$c \cdot c = c2$
67	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
68	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
69	$L15 = d \cdot d$	$d \cdot d = L15$
70	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
71	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
72	$L16 = f \cdot f$	$f \cdot f = L16$
73	$f_square_minus_one = L16 - 1$	$f_square_minus_one + 1 = L16$
74	$R16 = A \cdot f_square_minus_one$	$A \cdot f_square_minus_one = R16$
75	$of = o \cdot f$	$o \cdot f = of$
76	$dof = of - L15$	$dof + L15 = of$
77	$vplus1 = dof + 1$	$dof + 1 = vplus1$
78	$L17 = dof \cdot vplus1$	$dof \cdot vplus1 = L17$
79	$Gplus1 = ic22 - 1$	$Gplus1 + 1 = ic22$
80	$G2m1 = ic22 \cdot Gplus1$	$ic22 \cdot Gplus1 = G2m1$
81	$jc = j \cdot c$	$j \cdot c = jc$
82	$H17 = tr1 + jc$	$tr1 + jc = H17$
83	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
84	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
85	$amb = a - th$	$amb + th = a$
86	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
87	$qk = q + kamb$	$q + kamb = qk$
88	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
89	$Mod = A - amb2$	$Mod + amb2 = A$
90	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
91	$R18 = qk + rM$	$qk + rM = R18$
92	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
93	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
94	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$

No.	Assignment	Addition/multiplication check
95	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
96	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
97	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & q2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

U The 96-instruction linear-radix certificate

Use the rebuilt index of Section 28, including the paired zero targets, both unit tests and their carry-reset padding. The supplied parameters are x, V, H, T_L , with $T_L = \psi_4(L)$. There are 35 positive unknowns and 23 equations; the additional unknown is α_2 , printed as *al2* in the table.

The radix is $B = Hb$, and the third packed block is $S_3 = (\lambda - e)(q - C^2)$ with $C = x + g$. The new positive bound $\ell + \alpha_2 = q$ removes the quotient alias. Theorem 28.1 proves the full encoding and positive-witness equivalence. Its unit construction is part of the fixed coefficient index and adds no arithmetic instruction to this table.

The standalone checker `round30_1980_linear_radix_certificate.py` verifies every primitive, all 23 source residuals and the three acyclic corrections. The geometric and second-exponent corrections use the independently exact $\theta + 4 = B$ equation; the signed auxiliary correction uses the exact relaxed norm. There are 52 multiplications and 44 additions. The separate sparse encoding checker verifies finite coefficient and carry regressions; it does not replace the general proof.

This table is generated by `round30_1980_schedule_table.py`, which reruns the complete arithmetic checker. Fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q^2 = q \cdot q$	$q \cdot q = q^2$
2	$q^4 = q^2 \cdot q^2$	$q^2 \cdot q^2 = q^4$
3	$q^8 = q^4 \cdot q^4$	$q^4 \cdot q^4 = q^8$
4	$B = H \cdot b$	$H \cdot b = B$
5	$bx = x + \text{beta}$	$x + \text{beta} = bx$
6	$\text{theta_lambda} = \text{th} \cdot \text{la}$	$\text{th} \cdot \text{la} = \text{theta_lambda}$
7	$\text{Tcoef} = \text{theta_lambda} + 1$	$\text{theta_lambda} + 1 = \text{Tcoef}$
8	$\text{three_lambda} = 3 \cdot \text{la}$	$3 \cdot \text{la} = \text{three_lambda}$
9	$R2 = \text{Tcoef} + \text{three_lambda}$	$\text{Tcoef} + \text{three_lambda} = R2$
10	$t^2 = \text{la} - e$	$t^2 + e = \text{la}$
11	$L3 = \text{th} + 4$	$\text{th} + 4 = L3$
12	$eq = e \cdot q$	$e \cdot q = eq$
13	$S2 = l + eq$	$l + eq = S2$
14	$\text{indicator_bound} = l + \text{al2}$	$l + \text{al2} = \text{indicator_bound}$
15	$\text{pack_tth} = t \cdot \text{th}$	$t \cdot \text{th} = \text{pack_tth}$
16	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
17	$C = x + g$	$x + g = C$
18	$C^2 = C \cdot C$	$C \cdot C = C^2$
19	$L1 = S2 + \text{al}$	$S2 + \text{al} = L1$
20	$\text{code_gap} = q - C^2$	$\text{code_gap} + C^2 = q$
21	$S3 = t^2 \cdot \text{code_gap}$	$t^2 \cdot \text{code_gap} = S3$
22	$S3q^4 = S3 \cdot q^2$	$S3 \cdot q^2 = S3q^4$
23	$\text{Sin} = S2 + S3q^4$	$S2 + S3q^4 = \text{Sin}$
24	$Sq^4 = \text{Sin} \cdot q^2$	$\text{Sin} \cdot q^2 = Sq^4$
25	$S = g + Sq^4$	$g + Sq^4 = S$
26	$n^2 = n \cdot n$	$n \cdot n = n^2$
27	$n2n = n^2 - n$	$n2n + n = n^2$
28	$SA = S \cdot n2n$	$S \cdot n2n = SA$
29	$Tq^4 = \text{Tcoef} \cdot q^2$	$\text{Tcoef} \cdot q^2 = Tq^4$
30	$\text{bm1} = b - 1$	$\text{bm1} + 1 = b$
31	$\text{lbm1} = l \cdot \text{bm1}$	$l \cdot \text{bm1} = \text{lbm1}$
32	$T2 = Tq^4 - \text{lbm1}$	$T2 + \text{lbm1} = Tq^4$
33	$\text{packed_target_shift} = l \cdot q^4$	$l \cdot q^4 = \text{packed_target_shift}$
34	$\text{mask_term} = \text{th} \cdot \text{packed_target_shift}$	$\text{th} \cdot \text{packed_target_shift} = \text{mask_term}$

No.	Assignment	Addition/multiplication check
35	$T = T2 + \text{mask_term}$	$T2 + \text{mask_term} = T$
36	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
37	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
38	$R7 = SA + TB$	$SA + TB = R7$
39	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
40	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
41	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
42	$R12 = UM + sn2$	$UM + sn2 = R12$
43	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
44	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
45	$\text{scaled_norm_coefficient} = UM2 + wn2$	$UM2 + wn2 = \text{scaled_norm_coefficient}$
46	$\text{ratio_product2} = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = \text{ratio_product2}$
47	$L9 = \text{scaled_norm_coefficient} \cdot \text{ratio_product2}$	$\text{scaled_norm_coefficient} \cdot \text{ratio_product2} = L9$
48	$\text{tauplus1} = \text{tau} + 1$	$\text{tau} + 1 = \text{tauplus1}$
49	$R9 = \text{tau} \cdot \text{tauplus1}$	$\text{tau} \cdot \text{tauplus1} = R9$
50	$R10a = ksn2 + \text{eta}$	$ksn2 + \text{eta} = R10a$
51	$R10b = \text{eta} + \text{zeta}$	$\text{eta} + \text{zeta} = R10b$
52	$r1 = r + 1$	$r + 1 = r1$
53	$\text{hpm1} = h \cdot UM$	$h \cdot UM = \text{hpm1}$
54	$R11 = r1 + \text{hpm1}$	$r1 + \text{hpm1} = R11$
55	$\text{tr1} = r1 + r$	$r1 + r = \text{tr1}$
56	$R13 = ka + \text{phi}$	$ka + \text{phi} = R13$
57	$\text{cam2} = c \cdot a$	$c \cdot a = \text{cam2}$
58	$D1 = wn2 + \text{cam2}$	$wn2 + \text{cam2} = D1$
59	$a4 = 8 \cdot a$	$8 \cdot a = a4$
60	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
61	$\text{gam} = ga \cdot a4m5$	$ga \cdot a4m5 = \text{gam}$
62	$R14 = D1 + \text{gam}$	$D1 + \text{gam} = R14$
63	$a_square = a \cdot a$	$a \cdot a = a_square$
64	$A = a_square + a4m5$	$a_square + a4m5 = A$
65	$c2 = c \cdot c$	$c \cdot c = c2$
66	$\text{Ac2} = A \cdot c2$	$A \cdot c2 = \text{Ac2}$
67	$R15 = \text{Ac2} + 1$	$\text{Ac2} + 1 = R15$
68	$L15 = d \cdot d$	$d \cdot d = L15$
69	$\text{ic2} = i \cdot c2$	$i \cdot c2 = \text{ic2}$
70	$\text{ic22} = \text{ic2} \cdot \text{ic2}$	$\text{ic2} \cdot \text{ic2} = \text{ic22}$
71	$L16 = f \cdot f$	$f \cdot f = L16$
72	$\text{f_square_minus_one} = L16 - 1$	$\text{f_square_minus_one} + 1 = L16$
73	$R16 = A \cdot \text{f_square_minus_one}$	$A \cdot \text{f_square_minus_one} = R16$
74	$\text{of} = o \cdot f$	$o \cdot f = \text{of}$
75	$\text{dof} = \text{of} - L15$	$\text{dof} + L15 = \text{of}$
76	$\text{vplus1} = \text{dof} + 1$	$\text{dof} + 1 = \text{vplus1}$
77	$L17 = \text{dof} \cdot \text{vplus1}$	$\text{dof} \cdot \text{vplus1} = L17$
78	$\text{Gplus1} = \text{ic22} - 1$	$\text{Gplus1} + 1 = \text{ic22}$
79	$\text{G2m1} = \text{ic22} \cdot \text{Gplus1}$	$\text{ic22} \cdot \text{Gplus1} = \text{G2m1}$
80	$\text{jc} = j \cdot c$	$j \cdot c = \text{jc}$
81	$\text{H17} = \text{tr1} + \text{jc}$	$\text{tr1} + \text{jc} = \text{H17}$
82	$\text{H2} = \text{H17} \cdot \text{H17}$	$\text{H17} \cdot \text{H17} = \text{H2}$
83	$\text{P17} = \text{G2m1} \cdot \text{H2}$	$\text{G2m1} \cdot \text{H2} = \text{P17}$
84	$\text{amb} = a - \text{th}$	$\text{amb} + \text{th} = a$
85	$\text{kamb} = ka \cdot \text{amb}$	$ka \cdot \text{amb} = \text{kamb}$
86	$\text{qk} = q + \text{kamb}$	$q + \text{kamb} = \text{qk}$
87	$\text{amb2} = \text{amb} \cdot \text{amb}$	$\text{amb} \cdot \text{amb} = \text{amb2}$

No.	Assignment	Addition/multiplication check
88	$Mod = A - amb2$	$Mod + amb2 = A$
89	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
90	$R18 = qk + rM$	$qk + rM = R18$
91	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
92	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
93	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
94	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
95	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
96	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-three free equality tests are

$$\begin{aligned}
L1 &= q2, & b &= bx, & q2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega, & indicator_bound &= q
\end{aligned}$$

V The 95-instruction affine-radix certificate

Use the rebuilt index of Section 29. The fixed parameter H is now $H_0 + 1$, where H_0 is a sufficiently large power of two. The supplied parameters are x, V, H, T_L , with $T_L = \psi_4(L)$. There are 34 positive unknowns and 22 equations.

Relative to Appendix U, combine the two code bounds as $\ell + e + \alpha = q$ and remove α_2 . This still costs two additions. Replace $B = Hb$ by $B = H + b$, and replace the coefficient $b - 1$ in the first mask by b . The resulting first mask forbids exactly the H_0 bit at each coordinate position. The three-way coordinate representation, three-digit target windows, reset padding, and independent five- and seven-fold padding carries all belong to the fixed index. Theorem 29.1 proves the complete equivalence.

The standalone checker `round32_1980_affine_radix_certificate.py` verifies all 95 primitive instructions and all 22 source residuals, including their three acyclic corrections. The geometric and second-exponent corrections use the independently exact $\theta + 4 = B$ equation; the signed auxiliary correction uses the exact relaxed norm. There are 51 multiplications and 44 additions.

The table is generated by `round32_1980_schedule_table.py`, which reruns the complete arithmetic checker. Fixed numerals and equality tests are free. The register name *lbm1* is retained from the preceding schedules; its new value is ℓb .

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$B = H + b$	$H + b = B$
5	$bx = x + beta$	$x + beta = bx$
6	$theta_lambda = th \cdot la$	$th \cdot la = theta_lambda$
7	$Tcoef = theta_lambda + 1$	$theta_lambda + 1 = Tcoef$
8	$three_lambda = 3 \cdot la$	$3 \cdot la = three_lambda$
9	$R2 = Tcoef + three_lambda$	$Tcoef + three_lambda = R2$
10	$t2 = la - e$	$t2 + e = la$
11	$L3 = th + 4$	$th + 4 = L3$
12	$eq = e \cdot q$	$e \cdot q = eq$
13	$S2 = l + eq$	$l + eq = S2$
14	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
15	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
16	$C = x + g$	$x + g = C$
17	$C2 = C \cdot C$	$C \cdot C = C2$
18	$bound_sum = l + e$	$l + e = bound_sum$
19	$L1 = bound_sum + al$	$bound_sum + al = L1$
20	$code_gap = q - C2$	$code_gap + C2 = q$
21	$S3 = t2 \cdot code_gap$	$t2 \cdot code_gap = S3$
22	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
23	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
24	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
25	$S = g + Sq4$	$g + Sq4 = S$
26	$n2 = n \cdot n$	$n \cdot n = n2$
27	$n2n = n2 - n$	$n2n + n = n2$
28	$SA = S \cdot n2n$	$S \cdot n2n = SA$
29	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
30	$lbm1 = l \cdot b$	$l \cdot b = lbm1$
31	$T2 = Tq4 - lbm1$	$T2 + lbm1 = Tq4$

No.	Assignment	Addition/multiplication check
32	$packed_target_shift = l \cdot q4$	$l \cdot q4 = packed_target_shift$
33	$mask_term = th \cdot packed_target_shift$	$th \cdot packed_target_shift = mask_term$
34	$T = T2 + mask_term$	$T2 + mask_term = T$
35	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
36	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
37	$R7 = SA + TB$	$SA + TB = R7$
38	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
39	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
40	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
41	$R12 = UM + sn2$	$UM + sn2 = R12$
42	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
43	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
44	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
45	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
46	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
47	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
48	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
49	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
50	$R10b = eta + zeta$	$eta + zeta = R10b$
51	$r1 = r + 1$	$r + 1 = r1$
52	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
53	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
54	$tr1 = r1 + r$	$r1 + r = tr1$
55	$R13 = ka + phi$	$ka + phi = R13$
56	$cam2 = c \cdot a$	$c \cdot a = cam2$
57	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
58	$a4 = 8 \cdot a$	$8 \cdot a = a4$
59	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
60	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
61	$R14 = D1 + gam$	$D1 + gam = R14$
62	$a_square = a \cdot a$	$a \cdot a = a_square$
63	$A = a_square + a4m5$	$a_square + a4m5 = A$
64	$c2 = c \cdot c$	$c \cdot c = c2$
65	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
66	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
67	$L15 = d \cdot d$	$d \cdot d = L15$
68	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
69	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
70	$L16 = f \cdot f$	$f \cdot f = L16$
71	$f_square_minus_one = L16 - 1$	$f_square_minus_one + 1 = L16$
72	$R16 = A \cdot f_square_minus_one$	$A \cdot f_square_minus_one = R16$
73	$of = o \cdot f$	$o \cdot f = of$
74	$dof = of - L15$	$dof + L15 = of$
75	$vplus1 = dof + 1$	$dof + 1 = vplus1$
76	$L17 = dof \cdot vplus1$	$dof \cdot vplus1 = L17$
77	$Gplus1 = ic22 - 1$	$Gplus1 + 1 = ic22$
78	$G2m1 = ic22 \cdot Gplus1$	$ic22 \cdot Gplus1 = G2m1$
79	$jc = j \cdot c$	$j \cdot c = jc$
80	$H17 = tr1 + jc$	$tr1 + jc = H17$
81	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
82	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
83	$amb = a - th$	$amb + th = a$
84	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$

No.	Assignment	Addition/multiplication check
85	$qk = q + kamb$	$q + kamb = qk$
86	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
87	$Mod = A - amb2$	$Mod + amb2 = A$
88	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
89	$R18 = qk + rM$	$qk + rM = R18$
90	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
91	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
92	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
93	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
94	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
95	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q, & b &= bx, & q2 &= R2, \\
L3 &= B, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

W The 94-instruction shifted-index certificate

Translate the fixed index of Appendix V by replacing H_{old} with the supplied positive integer $H = H_{\text{old}} - 4 = H_0 - 3$. The actual radix $B = H + b + 4$ and all 34 positive unknowns are unchanged. There are still 22 equations and three fixed index parameters besides the input. The complete positive-witness bijection is proved in Theorem 30.1.

The old additions $B_{\text{reg}} = H_{\text{old}} + b$ and $L_3 = \theta + 4$ served only the equality $L_3 = B_{\text{reg}}$. Retain the addition $\text{theta_sum} = H + b$ and compare it directly with θ . The mathematical radix is never separately constructed. Every other instruction and equality test is unchanged, so the complete table has 51 multiplications and 43 additions.

The standalone checker `round33_1980_shifted_affine_certificate.py` verifies every primitive and source residual, the complete fixed-parameter substitution, and the absence of downstream uses of the deleted register. The geometric and second-exponent corrections use the exact translated radix equation; the signed correction uses the exact relaxed norm. The 95-operation coefficient and carry regression applies without change.

This table is generated by `round33_1980_schedule_table.py`. Fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$\text{theta_sum} = H + b$	$H + b = \text{theta_sum}$
5	$bx = x + \text{beta}$	$x + \text{beta} = bx$
6	$\text{theta_lambda} = th \cdot la$	$th \cdot la = \text{theta_lambda}$
7	$\text{Tcoef} = \text{theta_lambda} + 1$	$\text{theta_lambda} + 1 = \text{Tcoef}$
8	$\text{three_lambda} = 3 \cdot la$	$3 \cdot la = \text{three_lambda}$
9	$R2 = \text{Tcoef} + \text{three_lambda}$	$\text{Tcoef} + \text{three_lambda} = R2$
10	$t2 = la - e$	$t2 + e = la$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$\text{pack_tth} = t \cdot th$	$t \cdot th = \text{pack_tth}$
14	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
15	$C = x + g$	$x + g = C$
16	$C2 = C \cdot C$	$C \cdot C = C2$
17	$\text{bound_sum} = l + e$	$l + e = \text{bound_sum}$
18	$L1 = \text{bound_sum} + al$	$\text{bound_sum} + al = L1$
19	$\text{code_gap} = q - C2$	$\text{code_gap} + C2 = q$
20	$S3 = t2 \cdot \text{code_gap}$	$t2 \cdot \text{code_gap} = S3$
21	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
22	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
23	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
24	$S = g + Sq4$	$g + Sq4 = S$
25	$n2 = n \cdot n$	$n \cdot n = n2$
26	$n2n = n2 - n$	$n2n + n = n2$
27	$SA = S \cdot n2n$	$S \cdot n2n = SA$
28	$Tq4 = \text{Tcoef} \cdot q2$	$\text{Tcoef} \cdot q2 = Tq4$
29	$\text{lbm1} = l \cdot b$	$l \cdot b = \text{lbm1}$
30	$T2 = Tq4 - \text{lbm1}$	$T2 + \text{lbm1} = Tq4$
31	$\text{packed_target_shift} = l \cdot q4$	$l \cdot q4 = \text{packed_target_shift}$
32	$\text{mask_term} = th \cdot \text{packed_target_shift}$	$th \cdot \text{packed_target_shift} = \text{mask_term}$
33	$T = T2 + \text{mask_term}$	$T2 + \text{mask_term} = T$

No.	Assignment	Addition/multiplication check
34	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
35	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
36	$R7 = SA + TB$	$SA + TB = R7$
37	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
38	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
39	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
40	$R12 = UM + sn2$	$UM + sn2 = R12$
41	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
42	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
43	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
44	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
45	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
46	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
47	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
48	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
49	$R10b = eta + zeta$	$eta + zeta = R10b$
50	$r1 = r + 1$	$r + 1 = r1$
51	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
52	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
53	$tr1 = r1 + r$	$r1 + r = tr1$
54	$R13 = ka + phi$	$ka + phi = R13$
55	$cam2 = c \cdot a$	$c \cdot a = cam2$
56	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
57	$a4 = 8 \cdot a$	$8 \cdot a = a4$
58	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
59	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
60	$R14 = D1 + gam$	$D1 + gam = R14$
61	$a_square = a \cdot a$	$a \cdot a = a_square$
62	$A = a_square + a4m5$	$a_square + a4m5 = A$
63	$c2 = c \cdot c$	$c \cdot c = c2$
64	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
65	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
66	$L15 = d \cdot d$	$d \cdot d = L15$
67	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
68	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
69	$L16 = f \cdot f$	$f \cdot f = L16$
70	$f_square_minus_one = L16 - 1$	$f_square_minus_one + 1 = L16$
71	$R16 = A \cdot f_square_minus_one$	$A \cdot f_square_minus_one = R16$
72	$of = o \cdot f$	$o \cdot f = of$
73	$dof = of - L15$	$dof + L15 = of$
74	$vplus1 = dof + 1$	$dof + 1 = vplus1$
75	$L17 = dof \cdot vplus1$	$dof \cdot vplus1 = L17$
76	$Gplus1 = ic22 - 1$	$Gplus1 + 1 = ic22$
77	$G2m1 = ic22 \cdot Gplus1$	$ic22 \cdot Gplus1 = G2m1$
78	$jc = j \cdot c$	$j \cdot c = jc$
79	$H17 = tr1 + jc$	$tr1 + jc = H17$
80	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
81	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
82	$amb = a - th$	$amb + th = a$
83	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
84	$qk = q + kamb$	$q + kamb = qk$
85	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
86	$Mod = A - amb2$	$Mod + amb2 = A$

No.	Assignment	Addition/multiplication check
87	$rM = \rho \cdot Mod$	$\rho \cdot Mod = rM$
88	$R18 = qk + rM$	$qk + rM = R18$
89	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
90	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
91	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
92	$L19 = \mu \cdot \mu$	$\mu \cdot \mu = L19$
93	$Dam1 = \Delta \cdot a$	$\Delta \cdot a = Dam1$
94	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q, & b &= bx, & q2 &= R2, \\
th &= theta_sum, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & \mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

X The 93-instruction factored-mask certificate

Use exactly the system and fixed index of Appendix W. Theorem 31.1 replaces its six-instruction packed-mask block by five instructions, using

$$T_+ = q^2 T_{\text{coef}} + \ell(\theta q^4 - b).$$

All other registers and all 22 equality tests have their old values. There are 50 multiplications and 43 additions, with the same 34 positive unknowns and the same three fixed index parameters besides the input. The positive-witness map is the identity.

The complete checker `round34_1980_factored_mask_certificate.py` verifies every primitive, all 22 unchanged source residuals and their acyclic corrections, equality of every retained register, and the absence of outside uses for the deleted temporaries. The table is generated by `round34_1980_schedule_table.py`. Its literal numerals are exactly 1, 3, 8, 15; fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$\text{theta_sum} = H + b$	$H + b = \text{theta_sum}$
5	$bx = x + \text{beta}$	$x + \text{beta} = bx$
6	$\text{theta_lambda} = \text{th} \cdot \text{la}$	$\text{th} \cdot \text{la} = \text{theta_lambda}$
7	$\text{Tcoef} = \text{theta_lambda} + 1$	$\text{theta_lambda} + 1 = \text{Tcoef}$
8	$\text{three_lambda} = 3 \cdot \text{la}$	$3 \cdot \text{la} = \text{three_lambda}$
9	$R2 = \text{Tcoef} + \text{three_lambda}$	$\text{Tcoef} + \text{three_lambda} = R2$
10	$t2 = \text{la} - e$	$t2 + e = \text{la}$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$\text{pack_tth} = t \cdot \text{th}$	$t \cdot \text{th} = \text{pack_tth}$
14	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
15	$C = x + g$	$x + g = C$
16	$C2 = C \cdot C$	$C \cdot C = C2$
17	$\text{bound_sum} = l + e$	$l + e = \text{bound_sum}$
18	$L1 = \text{bound_sum} + \text{al}$	$\text{bound_sum} + \text{al} = L1$
19	$\text{code_gap} = q - C2$	$\text{code_gap} + C2 = q$
20	$S3 = t2 \cdot \text{code_gap}$	$t2 \cdot \text{code_gap} = S3$
21	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
22	$\text{Sin} = S2 + S3q4$	$S2 + S3q4 = \text{Sin}$
23	$Sq4 = \text{Sin} \cdot q2$	$\text{Sin} \cdot q2 = Sq4$
24	$S = g + Sq4$	$g + Sq4 = S$
25	$n2 = n \cdot n$	$n \cdot n = n2$
26	$n2n = n2 - n$	$n2n + n = n2$
27	$SA = S \cdot n2n$	$S \cdot n2n = SA$
28	$Tq4 = \text{Tcoef} \cdot q2$	$\text{Tcoef} \cdot q2 = Tq4$
29	$\text{theta_q4} = \text{th} \cdot q4$	$\text{th} \cdot q4 = \text{theta_q4}$
30	$\text{mask_gap} = \text{theta_q4} - b$	$\text{mask_gap} + b = \text{theta_q4}$
31	$\text{indicator_mask} = l \cdot \text{mask_gap}$	$l \cdot \text{mask_gap} = \text{indicator_mask}$
32	$T = Tq4 + \text{indicator_mask}$	$Tq4 + \text{indicator_mask} = T$
33	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
34	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
35	$R7 = SA + TB$	$SA + TB = R7$
36	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$

No.	Assignment	Addition/multiplication check
37	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
38	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
39	$R12 = UM + sn2$	$UM + sn2 = R12$
40	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
41	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
42	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
43	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
44	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
45	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
46	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
47	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
48	$R10b = eta + zeta$	$eta + zeta = R10b$
49	$r1 = r + 1$	$r + 1 = r1$
50	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
51	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
52	$tr1 = r1 + r$	$r1 + r = tr1$
53	$R13 = ka + phi$	$ka + phi = R13$
54	$cam2 = c \cdot a$	$c \cdot a = cam2$
55	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
56	$a4 = 8 \cdot a$	$8 \cdot a = a4$
57	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
58	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
59	$R14 = D1 + gam$	$D1 + gam = R14$
60	$a_square = a \cdot a$	$a \cdot a = a_square$
61	$A = a_square + a4m5$	$a_square + a4m5 = A$
62	$c2 = c \cdot c$	$c \cdot c = c2$
63	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
64	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
65	$L15 = d \cdot d$	$d \cdot d = L15$
66	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
67	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
68	$L16 = f \cdot f$	$f \cdot f = L16$
69	$f_square_minus_one = L16 - 1$	$f_square_minus_one + 1 = L16$
70	$R16 = A \cdot f_square_minus_one$	$A \cdot f_square_minus_one = R16$
71	$of = o \cdot f$	$o \cdot f = of$
72	$dof = of - L15$	$dof + L15 = of$
73	$vplus1 = dof + 1$	$dof + 1 = vplus1$
74	$L17 = dof \cdot vplus1$	$dof \cdot vplus1 = L17$
75	$Gplus1 = ic22 - 1$	$Gplus1 + 1 = ic22$
76	$G2m1 = ic22 \cdot Gplus1$	$ic22 \cdot Gplus1 = G2m1$
77	$jc = j \cdot c$	$j \cdot c = jc$
78	$H17 = tr1 + jc$	$tr1 + jc = H17$
79	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
80	$P17 = G2m1 \cdot H2$	$G2m1 \cdot H2 = P17$
81	$amb = a - th$	$amb + th = a$
82	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
83	$qk = q + kamb$	$q + kamb = qk$
84	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
85	$Mod = A - amb2$	$Mod + amb2 = A$
86	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
87	$R18 = qk + rM$	$qk + rM = R18$
88	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
89	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$

No.	Assignment	Addition/multiplication check
90	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
91	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
92	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
93	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q, & b &= bx, & q2 &= R2, \\
th &= theta_sum, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega
\end{aligned}$$

Y The 92-instruction smaller-parameter Pell certificate

Use the fixed index and coefficient encoding of Appendix X. Theorem 32.1 replaces the signed auxiliary norm with

$$u = c + of, \quad K(u^2 - y_{\text{aux}}^2) = 1 - y_{\text{aux}}^2, \quad u = 2r + 1 + jc, \quad K = (ic^2)^2.$$

The computed u is named **H17** in the schedule. The new supplied unknown y_{aux} is positive; the auxiliary witnesses o, j are chosen anew in necessity. Every fixed parameter and all other canonical witnesses are unchanged. The full proof establishes the required index before applying the retained exponent and decoding arguments.

There are 35 positive unknowns, 23 equations, 49 multiplications and 43 additions. The differences $u^2 - y_{\text{aux}}^2$ and $1 - y_{\text{aux}}^2$ are negative integer registers on solutions; their subtractions are certified by reversed additions.

The complete checker `round35_1980_half_parameter_pell_certificate.py` verifies every primitive, all 23 source residuals, and the retained geometric and second-exponent corrections. The new auxiliary correction is $F_{16}(u^2 - y_{\text{aux}}^2)$, where F_{16} is the independently exact relaxed norm. The new linear congruence is exact. The separate focused polynomial, modular and positive-witness regression corroborates the general proof. The table is generated by `round35_1980_schedule_table.py`. Its literal numerals remain 1, 3, 8, 15; fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q^2 = q \cdot q$	$q \cdot q = q^2$
2	$q^4 = q^2 \cdot q^2$	$q^2 \cdot q^2 = q^4$
3	$q^8 = q^4 \cdot q^4$	$q^4 \cdot q^4 = q^8$
4	$\text{theta_sum} = H + b$	$H + b = \text{theta_sum}$
5	$bx = x + \text{beta}$	$x + \text{beta} = bx$
6	$\text{theta_lambda} = th \cdot la$	$th \cdot la = \text{theta_lambda}$
7	$\text{Tcoef} = \text{theta_lambda} + 1$	$\text{theta_lambda} + 1 = \text{Tcoef}$
8	$\text{three_lambda} = 3 \cdot la$	$3 \cdot la = \text{three_lambda}$
9	$R2 = \text{Tcoef} + \text{three_lambda}$	$\text{Tcoef} + \text{three_lambda} = R2$
10	$t2 = la - e$	$t2 + e = la$
11	$eq = e \cdot q$	$e \cdot q = eq$
12	$S2 = l + eq$	$l + eq = S2$
13	$\text{pack_tth} = t \cdot th$	$t \cdot th = \text{pack_tth}$
14	$\text{pack_rhs} = V + \text{pack_tth}$	$V + \text{pack_tth} = \text{pack_rhs}$
15	$C = x + g$	$x + g = C$
16	$C2 = C \cdot C$	$C \cdot C = C2$
17	$\text{bound_sum} = l + e$	$l + e = \text{bound_sum}$
18	$L1 = \text{bound_sum} + al$	$\text{bound_sum} + al = L1$
19	$\text{code_gap} = q - C2$	$\text{code_gap} + C2 = q$
20	$S3 = t2 \cdot \text{code_gap}$	$t2 \cdot \text{code_gap} = S3$
21	$S3q^4 = S3 \cdot q^2$	$S3 \cdot q^2 = S3q^4$
22	$\text{Sin} = S2 + S3q^4$	$S2 + S3q^4 = \text{Sin}$
23	$Sq^4 = \text{Sin} \cdot q^2$	$\text{Sin} \cdot q^2 = Sq^4$
24	$S = g + Sq^4$	$g + Sq^4 = S$
25	$n^2 = n \cdot n$	$n \cdot n = n^2$
26	$n2n = n^2 - n$	$n2n + n = n^2$
27	$SA = S \cdot n2n$	$S \cdot n2n = SA$
28	$Tq^4 = \text{Tcoef} \cdot q^2$	$\text{Tcoef} \cdot q^2 = Tq^4$
29	$\text{theta_q}^4 = th \cdot q^4$	$th \cdot q^4 = \text{theta_q}^4$
30	$\text{mask_gap} = \text{theta_q}^4 - b$	$\text{mask_gap} + b = \text{theta_q}^4$

No.	Assignment	Addition/multiplication check
31	$indicator_mask = l \cdot mask_gap$	$l \cdot mask_gap = indicator_mask$
32	$T = Tq4 + indicator_mask$	$Tq4 + indicator_mask = T$
33	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
34	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
35	$R7 = SA + TB$	$SA + TB = R7$
36	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
37	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
38	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
39	$R12 = UM + sn2$	$UM + sn2 = R12$
40	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
41	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
42	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
43	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
44	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
45	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
46	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
47	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
48	$R10b = eta + zeta$	$eta + zeta = R10b$
49	$r1 = r + 1$	$r + 1 = r1$
50	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
51	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
52	$tr1 = r1 + r$	$r1 + r = tr1$
53	$R13 = ka + phi$	$ka + phi = R13$
54	$cam2 = c \cdot a$	$c \cdot a = cam2$
55	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
56	$a4 = 8 \cdot a$	$8 \cdot a = a4$
57	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
58	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
59	$R14 = D1 + gam$	$D1 + gam = R14$
60	$a_square = a \cdot a$	$a \cdot a = a_square$
61	$A = a_square + a4m5$	$a_square + a4m5 = A$
62	$c2 = c \cdot c$	$c \cdot c = c2$
63	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
64	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
65	$L15 = d \cdot d$	$d \cdot d = L15$
66	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
67	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
68	$L16 = f \cdot f$	$f \cdot f = L16$
69	$f_square_minus_one = L16 - 1$	$f_square_minus_one + 1 = L16$
70	$R16 = A \cdot f_square_minus_one$	$A \cdot f_square_minus_one = R16$
71	$of = o \cdot f$	$o \cdot f = of$
72	$aux_u_rhs = c + of$	$c + of = aux_u_rhs$
73	$jc = j \cdot c$	$j \cdot c = jc$
74	$H17 = tr1 + jc$	$tr1 + jc = H17$
75	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
76	$aux_y2 = y_aux \cdot y_aux$	$y_aux \cdot y_aux = aux_y2$
77	$aux_square_gap = H2 - aux_y2$	$aux_square_gap + aux_y2 = H2$
78	$L17 = ic22 \cdot aux_square_gap$	$ic22 \cdot aux_square_gap = L17$
79	$P17 = 1 - aux_y2$	$P17 + aux_y2 = 1$
80	$amb = a - th$	$amb + th = a$
81	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
82	$qk = q + kamb$	$q + kamb = qk$
83	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$

No.	Assignment	Addition/multiplication check
84	$Mod = A - amb2$	$Mod + amb2 = A$
85	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
86	$R18 = qk + rM$	$qk + rM = R18$
87	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
88	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
89	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
90	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
91	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
92	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-three free equality tests are

$$\begin{aligned}
L1 &= q, & b &= bx, & q2 &= R2, \\
th &= theta_sum, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
t2 &= Omega, & H17 &= aux_u_rhs
\end{aligned}$$

Z The 91-instruction product-bound certificate

Use the coefficient index of Theorem 33.1 and the half-parameter Pell equations of Appendix Y. The two changed coding equations are

$$\sigma = (e - \ell)C^2, \quad \ell + \sigma + \alpha = q, \quad C = x + g.$$

The computed difference $e - \ell$ is positive because $\sigma > 0$ and $C > 0$. No supplied Ω or separate equality for it is needed. The new fixed index uses two encoded copies of the input to make all extra negative-coefficient tests exact. The full proof verifies the carry bounds, both directions of universality, and the parity needed by the retained auxiliary Pell construction.

There are 34 positive unknowns, 22 equations, 49 multiplications and 42 additions. Relative to Appendix Y, compute $\mathbf{t2}$ as $e - \ell$, compute $\mathbf{bound_sum}$ as $\ell + \sigma$, and multiply $\mathbf{t2}$ by the existing C^2 to obtain $\mathbf{S3}$. Delete the subtraction $q - C^2$. The free equality $\mathbf{t2}=\Omega$ disappears with its supplied unknown.

The complete checker `round36_1980_product_bound_certificate.py` verifies every primitive and all 22 fresh source residuals, including the inherited geometric, auxiliary-norm and second-exponent corrections. A separate sparse compiler regression corroborates the general proof. The table is generated by `round36_1980_schedule_table.py`. Literal numerals remain 1, 3, 8, 15; fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$\mathit{theta_sum} = H + b$	$H + b = \mathit{theta_sum}$
5	$\mathit{bx} = x + \mathit{beta}$	$x + \mathit{beta} = \mathit{bx}$
6	$\mathit{theta_lambda} = \mathit{th} \cdot \mathit{la}$	$\mathit{th} \cdot \mathit{la} = \mathit{theta_lambda}$
7	$\mathit{Tcoef} = \mathit{theta_lambda} + 1$	$\mathit{theta_lambda} + 1 = \mathit{Tcoef}$
8	$\mathit{three_lambda} = 3 \cdot \mathit{la}$	$3 \cdot \mathit{la} = \mathit{three_lambda}$
9	$\mathit{R2} = \mathit{Tcoef} + \mathit{three_lambda}$	$\mathit{Tcoef} + \mathit{three_lambda} = \mathit{R2}$
10	$\mathit{t2} = e - l$	$\mathit{t2} + l = e$
11	$\mathit{eq} = e \cdot q$	$e \cdot q = \mathit{eq}$
12	$\mathit{S2} = l + \mathit{eq}$	$l + \mathit{eq} = \mathit{S2}$
13	$\mathit{pack_tth} = t \cdot \mathit{th}$	$t \cdot \mathit{th} = \mathit{pack_tth}$
14	$\mathit{pack_rhs} = V + \mathit{pack_tth}$	$V + \mathit{pack_tth} = \mathit{pack_rhs}$
15	$C = x + g$	$x + g = C$
16	$\mathit{C2} = C \cdot C$	$C \cdot C = \mathit{C2}$
17	$\mathit{bound_sum} = l + \mathit{sigma}$	$l + \mathit{sigma} = \mathit{bound_sum}$
18	$\mathit{L1} = \mathit{bound_sum} + \mathit{al}$	$\mathit{bound_sum} + \mathit{al} = \mathit{L1}$
19	$\mathit{S3} = \mathit{t2} \cdot \mathit{C2}$	$\mathit{t2} \cdot \mathit{C2} = \mathit{S3}$
20	$\mathit{S3q4} = \mathit{S3} \cdot q2$	$\mathit{S3} \cdot q2 = \mathit{S3q4}$
21	$\mathit{Sin} = \mathit{S2} + \mathit{S3q4}$	$\mathit{S2} + \mathit{S3q4} = \mathit{Sin}$
22	$\mathit{Sq4} = \mathit{Sin} \cdot q2$	$\mathit{Sin} \cdot q2 = \mathit{Sq4}$
23	$\mathit{S} = g + \mathit{Sq4}$	$g + \mathit{Sq4} = \mathit{S}$
24	$\mathit{n2} = n \cdot n$	$n \cdot n = \mathit{n2}$
25	$\mathit{n2n} = \mathit{n2} - n$	$\mathit{n2n} + n = \mathit{n2}$
26	$\mathit{SA} = \mathit{S} \cdot \mathit{n2n}$	$\mathit{S} \cdot \mathit{n2n} = \mathit{SA}$
27	$\mathit{Tq4} = \mathit{Tcoef} \cdot q2$	$\mathit{Tcoef} \cdot q2 = \mathit{Tq4}$
28	$\mathit{theta_q4} = \mathit{th} \cdot q4$	$\mathit{th} \cdot q4 = \mathit{theta_q4}$
29	$\mathit{mask_gap} = \mathit{theta_q4} - b$	$\mathit{mask_gap} + b = \mathit{theta_q4}$
30	$\mathit{indicator_mask} = l \cdot \mathit{mask_gap}$	$l \cdot \mathit{mask_gap} = \mathit{indicator_mask}$

No.	Assignment	Addition/multiplication check
31	$T = Tq4 + \text{indicator_mask}$	$Tq4 + \text{indicator_mask} = T$
32	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
33	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
34	$R7 = SA + TB$	$SA + TB = R7$
35	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
36	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
37	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
38	$R12 = UM + sn2$	$UM + sn2 = R12$
39	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
40	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
41	$\text{scaled_norm_coefficient} = UM2 + wn2$	$UM2 + wn2 = \text{scaled_norm_coefficient}$
42	$\text{ratio_product2} = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = \text{ratio_product2}$
43	$L9 = \text{scaled_norm_coefficient} \cdot \text{ratio_product2}$	$\text{scaled_norm_coefficient} \cdot \text{ratio_product2} = L9$
44	$\text{tauplus1} = \text{tau} + 1$	$\text{tau} + 1 = \text{tauplus1}$
45	$R9 = \text{tau} \cdot \text{tauplus1}$	$\text{tau} \cdot \text{tauplus1} = R9$
46	$R10a = ksn2 + \text{eta}$	$ksn2 + \text{eta} = R10a$
47	$R10b = \text{eta} + \text{zeta}$	$\text{eta} + \text{zeta} = R10b$
48	$r1 = r + 1$	$r + 1 = r1$
49	$\text{hpm1} = h \cdot UM$	$h \cdot UM = \text{hpm1}$
50	$R11 = r1 + \text{hpm1}$	$r1 + \text{hpm1} = R11$
51	$\text{tr1} = r1 + r$	$r1 + r = \text{tr1}$
52	$R13 = ka + \text{phi}$	$ka + \text{phi} = R13$
53	$\text{cam2} = c \cdot a$	$c \cdot a = \text{cam2}$
54	$D1 = wn2 + \text{cam2}$	$wn2 + \text{cam2} = D1$
55	$a4 = 8 \cdot a$	$8 \cdot a = a4$
56	$a4m5 = a4 + 15$	$a4 + 15 = a4m5$
57	$\text{gam} = ga \cdot a4m5$	$ga \cdot a4m5 = \text{gam}$
58	$R14 = D1 + \text{gam}$	$D1 + \text{gam} = R14$
59	$a_square = a \cdot a$	$a \cdot a = a_square$
60	$A = a_square + a4m5$	$a_square + a4m5 = A$
61	$c2 = c \cdot c$	$c \cdot c = c2$
62	$\text{Ac2} = A \cdot c2$	$A \cdot c2 = \text{Ac2}$
63	$R15 = \text{Ac2} + 1$	$\text{Ac2} + 1 = R15$
64	$L15 = d \cdot d$	$d \cdot d = L15$
65	$\text{ic2} = i \cdot c2$	$i \cdot c2 = \text{ic2}$
66	$\text{ic22} = \text{ic2} \cdot \text{ic2}$	$\text{ic2} \cdot \text{ic2} = \text{ic22}$
67	$L16 = f \cdot f$	$f \cdot f = L16$
68	$\text{f_square_minus_one} = L16 - 1$	$\text{f_square_minus_one} + 1 = L16$
69	$R16 = A \cdot \text{f_square_minus_one}$	$A \cdot \text{f_square_minus_one} = R16$
70	$\text{of} = o \cdot f$	$o \cdot f = \text{of}$
71	$\text{aux_u_rhs} = c + \text{of}$	$c + \text{of} = \text{aux_u_rhs}$
72	$\text{jc} = j \cdot c$	$j \cdot c = \text{jc}$
73	$H17 = \text{tr1} + \text{jc}$	$\text{tr1} + \text{jc} = H17$
74	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
75	$\text{aux_y2} = y_aux \cdot y_aux$	$y_aux \cdot y_aux = \text{aux_y2}$
76	$\text{aux_square_gap} = H2 - \text{aux_y2}$	$\text{aux_square_gap} + \text{aux_y2} = H2$
77	$L17 = \text{ic22} \cdot \text{aux_square_gap}$	$\text{ic22} \cdot \text{aux_square_gap} = L17$
78	$P17 = 1 - \text{aux_y2}$	$P17 + \text{aux_y2} = 1$
79	$\text{amb} = a - \text{th}$	$\text{amb} + \text{th} = a$
80	$\text{kamb} = ka \cdot \text{amb}$	$ka \cdot \text{amb} = \text{kamb}$
81	$\text{qk} = q + \text{kamb}$	$q + \text{kamb} = \text{qk}$
82	$\text{amb2} = \text{amb} \cdot \text{amb}$	$\text{amb} \cdot \text{amb} = \text{amb2}$
83	$\text{Mod} = A - \text{amb2}$	$\text{Mod} + \text{amb2} = A$

No.	Assignment	Addition/multiplication check
84	$rM = \rho \cdot Mod$	$\rho \cdot Mod = rM$
85	$R18 = qk + rM$	$qk + rM = R18$
86	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
87	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
88	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
89	$L19 = \mu \cdot \mu$	$\mu \cdot \mu = L19$
90	$Dam1 = \Delta \cdot a$	$\Delta \cdot a = Dam1$
91	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q, & b &= bx, & q2 &= R2, \\
th &= theta_sum, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & \mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
H17 &= aux_u_rhs
\end{aligned}$$

AA The 90-instruction binary product-bound certificate

Use the binary coefficient index of Theorem 34.1: $H = H_0 - 1$, $B = H_0 + 1 + b = H + b + 2$, and $(V, H, T_L) = (\ell_0(2) + 2^L e_0(2), H, \psi_2(L))$. The positive unknowns are those of Appendix Z, with the changed Pell base $A = a + 2$. There are 34 positive unknowns, 22 equations, 48 multiplications and 42 additions.

The product and bound remain $\sigma = (e - \ell)(x + g)^2$ and $\ell + \sigma + \alpha = q$. The new reverse helper rows and two unit tests make the fixed codes binary, permitting $\theta = B - 2$. The geometric identity becomes $q^2 = T_{\text{coef}} + \lambda$ and the instruction 3λ disappears. The first exponent modulus is $4a + 3$ and the discriminant is $a^2 + 4a + 3$; they share the same three instructions as their predecessors. The complete proof in Section 34 includes the changed Pell bootstrap, exponent and index recovery, exact rounding, and all positive witnesses.

The complete checker `round37_1980_binary_product_certificate.py` verifies all 90 primitives and all 22 fresh source residuals, including the inherited triangular corrections. The table is generated by `round37_1980_schedule_table.py`. Literal numerals are exactly 1, 3, 4; fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q2 = q \cdot q$	$q \cdot q = q2$
2	$q4 = q2 \cdot q2$	$q2 \cdot q2 = q4$
3	$q8 = q4 \cdot q4$	$q4 \cdot q4 = q8$
4	$theta_sum = H + b$	$H + b = theta_sum$
5	$bx = x + beta$	$x + beta = bx$
6	$theta_lambda = th \cdot la$	$th \cdot la = theta_lambda$
7	$Tcoef = theta_lambda + 1$	$theta_lambda + 1 = Tcoef$
8	$R2 = Tcoef + la$	$Tcoef + la = R2$
9	$t2 = e - l$	$t2 + l = e$
10	$eq = e \cdot q$	$e \cdot q = eq$
11	$S2 = l + eq$	$l + eq = S2$
12	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
13	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
14	$C = x + g$	$x + g = C$
15	$C2 = C \cdot C$	$C \cdot C = C2$
16	$bound_sum = l + sigma$	$l + sigma = bound_sum$
17	$L1 = bound_sum + al$	$bound_sum + al = L1$
18	$S3 = t2 \cdot C2$	$t2 \cdot C2 = S3$
19	$S3q4 = S3 \cdot q2$	$S3 \cdot q2 = S3q4$
20	$Sin = S2 + S3q4$	$S2 + S3q4 = Sin$
21	$Sq4 = Sin \cdot q2$	$Sin \cdot q2 = Sq4$
22	$S = g + Sq4$	$g + Sq4 = S$
23	$n2 = n \cdot n$	$n \cdot n = n2$
24	$n2n = n2 - n$	$n2n + n = n2$
25	$SA = S \cdot n2n$	$S \cdot n2n = SA$
26	$Tq4 = Tcoef \cdot q2$	$Tcoef \cdot q2 = Tq4$
27	$theta_q4 = th \cdot q4$	$th \cdot q4 = theta_q4$
28	$mask_gap = theta_q4 - b$	$mask_gap + b = theta_q4$
29	$indicator_mask = l \cdot mask_gap$	$l \cdot mask_gap = indicator_mask$
30	$T = Tq4 + indicator_mask$	$Tq4 + indicator_mask = T$
31	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
32	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
33	$R7 = SA + TB$	$SA + TB = R7$
34	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$

No.	Assignment	Addition/multiplication check
35	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
36	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
37	$R12 = UM + sn2$	$UM + sn2 = R12$
38	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$
39	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
40	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
41	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
42	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
43	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
44	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
45	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
46	$R10b = eta + zeta$	$eta + zeta = R10b$
47	$r1 = r + 1$	$r + 1 = r1$
48	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
49	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
50	$tr1 = r1 + r$	$r1 + r = tr1$
51	$R13 = ka + phi$	$ka + phi = R13$
52	$cam2 = c \cdot a$	$c \cdot a = cam2$
53	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
54	$a4 = 4 \cdot a$	$4 \cdot a = a4$
55	$a4m5 = a4 + 3$	$a4 + 3 = a4m5$
56	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
57	$R14 = D1 + gam$	$D1 + gam = R14$
58	$a_square = a \cdot a$	$a \cdot a = a_square$
59	$A = a_square + a4m5$	$a_square + a4m5 = A$
60	$c2 = c \cdot c$	$c \cdot c = c2$
61	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
62	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
63	$L15 = d \cdot d$	$d \cdot d = L15$
64	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
65	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
66	$L16 = f \cdot f$	$f \cdot f = L16$
67	$f_square_minus_one = L16 - 1$	$f_square_minus_one + 1 = L16$
68	$R16 = A \cdot f_square_minus_one$	$A \cdot f_square_minus_one = R16$
69	$of = o \cdot f$	$o \cdot f = of$
70	$aux_u_rhs = c + of$	$c + of = aux_u_rhs$
71	$jc = j \cdot c$	$j \cdot c = jc$
72	$H17 = tr1 + jc$	$tr1 + jc = H17$
73	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
74	$aux_y2 = y_aux \cdot y_aux$	$y_aux \cdot y_aux = aux_y2$
75	$aux_square_gap = H2 - aux_y2$	$aux_square_gap + aux_y2 = H2$
76	$L17 = ic22 \cdot aux_square_gap$	$ic22 \cdot aux_square_gap = L17$
77	$P17 = 1 - aux_y2$	$P17 + aux_y2 = 1$
78	$amb = a - th$	$amb + th = a$
79	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
80	$qk = q + kamb$	$q + kamb = qk$
81	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
82	$Mod = A - amb2$	$Mod + amb2 = A$
83	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
84	$R18 = qk + rM$	$qk + rM = R18$
85	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
86	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
87	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$

No.	Assignment	Addition/multiplication check
88	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
89	$Dam1 = Delta \cdot a$	$Delta \cdot a = Dam1$
90	$R20 = Dam1 + Tindex$	$Dam1 + Tindex = R20$

The twenty-two free equality tests are

$$\begin{aligned}
L1 &= q, & b &= bx, & q2 &= R2, \\
th &= theta_sum, & S2 &= pack_rhs, & n &= q8, \\
r &= R7, & L9 &= R9, & c &= R10a, \\
k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
H17 &= aux_u_rhs
\end{aligned}$$

AB The 89-instruction narrow binary certificate

Use exactly the fixed binary coefficient index and 34 positive unknowns of Appendix AA. The new source has 22 equations and uses 47 multiplications and 42 additions. Its three packed fields have widths q, q, q^2 , and $n = q^4$. Removing the baseline from the mask permits this narrower power chain without an additional arithmetic operation.

Section 35 proves the wider preliminary Pell range, recovers the narrower mask bound after the exponent equations, excludes the possible spill into the fixed-code field, and supplies fresh positive witnesses at the new packed index. The unchanged raw query and fixed index still represent the same recursively enumerable set.

The complete checker `round38_1980_binary_product_certificate.py` verifies every primitive and all 22 fresh polynomial residuals. The table is generated by `round38_1980_schedule_table.py`. Literal numerals remain 1, 3, 4; fixed numerals and equality tests are free.

No.	Assignment	Addition/multiplication check
1	$q^2 = q \cdot q$	$q \cdot q = q^2$
2	$q^4 = q^2 \cdot q^2$	$q^2 \cdot q^2 = q^4$
3	$theta_sum = H + b$	$H + b = theta_sum$
4	$bx = x + beta$	$x + beta = bx$
5	$theta_lambda = th \cdot la$	$th \cdot la = theta_lambda$
6	$Tcoef = theta_lambda + 1$	$theta_lambda + 1 = Tcoef$
7	$R2 = Tcoef + la$	$Tcoef + la = R2$
8	$t2 = e - l$	$t2 + l = e$
9	$eq = e \cdot q$	$e \cdot q = eq$
10	$S2 = l + eq$	$l + eq = S2$
11	$pack_tth = t \cdot th$	$t \cdot th = pack_tth$
12	$pack_rhs = V + pack_tth$	$V + pack_tth = pack_rhs$
13	$C = x + g$	$x + g = C$
14	$C2 = C \cdot C$	$C \cdot C = C2$
15	$bound_sum = l + sigma$	$l + sigma = bound_sum$
16	$L1 = bound_sum + al$	$bound_sum + al = L1$
17	$S3 = t2 \cdot C2$	$t2 \cdot C2 = S3$
18	$S3q4 = S2 \cdot q$	$S2 \cdot q = S3q4$
19	$Sin = S3 + S3q4$	$S3 + S3q4 = Sin$
20	$Sq4 = Sin \cdot q$	$Sin \cdot q = Sq4$
21	$S = g + Sq4$	$g + Sq4 = S$
22	$n2 = n \cdot n$	$n \cdot n = n2$
23	$n2n = n2 - n$	$n2n + n = n2$
24	$SA = S \cdot n2n$	$S \cdot n2n = SA$
25	$Tq4 = theta_lambda \cdot q^2$	$theta_lambda \cdot q^2 = Tq4$
26	$theta_q4 = th \cdot q$	$th \cdot q = theta_q4$
27	$mask_gap = theta_q4 - b$	$mask_gap + b = theta_q4$
28	$indicator_mask = l \cdot mask_gap$	$l \cdot mask_gap = indicator_mask$
29	$T = Tq4 + indicator_mask$	$Tq4 + indicator_mask = T$
30	$n2m1 = n2 - 1$	$n2m1 + 1 = n2$
31	$TB = T \cdot n2m1$	$T \cdot n2m1 = TB$
32	$R7 = SA + TB$	$SA + TB = R7$
33	$wn2 = w \cdot n2$	$w \cdot n2 = wn2$
34	$sn2 = s \cdot n2$	$s \cdot n2 = sn2$
35	$UM = wn2 \cdot sn2$	$wn2 \cdot sn2 = UM$
36	$R12 = UM + sn2$	$UM + sn2 = R12$
37	$ksn2 = k \cdot sn2$	$k \cdot sn2 = ksn2$

No.	Assignment	Addition/multiplication check
38	$UM2 = UM \cdot UM$	$UM \cdot UM = UM2$
39	$scaled_norm_coefficient = UM2 + wn2$	$UM2 + wn2 = scaled_norm_coefficient$
40	$ratio_product2 = ksn2 \cdot ksn2$	$ksn2 \cdot ksn2 = ratio_product2$
41	$L9 = scaled_norm_coefficient \cdot ratio_product2$	$scaled_norm_coefficient \cdot ratio_product2 = L9$
42	$tauplus1 = tau + 1$	$tau + 1 = tauplus1$
43	$R9 = tau \cdot tauplus1$	$tau \cdot tauplus1 = R9$
44	$R10a = ksn2 + eta$	$ksn2 + eta = R10a$
45	$R10b = eta + zeta$	$eta + zeta = R10b$
46	$r1 = r + 1$	$r + 1 = r1$
47	$hpm1 = h \cdot UM$	$h \cdot UM = hpm1$
48	$R11 = r1 + hpm1$	$r1 + hpm1 = R11$
49	$tr1 = r1 + r$	$r1 + r = tr1$
50	$R13 = ka + phi$	$ka + phi = R13$
51	$cam2 = c \cdot a$	$c \cdot a = cam2$
52	$D1 = wn2 + cam2$	$wn2 + cam2 = D1$
53	$a4 = 4 \cdot a$	$4 \cdot a = a4$
54	$a4m5 = a4 + 3$	$a4 + 3 = a4m5$
55	$gam = ga \cdot a4m5$	$ga \cdot a4m5 = gam$
56	$R14 = D1 + gam$	$D1 + gam = R14$
57	$a_square = a \cdot a$	$a \cdot a = a_square$
58	$A = a_square + a4m5$	$a_square + a4m5 = A$
59	$c2 = c \cdot c$	$c \cdot c = c2$
60	$Ac2 = A \cdot c2$	$A \cdot c2 = Ac2$
61	$R15 = Ac2 + 1$	$Ac2 + 1 = R15$
62	$L15 = d \cdot d$	$d \cdot d = L15$
63	$ic2 = i \cdot c2$	$i \cdot c2 = ic2$
64	$ic22 = ic2 \cdot ic2$	$ic2 \cdot ic2 = ic22$
65	$L16 = f \cdot f$	$f \cdot f = L16$
66	$f_square_minus_one = L16 - 1$	$f_square_minus_one + 1 = L16$
67	$R16 = A \cdot f_square_minus_one$	$A \cdot f_square_minus_one = R16$
68	$of = o \cdot f$	$o \cdot f = of$
69	$aux_u_rhs = c + of$	$c + of = aux_u_rhs$
70	$jc = j \cdot c$	$j \cdot c = jc$
71	$H17 = tr1 + jc$	$tr1 + jc = H17$
72	$H2 = H17 \cdot H17$	$H17 \cdot H17 = H2$
73	$aux_y2 = y_aux \cdot y_aux$	$y_aux \cdot y_aux = aux_y2$
74	$aux_square_gap = H2 - aux_y2$	$aux_square_gap + aux_y2 = H2$
75	$L17 = ic22 \cdot aux_square_gap$	$ic22 \cdot aux_square_gap = L17$
76	$P17 = 1 - aux_y2$	$P17 + aux_y2 = 1$
77	$amb = a - th$	$amb + th = a$
78	$kamb = ka \cdot amb$	$ka \cdot amb = kamb$
79	$qk = q + kamb$	$q + kamb = qk$
80	$amb2 = amb \cdot amb$	$amb \cdot amb = amb2$
81	$Mod = A - amb2$	$Mod + amb2 = A$
82	$rM = rho \cdot Mod$	$rho \cdot Mod = rM$
83	$R18 = qk + rM$	$qk + rM = R18$
84	$ka2 = ka \cdot ka$	$ka \cdot ka = ka2$
85	$Aka2 = A \cdot ka2$	$A \cdot ka2 = Aka2$
86	$R19 = Aka2 + 1$	$Aka2 + 1 = R19$
87	$L19 = mu \cdot mu$	$mu \cdot mu = L19$
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The twenty-two free equality tests are

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L1 &= q, & b &= bx, & q2 &= R2, \\
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k &= R10b, & k &= R11, & a &= R12, \\
c &= R13, & d &= R14, & L15 &= R15, \\
ic22 &= R16, & L17 &= P17, & mu &= R18, \\
L19 &= R19, & ka &= R20, & S3 &= sigma, \\
H17 &= aux_u_rhs
\end{aligned}$$